Randomized Linear Algebra and Applications
Lecture Notes

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This lecture notes are meant to be supplementary material for Math 718: Randomized Linear Algebra and Applications at UW-Madison. It is likely that there are numerous typos and errors. It would be greatly appreciated if you let the author know of any.

CHAPTER 1

Introduction

Random solvers have been playing increasingly crucial roles in the modern computational tasks. The recent breakthroughs in applied and computational linear algebra that incorporate techniques of randomization have proven to be of great importance in modern applied math, computational sciences and data science, such as inverse problems, machine learning and scientific computing. The guiding principle is that one may greatly reduce computational and storage expenses at the cost of a small probability of failure.

This course will provide a systematic study of these modern methods of randomized linear algebra solvers, presenting mathematical backgrounds, algorithms, and concrete applications. Core theoretical topics include randomized Kaczmarz and its generalization to stochastic gradient descent, randomized singular value decomposition, random sketching, matrix completion, and other applications. We will leverage various probabilistic tools such as matrix concentration inequalities.

1.1. Preliminaries

1.1.1. Notations. Let $\mathbf{A} = (a_{ij})_{i,j} \in \mathbb{C}^{m\times n}$ denote an $m \times n$ matrix of complex entries. $\mathbf{A}^* = \overline{\mathbf{A}^\top}$ denotes the complex transpose of $\mathbf{A}$. A square matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is *self-adjoint* if $\mathbf{A} = \mathbf{A}^* := \overline{\mathbf{A}^\top}$. Let $\sigma_i(\mathbf{A})$ denote the ith *singular value* of $\mathbf{A}$ (i.e., the square root of the ith largest eigenvalue of the self-adjoint matrix $\mathbf{A}^*\mathbf{A}$).

A self-adjoint matrix is *positive semi-definite* (PSD) if its eigenvalues are nonnegative (equivalently, $\mathbf{x}^*\mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x}$); a self-adjoint matrix is *positive definite* (PD) if its eigenvalues are positive (equivalently, $\mathbf{x}^*\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x}$). The symbol $\mathbf{A} \preceq \mathbf{B}$ refers to the PSD (Loewner) order: $\mathbf{A} \preceq \mathbf{B}$ if and only if $\mathbf{B} - \mathbf{A}$ is psd.

1.1.2. Matrix norms. For any vector norm $\|\cdot\|$, the induced matrix norm $\|\mathbf{A}\|$ is defined as:

$$\|\mathbf{A}\| := \sup_{\mathbf{x}\neq 0} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|}.$$

For specific choices of vector norms, we get different induced norms:

$$(\triangleright \text{ Induced } \ell_1\text{-norm}) \qquad \|\mathbf{A}\|_1 = \max_{1\leq j\leq n} \sum_{i=1}^{m} |a_{ij}| = \text{maximum absolute column sum of } \mathbf{A},$$

$$(\triangleright \text{ Induced } \ell_\infty\text{-norm}) \qquad \|\mathbf{A}\|_\infty = \max_{1\leq i\leq m} \sum_{j=1}^{n} |a_{ij}| = \text{maximum absolute row sum of } \mathbf{A}.$$

When $\|\cdot\|$ is the Euclidean (ℓ_2-)norm, the induced matrix norm is called the *spectral norm* (or ℓ_2-norm). It satisfies

$$(\triangleright \text{ Induced } \ell_2\text{-norm}) \qquad \|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}(\mathbf{A}^*\mathbf{A})} = \sigma_{\max}(\mathbf{A}),$$

where $\lambda_{\max}(\mathbf{A}^*\mathbf{A})$ denotes the largest eigenvalue of the self-adjoint matrix $\mathbf{A}^*\mathbf{A}$, which equals the largest singular value of $\mathbf{A}$, denoted as $\sigma_{\max}(\mathbf{A})$. If $\mathbf{A}$ itself is self-adjoint, then

$$\|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}(\mathbf{A}^2)} = \sqrt{\max\{\lambda^2 : \lambda \text{ evals of } \mathbf{A}\}} = \max\{\lambda_{\max}(\mathbf{A}), \lambda_{\max}(-\mathbf{A})\}. \tag{1}$$

Specral norms are invariant under conjugation:

$$\|\mathbf{A}^*\|_2 = \|\mathbf{A}\|_2.$$

This is because $\mathbf{A}^*$ and $\mathbf{A}$ have the same singular values (can be checked by writing SVDs).

The *nuclear norm* (also known as the *trace norm* or the *Schatten 1-norm*) of a matrix $\mathbf{A}$ is defined as the sum of its singular values:

$$\|\mathbf{A}\|_* := \sum_{i=1}^{m\wedge n} \sigma_i(\mathbf{A})$$

The *max norm* of a matrix $\mathbf{A}$ is defined as the maximum absolute value of its entries:

$$\|\mathbf{A}\|_{\max} := \max_{i,j} |a_{ij}|.$$

While not a true norm, the ℓ_0 "norm" counts the number of non-zero elements in the matrix:

$$\|\mathbf{A}\|_0 = \sum_{i,j} \mathbf{1}(a_{ij} \neq 0).$$

The *Frobenius norm* of a matrix A is defined as the square root of the sum of the absolute squares of its elements:

$$\|\mathbf{A}\|_F := \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2} = \sqrt{\operatorname{trace}(\mathbf{A}^*\mathbf{A})} = \sqrt{\sum_i \sigma_i(\mathbf{A})^2}.$$

In general, the *Shatten p-norm* of $\mathbf{A}$ is defined as

$$\|\mathbf{A}\|_p := \left(\sum_i \sigma_i(\mathbf{A})^p\right)^{1/p}. \tag{2}$$

Shatten norms share the notations with the induced and entry-wise p-norms, but they are different.

Exercise 1.1.1. Show that the induced ℓ_2-norm $\|\cdot\|_2$ satisfies $\|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}(\mathbf{A}^*\mathbf{A})}$.

Exercise 1.1.2 (Monotonicity of eigenvalues)**.** Let X, Y be symmetric matrices with $\mathbf{X} \preceq \mathbf{Y}$. Show that $\lambda_i(\mathbf{X}) \leq \lambda_i(\mathbf{Y})$ for each i. (*Hint*: Use Courant-Fisher theorem to pick a unit u s.t. $u^\top \mathbf{Y} u = \lambda_1(\mathbf{Y})$. Since $Y - X$ is PSD, $u^\top \mathbf{X} u \leq u^\top \mathbf{Y} u = \lambda_1(\mathbf{Y})$. By Courant-Fisher on $\mathbf{X}$, this shows $\lambda_1(\mathbf{X}) \leq \lambda_1(\mathbf{Y})$. Proceed on orthogonal complements.)

1.2. Linear Systems: Randomized Kaczmarz

We start the discussion on randomized linear algebra by studying the celebrated *Randomized Kaczmarz* algorithm due to Strohmer and Vershynin in 2008 [SV09] for solving exact linear systems. Namely, the goal is to solve the linear equations of the form

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \tag{3}$$

where $\mathbf{A}$ is a full-rank $m \times n$ matrix with $m \geq n$, and $\mathbf{b} \in \mathbb{C}^m$. Here we will think of m and n as

$$m = \text{number of linear constraints,}$$
$$n = \text{number of variables (coordinates of } \mathbf{x} \in \mathbb{R}^n).$$

So $m \geq n$ means the system is "overdetermined" (e.g., exact linear regression with high-dimensional data).

The linear algebraic way to solve this would be multiplying both sides by $\mathbf{A}^*$ and then multiply again by the inverse $(\mathbf{A}^*\mathbf{A})^{-1}$ (which exists since $\mathbf{A}$ is full-rank):

$$\mathbf{A}\mathbf{x} = \mathbf{b} \iff \mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b} \iff \mathbf{x} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}.$$

The above solution is perfectly fine, there are several practical limitations. First, the direct computation of the solution involves

- Computing $\mathbf{A}^*\mathbf{A}$, which requires $O(mn^2)$ operations (why?).
- Inverting $\mathbf{A}^*\mathbf{A}$, which typically involves $O(n^3)$ operations.
- Multiplying the result by $\mathbf{A}^*\mathbf{b}$, requiring $O(n^2 m)$ operations.

The overall complexity can be substantial, especially for large matrices. Also, the matrix $\mathbf{A}^*\mathbf{A}$ can be ill-conditioned if the columns of $\mathbf{A}$ are nearly linearly dependent. In such cases, $\mathbf{A}^*\mathbf{A}$ may have a large condition number, leading to numerical instability during inversion. This instability can result in significant errors due to amplification of rounding errors, affecting the accuracy of the solution.

A widely-used method for solving such overdetermined systems is the Kaczmarz method (1937) [Kar37], which is a type of alternating projection technique. This method is also known as the Algebraic Reconstruction Technique (ART) in the field of computer tomography, and it was actually implemented in the first medical scanner.

Let the rows of $\mathbf{A}$ be denoted as $\mathbf{a}_1,\dots,\mathbf{a}_m$, and let $\mathbf{b} = (b_1,\dots,b_m)^T$. The traditional Kaczmarz method iteratively cycles through the rows of $\mathbf{A}$, projecting the current estimate orthogonally onto the solution hyperplane defined by $\langle \mathbf{a}_i, \mathbf{x}\rangle = b_i$, and uses this projection as the next estimate (see Fig. 1.2.1). Starting with an initial guess x_0, the algorithm updates the solution as follows:

$$\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k + \frac{b_i - \langle \mathbf{a}_i, \mathbf{x}_k\rangle}{\|\mathbf{a}_i\|^2}\mathbf{a}_i, \tag{4}$$

where $i = k \pmod{m+1}$, and $\|\cdot\|$ represents the Euclidean norm in $\mathbb{C}^n$. In this context, one projection is considered a single iteration, so a complete cycle through all m rows of A consists of m iterations.

Exercise 1.2.1. Show that the orthogonal projection of $\mathbf{x}_k$ onto the hyperplane $\langle \mathbf{a}_i, \mathbf{x}\rangle = b_i$ is exactly $\mathbf{x}_{k+1}$ in (4).

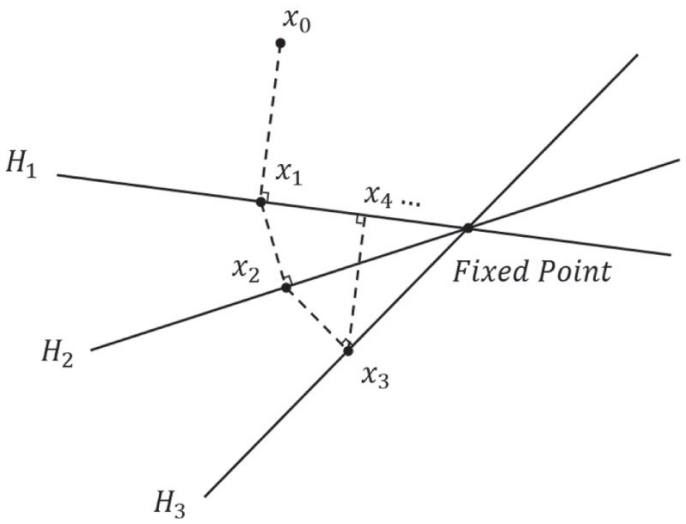


FIGURE 1.2.1. Illustration of the Kaczmarz method as iterative projection on the hyperplanes given by the linear constraints.

Definition 1.2.2 (Condition number)**.** The *condition number* of $\mathbf{A}$ is defined as

$$\kappa_2(A) := \|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2 = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})}.$$

The *scaled condition number* introduced by Demmel [Dem88]:

$$\kappa(A) := \|\mathbf{A}\|_F\|\mathbf{A}^{-1}\|_2.$$

One can easily verify that:

$$1 \le \frac{\kappa(\mathbf{A})}{\sqrt{n}} \le \kappa_2(\mathbf{A}).$$

It has been observed in numerical simulations that the convergence rate of the Kaczmarz method can be significantly improved when the algorithm (4) sweeps through the rows of $\mathbf{A}$ in a random manner,

rather than sequentially in the given order. In fact, the improvement in convergence can be quite dramatic. Here, we propose a specific version of this randomized Kaczmarz method, which chooses each row of $\mathbf{A}$ with probability proportional to its relevancemore precisely, proportional to the square of its Euclidean norm.

Algorithm 1 Randomized Kaczmarz

1: **Goal:** Given $\mathbf{A}$ and $\mathbf{b}$, find $\mathbf{x}^*$ s.t. $\mathbf{A}\mathbf{x}^* = \mathbf{b}$
2: **Input:** $\mathbf{x}_0$ (initial estimate); T (number of iterations)
3: **for** $t = 1, \dots, T$ **do**:
4: Sample $I_t \in [m]$ independently with prob. proportional to the squared norm of the rows: $\mathbb{P}(I_t = i) = \frac{\|\mathbf{a}_i\|^2}{\|\mathbf{A}\|_F^2}$
5: $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k + \dfrac{b_{I_t} - \langle \mathbf{a}_{I_t}, \mathbf{x}_k \rangle}{\|\mathbf{a}_{I_t}\|^2} \mathbf{a}_{I_t}$,
6: **end for**
7: **output:** $\mathbf{x}_T$

The following result, due to Strohmer and Vershynin [SV09], states that x_k obtained by Randomized Kaczmarz (Alg. 1) converges exponentially fast to the solution of (3), and the rate of convergence depends only on the scaled condition number $\kappa(\mathbf{A})$.

Theorem 1.2.3 (Exponential convergence of randomized Kaczmarz)**.** *Let $\mathbf{x}^*$ be the solution of (3). Then Algorithm 1 converges to $\mathbf{x}^*$ in expectation, with the average error*

$$\mathbb{E}\left[\|\mathbf{x}_k - \mathbf{x}^*\|^2\right] \le \left(1 - \frac{1}{\kappa(\mathbf{A})^2}\right)^k \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

PROOF. Define the error vector $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$. Substituting $b_{I_k} = \langle \mathbf{a}_{I_k}, \mathbf{x}^* \rangle$, we get:

$$\mathbf{e}_{k+1} = \mathbf{x}_{k+1} - \mathbf{x}^* = \mathbf{x}_k - \mathbf{x}^* + \frac{\langle \mathbf{a}_{I_k}, \mathbf{x}^* \rangle - \langle \mathbf{a}_{I_k}, \mathbf{x}_k \rangle}{\|\mathbf{a}_{I_k}\|^2} \mathbf{a}_{I_k}.$$

Simplifying, we have:

$$\mathbf{e}_{k+1} = \mathbf{e}_k - \frac{\langle \mathbf{a}_{I_k}, \mathbf{e}_k \rangle}{\|\mathbf{a}_{I_k}\|^2} \mathbf{a}_{I_k}.$$

Consider the squared norm of the error:

$$\|\mathbf{e}_{k+1}\|^2 = \|\mathbf{e}_k\|^2 - 2\frac{\langle \mathbf{a}_{I_k}, \mathbf{e}_k \rangle^2}{\|\mathbf{a}_{I_k}\|^2} + \frac{\langle \mathbf{a}_{I_k}, \mathbf{e}_k \rangle^2}{\|\mathbf{a}_{I_k}\|^2} = \|\mathbf{e}_k\|^2 - \frac{\langle \mathbf{a}_{I_k}, \mathbf{e}_k \rangle^2}{\|\mathbf{a}_{I_k}\|^2}.$$

Taking the conditional expectation w.r.t. the filtration $\mathscr{F}_k := \sigma(I_1, \dots, I_{k-1})$,

$$\begin{aligned}
\mathbb{E}[\|\mathbf{e}_{k+1}\|^2 \mid \mathscr{F}_k] &= \|\mathbf{e}_k\|^2 - \mathbb{E}\left[\frac{\langle \mathbf{a}_{I_k}, \mathbf{e}_k \rangle^2}{\|\mathbf{a}_{I_k}\|^2} \,\middle|\, \mathscr{F}_k\right] \\
&= \|\mathbf{e}_k\|^2 - \sum_{i=1}^m \frac{\|\mathbf{a}_i\|^2}{\|\mathbf{A}\|_F^2} \frac{\langle \mathbf{a}_i, \mathbf{e}_k \rangle^2}{\|\mathbf{a}_i\|^2} \\
&= \|\mathbf{e}_k\|^2 - \frac{1}{\|\mathbf{A}\|_F^2} \sum_{i=1}^m \langle \mathbf{a}_i, \mathbf{e}_k \rangle^2.
\end{aligned}$$

Next, note that

$$\sum_{i=1}^m \langle \mathbf{a}_i, \mathbf{e}_k \rangle^2 = \|\mathbf{A}\mathbf{e}_k\|^2 = \|\mathbf{e}_k^* \mathbf{A}^* \mathbf{A} \mathbf{e}_k\| \ge \sigma_{\min}^2(\mathbf{A}) \|\mathbf{e}_k\|^2.$$

This yields

$$\mathbb{E}[\|\mathbf{e}_{k+1}\|^2 \mid \mathscr{F}_k] \le \left(1 - \frac{\sigma_{\min}^2(\mathbf{A})}{\|\mathbf{A}\|_F^2}\right) \|\mathbf{e}_k\|^2 = \left(1 - \frac{1}{\kappa(\mathbf{A})^2}\right) \|\mathbf{e}_k\|^2.$$

Taking the total expectation, we obtain the following contractive recursion

$$\mathbb{E}[\|\mathbf{e}_{k+1}\|^2] \le \left(1 - \frac{1}{\kappa(\mathbf{A})^2}\right)\mathbb{E}[\|\mathbf{e}_k\|^2].$$

Then the assertion follows by an induction. □

Exercise 1.2.4. SELECT algorithm for sampling a row index in one pass without reading the entire matrix **A**

1.3. Optimization basics

1.3.1. Strong convexity and Smoothness. Convex optimization is a fundamental area in mathematical optimization where the objective function is convex. This framework is crucial in many fields such as machine learning, signal processing, and operations research. In this section, we will cover key concepts in convex optimization including:

- Convexity and μ-strong convexity
- L-smoothness
- Relation between the spectrum of the Hessian and strong convexity and smoothness

Definition 1.3.1 (Convexity)**.** A set $\mathbf{\Theta} \subseteq \mathbb{R}^p$ is *convex* if the line segment between any two points in $\mathbf{\Theta}$ is contained in $\mathbf{\Theta}$. That is, $\{\lambda\mathbf{x} + (1-\lambda)\mathbf{y} : \lambda \in [0,1], \mathbf{x}, \mathbf{y} \in \mathbf{\Theta}\} \subseteq \mathbf{\Theta}$ for all $\mathbf{x}, \mathbf{y} \in \mathbf{\Theta}$. A function $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ is said to be *convex* if its domain $\mathbf{\Theta}$ is a convex and for all $x, y \in \mathrm{dom}(f)$ and $\lambda \in [0,1]$, the following inequality holds:

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y).$$

Intuitively, a function f is convex if the line segment connecting any two points on the graph of f lies above the graph. If f is differentiable, then we can formulate convexity using tangent lines instead of line segments.

Exercise 1.3.2. Suppose $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ is differentiable and $\mathbf{\Theta}$ is convex. Show that f is convex if and only if

$$f(\mathbf{y}) - f(\mathbf{x}) \ge \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \qquad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbf{\Theta}.$$

Furthermore, show that if $\mathbf{x}^*$ is a stationary point of f over $\mathbf{\Theta}$, then $\mathbf{x}^* \in \arg\min_{\mathbf{x} \in \mathbf{\Theta}} f(\mathbf{x})$.

Definition 1.3.3 (Strong convexity)**.** A differentiable function $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ is *μ-strongly convex* for some constant $\mu \ge 0$ if

$$f(\mathbf{y}) \ge f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2 \qquad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbf{\Theta}. \tag{5}$$

Lemma 1.3.4. *Let $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ be a μ-strongly convex function for some $\mu > 0$. Then*

$$f(\mathbf{x}) - f(\mathbf{y}) \le \frac{\mu}{2}\|\nabla f(\mathbf{x})\|^2 \qquad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbf{\Theta}. \tag{6}$$

In particular, the following Polyak-Łojasiewicz condition holds:

$$f(\mathbf{x}) - \inf_{\mathbf{y} \in \mathbf{\Theta}} f(\mathbf{y}) \le \frac{1}{2\mu}\left\|\nabla f(\mathbf{x})\right\|^2. \tag{7}$$

In particular, every critical point is a global minimizer.

PROOF. Note that by Cauchy-Schwarz inequality and Young's inequality ($ab \le \frac{ta^2}{2} + \frac{b^2}{2t^2}$ for $a, b \ge 0$ and $t > 0$),

$$\langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2 \ge -\|\nabla f(\mathbf{x})\| \cdot \|\mathbf{y} - \mathbf{x}\| + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2 \ge -\frac{\mu}{2}\|\nabla f(\mathbf{x})\|^2.$$

Combining with the μ-strong convexity (5), we can deduce (6). Then minimizing the left-hand side of (6) over $\mathbf{y} \in \mathbf{\Theta}$ shows the PL-inequality (7). □

Definition 1.3.5 (Smoothness)**.** A differentiable function $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ is *L-smooth* for some constant $L > 0$ if

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2 \qquad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbf{\Theta}.$$

Intuitively, a function is smooth if its gradient does not change very rapidly. In that case, the function should not deviate too much from the tangent line. The L-smoothness above says that the "error term" in the first-order Taylor expansion is at most a fixed quadratic function.

Proposition 1.3.6 (Neterov, Lem. 1.2.3 in [Nes98])**.** *Suppose a differentiable function $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ has L-Lipschitz continuous gradient for some $L > 0$:*

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\| \qquad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbf{\Theta}.$$

Then f is L-smooth.

PROOF. Let $\gamma : [0,1] \to \mathbf{\Theta}$ be the parameterized line segment from $\mathbf{x}$ to $\mathbf{y}$ in $\mathbf{\Theta}$: $\gamma(s) := (1-s)\mathbf{x} + s\mathbf{y}$. Then $(f(\gamma(s)))' = \langle \nabla f(\gamma(s)), \gamma'(s) \rangle = \langle \nabla f(\gamma(s)), \mathbf{y} - \mathbf{x} \rangle$, so by the fundamental theorem of calculus,

$$f(\mathbf{y}) - f(\mathbf{x}) = f(\gamma(1)) - f(\gamma(0)) = \int_0^1 \left\langle \nabla f(\gamma(s)), \mathbf{y} - \mathbf{x} \right\rangle ds.$$

Also, we may write

$$\langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle = \int_0^1 \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \, ds.$$

Then by Cauchy-Schwarz inequality and L-Lipscthizness of ∇f,

$$\begin{aligned}
\left| f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \right| &= \left| \int_0^1 \left\langle \nabla f(\gamma(s)) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \right\rangle ds \right| \\
&\leq \int_0^1 \|\nabla f(\gamma(s)) - \nabla f(\mathbf{x})\| \cdot \|\mathbf{y} - \mathbf{x}\| \, ds \\
&\leq \int_0^1 L \|\underbrace{\gamma(s) - \mathbf{x}}_{= s(\mathbf{y} - \mathbf{x})}\| \cdot \|\mathbf{y} - \mathbf{x}\| \, ds \\
&= L\|\mathbf{y} - \mathbf{x}\|^2 \int_0^1 s \, ds \\
&= \frac{L}{2}\|\mathbf{x} - \mathbf{y}\|^2.
\end{aligned}$$

This is enough to conclude. □

If the function f is twice-differentiable, then the parameters for strong convexity and smoothness are closely related to the eigenvalues of the Hessian.

Proposition 1.3.7 (Hessian eigenvalues and strong convexity and smoothness)**.** *Suppose a function $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ is twice continuously differentiable. Suppose $\mathbf{\Theta}$ has a non-empty interior in $\mathbb{R}^p$. Then*

$$\begin{aligned}
f \text{ is } \mu\text{-strongly convex} &\iff \nabla_{\mathbf{x}} \nabla_{\mathbf{x}^\top} f(\mathbf{x}) \succeq \mu \mathbf{I} \qquad \text{for all } \mathbf{x} \in \mathbf{\Theta}, \\
f \text{ is } L\text{-strongly convex} &\iff \nabla_{\mathbf{x}} \nabla_{\mathbf{x}^\top} f(\mathbf{x}) \preceq L \mathbf{I} \qquad \text{for all } \mathbf{x} \in \mathbf{\Theta}.
\end{aligned}$$

PROOF. The proof follows immediately from Taylor's theorem. Namely, denoting the Hessian of f by H, for each $\mathbf{x}, \mathbf{y} \in \mathbf{\Theta}$, there exists $z \in \mathbf{\Theta}$ somewhere in the second line between $\mathbf{x}$ and $\mathbf{y}$ such that

$$f(\mathbf{y}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{1}{2}(\mathbf{y} - \mathbf{x})^T H(\mathbf{z})(\mathbf{y} - \mathbf{x}).$$

If $H(\mathbf{z}) \succeq \mu\mathbf{I}$, then by definition $\mathbf{v}^\top H(\mathbf{z})\mathbf{v} \geq \mu\|\mathbf{v}\|^2$ for all $\mathbf{v} \in \mathbb{R}^p$, so the above yields μ-strong convexity of f immediately. Conversely, suppose f is μ-strongly convex:

$$f(\mathbf{y}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y}-\mathbf{x}\rangle \geq \frac{\mu}{2}\|\mathbf{y}-\mathbf{x}\|^2.$$

Subtracting both inequalities,

$$\frac{1}{2}(\mathbf{y}-\mathbf{x})^T H(\mathbf{z})(\mathbf{y}-\mathbf{x}) - \frac{\mu}{2}\|\mathbf{y}-\mathbf{x}\|^2 \geq 0 \quad \text{for all } \mathbf{x},\mathbf{y} \in \boldsymbol{\Theta},$$

rewriting,

$$(\mathbf{y}-\mathbf{x})^T \left(H(\mathbf{z}) - \mu\mathbf{I}\right)(\mathbf{y}-\mathbf{x}) \geq 0 \quad \text{for all } \mathbf{x},\mathbf{y} \in \boldsymbol{\Theta}.$$

This shows $H(\mathbf{z}) \succeq \mu\mathbf{I}$. The proof of L-smoothness is similar. □

Example 1.3.8 (Least squares problem)**.** Consider the least squares problem:

$$\min_{\mathbf{x}\in\mathbb{R}^p} f(\mathbf{x}) := \|\mathbf{A}\mathbf{x}-\mathbf{b}\|^2,$$

where $A \in \mathbb{C}^{m\times n}$ and $b \in \mathbb{C}^m$. Note that

$$f(\mathbf{x}) = (\mathbf{A}\mathbf{x}-\mathbf{b})^*(\mathbf{A}\mathbf{x}-\mathbf{b}) = \mathbf{x}^T\mathbf{A}^*\mathbf{A}\mathbf{x} - 2\mathbf{x}^*\mathbf{A}^*\mathbf{b} + \mathbf{b}^*\mathbf{b}.$$

It follows that $\nabla f(\mathbf{x}) = \mathbf{A}^*\mathbf{A}\mathbf{x} - 2\mathbf{A}^*\mathbf{b} = \mathbf{A}^*(\mathbf{A}\mathbf{x}-\mathbf{b})$ and f has the constant Hessian

$$\nabla_{\mathbf{x}}\nabla_{\mathbf{x}^\top} f(\mathbf{x}) = \mathbf{A}^*\mathbf{A}.$$

By Prop. 1.3.7, it follows that f is $\lambda_{\min}(\mathbf{A}^*\mathbf{A}) = \sigma_{\min}(\mathbf{A})^2$-strongly convex and $\lambda_{\max}(\mathbf{A}^*\mathbf{A}) = \sigma_{\max}(\mathbf{A})^2$-smooth. ▲

Exercise 1.3.9 (Matrix Least Squares)**.** Suppose we have matrices $\mathbf{Y} \in \mathbb{R}^{d\times n}$ and $\mathbf{X} \in \mathbb{R}^{d\times r}$. We seek to find a matrix $\hat{\mathbf{B}} \in \mathbb{R}^{r\times n}$ where

$$\hat{\mathbf{B}} = \operatorname*{argmin}_{\mathbf{B}\in\mathbb{R}^{r\times n}} \left(f(\mathbf{B}) := \|\mathbf{Y}-\mathbf{X}\mathbf{B}\|_F^2 + \lambda\|\mathbf{B}\|_F^2\right).$$

Here $\lambda \geq 0$ is called the L_2-regularization parameter. (This is an instance of the unconstrained quadratic optimization problem.)

(i) Show that (use Exercise 1.3.10)

$$\nabla_{\boldsymbol{\theta}} f(\mathbf{B}) = 2(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\mathbf{B} - 2\mathbf{X}^\top\mathbf{Y}.$$

(ii) For each $\mathbf{x} = [x_1,\ldots,x_n] \in \mathbb{R}^{a\times b}$, let $\mathbf{x}^{\text{vec}} = [x_1^\top,\ldots,x_n^\top]^\top \in \mathbb{R}^{ab}$ denote its vectorization. Show that $(\mathbf{X}\mathbf{B})^{\text{vec}} = \mathbf{X}^{n\times n}\mathbf{B}^{\text{vec}}$, where

$$\mathbf{X}^{n\times n} := \overbrace{\begin{bmatrix} \mathbf{X} & O & O & \cdots & O \\ O & \mathbf{X} & O & \cdots & O \\ \vdots & \vdots & & & \vdots \\ O & \cdots & O & \mathbf{X} & O \\ O & O & \cdots & O & \mathbf{X} \end{bmatrix}}^{n\times n \text{ block matrix}} \in \mathbb{R}^{dn\times rn}.$$

Show that $f(\mathbf{B}^{\text{vec}}) = \|\mathbf{Y}^{\text{vec}} - \mathbf{X}^{n\times n}\mathbf{B}^{\text{vec}}\|^2 + \lambda\|\mathbf{B}^{\text{vec}}\|^2$ and

$$\nabla_{\mathbf{B}^{\text{vec}}}\nabla_{(\mathbf{B}^{\text{vec}})^\top} f(\mathbf{B}^{\text{vec}}) = 2\overbrace{\begin{bmatrix} (\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}) & O & O & \cdots & O \\ O & (\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}) & O & \cdots & O \\ \vdots & \vdots & & & \vdots \\ O & \cdots & O & (\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}) & O \\ O & O & \cdots & O & (\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}) \end{bmatrix}}^{n\times n \text{ block matrix}} \in \mathbb{R}^{rn\times rn}.$$

(iii) Deduce that

$$\lambda_{\min}(\mathbf{X}^\top\mathbf{X}) + \lambda \preceq \nabla_{\mathbf{B}^{\text{vec}}} \nabla_{(\mathbf{B}^{\text{vec}})^\top} f(\mathbf{B}) \preceq \lambda_{\max}(\mathbf{X}^\top\mathbf{X}) + \lambda.$$

Show that f is μ-stongly convex and L-smooth with $\mu = \lambda_{\min}(\mathbf{X}^\top\mathbf{X}) + \lambda$ and $L = \lambda_{\max}(\mathbf{X}^\top\mathbf{X}) + \lambda$.

(iv) Suppose $\lambda > 0$. Then argue that $\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$ is always invertible[1], and the quadratic function f has a unique global minimizer $\hat{\mathbf{B}} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{Y}$.

(v) Suppose $\lambda = 0$. Let $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ denote its SVD. The Moore-Penrose inverse of $\mathbf{X}$ is $\mathbf{X}^\dagger := \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^\top$, where $\mathbf{\Sigma}^\dagger$ obtained from $\mathbf{\Sigma}$ by inverting its nonzero diagonal entries of $\mathbf{\Sigma}$. Show that

$$\hat{\mathbf{B}} := \mathbf{X}^\dagger\mathbf{Y} = (\mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U})\mathbf{Y} = (\mathbf{X}^\top\mathbf{X})^\dagger\mathbf{X}\mathbf{Y}$$

is a critical point of f (i.e., $\nabla_{\boldsymbol{\theta}} f(\hat{\mathbf{B}}) = \mathbf{O}$). Conclude that $\hat{\mathbf{B}}$ is a (possibly non-unique) global minimizer of f. (c.f. $\hat{\mathbf{B}}$ is the minimum L^2-norm minimizer of f.)

Exercise 1.3.10 (Matrix derivatives)**.** Show the following matrix derivatives: (Ref: The Matrix Cookbook)

(i) (First order) $\dfrac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{B}\mathbf{X}\mathbf{C}) = \mathbf{B}^\top\mathbf{C}^\top, \qquad \dfrac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{B}\mathbf{X}^\top\mathbf{C}) = \mathbf{C}\mathbf{B}.$

(ii) (Second order) $\dfrac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{X}^\top\mathbf{B}\mathbf{X}) = \mathbf{B}\mathbf{X} + \mathbf{B}^\top\mathbf{X}, \qquad \dfrac{\partial}{\partial \mathbf{X}} \operatorname{tr}(\mathbf{B}^\top\mathbf{X}^\top\mathbf{C}\mathbf{X}\mathbf{B}) = \mathbf{C}^\top\mathbf{X}\mathbf{B}\mathbf{B}^\top + \mathbf{C}\mathbf{X}\mathbf{B}\mathbf{B}^\top.$

Exercise 1.3.11 (First-order optimality)**.** Suppose $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ is differentiable and $\mathbf{\Theta}$ is convex. $\mathbf{x} \in \mathbf{\Theta}$ is said to be a *critical point* of f if $\nabla f(\mathbf{x}) = \mathbf{0}$ and a *stationary point of* f over $\mathbf{\Theta}$ if

$$\langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \geq 0 \qquad \text{for all } \mathbf{y} \in \mathbf{\Theta}.$$

(i) Show that critical points are stationary points. Also, show that stationary points in the interior of $\mathbf{\Theta}$ are critical points.

(ii) Show that if $\mathbf{x}^* \in \arg\min_{\mathbf{x}\in\mathbf{\Theta}} f(\mathbf{x})$, then it must be a stationary point of f over $\mathbf{\Theta}$. Does $\mathbf{x}^*$ need to be a critical point? Does the converse also hold?

1.3.2. Elementary convergence analysis of gradient descent. In this section, we state and prove linear convergence of projected (stochastic) gradient descent algorithms for minimizing a strongly convex and smooth objective function within a convex constraint set $\mathbf{\Theta} \subseteq \mathbb{R}^p$. We will denote the orthogonal projection operator $\Pi_{\mathbf{\Theta}} : \mathbb{R}^p \to \mathbf{\Theta}$ as

$$\Pi_{\mathbf{\Theta}}(\mathbf{x}) = \operatorname*{arg\,min}_{y\in\mathbf{\Theta}} \|\mathbf{y} - \mathbf{x}\|.$$

We first study the linear convergence of the (deterministic) PGD algorithm.

Proposition 1.3.12 (Convergence rate of PGD for strongly convex and smooth objectives)**.** *Let $f : \mathbf{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ be a L-smooth and μ-strongly convex function for some $\mu, L > 0$. Suppose $f^* := \inf_{\boldsymbol{\theta}\in\mathbf{\Theta}} f(\boldsymbol{\theta}) > -\infty$ and denote $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}\in\mathbf{\Theta}} f(\boldsymbol{\theta})$. Consider the following projected gradient descent (PSD) iterates:*

$$\boldsymbol{\theta}_{t+1} \leftarrow \Pi_{\mathbf{\Theta}}\left(\boldsymbol{\theta}_t - \alpha_t \nabla f(\boldsymbol{\theta}_t)\right).$$

Suppose f is twice differentiable and choose constant stepsize $\alpha_t \equiv \alpha < 2/L$. Denote $\rho(\alpha) := \max\{|1 - \alpha L|, |1 - \alpha\mu|\} \in (0, 1)$. Then for all $t \geq 1$,

$$\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\| \leq \rho(\alpha)^t \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|.$$

The bound $\rho(\alpha)$ on the linear convergence rate is minimized at $\alpha = \frac{2}{\mu+L}$ with minimum value $\frac{L-\mu}{L+\mu}$.

[1] Hint: Show that $\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$ is positive definite if $\lambda > 0$. Use the fact that the eigenvalues of a positive definite matrix $\mathbf{A}$ has to be positive (why?), so $\mathbf{A}\mathbf{y} \neq$ for any $\mathbf{y}$ (why?) so $\mathbf{A}$ is invertible (why?).

PROOF. Suppose $\alpha_t \equiv \alpha > 0$. Note that $\boldsymbol{\theta}^*$ is a unique minimizer of f over $\boldsymbol{\Theta}$ because of strong convexity. Since $-\nabla f(\boldsymbol{\theta}^*)$ lies in the normal cone of $\boldsymbol{\Theta}$ at $\boldsymbol{\theta}^*$, we have $\boldsymbol{\theta}^* = \Pi_{\boldsymbol{\Theta}}(\boldsymbol{\theta}^* - \alpha\nabla f(\boldsymbol{\theta}^*))$ (Draw a picture!). By non-expansiveness of convex projection $\Pi_{\boldsymbol{\Theta}}$,

$$\begin{aligned}\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\| &= \|\Pi_{\boldsymbol{\Theta}}(\boldsymbol{\theta}_t - \alpha\nabla f(\boldsymbol{\theta}_t)) - \Pi_{\boldsymbol{\Theta}}(\boldsymbol{\theta}^* - \alpha\nabla f(\boldsymbol{\theta}^*))\| \\ &\le \|(\boldsymbol{\theta}_t - \alpha\nabla f(\boldsymbol{\theta}_t)) - (\boldsymbol{\theta}^* - \alpha\nabla f(\boldsymbol{\theta}^*))\|.\end{aligned}$$

Define a function $F(\boldsymbol{\theta}) = \boldsymbol{\theta} - \alpha\nabla f(\boldsymbol{\theta}) : \boldsymbol{\Theta} \to \mathbb{R}^p$ and let $\gamma(s) := (1-s)\boldsymbol{\theta}_t + s\boldsymbol{\theta}^*$ be the secant line from $\boldsymbol{\theta}_t$ to $\boldsymbol{\theta}^*$ (which is contained in the convex set $\boldsymbol{\Theta}$). Then by chain rule, we compute the Jacobian of $F \circ \gamma$ as

$$\begin{aligned}J_{F\circ\gamma}(s) &= J_F(\gamma(s))J_\gamma(s) \\ &= (\nabla_{\boldsymbol{\theta}}(F(\gamma(s))^\top))^\top(\boldsymbol{\theta}^* - \boldsymbol{\theta}_t) \\ &= (\mathbf{I} - \alpha\nabla^2 f(\gamma(s)))(\boldsymbol{\theta}^* - \boldsymbol{\theta}_t).\end{aligned}$$

Hence by the fundamental theorem of calculus,

$$\begin{aligned}\|(\boldsymbol{\theta}_t - \alpha\nabla f(\boldsymbol{\theta}_t)) - (\boldsymbol{\theta}^* - \alpha\nabla f(\boldsymbol{\theta}^*))\| &= \|F(\gamma(0)) - F(\gamma(1))\| \\ &= \left\|\int_0^1 (\mathbf{I} - \alpha\nabla^2 f(\gamma(s)))(\boldsymbol{\theta}^* - \boldsymbol{\theta}_t)\, ds\right\| \\ &\le \sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \left\|\mathbf{I} - \alpha\nabla^2 f(\boldsymbol{\theta})\right\|_2 \|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| \\ &\le \underbrace{\max\{|1-\alpha L|, |1-\alpha\mu|\}}_{=:\rho(\alpha)} \|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\|.\end{aligned}$$

For the last inequality, note that the eigenvalues of $\nabla^2 f(\boldsymbol{\theta})$ are contained in the interval $[\mu, L]$ (see Prop. 1.3.7), so the eigenvalues of $\mathbf{I} - \alpha\nabla^2 f(\boldsymbol{\theta})$ are between $\min(1-\alpha L, 1-\alpha\mu)$ and $\max(1-\alpha L, 1-\alpha\mu)$, so they are at most $\rho(\alpha)$. It follows that for any $t \ge 0$,

$$\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\| \le \rho(\alpha)^{t+1}\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|.$$

Note that $\rho(\alpha)$ is minimized at $\alpha = \frac{2}{\mu+L}$ with minimum value $\frac{L-\mu}{L+\mu}$ (Draw picture!). □

Next, we study the linear convergence of stochastic PGD.

Proposition 1.3.13 (Convergence rate of PSGD for strongly convex objectives)**.** *Let $f : \boldsymbol{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ be a L-smooth and μ-strongly convex function for some $\mu, L > 0$. Suppose $f^* := \inf_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} f(\boldsymbol{\theta}) > -\infty$ and denote $\boldsymbol{\theta}^* := \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} f(\boldsymbol{\theta})$. Consider the following projected stochastic gradient descent (PSGD) iterates:*

$$\boldsymbol{\theta}_{t+1} \leftarrow \Pi_{\boldsymbol{\Theta}}\left(\boldsymbol{\theta}_t - \alpha_t G(\boldsymbol{\theta}_t, \xi_t)\right),$$

where

(i) *(Unbiased gradient estimation) $G(\boldsymbol{\theta}, \xi)$ is an unbiased stochastic gradient estimator for f: $\mathbb{E}_\xi[G(\boldsymbol{\theta}, \xi)] = \nabla f(\boldsymbol{\theta})$ for all $x \in \boldsymbol{\Theta}$;*

(ii) *(Bounded variance of gradient estimation) The variance of the gradient estimation error is uniformly bounded: $\mathbb{E}[\|G(\boldsymbol{\theta}, \xi) - \nabla f(\boldsymbol{\theta})\|^2] \le \sigma^2$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$.*

Let $\rho := \frac{2\mu L}{\mu+L}$. Suppose $\alpha_t \equiv \alpha \le 1/(\mu+L)$. Then for all $t \ge 0$,

$$\mathbb{E}[\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2] \le (1-\rho\alpha)^t\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|^2 + \frac{\alpha(\sigma^2 + 2\|\nabla f(\boldsymbol{\theta}^*)\|^2)}{\rho}.$$

PROOF. Let $\bar{\boldsymbol{\theta}}_t := \boldsymbol{\theta}_t - \alpha G(\boldsymbol{\theta}_t; \xi_t)$. Then

$$\begin{aligned}\|\bar{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|^2 &= \|\bar{\boldsymbol{\theta}}_t - \boldsymbol{\theta}_t\|^2 + \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 + 2\langle\bar{\boldsymbol{\theta}}_t - \boldsymbol{\theta}_t, \boldsymbol{\theta}_t - \boldsymbol{\theta}^*\rangle \\ &= \alpha_t^2\|G(\boldsymbol{\theta}_t; \xi_t)\|^2 + \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 - 2\alpha_t\langle\boldsymbol{\theta}_t - \boldsymbol{\theta}^*, G(\boldsymbol{\theta}_t; \xi_t)\rangle \\ &\le \alpha_t^2\|G(\boldsymbol{\theta}_t; \xi_t)\|^2 + \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 - 2\alpha_t\langle\boldsymbol{\theta}_t - \boldsymbol{\theta}^*, G(\boldsymbol{\theta}_t; \xi_t) - \nabla f(\boldsymbol{\theta}^*)\rangle,\end{aligned}$$

where the inequality uses that $\langle \nabla f(\boldsymbol{\theta}^*), \boldsymbol{\theta}_t - \boldsymbol{\theta}^* \rangle \geq 0$ since $\boldsymbol{\theta}^*$ is a stationary point of f over $\boldsymbol{\Theta}$. Let $\mathscr{F}_t = \sigma(\xi_1, \ldots, \xi_{t-1})$ denote the σ-algebra generated by the random variables $\xi_1, \ldots, \xi_{t-1}$. Then $\boldsymbol{\theta}_t$ is measurable w.r.t. $\mathscr{F}_t$, so taking conditional expectation with respect to $\mathscr{F}_t$ (assuming $\alpha_t \in \mathscr{F}_t$),

$$\mathbb{E}[\|\bar{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|^2 \,|\, \mathscr{F}_t] \leq \alpha_t^2 \sigma^2 + \alpha_t^2 \|\nabla f(\boldsymbol{\theta}_t)\|^2 + \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 - 2\alpha_t \langle \boldsymbol{\theta}_t - \boldsymbol{\theta}^*, \nabla f(\boldsymbol{\theta}_t) - \nabla f(\boldsymbol{\theta}^*) \rangle.$$

Then using the co-coercivity of strongly convex functions and smooth functions (see Exc. 1.3.14)

$$\langle \boldsymbol{\theta}_t - \boldsymbol{\theta}^*, \nabla f(\boldsymbol{\theta}_t) - \nabla f(\boldsymbol{\theta}^*) \rangle \geq \frac{\mu L}{\mu + L} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 + \frac{1}{\mu + L} \|\nabla f(\boldsymbol{\theta}_t) - \nabla f(\boldsymbol{\theta}^*)\|^2.$$

By Young's inequality,

$$\|\nabla f(\boldsymbol{\theta}_t)\|^2 \leq 2\|\nabla f(\boldsymbol{\theta}_t) - \nabla f(\boldsymbol{\theta}^*)\|^2 + 2\|\nabla f(\boldsymbol{\theta}^*)\|^2.$$

Combining the above, we get

$$\begin{aligned}
\mathbb{E}[\|\bar{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|^2 \,|\, \mathscr{F}_t] &\leq \alpha_t^2 \sigma^2 + \left(1 - \frac{2\alpha_t \mu L}{\mu + L}\right) \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 + 2\alpha_t^2 \|\nabla f(\boldsymbol{\theta}^*)\|^2 \\
&\quad + 2\alpha_t \underbrace{\left(\alpha_t - \frac{1}{\mu + L}\right)}_{\leq 0} \|\nabla f(\boldsymbol{\theta}_t) - \nabla f(\boldsymbol{\theta}^*)\|^2 \\
&\leq \alpha_t^2 \sigma^2 + \left(1 - \rho \alpha_t\right) \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 + 2\alpha_t^2 \|\nabla f(\boldsymbol{\theta}^*)\|^2,
\end{aligned}$$

given that $\alpha_t \leq 1/(\mu + L)$. Then by taking the full expectation and using the non-expansivity of convex projection,

$$\mathbb{E}[\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\|^2] \leq \mathbb{E}[\|\bar{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|^2] \leq \left(1 - \alpha_t \rho\right) \mathbb{E}[\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2] + \alpha_t^2 \underbrace{\left(\sigma^2 + 2\|\nabla f(\boldsymbol{\theta}^*)\|^2\right)}_{:= \sigma_0^2}, \tag{8}$$

where we have denoted $\rho := \frac{2\mu L}{\mu + L}$.

Now applying (8) with $\alpha_t \equiv \alpha \leq \frac{1}{\mu + L}$ recursively, we have

$$\begin{aligned}
\mathbb{E}[\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2] &\leq (1 - \rho\alpha)^t \|\boldsymbol{\theta}^0 - \boldsymbol{\theta}^*\|^2 + \sigma_0^2 \sum_{j=0}^{t} (1 - \rho\alpha)^j \alpha^2 \\
&\leq (1 - \rho\alpha)^t \|\boldsymbol{\theta}^0 - \boldsymbol{\theta}^*\|^2 + \frac{\alpha \sigma_0^2}{\rho}.
\end{aligned}$$

This shows the assertion. □

Exercise 1.3.14 (Co-coercivity)**.** Let $f : \boldsymbol{\Theta} \subseteq \mathbb{R}^p \to \mathbb{R}$ be a L-smooth and μ-strongly convex function for some $\mu, L > 0$. Show that

$$\langle \mathbf{x} - \mathbf{y}, \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}) \rangle \geq \frac{\mu L}{\mu + L} \|\mathbf{x} - \mathbf{y}\|^2 + \frac{1}{\mu + L} \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|^2.$$

Example 1.3.15 (Randomized Kaczmarz as SGD)**.** Consider the linear system of equations

$$\mathbf{A}\mathbf{x} = \mathbf{b},$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{b} \in \mathbb{R}^m$. We aim to find $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{b}$. This can be formulated as the least squares minimization problem:

$$\min_{\mathbf{x} \in \mathbb{R}^n} \left(f(\mathbf{x}) = \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 = \frac{1}{2} \sum_{i=1}^{m} (\langle \mathbf{a}_i, \mathbf{x} \rangle - b_i)^2 \right).$$

Then randomized Kaczmarz (Alg. 1) can be viewed as SGD with stochastic gradient estimator

$$G(\mathbf{x}_k; I_k) = \nabla f_{I_k}(\mathbf{x}_k) = (\langle \mathbf{a}_{I_k}, \mathbf{x} \rangle - b_{I_k}) \mathbf{a}_{I_k}$$

with random stepsize

$$\alpha_k = 1/\|\mathbf{a}_{I_k}\|^2$$

where the random row intex I_k is sampled independently from $\mathscr{F}_k = \sigma(I_0, \ldots, I_{k-1})$ according to the following non-uniform distribution $\mathbb{P}(I_k = i) \propto \|\mathbf{a}_i\|^2$. Note that

$$\mathbb{E}\left[\nabla f_{I_k}(\mathbf{x}_k) \,|\, \mathscr{F}_k\right] = \sum_{i=1}^{m} \frac{\|\mathbf{a}_i\|^2}{\|\mathbf{A}\|_F^2}(\langle \mathbf{a}_{I_k}, \mathbf{x}\rangle - b_{I_k})\mathbf{a}_{I_k} \neq \sum_{i=1}^{m}(\langle \mathbf{a}_{I_k}, \mathbf{x}\rangle - b_{I_k})\mathbf{a}_{I_k} = \nabla f(\mathbf{x}),$$

so $G(\mathbf{x}_k; I_k)$ is *not* an unbiased estimator for the full gradient $\nabla f(\mathbf{x}_k)$. Instead, note that

$$\mathbb{E}\left[\alpha_k \nabla f_{I_k}(\mathbf{x}_k) \,|\, \mathscr{F}_k\right] = \sum_{i=1}^{m} \frac{\|\mathbf{a}_i\|^2}{\|\mathbf{A}\|_F^2} \frac{1}{\|\mathbf{a}_i\|^2}(\langle \mathbf{a}_{I_k}, \mathbf{x}\rangle - b_{I_k})\mathbf{a}_{I_k} = \frac{1}{\|\mathbf{A}\|_F^2}\nabla f(\mathbf{x}).$$

So the stochastic gradient times the random step size has conditional expectation equals to the full gradient times scalar $1/\|\mathbf{A}\|_F^2$. So on average, randomized Kaczmarz works like full gradient descent with constant step size $1/\|\mathbf{A}\|_F^2$. (What about the variance?) ▲

CHAPTER 2

Matrix Concentration Inequalities

The core mathmatical contents in this chapter are based on [Tro19, Ch. 1].

2.1. Introduction

2.1.1. What is random matrix concentration? Matrix concentration inequalities provide a unified probabilistic framework of analyzing numerous problems in linear algebra, statistics, and machine learning. In many problem instances, we often end up desiring to show certain random matrix $\mathbf{X}$ being close to its expectation $\mathbb{E}[\mathbf{X}]$ in the spectral norm with high probability:

$$\|\mathbf{X} - \mathbb{E}[\mathbf{X}]\|_2 \quad \text{is small with high probability.}$$

Bounds of this form give us a lot of information about how the random matrix $\mathbf{Y}$ is related to its expectation $\mathbb{E}[\mathbf{Y}]$. In particular:

- Each linear functional of $\mathbf{Y}$ is close to the same linear functional of $\mathbb{E}[\mathbf{Y}]$.
- Each eigenvalue of $\mathbf{Y}$ is close to the corresponding eigenvalue of $\mathbb{E}[\mathbf{Y}]$.
- Each eigenvector of $\mathbf{Y}$ is close to the corresponding eigenvector of $\mathbb{E}[\mathbf{Y}]$ when the eigenvalue is isolated from the rest of the spectrum.
- We can bound the expected norm of the random matrix:

$$\|\mathbf{Y}\| = \|\mathbb{E}[\mathbf{Y}]\| \pm \|Y - \mathbb{E}[\mathbf{Y}]\|.$$

Without additional information about the random matrix $\mathbf{Y}$, we cannot hope to say anything interesting. This work treats two basic, but very fruitful, models for the random matrix.

Definition 2.1.1 (Random matrix models)**.** We assume that our random matrix $\mathbf{Y}$ is the sum of many, potentially simpler, random matrices $\mathbf{X}_i$:

$$\mathbf{Y} = \sum_{i=1}^{n} \mathbf{X}_i.$$

For the *independent sum model*, we assume that $\mathbf{X}_i$'s are independent but not necessarily identically distributed. For the *matrix martingale model*, we assume the partial sums $\mathbf{Y}_k = \sum_{i=1}^{k} \mathbf{X}_i$ form a matrix matringale:

$$\mathbb{E}[\mathbf{Y}_k \,|\, \mathscr{F}_{k-1}] = \mathbf{Y}_{k-1}.$$

2.1.2. Why do we care? Here are some examples where we desire concentration of random matrices to their expectation:

Example 2.1.2 (Empirical Hessian of the Log-Likelihood Function)**.** In statistics and machine learning, the empirical Hessian H_{emp} is the second derivative of the log-likelihood function based on a sample. It approximates the true (population) Hessian H_{pop}, which describes the curvature of the population-level likelihood.

- **Applications**:
  - *Optimization*: Used in iterative methods like Newton's method to update parameter estimates.
  - *Statistical Inference*: Approximates the Fisher information matrix, used in confidence intervals and hypothesis testing.

  - *Machine Learning*: Appears in large-scale optimization, using stochastic approximations for faster computation.

▲

Example 2.1.3 (Adjacency Matrix of a Random Graph Model)**.** The adjacency matrix $\mathbf{A}$ of a random graph has entries that represent the presence or weight of an edge between nodes. We want $\mathbf{A}$ to behave like the expected adjacency matrix $\mathbb{E}[\mathbf{A}]$, which describes the average graph structure.

- **Applications**:
  - *Clustering*: Spectral clustering algorithms rely on the eigenvalues and eigenvectors of $\mathbf{A}$, approximating those of $\mathbb{E}[\mathbf{A}]$.
  - *Network Analysis*: Used in community detection, where $\mathbf{A}$ approximates the underlying modular structure.

▲

Example 2.1.4 (Covariance Matrix in High-Dimensional Statistics)**.** The sample covariance matrix $\hat{\Sigma} = \frac{1}{n}\mathbf{X}^T\mathbf{X}$, where $\mathbf{X}$ is a data matrix, approximates the true covariance matrix $\mathbf{\Sigma}$. The spectral properties of $\hat{\mathbf{\Sigma}}$ are essential for understanding data structure in high-dimensional settings.

- **Applications**:
  - *Principal Component Analysis (PCA)*: Finds the eigenvectors of $\hat{\mathbf{\Sigma}}$, approximating the population covariance principal components.
  - *Regression and Classification*: Helps estimate parameter variability in regularized models like Ridge or Lasso.

▲

Example 2.1.5 (Empirical Kernel Matrix (Gram Matrix))**.** The empirical kernel matrix K is constructed using a kernel function $k(x_i, x_j)$ on data points x_i. K approximates the population kernel matrix.

- **Applications**:
  - *Support Vector Machines (SVMs)*: The Gram matrix helps determine support vectors and the decision boundary.
  - *Kernel PCA*: Nonlinear features are extracted using K, similar to traditional PCA in feature space.

▲

Example 2.1.6 (Randomized Low-Rank Matrix Approximation)**.** In randomized linear algebra, matrices are approximated by multiplying with a random matrix to reduce dimension while preserving key properties. The goal is to approximate the original matrix $\mathbf{A}$ with a low-rank matrix $\mathbf{B}$, so that $\mathbf{B} \approx \mathbf{A}$ in terms of spectral properties.

- **Applications**:
  - *Matrix Factorization*: Used in recommendation systems (e.g., Netflix problem) for approximating user-item matrices.
  - *Latent Semantic Analysis (LSA)*: Applied in NLP to extract hidden structure from large text corpora.

▲

Example 2.1.7 (Empirical Transition Matrix for Markov Chains)**.** The empirical transition matrix $\hat{P}$ is constructed from observed transitions in a Markov chain, approximating the true transition matrix P that governs population-level dynamics.

- **Applications**:
  - *Reinforcement Learning*: Used in algorithms to approximate system behavior and derive optimal policies.

– *Time Series Modeling*: Appears in hidden Markov models (HMM) for sequence prediction and classification.

▲

2.2. Basic tail bounds for random variables

In this section, we review some of the basic concentration inequalities for random variables.

Proposition 2.2.1 (Markov's inequality)**.** *Let $X \geq 0$ be a nonnegative RV with finite expectation. Then for any $a > 0$, we have*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

PROOF. Consider an auxiliary RV Y define as follows:

$$Y = \begin{cases} a & \text{if } X \geq a \\ 0 & \text{if } X < a. \end{cases}$$

Note that we always have $Y \leq X$. Hence we should have $\mathbb{E}[Y] \leq \mathbb{E}[X]$. But since $\mathbb{E}[Y] = a\mathbb{P}(X \geq a)$, we have

$$\lambda\mathbb{P}(X \geq a) \leq \mathbb{E}[X].$$

Dividing both sides by $a > 0$ gives the assertion. □

Example 2.2.2. We will show that, for any RV Z, $\mathbb{E}[Z^2] = 0$ implies $\mathbb{P}(Z = 0) = 1$. Indeed, Markov's inequality gives that for any $a > 0$,

$$\mathbb{P}(Z^2 \geq a) \leq \frac{\mathbb{E}[Z^2]}{a} = 0.$$

By continuity of measure, it follows that $\mathbb{P}(Z^2 = 0) = \lim_{\varepsilon \searrow 0} \mathbb{P}(Z^2 < \varepsilon) = 1$, so $\mathbb{P}(Z = 0) = 1$. ▲

Proposition 2.2.3 (Chebyshef's inequality)**.** *Let X be any RV with $\mathbb{E}[X] = \mu < \infty$ and $\mathrm{Var}(X) < \infty$. Then for any $a > 0$, we have*

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

PROOF. Applying Markov's inequality for the nonnegative RV $(X - \mu)^2$, we get

$$\mathbb{P}(|X - \mu| \geq a) = \mathbb{P}((X - \mu)^2 \geq a^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}.$$

□

Example 2.2.4. Let $X \sim \mathrm{Exp}(\lambda)$. Since $\mathbb{E}[X] = 1/\lambda$, for any $a > 0$, the Markov's inequality gives

$$\mathbb{P}(X \geq a) \leq \frac{1}{a\lambda},$$

while the true probability is

$$\mathbb{P}(X \geq a) = e^{-\lambda a}.$$

On the other hand, $\mathrm{Var}(X) = 1/\lambda^2$ so Chebyshef's inequality gives

$$\mathbb{P}(|X - 1/\lambda| \geq a) = \frac{1}{a^2\lambda^2}.$$

If $1/\lambda \leq a$, the true probability is

$$\begin{aligned} \mathbb{P}(|X - 1/\lambda| \geq a) &= \mathbb{P}(X \geq a + 1/\lambda) + \mathbb{P}(X \leq -a + 1/\lambda) \\ &= \mathbb{P}(X \geq a + 1/\lambda) = e^{-\lambda(a+1/\lambda)} = e^{-1-\lambda a}. \end{aligned}$$

As we can see, both Markov's and Chebyshef's inequalities give loose estimates, but the latter gives a slightly stronger bound. ▲

Example 2.2.5 (Chebyshef's inequality for bounded RVs)**.** Let X be a RV taking values from the interval $[a,b]$. Suppose we don't know anything else about X. Can we say anything useful about tail probability $\mathbb{P}(X \geq \lambda)$? If we were to use Markov's inequality, then certainly $a \leq \mathbb{E}[X] \leq b$ and in the worst case $\mathbb{E}[X] = b$. Hence we can at least conclude

$$\mathbb{P}(X \geq \lambda) \leq \frac{b}{\lambda}.$$

On the other hand, let's get a bound on $\mathrm{Var}(X)$ and use Chebyshef's inequality instead. We claim that

$$\mathrm{Var}(X) \leq \frac{(b-a)^4}{4},$$

which would yield by Chebyshef's inequality that

$$\mathbb{P}(|X - \mathbb{E}[X]| \leq \lambda) \leq \frac{(b-a)^2}{4\lambda^2}.$$

Intuitively speaking, $\mathrm{Var}(X)$ is the largest when the value of X is as much spread out as possible at the two extreme values, a and b. Hence the largest variance will be achieved when X takes a and b with equal probabilities. In this case, $\mathbb{E}[X] = (a+b)/2$ so

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{a^2+b^2}{2} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{4}.$$

▲

Exercise 2.2.6. Let X be a RV taking values from the interval $[a,b]$.

(i) Use the usual 'completing squares' trick for a second moment to show that

$$0 \leq \mathbb{E}[(X-t)^2] = (t - \mathbb{E}[X])^2 + \mathrm{Var}(X) \quad \forall\, t \in \mathbb{R}.$$

(ii) Conclude that $\mathbb{E}[(X-t)^2]$ is minimized when $t = \mathbb{E}[X]$ and the minimum is $\mathrm{Var}(X)$.

(iii) By plugging in $t = (a+b)/2$ in (2.2.6), show that

$$\mathrm{Var}(X) = \mathbb{E}[(X-a)(X-b)] + \frac{(b-a)^2}{4} - \left(\mathbb{E}[X] - \frac{a+b}{2}\right)^2.$$

(iv) Show that $\mathbb{E}[(X-a)(X-b)] \leq 0$.

(v) Conclude that $\mathrm{Var}(X) \leq (b-a)^2/4$, where the equality holds if and only if X takes the extreme values a and b with equal probabilities.

Exercise 2.2.7 (Paley-Zigmund inequality)**.** Let X be a nonnegative RV with $\mathbb{E}[X^2] < \infty$. Fix a constant $\theta \geq 0$. We prove the Paley-Zigmund inequality, which gives a lower bound on the tail probabilities and also implies the so-called 'second moment method'.

(i) Write $X = X\mathbf{1}(X > \theta\mathbb{E}[X]) + X\mathbf{1}(X \leq \theta\mathbb{E}[X])$. Show that

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[X\mathbf{1}(X \leq \theta\mathbb{E}[X])] + \mathbb{E}[X\mathbf{1}(X > \theta\mathbb{E}[X])] \\ &\leq \theta\mathbb{E}[X] + \mathbb{E}[X\mathbf{1}(X > \theta\mathbb{E}[X])].\end{aligned}$$

(ii) Use Cauchy-Schwartz inequality to show

$$\begin{aligned}(\mathbb{E}[X\mathbf{1}(X > \theta\mathbb{E}[X])])^2 &\leq \mathbb{E}[X^2]\mathbb{E}[\mathbf{1}(X > \theta\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2]\mathbb{E}[\mathbf{1}(X > \theta\mathbb{E}[X])] \\ &= \mathbb{E}[X^2]\mathbb{P}(X > \theta\mathbb{E}X).\end{aligned}$$

(iii) From (i) and (ii), derive

$$\mathbb{E}[X] \le \theta\mathbb{E}[X] + \sqrt{\mathbb{E}[X^2]\mathbb{P}(X > \theta\mathbb{E}X)}.$$

(Note that since $X \ge 0$, the above inequality is meaningful only when $\theta \le 1$.) Conclude that, for $\theta \in [0,1]$,

$$\mathbb{P}(X > \theta\mathbb{E}X) \ge \frac{(1-\theta)^2\mathbb{E}[X]^2}{\mathbb{E}[X^2]}.$$

(iv) (Second moment method) From (iii), conclude that

$$\mathbb{P}(X > 0) \ge \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}.$$

Alternatively, use the Cauchy-Schwarz inequality to the RV $X\mathbf{1}(X > 0)$ to deduce the above result.

The following Chernoff bound is essentially obtained by applying Markov's inequality after applying the exponential function $x \mapsto \exp(tx)$ for suitable $t \ge 0$. This is also called the Laplace transform method. The resulting generic tail bound (9) will later be specialized by using further knowledge on the tail of the RVs.

Proposition 2.2.8 (Chernoff bound)**.** *Let $X_n := \sum_{i=1}^n \xi_i$ where $\xi_1, \dots, \xi_n$ are independent RVs such that $\mathbb{E}[\exp(\theta\xi_i)] < \infty$ for all $\theta \in [0, c)$ and $1 \le i \le n$. Denote the log moment generating function of ξ_i as $\varphi_i(\theta) = \log\mathbb{E}[\exp(\theta\xi_i)]$. Then*

$$\mathbb{P}(X_n \ge t) \le \exp\left(-\sup_{\theta\in[0,c)}\left(\theta t - \sum_{i=1}^n \varphi_i(\theta)\right)\right). \tag{9}$$

Furthermore, assume ξ_i's are i.i.d. and $\mathbb{E}[\xi_1] = 0$. Then $L := \sup_{0\le\theta\le c/2}|\varphi''(\theta)| < \infty$ and

$$\mathbb{P}(X_n \ge t) \le \begin{cases} \exp\left(-\frac{Lt^2}{2}\right) & \text{if } t \le \frac{cnL}{2} \quad (\textit{Gaussian tail}) \\ \exp\left(-\frac{ct}{2L} + \frac{c^2}{8L}\right) & \text{if } t > \frac{cnL}{2} \quad (\textit{Exponential tail}). \end{cases}$$

PROOF. For any parameter $\theta \in [0, c)$, by exponentiating and taking Markov's inequality,

$$\begin{aligned} \mathbb{P}(X_n \ge t) &\le \mathbb{P}(\exp(\theta X_n) \ge \exp(\theta t)) \\ &\le \exp(-\theta t)\,\mathbb{E}[\exp(\theta X_n)] \\ &= \exp(-\theta t)\,\mathbb{E}[\exp(\theta\xi_1)\cdots\exp(\theta\xi_n)] \qquad \textit{(Multiplicativity of scalar exponential)} \\ &= \exp\left(-\theta t + n\varphi(\theta)\right), \end{aligned}$$

where the identity above uses the independence between the increments ξ_i. This holds for all $\theta \in [0, c)$. This shows (9).

Next, denote the moment generating function of ξ_1 as $\psi(\theta) = \mathbb{E}[\exp(\theta\xi_1)]$. Recall that it is a power series with radius of convergence $\ge c$ by the hypothesis. Hence we can differentiate it term-by-term when $\theta \in (-c, c)$. In particular, this gives $\psi(0) = 1$, $\psi'(0) = \mathbb{E}[\xi_1] = 0$, and $\psi''(0) = \mathbb{E}[\xi_1^2] \ge 0$. It follows that $\varphi(0) = 0$, $\varphi'(0) = \psi'(0) = 0$, and $\varphi''(0) = \psi''(0) \ge 0$. Then by Taylor's theorem,

$$\varphi(\theta) \le \varphi(0) + \varphi'(0)\theta + \frac{L}{2}\theta^2 = \frac{L}{2}\theta^2 \quad \text{for all } \theta \in [0, c/2].$$

This and (9) gives

$$\mathbb{P}(X_n \ge t) \le \exp\left(-\sup_{\theta\in[0,c/2]}\theta t - \frac{nL}{2}\theta^2\right).$$

The quadratic function $\theta \mapsto \theta t - \frac{nL}{2}\theta^2$ is globally minimized at $\theta = \frac{t}{nL}$ with minimum value $\frac{nL\theta^2}{2}$. Hence when $t < \frac{cnL}{2}$, the supremum in the above bound is attained at $\theta = \frac{t}{nL}$. If $t \ge \frac{cnL}{2}$, the same quadratic function is increasing in $[0, c/2nL]$ so the constrained maximization is solved at $\theta = \frac{c}{2nL}$. □

Exercise 2.2.9 (Hoeffding's lemma)**.** Let X be a bounded random variable such that $a \le X \le b$, and let $\mathbb{E}[X] = \mu$. Show that, for any $\lambda \in \mathbb{R}$,

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \le \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Proposition 2.2.10 (Hoeffding's inequlaity)**.** *Let $X_n := \sum_{i=1}^n \xi_i$ where $\xi_1, \dots, \xi_n$ are independent bounded random variables such that $\xi_i \in [a_i, b_i]$ for all i. Then for any $t > 0$,*

$$\mathbb{P}(X_n - \mathbb{E}[X_n] \ge t) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Similarly,

$$\mathbb{P}(X_n - \mathbb{E}[X_n] \le -t) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

PROOF. We combine the generic Chernoff bound (9) with Hoeffding's lemma (Exc. 2.2.9). Namely, by Hoeffding's lemma,

$$\varphi_i(\theta) = \log \mathbb{E}[\exp(\theta \xi_i)] \le \frac{\theta^2 (b_i - a_i)^2}{8}.$$

Since the RVs are bounded, their MGFs are defined for all $\theta \ge 0$. By the generic Chernoff bound,

$$\begin{aligned}\mathbb{P}(X_n \ge t) &\le \exp\left(-\sup_{\theta \ge 0}\left(\theta t - \sum_{i=1}^n \varphi_i(\theta)\right)\right) \\ &\le \exp\left(-\sup_{\theta \ge 0}\left(\theta t - \theta^2 \sum_{i=1}^n \frac{(b_i - a_i)^2}{8}\right)\right).\end{aligned}$$

The quadratic function in θ above is minimized at $\theta = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}$ and the minimum value is $-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}$. This shows the first inequality in the statement. The seoncd one can be deduced by considering $-\xi_i$'s instead of ξ_i's. □

Proposition 2.2.11 (Azuma-Hoeffding inequality)**.** *Let $(X_n)_{n \ge 0}$ be a martingale with respect to a filtration $(\mathscr{F}_n)_{n \ge 0}$, and let a_n be a non-negative sequence such that for each n,*

$$|X_{n+1} - X_n| \le a_n \quad \text{almost surely.}$$

If X_0 is bounded, say $X_0 \le b$ for some constant b, then for any $t > 0$,

$$\mathbb{P}(X_n - X_0 \ge t) \le \exp\left(-\frac{t^2}{2\sum_{k=0}^{n-1} a_k^2}\right)$$

and

$$\mathbb{P}(X_n - X_0 \le -t) \le \exp\left(-\frac{t^2}{2\sum_{k=0}^{n-1} a_k^2}\right).$$

Proposition 2.2.12 (Bounded difference inequality (McDiarmid's inequality))**.** *Let $f : \mathbb{R}^m \to \mathbb{R}$ be a function that satisfies the bounded differences condition. That is, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$ such that $\mathbf{x}_i = \mathbf{y}_i$ for all $i \ne j$ (i.e., $\mathbf{x}$ and $\mathbf{y}$ differ only in the j-th coordinate), we have:*

$$|f(\mathbf{x}) - f(\mathbf{y})| \le c_j$$

for some constant c_j that depends on j. Then, if $X_1, X_2, \dots, X_m$ are independent random variables,

$$\mathbb{P}\left(f(X_1, X_2, \dots, X_m) - \mathbb{E}[f(X_1, X_2, \dots, X_m)] \ge t\right) \le \exp\left(-\frac{t^2}{2\sum_{j=1}^m c_j^2}\right).$$

Similarly, for the lower tail, we have:

$$\mathbb{P}\left(f(X_1,X_2,\dots,X_m)-\mathbb{E}[f(X_1,X_2,\dots,X_m)]\le -t\right)\le \exp\left(-\frac{t^2}{2\sum_{j=1}^m c_j^2}\right).$$

Sketch of Proof. The idea is to reduce randomness by revealing the values of $X_1,\dots,X_m$ one after the other, which is called an 'exposure martingale'. Namely, consider

$$Y_k := \mathbb{E}[f(X_1,\dots,X_m)\,|\,\mathscr{F}_k]$$

where $\mathscr{F}_{k-1}$ is the σ-algebra generated by $X_1,\dots,X_k$ and $\mathscr{F}_0$ is the trivial σ-algebra. Y_k is a random variable depending on the non-averaged coordinates $X_{k+1},\dots,X_m$. Then $Y_1,\dots,Y_m$ form a martingale w.r.t. the filtration $\mathscr{F}_k$ by iterated expectation:

$$\mathbb{E}[Y_k\,|\,\mathscr{F}_{k-1}] = Y_{k-1}.$$

Then we can write the desired difference $Y_m - Y_0$ by the following telescoping sum (denoting $Y_0 = \mathbb{E}[f]$):

$$Y_m - Y_0 = \sum_{k=1}^m Y_k - Y_{k-1}.$$

Note that by the bounded difference assumption $|Y_k - Y_{k-1}| \le c_k$. So we are in a situation very similar to Hoeffding's inequality but for sums of martingale differences, which are only conditionally independent but not independent. We can still use Laplace method by using conditional MGFs and replicate the argument for proving Hoeffding's inequliaty (Margingle version of Hoeffding is called Azuma's inequality). (*Q: Where is independence of X_i's used?*) □

2.3. Sub-Gaussian random variables

Which random variables ξ_i must obey a concentration inequality like Hoeffding's inequality (Prop. 2.2.10) of the form

$$\text{for all } t\ge 0: \qquad \mathbb{P}\left(\left|\sum_{i=1}^n a_i\xi_i\right|\ge t\right)\le 2\exp\left(-\frac{ct^2}{\|\mathbf{a}\|^2}\right)?$$

If the sum $\sum_{i=1}^n a_i\xi_i$ consists of a single term ξ_i, this inequality reads as

$$\mathbb{P}(|\xi_i|>t)\le 2\exp\left(-ct^2\right).$$

Therefore, necessarily the RVs ξ_i must have sub-Gaussian tails.

Having a sub-Gaussian tail can be equivalently formulated in many different ways.

Proposition 2.3.1 (Sub-gaussian properties)**.** *Let X be a random variable. Then the following properties are equivalent; the parameters $K_i > 0$ appearing in these properties differ from each other by at most an absolute constant factor.*

(i) There exists $K_1 > 0$ such that the tails of X satisfy

$$\mathbb{P}(|X|\ge t)\le 2\exp\left(-\frac{t^2}{K_1^2}\right) \quad \text{for all } t\ge 0.$$

(ii) There exists $K_2 > 0$ such that the moments of X satisfy

$$\|X\|_p = \left(\mathbb{E}|X|^p\right)^{1/p} \le K_2\sqrt{p} \quad \text{for all } p\ge 1.$$

(iii) There exists $K_3 > 0$ such that the moment generating function (MGF) of X^2 satisfies

$$\mathbb{E}\exp(\lambda^2 X^2)\le \exp(K_3^2\lambda^2) \quad \text{for all } \lambda \text{ such that } |\lambda|\le \frac{1}{K_3}.$$

(iv) There exists $K_4 > 0$ such that the MGF of X^2 is bounded at some point, namely

$$\mathbb{E}\exp\left(\frac{X^2}{K_4^2}\right)\le 2.$$

(v) Moreover, if $\mathbb{E}[X] = 0$, *then properties (i)-(iv) are also equivalent to the following one: There exists* $K_5 > 0$ *such that the MGF of* X *satisfies*

$$\mathbb{E}\exp(\lambda X) \le \exp(K_5^2 \lambda^2) \quad \textit{for all } \lambda \in \mathbb{R}.$$

PROOF. Omited. □

Definition 2.3.2 (Sub-gaussian random variables)**.** A random variable X that satisfies one of the equivalent properties (i)-(iv) in Proposition 2.3.1 is called a *sub-gaussian random variable*. The *sub-gaussian norm* of X, denoted $\|X\|_{\psi_2}$, is defined to be the smallest K_4 in property (iv). In other words, we define

$$\|X\|_{\psi_2} := \inf\left\{t > 0 : \mathbb{E}\left[\exp\left(\frac{X^2}{t^2}\right)\right] \le 2\right\}.$$

Example 2.3.3 (Examples of sub-Gaussian distributions)**.** Here are some classical examples of sub-gaussian distributions.

(a) *(Gaussian):* As we already noted, $X \sim N(0,1)$ is a sub-gaussian random variable with $\|X\|_{\psi_2} \le C$, where C is an absolute constant. More generally, if $X \sim N(0,\sigma^2)$, then X is sub-gaussian with

$$\|X\|_{\psi_2} \le C\sigma.$$

(Why?)

(b) *(Bernoulli):* Let X be a random variable with symmetric Bernoulli distribution (recall Definition 2.2.1). Since $|X| = 1$, it follows that X is a sub-gaussian random variable with

$$\|X\|_{\psi_2} = \frac{1}{\sqrt{\ln 2}}.$$

(c) *(Bounded):* More generally, any bounded random variable X is sub-gaussian with

$$\|X\|_{\psi_2} \le \frac{1}{\sqrt{\ln 2}}.$$

Proposition 2.3.4 (General Hoeffding's inequlaity)**.** *Let* $X_n := \sum_{i=1}^n \xi_i$ *where* $\xi_1, \dots, \xi_n$ *are independent sub-Gaussian random variables. Then there exists an absolute constant* $c > 0$ *such that for any* $t > 0$,

$$\mathbb{P}(S_n - \mu_n \ge t) \le \exp\left(-\frac{ct^2}{\sum_{i=1}^n \|\xi_i\|_{\psi_2}^2}\right).$$

PROOF. By centering, we may assume $\mathbb{E}[\xi_i] = 0$ for all $1 \le i \le n$. By the sub-Gaussian property (v) in Prop. 2.3.1,

$$\varphi_i(\theta) = \log \mathbb{E}[\exp(\theta \xi_i)] \le C\|\xi_i\|_{\psi_2}^2 \theta^2 \qquad \text{for all } \theta \in \mathbb{R},$$

where $C > 0$ is an absolute constant. Hence by the generic Chernoff bound (9),

$$\begin{aligned}\mathbb{P}(X_n \ge t) &\le \exp\left(-\sup_{\theta \ge 0}\left(\theta t - \sum_{i=1}^n \varphi_i(\theta)\right)\right) \\ &\le \exp\left(-\sup_{\theta \ge 0}\left(\theta t - \theta^2 \sum_{i=1}^n C\|\xi_i\|_{\psi_2}^2\right)\right).\end{aligned}$$

The quadratic function in θ above is minimized at $\theta = \frac{t}{2C\sum_{i=1}^n \|\xi_i\|_{\psi_2}^2}$ and the minimum value is $-\frac{t^2}{4C\sum_{i=1}^n \|\xi_i\|_{\psi_2}^2}$. □

2.4. Sub-exponential random variables

In this section, we introduce the notion of sub-exponential and sub-Gaussian random variables and discuss their concentration properties.

Proposition 2.4.1 (Sub-exponential properties)**.** *Let X be a random variable. Then the following properties are equivalent; the parameters $K_i > 0$ appearing in these properties differ from each other by at most an absolute constant factor.*[1]

(a) The tails of X satisfy

$$\mathbb{P}(|X| \geq t) \leq 2\exp\left(-\frac{t}{K_1}\right) \quad \text{for all } t \geq 0.$$

(b) The moments of X satisfy

$$\|X\|_{L_p} = \left(\mathbb{E}|X|^p\right)^{1/p} \leq K_2 p \quad \text{for all } p \geq 1.$$

(c) The moment generating function (MGF) of $|X|$ satisfies

$$\mathbb{E}\exp(\lambda|X|) \leq \exp(K_3\lambda) \quad \text{for all } \lambda \text{ such that } 0 \leq \lambda \leq \frac{1}{K_3}.$$

(d) The MGF of $|X|$ is bounded at some point, namely

$$\mathbb{E}\exp\left(\frac{|X|}{K_4}\right) \leq 2.$$

(e) Moreover, if $\mathbb{E}[X] = 0$, then properties (a)-(d) are also equivalent to the following one: The MGF of X satisfies

$$\mathbb{E}\exp(\lambda X) \leq \exp(K_5^2\lambda^2) \quad \text{for all } \lambda \text{ such that } |\lambda| \leq \frac{1}{K_5}.$$

PROOF. Omitted. □

Definition 2.4.2 (Sub-exponential tail)**.** A random variable X that satisfies one of the equivalent properties (a)-(d) in Proposition 2.4.1 is called a *sub-exponential random variable*. The *sub-exponential norm* of X, denoted $\|X\|_{\psi_1}$, is defined to be the smallest K_3 in property (c). In other words,

$$\|X\|_{\psi_1} := \inf\left\{t > 0 : \mathbb{E}\left[\exp\left(\frac{|X|}{t}\right)\right] \leq 2\right\}.$$

Exercise 2.4.3 (Sub-exponential is sub-gaussian squared)**.** Show that a random variable X is sub-gaussian if and only if X^2 is sub-exponential. Moreover,

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2.$$

Exercise 2.4.4 (Product of sub-gaussians is sub-exponential)**.** Let X and Y be sub-gaussian random variables. Show that XY is sub-exponential. Moreover,

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2}\|Y\|_{\psi_2}.$$

Proposition 2.4.5 (Bernstein's inequality)**.** *Let $\xi_1, \ldots, \xi_n$ be independent, mean zero, sub-exponential random variables. Then, for every $t \geq 0$, we have*

$$\mathbb{P}\left(\sum_{i=1}^n \xi_i \geq t\right) \leq 2\exp\left(-c\min\left(\frac{t^2}{\sum_{i=1}^n \|\xi_i\|_{\psi_1}^2}, \frac{t}{\max_i \|\xi_i\|_{\psi_1}}\right)\right),$$

where $c > 0$ is an absolute constant.

[1] Here we ignored the pre-factor $1/t$, which does not make much effect on the exponent. The precise meaning of this equivalence is the following: There exists an absolute constant C such that property i implies property j with parameter $K_j \leq CK_i$ for any two properties $i, j = 1, 2, 3, 4$.

PROOF. By the sub-exponential property (e) in Prop. 2.4.1,

$$\varphi_i(\theta) = \log \mathbb{E}[\exp(\theta \xi_i)] \le C\|\xi_i\|_{\psi_1}^2 \theta^2 \qquad \text{if } |\theta| \le \theta_0 := \frac{C}{\max_i \|\xi_i\|_{\psi_1}},$$

where $C > 0$ is an absolute constant. Hence by the generic Chernoff bound (9),

$$\begin{aligned}\mathbb{P}(X_n \ge t) &\le \exp\left(-\sup_{\theta\in[-\theta_0,\theta_0]}\left(\theta t - \sum_{i=1}^n \varphi_i(\theta)\right)\right) \\ &\le \exp\left(-\sup_{\theta\in[-\theta_0,\theta_0]}\left(\theta t - \theta^2 \sum_{i=1}^n C\|\xi_i\|_{\psi_1}^2\right)\right).\end{aligned}$$

The quadratic function in θ above is minimized at $\theta^* = \frac{t}{2C\sum_{i=1}^n \|\xi_i\|_{\psi_1}^2}$ and the minimum value is $-\frac{t^2}{4C\sum_{i=1}^n \|\xi_i\|_{\psi_1}^2}$. However, we are now only allowed to look for θ in the interval $[-\theta_0, \theta_0]$ and the global minimizer θ^* may not be in this constraint set. It is easy to check that the maximizer of the quadratic within the interval is given by $\theta_0 \wedge \theta^*$. Plugging this value back in above give the desired result. □

2.5. The Matrix Laplace Transform Method

In the scalar setting, the Laplace transform method allows us to obtain tail bounds for a random variable in terms of its moment generating function (MGF) (e.g., see the proof of Chernoff bound in Prop. 2.2.8). The key idea is to exponentiate both sides of the tail bounds, use Markov's inequality, and then optimize for the bound that involves the MGF. Indeed, we can execute the exact same idea in the matrix setting. This gives a control on the tail probabilities of the extreme eigenvalues of a random matrix via a bound for the trace of the matrix MGF.

Definition 2.5.1 (Matrix exponential and logarithm)**.** The *matrix exponential* of a square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is defined as

$$e^{\mathbf{A}} = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!},$$

where the series converges for all square matrices $\mathbf{A}$. The *matrix logarithm* $\log(\mathbf{A})$ is defined as the inverse function of the matrix exponential, such that

$$\mathbf{A} = e^{\log(\mathbf{A})}.$$

Note that that $\log(\mathbf{A})$ is well-defined for matrices that are invertible and have no eigenvalues on the non-positive real axis. The logarithm of a positive number a is given by the integral

$$\log a = \int_0^\infty \left(\frac{1}{1+u} - \frac{1}{a+u}\right) du.$$

Similarly, the logarithm of a PD matrix A is given by the integral

$$\log \mathbf{A} = \int_0^\infty \left((1+u)^{-1}\mathbf{I} - (\mathbf{A}+u\mathbf{I})^{-1}\right) du.$$

Exercise 2.5.2 (Monotonicity of $\mathrm{tr}\exp$ and $\log$)**.** Let $\mathbf{A}, \mathbf{B}$ be PSD matrices of the same dimension. Recall that the semidefinite (Loewner) ordering between PSD matrices:

$$\mathbf{A} \preceq \mathbf{B} \quad \overset{def}{\iff} \quad \mathbf{B} - \mathbf{A} \text{ is PSD} \quad \overset{def}{\iff} \quad \text{eigenvalues of } \mathbf{B} - \mathbf{A} \text{ are real and } \ge 0.$$

Show that the matrix trace exponential and logarithm are monotone w.r.t. the semidefinite order:

$$\begin{aligned}\mathbf{A} \preceq \mathbf{B} \quad &\Longrightarrow \quad \mathrm{tr}\exp(\mathbf{A}) \le \mathrm{tr}\exp(\mathbf{B}), \\ \mathbf{O} \prec \mathbf{A} \preceq \mathbf{B} \quad &\Longrightarrow \quad \log(\mathbf{A}) \preceq \log(\mathbf{B}).\end{aligned}$$

Next, we define matrix versions of the moment generating function (MGF) and cumulant generating function (CGF).

Definition 2.5.3 (Matrix MGF and CGF)**.** Let X be a random self-adjoint matrix. The *matrix moment generating function* (MGF) $M_{\mathbf{X}}(\theta)$ and the matrix *cumulant generating function* (CGF) $\Xi_{\mathbf{X}}(\theta)$ are given by

$$\mathbf{M}_{\mathbf{X}}(\theta) = \mathbb{E}[e^{\theta \mathbf{X}}] \quad \text{and} \quad \Xi_{\mathbf{X}}(\theta) = \log \mathbb{E}[e^{\theta \mathbf{X}}] \quad \text{for } \theta \in \mathbb{R}.$$

The expectations above may not exist for all values of θ.

Observe that the matrix MGF and CGF have formal power series expansions:

$$\mathbf{M}_{\mathbf{X}}(\theta) = I + \sum_{q=1}^{\infty} \frac{\theta^q}{q!} \mathbb{E}[\mathbf{X}^q] \quad \text{and} \quad \Xi_{\mathbf{X}}(\theta) = \sum_{q=1}^{\infty} \frac{\theta^q}{q!} \mathbf{\Psi}_q.$$

We call the coefficients $\mathbb{E}[\mathbf{X}^q]$ *matrix moments*, and we refer to $\mathbf{\Psi}_q$ as a *matrix cumulant*. The matrix cumulant $\mathbf{\Psi}_q$ has a formal expression as a (noncommutative) polynomial in the matrix moments up to order q. In particular, the first cumulant is the mean, and the second cumulant is the variance:

$$\mathbf{\Psi}_1 = \mathbb{E}[\mathbf{X}] \quad \text{and} \quad \mathbf{\Psi}_2 = \mathbb{E}[\mathbf{X}^2] - (\mathbb{E}[\mathbf{X}])^2.$$

Higher-order matrix cumulants are harder to write down and interpret.

The entry point to various matrix concentration inequalities is the matrix Laplace transform method introduced by Ahlswede and Winter [AW02] – take matrix exponential, use Markov, and then optimize the bound involving matrix MGF:

Proposition 2.5.4 (Matrix Laplace transform method)**.** *Let* $\mathbf{Y}$ *be a random self-adjoint matrix. For all* $t \in \mathbb{R}$*,*

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \geq t\right) \leq \inf_{\theta > 0} e^{-\theta t} \mathbb{E}[\operatorname{tr} e^{\theta \mathbf{Y}}], \tag{10}$$

$$\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \leq t\right) \leq \inf_{\theta < 0} e^{-\theta t} \mathbb{E}[\operatorname{tr} e^{\theta \mathbf{Y}}]. \tag{11}$$

PROOF. We begin with (10). Fix $\theta > 0$ and note that

$$\begin{aligned} \mathbb{P}(\lambda_{\max}(Y) \geq t) &\overset{(a)}{=} \mathbb{P}\left(e^{\theta \lambda_{\max}(Y)} \geq e^{\theta t}\right) \\ &\overset{(b)}{\leq} e^{-\theta t} \mathbb{E}\left[e^{\theta \lambda_{\max}(Y)}\right], \end{aligned}$$

where (a) follows since the function $x \mapsto e^{\theta x}$ is increasing and (b) follows from Markov's inequality. Next, note that

$$e^{\theta \lambda_{\max}(\mathbf{Y})} = e^{\lambda_{\max}(\theta \mathbf{Y})} = \lambda_{\max}(e^{\theta \mathbf{Y}}) \leq \operatorname{tr}(e^{\theta \mathbf{Y}}). \tag{12}$$

The first identity holds since the maximum eigenvalue is a positive-homogeneous map; The second follows from the spectral mapping theorem; The inequality follows because the exponential of a self-adjoint matrix is positive definite, and the maximum eigenvalue of a positive-definite matrix is bounded by the trace. Combining the above, we get

$$\mathbb{P}\left(\lambda_{\max}(Y) \geq t\right) \leq e^{-\theta t} \mathbb{E}\left[\operatorname{tr} e^{\theta Y}\right].$$

This inequality is valid for any $\theta > 0$, so we may take the infimum to achieve the tightest possible bound. This shows (10).

Next, to prove (11), we use a similar approach. Fix $\theta < 0$ and note that

$$\begin{aligned} \mathbb{P}(\lambda_{\min}(\mathbf{Y}) \leq t) = \mathbb{P}(e^{\theta \lambda_{\min}(\mathbf{Y})} \geq e^{\theta t}) &\leq e^{-\theta t} \mathbb{E}\left[e^{\theta \lambda_{\min}(\mathbf{Y})}\right] \\ &\overset{(a)}{=} e^{-\theta t} \mathbb{E}\left[e^{\lambda_{\max}(\theta \mathbf{Y})}\right] \\ &\overset{(b)}{\leq} e^{-\theta t} \mathbb{E}\left[\operatorname{tr}(e^{\theta \mathbf{Y}})\right]. \end{aligned}$$

The first equality follows since $x \mapsto e^{\theta x}$ is a *decreasing function* (since $\theta < 0$); (a) follows since the eigenvalues of $\theta \mathbf{Y}$ are θ times the eigenvalues of $\mathbf{Y}$ and $\theta < 0$; (b) follows from (12). Then optimizing the resulting bound for all $\theta < 0$ shows (11). □

Remark 2.5.5 (Bounding maximum eigenvalue by trace)**.** In the proof of Proposition 2.5.4, it might appear overly simplistic to bound the maximum eigenvalue using the trace. However, this approach only incurs a logarithmic factor loss in terms of the matrix dimension More details?. More importantly, this step enables us to take advantage of the remarkable convexity properties of the trace exponential due to Lieb [Lie73].

Exercise 2.5.6 (Expectation bounds for eigenvalues)**.** Let $\mathbf{Y}$ be a random self-adjoint matrix. Show that

$$\mathbb{E}[\lambda_{\max}(\mathbf{Y})] \le \inf_{\theta>0} \frac{1}{\theta} \log \mathbb{E}\left[\operatorname{tr} e^{\theta \mathbf{Y}}\right],$$
$$\mathbb{E}[\lambda_{\min}(\mathbf{Y})] \ge \sup_{\theta<0} \frac{1}{\theta} \log \mathbb{E}\left[\operatorname{tr} e^{\theta \mathbf{Y}}\right].$$

Note that the tail bounds in Prop. 2.5.4 is useful only if we have a good control on the matrix MGF $\mathbb{E}\left[\operatorname{tr} e^{\theta Y}\right]$. Without making some further structural assumption on $\mathbf{Y}$, this is hard to come by. We will first consider the independent sum model in Def. 2.1.1, whereby we assume that $\mathbf{Y}$ is the sum of many independent random matrices $\sum_{i=1}^{n} \mathbf{X}_i$.

For real-valued independent random variables $\xi_1, \dots, \xi_n$, the key property of MGFs we exploited in the proof of Chernoff bound (Prop. 2.2.8) was that the exponential of the sum is the product of the exponentials:

$$\exp\left(\sum_{i=1}^{n} \xi_i\right) = \prod_{i=1}^{n} \exp(\xi_i).$$

Taking expectation and using independence between ξ_i's, this gave us

$$\mathbb{E}\left[\exp\left(\sum_{i=1}^{n} \xi_i\right)\right] = \prod_{i=1}^{n} \mathbb{E}\left[\exp(\xi_i)\right].$$

In particular, the MGF of the sum of independent RVs is the product of the individual MGFs.

However, the situation is much more subtle for random matrices since the matrix products do not commute in general. In particular,

$$e^{\mathbf{A}+\mathbf{B}} \ne e^{\mathbf{A}} e^{\mathbf{B}} \qquad \text{unless } \mathbf{AB} = \mathbf{BA}.$$

Theorem 2.5.7 (Lieb, 1973 [Lie73])**.** *Fix a self-adjoint matrix* $\mathbf{H}$*. The function*

$$\mathbf{A} \mapsto \operatorname{tr}\left(\exp(\mathbf{H} + \log(\mathbf{A}))\right)$$

is concave on the set of all positive definite matrices with the same dimension as $\mathbf{H}$*.*

Corollary 2.5.8 (Tropp, [Tro12])**.** *Let* $\mathbf{H}$ *be a fixed self-adjoint matrix, and let* $\mathbf{X}$ *be a random self-adjoint matrix. Then*

$$\mathbb{E}\left[\operatorname{tr} \exp(\mathbf{H} + \mathbf{X})\right] \le \operatorname{tr} \exp\left(\mathbf{H} + \log \mathbb{E}\left[e^{\mathbf{X}}\right]\right).$$

PROOF. Define the random matrix $\mathbf{Y} = e^{\mathbf{X}}$. Then $\mathbf{Y}$ is always PD so $\log \mathbf{Y}$ is well-defined and equals to $\mathbf{X}$. Then using Lieb's theorem (Thm. 2.5.7) with Jensen's inequality,

$$\begin{aligned}
\mathbb{E}\left[\operatorname{tr} \exp(\mathbf{H} + \mathbf{X})\right] &= \mathbb{E}\left[\operatorname{tr} \exp(\mathbf{H} + \log \mathbf{Y})\right] \\
&\le \operatorname{tr} \exp\left(\mathbf{H} + \log \mathbb{E}[\mathbf{Y}]\right) \\
&= \operatorname{tr} \exp\left(\mathbf{H} + \log \mathbb{E}\left[e^{\mathbf{X}}\right]\right).
\end{aligned}$$

This shows the assertion. □

By applying Corollary 2.5.8 repeately, we can obtain the following very useful bound on the trace MGF of the sum of independent RMs in terms of the sum of the CGFs of individual RMs.

Lemma 2.5.9 (Bound on trace MGF)**.** *Let* $\mathbf{Y} = \sum_{i=1}^n \mathbf{X}_i$*, where* $\mathbf{X}_1, \dots, \mathbf{X}_n$ *are self-adjoint independent random matrices. Then*

$$\mathbb{E}\left[\operatorname{tr}\exp(\mathbf{Y})\right] \le \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\mathbf{X}_i}\right]\right).$$

PROOF. Denote $\mathscr{F}_k := \sigma(\mathbf{X}_1, \dots, \mathbf{X}_k)$ for $k = 1 \dots, n$ and let $\mathscr{F}_0$ be the trivial σ-algebra. Let $\mathbf{Y}_k := \sum_{i=1}^k \mathbf{X}_i$ for $k = 1, \dots, n$. We will prove the following slightly stronger statement by induction. For any fixed self-adjoint matrix $\mathbf{H}$,

$$\mathbb{E}\left[\operatorname{tr}\exp(\mathbf{H} + \mathbf{Y}_n)\right] \le \operatorname{tr}\exp\left(\mathbf{H} + \sum_{i=1}^n \log \mathbb{E}\left[e^{\mathbf{X}_i}\right]\right). \tag{13}$$

Then the assertion is recoverd by taking $\mathbf{H} = \mathbf{O}$.

If $n = 1$, then (13) holds immediately by Lieb's theorem (Thm. 2.5.7). For the induction step, note that $\mathbf{Y}_{n-1}$ is measurable (determistic) w.r.t. $\mathscr{F}_{n-1}$. So by Corollary 2.5.8, almost surely,

$$\begin{aligned}
\mathbb{E}\left[\operatorname{tr}\exp(\mathbf{H} + \mathbf{Y}_n) \,\middle|\, \mathscr{F}_{n-1}\right] &= \mathbb{E}\left[\operatorname{tr}\exp(\underbrace{\mathbf{H} + \mathbf{Y}_{n-1}}_{\in \mathscr{F}_{n-1}} + \mathbf{X}_n) \,\middle|\, \mathscr{F}_{n-1}\right] \\
&\le \operatorname{tr}\exp\left(\mathbf{H} + \mathbf{Y}_{n-1} + \log \mathbb{E}\left[e^{\mathbf{X}_n} \,\middle|\, \mathscr{F}_{n-1}\right]\right) \\
&= \operatorname{tr}\exp\left(\mathbf{H} + \mathbf{Y}_{n-1} + \log \mathbb{E}\left[e^{\mathbf{X}_n}\right]\right) \qquad (\because \mathbf{X}_n \perp \mathscr{F}_{n-1}).
\end{aligned}$$

Hence, by iterated expectation and the induction hypothesis (IH),

$$\begin{aligned}
\mathbb{E}\left[\operatorname{tr}\exp(\mathbf{H} + \mathbf{Y}_n)\right] &\le \mathbb{E}\left[\operatorname{tr}\exp\left(\left(\mathbf{H} + \log \mathbb{E}\left[e^{\mathbf{X}_n}\right]\right) + \mathbf{Y}_{n-1}\right)\right] \\
&\overset{IH}{\le} \operatorname{tr}\exp\left(\mathbf{H} + \log \mathbb{E}\left[e^{\mathbf{X}_n}\right] + \sum_{i=1}^{n-1} \log \mathbb{E}\left[e^{\mathbf{X}_i}\right]\right) \\
&= \operatorname{tr}\exp\left(\mathbf{H} + \sum_{i=1}^{n} \log \mathbb{E}\left[e^{\mathbf{X}_i}\right]\right).
\end{aligned}$$

This completes the induction. □

Now by the help of Lemma 2.5.9, we can transform the basic Laplace bounds on the spectrum of sums of independent matrices into more useful forms.

Theorem 2.5.10 (Master bound for a sum of independent random matrices)**.** *Let* $\mathbf{Y} = \sum_{i=1}^n \mathbf{X}_i$*, where* $\mathbf{X}_1, \dots, \mathbf{X}_n$ *are self-adjoint independent random matrices. Then for any* $t \ge 0$*,*

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge t\right) \le \inf_{\theta > 0} e^{-\theta t} \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right),$$

$$\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \le t\right) \le \inf_{\theta < 0} e^{-\theta t} \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right).$$

Furthermore,

$$\mathbb{E}[\lambda_{\max}(\mathbf{Y})] \le \inf_{\theta > 0} \frac{1}{\theta} \log \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right),$$

$$\mathbb{E}[\lambda_{\min}(\mathbf{Y})] \ge \sup_{\theta < 0} \frac{1}{\theta} \log \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right).$$

PROOF. The tail bounds on the extreme eigenvalues of $\mathbf{Y}$ follow immediately from Proposition 2.5.4 and Lemma 2.5.9. The bounds on their expectations follow immediately from Exc. 2.5.6 and Lemma 2.5.9. □

In order to make the bounds in Theorem 2.5.10 more applicable, we only need to find useful bounds on the sum of CGFs of $\mathbf{X}_i$. Typically, this can be done by Taylor-expanding exponential functions and using the resulting polynomial upper bounds. One example is the following Bernstein's bound on matrix CGF.

Lemma 2.5.11 (Bernstein's bound on matrix CGF)**.** *Suppose that* $\mathbf{X}$ *is a random self-adjoint matrix that satisfies* $\mathbb{E}[\mathbf{X}] = \mathbf{O}$ *and* $\|\mathbf{X}\|_2 \le 1$*. Then*

$$\log \mathbb{E}\left[e^{\theta \mathbf{X}}\right] \preceq \frac{\theta^2/2}{1-|\theta|/3}\mathbb{E}\left[\mathbf{X}^2\right] \quad \text{for} \quad |\theta| < 3.$$

PROOF. Fix $x \in [-1,1]$ and $\theta \in (-3,3)$. Using the Taylor series expansion of the exponential function:

$$\begin{aligned} e^{\theta x} &= 1 + \theta x + \sum_{p=2}^{\infty} \frac{\theta^p}{p!} x^p \\ &\le 1 + \theta x + \left(\sum_{p=2}^{\infty} \frac{|\theta|^p}{2 \cdot 3^{p-2}}\right) x^2 \\ &= 1 + \theta x + \frac{\theta^2/2}{1-|\theta|/3} x^2. \end{aligned}$$

Denote the quadratic polynomial in the last expression as $f(x)$. Note that both $e^{\theta \mathbf{X}}$ and $f(\mathbf{X})$ are self-adjoint random matrices with the same eigenvectors as $\mathbf{X}$. Since the eigenvalues of $\mathbf{X}$ lie in $[-1,1]$, the above inequality shows that the eigenvalues of $e^{\theta \mathbf{X}}$ are bounded above by the corresponding eigenvalues of $f(\mathbf{X})$. Hence, almost surely,

$$e^{\theta \mathbf{X}} \preceq \mathbf{I} + \theta \mathbf{X} + \frac{\theta^2/2}{1-|\theta|/3}\mathbf{X}^2.$$

Taking the expectation, we get:

$$\mathbb{E}\left[e^{\theta \mathbf{X}}\right] \preceq \mathbf{I} + \frac{\theta^2/2}{1-|\theta|/3}\mathbb{E}\left[\mathbf{X}^2\right].$$

The matrix on the left-hand side is positive definite, so its logarithm is well-defined. Since matrix logarithm is monotone w.r.t. semidefinite order (see Exc. 2.5.2),

$$\log \mathbb{E}\left[e^{\theta \mathbf{X}}\right] \preceq \log\left(\mathbf{I} + \frac{\theta^2/2}{1-|\theta|/3}\mathbb{E}\left[\mathbf{X}^2\right]\right) \preceq \frac{\theta^2/2}{1-|\theta|/3}\mathbb{E}\left[\mathbf{X}^2\right].$$

For the last domination, we used three facts: (1) $\log(1+x) \le x$ for all $x > -1$ so for a self-adjoint matrix $\mathbf{A}$ with $\lambda_{\min}(\mathbf{A}) > -1$, $\log(\mathbf{I}+\mathbf{A}) \preceq \mathbf{A}$; and (2) $\frac{\theta^2/2}{1-|\theta|/3} > 0$ if $|\theta| < 3$, and (3) $\mathbb{E}[\mathbf{X}^2]$ is PSD. □

We need one more trick called the "self-adjoint dilation". This is a simple trick to form a square

So far we have only considered concentration self-adjoint matrices. It is important to notice that concentration results for rectangular matrices follow as a formal consequence. The key to do so is the following self-adjoint dilation.

Exercise 2.5.12 (The self-adjoint dilation)**.** The *self-adjoint dilation* $\mathscr{H}(\mathbf{S})$ of a rectangular matrix $\mathbf{S} \in \mathbb{C}^{p \times q}$ is the self-adjoint matrix

$$\mathscr{H}(\mathbf{S}) := \begin{bmatrix} 0 & \mathbf{S} \\ \mathbf{S}^* & 0 \end{bmatrix} \in \mathbb{C}^{(p+q)\times(p+q)}.$$

Note that the map $\mathscr{H}$ is real-linear. Show that

$$\mathscr{H}(\mathbf{S})^2 = \begin{bmatrix} \mathbf{S}\mathbf{S}^* & 0 \\ 0 & \mathbf{S}^*\mathbf{S} \end{bmatrix}.$$

Also show that

$$\lambda_{\max}(\mathscr{H}(\mathbf{S})) = \|\mathscr{H}(\mathbf{S})\|_2 = \|\mathbf{S}\|_2.$$

Now we are able to establish the following, extremely versatile, matrix concentration inequality. The proof is based on the approach in [Tro12]. The tail bound in (15) was proved earlier by Recht in 2009 and published in [Rec11], making use of Golden-Thompson inequality instead of Lieb's theorem.

Theorem 2.5.13 (Matrix Bernstein inequality)**.** *Let* $\mathbf{Y} = \sum_{k=1}^n \mathbf{X}_k$*, where* $\{\mathbf{X}_k : 1 \le k \le n\}$ *are independent self-adjoint random matrices with dimension* d*. Suppose that*

$$\mathbb{E}[\mathbf{X}_k] = \mathbf{O} \quad \text{and} \quad \|\mathbf{X}_k\|_2 \le R \quad \text{for each index } k.$$

Define the matrix variance proxy:

$$\sigma^2 = \left\|\mathbb{E}[\mathbf{Y}^2]\right\|_2 = \left\| \sum_{k=1}^n \mathbb{E}\left[\mathbf{X}_k^2\right] \right\|_2. \tag{14}$$

Then, for all $t \ge 0$*,*

$$\begin{aligned} \mathbb{P}\left(\|\mathbf{Y}\|_2 \ge t\right) &\le 2d \, \exp\left(\frac{-t^2/2}{\sigma^2 + Rt/3}\right) \\ &\le \begin{cases} d \, \exp(-t^2/4\sigma^2) & \text{if } t \le \sigma^2/R, \\ d \, \exp(-t/4R) & \text{if } t > \sigma^2/R. \end{cases} \end{aligned} \tag{15}$$

Furthermore,

$$\mathbb{E}[\|\mathbf{Y}\|_2] \le \sqrt{2\sigma^2 \log(2d)} + \frac{1}{3} R \log(2d). \tag{16}$$

PROOF. First, rescale so that $R = 1$. The general form of the result follows from homogeneity arguments. The Bernstein CGF bound in Lem. 2.5.11 implies that

$$\log \mathbb{E}\left[e^{\theta \mathbf{X}_k}\right] \preceq g(\theta)\, \mathbb{E}\left[\mathbf{X}_k^2\right] \quad \text{where} \quad g(\theta) = \frac{\theta^2/2}{1 - |\theta|/3}. \tag{17}$$

Note that $g(\theta) \ge 0$ for all θ with $|\theta| < 3$. Note that for $|\theta| < 3$,

$$\begin{aligned} \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) &\overset{(a)}{\le} \operatorname{tr}\exp\left(g(\theta) \sum_k \mathbb{E}[\mathbf{X}_k^2]\right) \\ &\overset{(b)}{\le} d\,\lambda_{\max}\left(\exp\left(g(\theta)\mathbb{E}[\mathbf{Y}^2]\right)\right) \\ &\overset{(c)}{=} d \exp\left(g(\theta)\lambda_{\max}\left(\mathbb{E}[\mathbf{Y}^2]\right)\right) \\ &\overset{(d)}{=} d \exp\left(g(\theta)\sigma^2\right), \end{aligned}$$

where (a) uses the Bernstein CGF bound (17) along with the monotonicity of $\operatorname{tr}\exp$ (Exc. 2.5.2), (b) is from bounding the trace of a PSD matrix by the dimension times the maximum eigenvalue, and (c) uses homogeneity of eigenvalues, and (d) is the definition of σ^2 in (14).

Now, using the above and the master inequality in Thm. 2.5.10,

$$\begin{aligned} \mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge t\right) &\le \inf_{\theta > 0} e^{-\theta t} \operatorname{tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) \\ &\le d \inf_{0 < \theta < 3} \exp\left(-\theta t + g(\theta)\sigma^2\right). \end{aligned}$$

In order to find infemum in the last expression, we need to minimize the function $h(\theta) := -\theta t + g(\theta)\sigma^2$ over $\theta > 0$. We make the clever choice $\theta = \frac{t}{\sigma^2 + t/3} \in (0,3)$ (not optimal) to obtain

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge t\right) \le d \exp\left(\frac{-t^2/2}{\sigma^2 + t/3}\right).$$

Applying the same argument for $-\mathbf{Y}$, we get

$$\mathbb{P}\left(\lambda_{\max}(-\mathbf{Y}) \geq t\right) \leq d \exp\left(\frac{-t^2/2}{\sigma^2 + t/3}\right).$$

Since $\|\mathbf{Y}\|_2 = \max\{\lambda_{\max}(\mathbf{Y}), \lambda_{\max}(-\mathbf{Y})\}$, the desired bound (15) then follows from a union bound.

Lastly, we deduce the expectation bound in (16). For the upper bound on the mean max eigenvalue, Theorem 2.5.10 gives

$$\begin{aligned}
\mathbb{E}[\lambda_{\max}(\mathbf{Y})] &\leq \inf_{\theta>0} \frac{1}{\theta} \log \operatorname{tr} \exp\left(\sum_{i=1}^{n} \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) \\
&\leq \inf_{0<\theta<3} \frac{1}{\theta}\left[\log d \exp\left(g(\theta)\sigma^2\right)\right] \\
&= \inf_{0<\theta<3} \frac{1}{\theta}\left[(\log d) + g(\theta)\sigma^2\right] \\
&= \inf_{0<\theta<3} \left[\frac{(\log d)}{\theta} + \frac{3\theta}{2(3-\theta)}\sigma^2\right].
\end{aligned}$$

The function in the bracket is minimized at $\theta = \frac{3\sqrt{2\log d}}{3\sigma + \sqrt{2\log d}} \in (0,3)$ and the minimum value is $\sqrt{2\sigma^2 \log d} + \frac{1}{3}\log d$. Hence

$$\mathbb{E}[\lambda_{\max}(\mathbf{Y})] \leq \sqrt{2\sigma^2 \log d} + \frac{1}{3}\log d.$$

Now, in order to deduce the bound on the expected spectral norm of $\mathbf{Y}$, we apply the above bound with $\mathbf{Y}$ replaced with its self-adjoint dilation $\mathscr{H}(\mathbf{Y})$ (see Exc. 2.5.12). This may seem redundant since $\mathbf{Y}$ is already self-adjoint. The benefit is that

$$\lambda_{\max}(\mathscr{H}(\mathbf{Y})) = \|\mathbf{Y}\|_2.$$

Hence, noting that $\mathscr{H}(\mathbf{Y})$ is self-adjoint with dimension $2d$, we deduce

$$\mathbb{E}[\|\mathbf{Y}\|_2] = \mathbb{E}[\lambda_{\max}(\mathscr{H}(\mathbf{Y}))] \leq \sqrt{2\sigma^2 \log(2d)} + \frac{1}{3}\log(2d).$$

This comples the proof.

□

Next, we will consider sums of independent PSD matrices and obtain the matrix Chernoff inequality. The corresponding matrix CGF bound is given in the following lemma, which first appeared in the proof of [AW02, Thm. 19].

Lemma 2.5.14 (Chernoff CGF)**.** *Suppose that* $\mathbf{X}$ *is a random self-adjoint matrix that satisfies* $\mathbf{O} \preceq \mathbf{X} \preceq \mathbf{I}$*. Then*

$$\log \mathbb{E}\left[e^{\theta \mathbf{X}}\right] \preceq (e^{\theta} - 1)\mathbb{E}[\mathbf{X}] \quad \textit{for} \quad \theta \in \mathbb{R}.$$

PROOF. Fix $\theta > 0$. The function $x \mapsto e^{\theta x}$ is convex, so the graph lies below the secant line connecting the $(0,1)$ and $(1, e^{\theta x})$. This gives

$$e^{\theta x} \leq 1 + (e^{\theta} - 1)x \quad \text{for} \quad x \in [0,1].$$

The eigenvalues of $\mathbf{X}$ lie in the interval $[0,1]$ by the hypothesis and the matrices $\mathbf{X}$ and $\mathbf{I} + (e^{\theta} - 1)\mathbf{X}$ have the same eigenvectors. So almost surely,

$$e^{\theta \mathbf{X}} \preceq \mathbf{I} + (e^{\theta} - 1)\mathbf{X}.$$

Taking the expectation:

$$\mathbb{E}\left[e^{\theta \mathbf{X}}\right] \preceq \mathbf{I} + (e^{\theta} - 1)\mathbb{E}[\mathbf{X}].$$

The matrix on the left-hand side is positive definite, so its logarithm is well-defined. Since matrix logarithm is monotone w.r.t. semidefinite order (see Exc. 2.5.2),

$$\log \mathbb{E}\left[e^{\theta \mathbf{X}}\right] \preceq \log\left(\mathbf{I} + (e^{\theta}-1)\mathbb{E}[\mathbf{X}]\right) \preceq (e^{\theta}-1)\mathbb{E}[\mathbf{X}],$$

where the last domination we used three facts: (1) $\log(1+x) \le x$ for all $x > -1$ so for any self-adjoint matrix $\mathbf{A}$ with $\lambda_{\min}(\mathbf{A}) > -1$, $\log(\mathbf{I}+\mathbf{A}) \preceq \mathbf{A}$; and (2) $(e^{\theta}-1)\mathbb{E}[\mathbf{X}]$ is self-adjoint and has minimum eigenvalue > -1 since $e^{\theta}-1 > -1$ for all $\theta \in \mathbb{R}$ and $\lambda_{\max}(\mathbf{X}) \le 1$. □

Theorem 2.5.15 (Matrix Chernoff inequality)**.** *Let $\mathbf{Y} = \sum_{k=1}^{n} \mathbf{X}_k$, where $\{\mathbf{X}_k : 1 \le k \le n\}$ are independent random matrices with dimension d. Suppose that*

$$\mathbf{O} \preceq \mathbf{X}_k \preceq R\mathbf{I} \quad \textit{for each index } k.$$

Define the lower and upper eigenvalues of the expectation:

$$\mu_{\min} := \lambda_{\min}(\mathbb{E}[\mathbf{Y}]) \quad \textit{and} \quad \mu_{\max} := \lambda_{\max}(\mathbb{E}[\mathbf{Y}]).$$

Then we have

$$\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \le (1-\delta)\mu_{\min}\right) \le d\left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\mu_{\min}/R} \qquad \text{for } 0 < \delta \le 1, \tag{18}$$

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\delta)\mu_{\max}\right) \le d\left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu_{\max}/R} \qquad \text{for } \delta > 0. \tag{19}$$

(c.f. Note that $\log \frac{e^{\delta}}{(1+\delta)^{1+\delta}} = \delta - (1+\delta)\log(1+\delta) \le 0$*.)*

PROOF. Without loss of generality we assume $R > 0$ since otherwise there is nothing to show. By a rescaling argument, it will be enough to prove the assertion for the case $R = 1$. To see this, for example, suppose (19) holds for sums of independent PSD matrices with maximum eigenvalues uniformly bounded by one. In order to deduce (19) for $\mathbf{Y} = \sum_{k=1}^{n} \mathbf{X}_k$ with $R > 0$, consider the rescaled random matrix $\mathbf{Y}' := \mathbf{Y}/R = \sum_{k=1}^{n} \mathbf{X}_k/R$. Note that $\mu_{\max}(\mathbf{Y}') = \mu_{\max}(\mathbf{Y})/R$. Then by the result we just proved above,

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\delta)\mu_{\max}(\mathbf{Y})\right) = \mathbb{P}\left(R\lambda_{\max}(\mathbf{Y}') \ge (1+\delta)R\mu_{\max}(\mathbf{Y}')\right) \le d\left(\frac{e^{-\delta}}{(1+\delta)^{1+\delta}}\right)^{\underbrace{\mu_{\max}(\mathbf{Y}')}_{=\mu_{\max}(\mathbf{Y})/R}}.$$

Hence, without loss of generality, we will assume $R = 1$.

We begin with the tail bound for the maximum eigenvalue $\lambda_{\max}(\mathbf{Y})$ in (19). The Chernoff CGF bound (Lem. 2.5.14) implies

$$\log \mathbb{E}\left[e^{\theta \mathbf{X}_k}\right] \le g(\theta)\,\mathbb{E}[\mathbf{X}_k] \quad \text{where} \quad g(\theta) = e^{\theta} - 1 \qquad \text{for } \theta \in \mathbb{R}. \tag{20}$$

Note that $g(\theta) > 0$ for $\theta > 0$. Fix $\theta > 0$. We observe the following sequence of inequalities:

$$\begin{aligned}
\operatorname{tr}\exp\left(\sum_{i=1}^{n} \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) &\overset{(a)}{\le} \operatorname{tr}\exp\left(g(\theta)\sum_{k}\mathbb{E}[\mathbf{X}_k]\right) \\
&\overset{(b)}{\le} d\,\lambda_{\max}\left(\exp\left(g(\theta)\mathbb{E}[\mathbf{Y}]\right)\right) \\
&\overset{(c)}{=} d\exp\left(g(\theta)\lambda_{\max}\left(\mathbb{E}[\mathbf{Y}]\right)\right) \\
&\overset{(d)}{=} d\exp\left(g(\theta)\mu_{\max}\right),
\end{aligned}$$

where (a) uses the Chernoff CGF bound in (20) along with the monotonicity of tr exp (Exc. 2.5.2), (b) is from bounding the trace of a PSD matrix by the dimension times the maximum eigenvalue, and (c) uses homogeneity of eigenvalues and $g(\theta) > 0$ for $\theta > 0$, and (d) is the definition of $\mu_{\max}$ in the statement.

Now, using the above and the master inequality in Thm. 2.5.10,

$$\begin{aligned}\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \geq t\right) &\leq \inf_{\theta>0} e^{-\theta t} \operatorname{tr}\exp\left(\sum_{i=1}^{n} \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) \\ &\leq d \inf_{\theta>0} \exp\left(-\theta t + g(\theta)\mu_{\max}\right).\end{aligned}$$

Since $g(\theta)$ is a smooth increasing function, the unconstrained global minimizer of the last expression is achieved when $-t + e^{\theta^*}\mu_{\max} = 0$, which is at $\theta^* = \log(t/\mu_{\max})$. Note that $\theta^* > 0$ whenever $t > \mu_{\max}$. Hence the achieved infemum is

$$d \exp\left(-t\log(t/\mu_{\max}) + t - \mu_{\max}\right).$$

Plugging in $t = (1+\delta)\mu_{\max}$ for $\delta > 0$, this becomes

$$d \exp\left(-(1+\delta)\mu_{\max}\log(1+\delta) + \delta\mu_{\max}\right) = d\left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu_{\max}}.$$

This shows (19) with $R = 1$.

Next, we prove the lower tail bound for $\lambda_{\min}(\mathbf{Y})$ in (18). As before, rescale so that $R = 1$. Fix $\theta < 0$. We observe the following sequence of inequalities:

$$\begin{aligned}\operatorname{tr}\exp\left(\sum_{i=1}^{n} \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) &\overset{(a)}{\leq} \operatorname{tr}\exp\left(g(\theta)\sum_{k} \mathbb{E}[\mathbf{X}_k]\right) \\ &\overset{(b)}{\leq} d\,\lambda_{\max}\left(\exp\left(g(\theta)\mathbb{E}[\mathbf{Y}]\right)\right) \\ &\overset{(c)}{=} d \exp\left(g(\theta)\lambda_{\min}\left(\mathbb{E}[\mathbf{Y}]\right)\right) \\ &\overset{(d)}{=} d \exp\left(g(\theta)\mu_{\min}\right),\end{aligned}$$

where (a) uses the Chernoff CGF bound in (20) along with the monotonicity of $\operatorname{tr}\exp$ (Exc. 2.5.2), (b) is from bounding the trace of a PSD matrix by the dimension times the maximum eigenvalue, and (c) uses homogeneity of eigenvalues and $g(\theta) < 0$ for $\theta < 0$, and (d) is the definition of $\mu_{\min}$ in the statement.

Now, using the above and the master inequality in Thm. 2.5.10,

$$\begin{aligned}\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \geq t\right) &\leq \inf_{\theta<0} e^{-\theta t} \operatorname{tr}\exp\left(\sum_{i=1}^{n} \log \mathbb{E}\left[e^{\theta \mathbf{X}_i}\right]\right) \\ &\leq d \inf_{\theta<0} \exp\left(-\theta t + g(\theta)\mu_{\min}\right).\end{aligned}$$

The unconstrained global minimum of the last expression is achieved when $-t + e^{\theta^*}\mu_{\min} = 0$, which is at $\theta^* = \log(t/\mu_{\min})$. Note that $\theta^* < 0$ whenever $t < \mu_{\min}$. Taking $t = (1-\delta)\mu_{\min}$, we obtain (18). □

Exercise 2.5.16. Derive the following consequences of Theorem 2.5.15. For $\delta \in (0, 1]$,

$$\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \leq (1-\delta)\mu_{\min}\right) \leq d \cdot e^{-\delta^2\mu_{\min}/(2R)},$$

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \geq (1+\delta)\mu_{\max}\right) \leq d \cdot e^{-\delta^2\mu_{\max}/(3R)}.$$

These simplifications are often more tractable in practice.

Now we can deduce the following general matrix Bernstein inequality for sums of independent rectangular matrices. This result is perhaps one of the most widely applicable matrix concentration inequalities.

Theorem 2.5.17 (Matrix Bernstein inequality for rectangular matrices)**.** *Let* $\mathbf{Y} = \sum_{k=1}^{n} \mathbf{X}_k$*, where* $\{\mathbf{X}_k : 1 \leq k \leq n\}$ *are independent random matrices with dimension* $d_1 \times d_2$*. Suppose that*

$$\mathbb{E}[\mathbf{X}_k] = \mathbf{O} \quad \textit{and} \quad \|\mathbf{X}_k\|_2 \leq R \quad \textit{for each index } k. \tag{21}$$

Define the matrix variance proxy:

$$\sigma^2 = \max\left\{\|\mathbb{E}[\mathbf{Y}\mathbf{Y}^*]\|_2, \|\mathbb{E}[\mathbf{Y}^*\mathbf{Y}]\|_2\right\} = \max\left\{\sum_{k=1}^n \mathbb{E}[\mathbf{X}_k\mathbf{X}_k^*], \sum_{k=1}^n \mathbb{E}[\mathbf{X}_k^*\mathbf{X}_k]\right\}.$$

Then, for all $t \geq 0$,

$$\mathbb{P}\left(\|\mathbf{Y}\|_2 \geq t\right) \leq (d_1+d_2)\cdot\exp\left(\frac{-t^2/2}{\sigma^2 + Rt/3}\right).$$

Furthermore,

$$\mathbb{E}[\|\mathbf{Y}\|_2] \leq \sqrt{2\sigma^2\log(d_1+d_2)} + \frac{1}{3}R\log(d_1+d_2).$$

Exercise 2.5.18. Establish Theorem 2.5.17 by applying the maximum eigenvalue bound appearing in the proof of Theorem 2.5.13 to the self-adjoint dilation $\mathscr{H}(\mathbf{Y})$.

Exercise 2.5.19 (Matrix Chernoff: Sampling without replacement)**.** Show that the matrix Chernoff inequality (Thm. 2.5.15) continues to hold if $\mathbf{X}_1,\dots,\mathbf{X}_k$ are sampled from a finite collection $\mathscr{T}$ of PSD matrices satisfying (21) *without* replacement. (c.f. The point is that these matrices sampled without replacement are *not* independent.) (*Hint:* Suppose $\mathbf{Y}_1,\dots,\mathbf{Y}_k$ are i.i.d. uniform samples from $\mathscr{T}$. Gross and Nesme [GN10] show that $\mathbb{E}[\operatorname{tr}\exp(\theta\sum_i \mathbf{X}_i)] \leq \mathbb{E}[\operatorname{tr}\exp(\theta\sum_i \mathbf{Y}_i)]$. Use this inequality to replace the sum $\mathbf{Y} = \sum_i \mathbf{X}_i$ of dependent matrices by the sum $\tilde{\mathbf{Y}} = \sum_i \tilde{\mathbf{X}}_i$ of independent matrices in the Laplace bound in Prop. 2.5.4. Then follow the proof of matrix Chernoff.)

Exercise 2.5.20 (Matrix Chernoff: Expectation bound)**.** Let $\mathbf{Y} = \sum_{k=1}^n \mathbf{X}_k$, where $\{\mathbf{X}_k : 1 \leq k \leq n\}$ are independent random matrices with dimension d. Suppose that

$$\mathbf{O} \preceq \mathbf{X}_k \preceq R\mathbf{I} \quad \text{for each index } k.$$

Denote $\mu_{\max} = \lambda_{\max}(\mathbb{E}[\mathbf{Y}])$ and $\mu_{\min} = \lambda_{\min}(\mathbb{E}[\mathbf{Y}])$. Show that for $\theta > 0$,

$$\mathbb{E}[\lambda_{\max}(\mathbf{Y})] \leq \frac{e^\theta - 1}{\theta}\mu_{\max} + \frac{R}{\theta}\log d,$$
$$\mathbb{E}[\lambda_{\min}(\mathbf{Y})] \geq \frac{1 - e^{-\theta}}{\theta}\mu_{\min} - \frac{R}{\theta}\log d.$$

In particular, taking $\theta = 1$, this gives

$$\mathbb{E}[\lambda_{\max}(\mathbf{Y})] \leq 1.72\mu_{\max} + R\log d,$$
$$\mathbb{E}[\lambda_{\min}(\mathbf{Y})] \geq 0.63\mu_{\min} - R\log d.$$

CHAPTER 3

Random Sampling

In applied mathematics, we frequently face the challenge of approximating a complex object with one that has a simpler structure. In some cases, this can be addressed using a probabilistic method known as *empirical approximation*. The core idea is to create a "simple" random object whose expected value matches the target. Then the object of interest is the expectation of the random object, so it can be approximated by the sample mean of a number of independent samples of the random object. One can expect that by taking the mean of a large number of independent samples, the empirical estimator will get close to the population expectation by the virtue of limit theorems (e.g., laws of large numbers) and the error may be controled by using central limit theorem.

A common situation involves approximating a matrix with one that has more structure. For instance, we might want to approximate a matrix by a sparse matrix or a low-rank matrix. Empirical approximation offers a way to achieve this. The matrix Bernstein inequality (Thm. 2.5.13 and Thm. 2.5.17) is a valuable tool for evaluating the quality of these randomized matrix approximations.

3.1. Matrix Monte Carlo

Let $\mathbf{A}$ be a fixed target matrix that we hope to approximate by a more structured matrix. To that end, suppose we can represent the target as a sum of simple matrices:

$$\mathbf{A} = \sum_{i=1}^{N} \mathbf{A}_i, \tag{22}$$

where the total number $\mathbb{N}$ of components is regarded to be very large. The idea is to identify summands $\mathbf{A}_i$ with desirable properties (e.g., sparsity or low rank) that we want our approximation to inherit.

Along with the decomposition (22), we need to construct a set of sampling distribution:

$$p_i := \mathbb{P}(\mathbf{A}_i \textit{ is sampled}) \qquad \text{where} \tag{22}$$
$$\sum_{i=1}^{N} p_i = 1 \quad \text{and} \quad p_i > 0 \quad \text{for} \quad i = 1, \dots, N.$$

Notice that we are free to design the sampling probability distirbution $(p_1, \dots, p_N)$ as we want. Ideally we may assign larger probabilities to the “more important" summands. Here, quantifying what “important" means is the most challenging aspect of randomized matrix approximation. Choosing the right sampling distribution for a specific problem often requires insight and ingenuity. Nevertheless, the matrix Bernstein inequality gives a guiding principle about which distributions lead to the most accurate approximations – by trying to minimize the upper bound that appears in the Bernstein inequalities as a function of the sampling distribution.

Given the decomposition (22) and a fixed sampling distribution (3.1), define a single-shot estimator $\mathbf{X}$, which is a random matrix such that

$$\mathbb{P}(\mathbf{X} = p_i^{-1}\mathbf{A}_i) = p_i \qquad \text{for } 1 \le i \le N.$$

Note that $\mathbf{X}$ is an unbiased estimator of $\mathbf{A}$: $\mathbb{E}[\mathbf{X}] = \mathbf{A}$. Even so, the random matrix $\mathbf{X}$ offers a poor approximation of the target $\mathbf{A}$ because it has a lot more structure. To improve the quality of the approximation,

we average n independent copies of the random matrix $\mathbf{X}$. By doing so, we obtain the so-called 'matrix Monte Carlo'[1] estimator.

Definition 3.1.1 (Matrix Monte Carlo esimtator)**.** Let $\mathbf{A} \in \mathbb{C}^{d_1 \times d_2}$ be a fixed matrix. Suppose there is a random matrix $\mathbf{X}$ such that $\mathbf{A} = \mathbb{E}[\mathbf{X}]$ The n-sample *matrix Monte Carlo estimator* $\bar{\mathbf{X}}_n$ for $\mathbf{A}$ is defined as

$$\bar{\mathbf{X}}_n := n^{-1} \sum_{k=1}^{n} \mathbf{X}_k, \tag{24}$$

where $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. copies of $\mathbf{X}$.

When forming the matrix Monte Carlo estimator in (24), we do not want to use too many samples in the Monte Carlo estimate since we want the approximating matrix $\bar{\mathbf{X}}_n$ to maintain the desired structure (e.g., sparsity, low-rank) that we often inherit to individual samples $\mathbf{X}_1, \dots, \mathbf{X}_n$. Thus, we are interested in the following problem of of determining the sample complexity of matrix Monte Carlo estimator:

$$\textit{Determine the smallest sample size } n = n(\varepsilon) \textit{ such that } \mathbb{E}[\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2] \le \varepsilon. \tag{25}$$

We can address the above problem as an application of the matrix Bernstein inequality (Thm. 2.5.17).

Theorem 3.1.2 (Matrix Monte Carlo for rectangular matrices)**.** *Let $\mathbf{A} \in \mathbb{C}^{d_1 \times d_2}$ be a fixed matrix. Suppose there is a random matrix $\mathbf{X}$ such that $\mathbf{A} = \mathbb{E}[\mathbf{X}]$ and $\|\mathbf{X}\|_2 \le R$ almost surely. Denote*

$$m_2(\mathbf{X}) := \max\left\{\|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\|_2, \|\mathbb{E}[\mathbf{X}\mathbf{X}^*]\|_2\right\}.$$

Form the matrix Monte Carlo estimator for $\mathbf{A}$:

$$\bar{\mathbf{X}}_n := n^{-1} \sum_{k=1}^{n} \mathbf{X}_k,$$

where $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. copies of $\mathbf{X}$. Then we have the following error bound: for $t \ge 0$,

$$\mathbb{P}\left(\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2 \ge t\right) \le (d_1 + d_2) \exp\left(\frac{-nt^2/2}{m_2(\mathbf{X}) + 2Rt/3}\right). \tag{26}$$

Furthermore,

$$\mathbb{E}\left[\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2\right] \le \sqrt{\frac{2m_2(\mathbf{X})\log(d_1+d_2)}{n}} + \frac{2R\log(d_1+d_2)}{3n}. \tag{27}$$

PROOF. Since $\mathbf{X}$ is an unbiased estimator of the target matrix $\mathbf{A}$, we can write

$$\mathbf{Y} := \bar{\mathbf{X}}_n - \mathbf{A} = \sum_{k=1}^{n} \underbrace{n^{-1}(\mathbf{X}_k - \mathbb{E}[\mathbf{X}])}_{=:\mathbf{Y}_k}.$$

Note that the random matrices $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are i.i.d. and has mean zero. In order to apply the matrix Bernstein inequality (Thm. 2.5.17), we need to verify that these matrices have uniformly bounded spectral norm. Indeed, by triangle inequality and Jensen's inequality,

$$\|\mathbf{Y}_k\|_2 \le n^{-1}(\|\mathbf{X}_k\|_2 + \|\mathbb{E}[\mathbf{X}]\|_2) \le n^{-1}(\|\mathbf{X}_k\|_2 + \mathbb{E}[\|\mathbf{X}\|_2]) \le \frac{2R}{n}.$$

Next, to control the matrix variance $\sigma^2 = \max\left\{\|\mathbb{E}[\mathbf{Y}\mathbf{Y}^*]\|_2, \|\mathbb{E}[\mathbf{Y}^*\mathbf{Y}]\|_2\right\}$, first note that

$$\begin{aligned}
\sum_{k=1}^{n} \mathbb{E}[\mathbf{Y}_k\mathbf{Y}_k^*] &= n^{-2} \sum_{k=1}^{n} \mathbb{E}[(\mathbf{X}_k - \mathbb{E}[\mathbf{X}_k])(\mathbf{X}_k - \mathbb{E}[\mathbf{X}_k])^*] \\
&= n^{-1}\mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^*] \\
&= n^{-1}\left(\mathbb{E}[\mathbf{X}\mathbf{X}^*] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^*\right) \\
&\preceq n^{-1}\mathbb{E}[\mathbf{X}\mathbf{X}^*],
\end{aligned}$$

[1] Monte Carlo usually just means the sample mean of (nearly) i.i.d. random samples.

where the last domination is from the fact that $\mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^*$ is PSD. By a similar computation,

$$\sum_{k=1}^{n} \mathbb{E}[\mathbf{Y}_k^* \mathbf{Y}_k] \preceq n^{-1} \mathbb{E}[\mathbf{X}^* \mathbf{X}].$$

It follows that (using Exc. 3.1.3)

$$\sigma^2 \le n^{-1} m_2(\mathbf{X}).$$

We are now prepared to apply the matrix Bernstein inequality (Thm. 2.5.13) to the random matrix $\mathbf{Y}$. This gives the stated results. □

Exercise 3.1.3. Suppose $\mathbf{A} \preceq \mathbf{B}$ (i.e., $\mathbf{B} - \mathbf{A} \succeq \mathbf{O}$) for self-adjoint matrices $\mathbf{A}$ and $\mathbf{B}$. Show that $\|\mathbf{A}\|_2 \le \|\mathbf{B}\|_2$. (*Hint:* Use the relation between eigenvalues and Rayleigh quotient: For self-adjoint $\mathbf{A}$, $\lambda_{\min}(\mathbf{A}) \le \frac{\mathbf{v}^\top \mathbf{A} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}} \le \lambda_{\max}(\mathbf{A})$ for all $\mathbf{v}$.)

Remark 3.1.4 (Sample complexity)**.** Let us return to the motivating question (25) of determining the sample complexity of matrix Monte Carlo estimator (24). In order to make the expected spectral norm estimation error $\mathbb{E}[\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2]$ small, it is sufficient to choose n large enough so that the upper bound in (27) small. Its dominant term is the first one of order $O(n^{-1/2})$, which is expected from to the central limit theorem — the sample mean with n samples fluctuate around the population mean with typical order of $\sqrt{n}$. Thus, we need to have n at least order ε^{-2} (using Landau notation, this is to say $n = \Omega(\varepsilon^{-2})$). More precisely, by inspecting the upper bound in (27), we deduce the following implication:

$$n \ge \frac{2m_2(\mathbf{X}) \log(d_1 + d_2)}{\varepsilon^2} + \frac{2R \log(d_1 + d_2)}{3\varepsilon} \quad \Longrightarrow \quad \mathbb{E}[\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2] \le 2\varepsilon. \tag{28}$$

The required n grows linearly in per-sample second moment $m_2(\mathbf{X})$ and only logarithmically in the dimension of the matrix; Its dependance on the uniform bound R on the spectral norm of the random samples is linear but in the scale of ε^{-1}, which is much smaller compared to ε^{-2}.

If instead we wanted a high-probability bound on the estimation error $\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2$, then we can use the exponential tail bound in (26). Namely, suppose that for fixed $\varepsilon, \delta > 0$, we want to choose n large enough so that with probability at least $1 - \delta$ the spectral norm estimation error is at most ε:

$$\mathbb{P}\left(\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2 \le \varepsilon\right) \ge 1 - \delta.$$

Using (26), we have the following implication:

$$n \ge \frac{2m_2(\mathbf{X}) + 2R\varepsilon/3}{\varepsilon^2/2} \log\left((d_1 + d_2)\delta^{-1}\right) \qquad \Longrightarrow \qquad \mathbb{P}\left(\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2 \le \varepsilon\right) \ge 1 - \delta.$$

Due to the exponential concentration, we can improve very quickly on the significance level δ by increasing the number n of samples with logarithmic dependence $\log \delta^{-1}$. But it is more sample-hungry to meet the precision level ε. If ε is so small that $m_2(\mathbf{X}) \gg R\varepsilon/3$, then n grows as ε^{-2}; if ε is moderate so that $m_2(\mathbf{X}) \le R\varepsilon/3$, then n grows only as ε^{-1}.

Remark 3.1.5 (Designing sampling distribution)**.** As we discussed at the beginning of this section, a typical way to design the random unbiased esimator $\mathbf{X}$ for $\mathbf{A}$ is to first have a decomposition (22) of the target matrix $\mathbf{A}$ and then choose a sampling distribution $(p_1, \dots, p_N)$. As for the latter, Theorem 3.1.2 offers us some guidance on how to choose it. The theorem identifies two summary statistics that control the quality of an empirical approximation in (24). Indeed, we want to select the sampling distribution to ensure that the upper bound R and the per-sample second moment $m_2(\mathbf{X})$ are both as small as possible.

This observation also hints at the possibility of achieving a bias-variance tradeoff when approximating $\mathbf{A}$. Indeed, we might drop all of the "unimportant" terms in the decomposition (22), i.e., those whose sampling probabilities are small. Then we construct a random approximation $\mathbf{X}$ only for the "important" terms that remain.

Remark 3.1.6 (Utility of spectral norm error bound)**.** It is quite valuable that the error bounds in Theorem 3.1.2 involve the spectral norm. This type of estimate simultaneously controls the error in every linear function of the approximation:

$$\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2 \le \epsilon \qquad \Longrightarrow \qquad |\mathrm{tr}(\bar{\mathbf{X}}_n \mathbf{C}) - \mathrm{tr}(\mathbf{A}\mathbf{C})| \le \epsilon \quad \text{for } \|\mathbf{C}\|_1 \le 1,$$

where we have written $\|\cdot\|_1$ for the Schatten 1-norm (see (2)). These bounds also control the error in each eigenvalue $\lambda_j(\bar{\mathbf{X}}_n)$ of the approximation:

$$\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2 \le \epsilon \implies |\lambda_j(\bar{\mathbf{X}}_n) - \lambda_j(\mathbf{A})| \le \epsilon.$$

Furthermore, when there is a gap between two corresponding eigenvalues of $\mathbf{A}$, we can also obtain bounds for the discrepancy between the associated eigenvectors of $\bar{\mathbf{X}}_n$ and $\mathbf{A}$ using perturbation theory [Bha13, Ch. VII].

3.2. Randomized matrix multiplication

As an illustration of the matrix Monte Carlo method, we will develop a toy algorithm for approximating a highly redundant matrix product.

Problem 3.2.1 (Matrix Product)**.** *Compute* $\mathbf{A}\mathbf{B}^*$ *where* $\mathbf{A} \in \mathbb{C}^{m\times d}$ *and* $\mathbf{B} \in \mathbb{C}^{n\times d}$*, with the inner dimension* $d \gg m \wedge n$*. For simplicity, we assume for the rest of this section that* $\|\mathbf{A}\| = \|\mathbf{B}\| = 1$*.*

Key idea: The matrix product $\mathbf{A}\mathbf{B}^*$ can be written out the sum of the rank-one outer product of respective columns. Namely, witing out the columns as $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_d]$ and $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_d]$,

$$\mathbf{A}\mathbf{B}^* = \sum_{k=1}^{d} \mathbf{a}_k \mathbf{b}_k^*.$$

Then, apply Monte Carlo approximation to the sum in the right-hand side.

To form the Monte Carlo estimator, we need a sampling probability distribution $(p_1, \dots, p_d)$. Once such a choise is made, define the random matrix $\mathbf{X} \in \mathbb{C}^{m\times n}$ that satisfies

$$\mathbb{P}\left(\mathbf{X} = p_k^{-1} \mathbf{a}_k \mathbf{b}_k^*\right) = p_k \quad \text{for each } k = 1, \dots, d.$$

Note that $\mathbf{X}$ is an unbiased esimator of $\mathbf{A}\mathbf{B}^*$:

$$\mathbb{E}[\mathbf{X}] = \sum_{k=1}^{d} p_k \left(p_k^{-1} \mathbf{a}_k \mathbf{b}_k^*\right) = \sum_{k=1}^{d} \mathbf{a}_k \mathbf{b}_k^* = \mathbf{A}\mathbf{B}^*.$$

The associated matrix Monte Carlo estimator is the empirical mean of s samples:

$$\bar{\mathbf{X}}_s := \frac{1}{s} \sum_{i=1}^{s} \mathbf{X}_i, \tag{29}$$

where $\mathbf{X}_1, \dots, \mathbf{X}_d$ are i.i.d. copies of $\mathbf{X}$. This procedure produces an unbiased estimator of the matrix product:

$$\mathbb{E}\left[\bar{\mathbf{X}}_s\right] = \mathbf{A}\mathbf{B}^*.$$

The computational cost of constructing the approximation $\bar{\mathbf{X}}_s$ is just $O(smn)$. When the number of samples s is significantly smaller than d, the Monte Carlo method can be much faster than the naïve matrix multiplication, which requires $O(dmn)$ operations.

However, this speed-up comes at the cost of a notable approximation error. The exact number of samples s needed to reach a desired precision ϵ depends on the chosen sampling probabilities. We will explore two common strategies: uniform sampling and importance sampling.

3.2.1. Monte Carlo matrix products via uniform sampling. We start by considering the simplest approach, where each term in the matrix product is sampled with equal probability. That is, $p_k = d^{-1}$ for all $k = 1, \dots, d$. The advantage of uniform sampling is that the distribution is easy to compute and doesn't require access to the matrix factors $\mathbf{A}$ and $\mathbf{B}$.

To analyze uniform sampling, we need to define a measure that captures the uniformity of the columns of a matrix with respect to the standard basis.

Definition 3.2.2 (Coherence statistic)**.** Let $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_d] \in \mathbb{C}^{m\times d}$ be a matrix with the normalization $\|\mathbf{A}\|_2 = 1$, where $\mathbf{a}_k$ denotes the kth column of $\mathbf{A}$. The coherence statistic is defined as

$$\mu(\mathbf{A}) := d \max_k \|\mathbf{a}_k\|^2.$$

Exercise 3.2.3. For $\mathbf{A} \in \mathbb{C}^{m\times d}$ with $\|\mathbf{A}\|_2 = 1$, verify that $\mu(\mathbf{A}) \in [m, d]$. (*Hint*: The lower bound is attained when all the columns of C have equal norm; the upper bound is attained when one column is a standard basis vector.)

We can apply the matrix Monte Carlo theroem (Thm. 3.1.2) to obtain error bounds on the Monte Carlo approximation of matrix products in (29) with uniform sampling distribution.

Corollary 3.2.4 (Approximating matrix products by uniform sampling)**.** *Fix matrices $\mathbf{A} \in \mathbb{C}^{m\times d}$ and $\mathbf{B} \in \mathbb{C}^{n\times d}$, with the inner dimension $d \gg m \wedge n$. Suppose that $\|\mathbf{A}\|_2 = \|\mathbf{B}\|_2 = 1$. Let $\bar{\mathbf{X}}_s$ denote the Monte Carlo estimator for the product $\mathbf{AB}^*$ in (29) where the sampling distribution $(p_1, \dots, p_d)$ is the uniform probability distribution on $[d]$. Then we have the following error bound: for $t \geq 0$,*

$$\mathbb{P}\left(\|\bar{\mathbf{X}}_s - \mathbf{AB}^*\|_2 \geq t\right) \leq (m+n) \exp\left(\frac{-st^2/2}{(\mu(\mathbf{A}) \vee \mu(\mathbf{B})) + (2t/3)(\mu(\mathbf{A}) \vee \mu(\mathbf{B}))}\right).$$

Furthermore,

$$\mathbb{E}\left[\|\bar{\mathbf{X}}_s - \mathbf{AB}^*\|_2\right] \leq \sqrt{\frac{2(\mu(\mathbf{A}) \vee \mu(\mathbf{B}))\log(m+n)}{s}} + \frac{2(\mu(\mathbf{A}) \vee \mu(\mathbf{B}))\log(m+n)}{3s}.$$

In particular, we have the following implication:

$$s \geq \left(\varepsilon^{-2} \vee (2/3)\varepsilon^{-1}\right)(\mu(\mathbf{A}) \vee \mu(\mathbf{B}))\log(m+n) \quad \Longrightarrow \quad \mathbb{E}[\|\bar{\mathbf{X}}_s - \mathbf{AB}^*\|_2] \leq 2\varepsilon. \tag{30}$$

PROOF. In order to apply the Matrix Monte Carlo theorem (Thm. 3.1.2), we need to compute the per-sample second moment $m_2(\mathbf{X})$ and provide an almost sure upper bound $\|\mathbf{X}\| \leq R$ for the sample matrix $\mathbf{X}$ constructed with the uniform sampling distribution.

Recall that $\mathbf{X} = d^{-1}\mathbf{a}_k\mathbf{b}_k^*$ with probability d^{-1} for $1 \leq k \leq d$. Moreover, for column vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\|\mathbf{ab}^*\|_2^2 = \lambda_{\max}\left((\mathbf{ba}^*)(\mathbf{ab}^*)\right) = \|\mathbf{a}\|^2\lambda_{\max}(\mathbf{bb}^*) = \|\mathbf{a}\|^2\|\mathbf{b}\|^2,$$

where the last equality follows from the observation that $\mathbf{bb}^*$ is rank-one and it has an eigenvector $\mathbf{b}$ with associated eigenvalue $\|\mathbf{b}\|^2$. Hence, almost surely,

$$\begin{aligned}
\|\mathbf{X}\|_2 &\leq \max_k \|d^{-1}\mathbf{a}_k\mathbf{b}_k^*\|_2 \\
&= \max_k d^{-1}\|\mathbf{a}_k\|\|\mathbf{b}_k^*\| \\
&\leq \left(\sqrt{d}\max_k \|\mathbf{a}_k\|\right)\left(\sqrt{d}\max_k \|\mathbf{b}_k\|\right) = \sqrt{\mu(\mathbf{A})\mu(\mathbf{B})} \leq \mu(\mathbf{A}) \vee \mu(\mathbf{B}).
\end{aligned}$$

Next, let us estimate the per-sample second moment. Note that

$$\mathbb{E}[\mathbf{X}\mathbf{X}^*] = \sum_{k=1}^{d} p_k \left(p_k^{-1}\mathbf{a}_k\mathbf{b}_k^*\right)\left(p_k^{-1}\mathbf{a}_k\mathbf{b}_k^*\right)^* = \sum_{k=1}^{d} p_k^{-1}\|\mathbf{b}_k\|^2\mathbf{a}_k\mathbf{a}_k^* \\ \preceq d\cdot\left(\max_k \|\mathbf{b}_k\|^2\right)\underbrace{\sum_{k=1}^{d}\mathbf{a}_k\mathbf{a}_k^*}_{=\mathbf{A}\mathbf{A}^*}.$$

It follows that

$$\|\mathbb{E}[\mathbf{X}\mathbf{X}^*]\|_2 \le \mu(\mathbf{B})\|\mathbf{A}\mathbf{A}^*\|_2 = \mu(\mathbf{B})\|\mathbf{A}\|_2^2 = \mu(\mathbf{B}),$$

where the last equality is from the hypothesis $\|\mathbf{A}\|_2 = 1$. A similar computation (with $\|\mathbf{B}\|_2 = 1$) shows that

$$\|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\|_2 \le \mu(\mathbf{A}).$$

Combining the above, we bet

$$m_2(\mathbf{X}) = \|\mathbb{E}[\mathbf{X}\mathbf{X}^*]\| \vee \|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\| \le \mu(\mathbf{A}) \vee \mu(\mathbf{B}).$$

Now invoking Theorem 3.1.2, we obtain the desired bounds. The last implication about the sample complexity follows immediately from (28) and the observation that $x \ge \sqrt{x}$ for $x \ge 1$ (see also Exc. 3.2.3). □

Remark 3.2.5 (Computational cost)**.** When $\mathbf{A}$ and $\mathbf{B}$ have uniformly small columns, it follows from Exercise 3.2.3 that $\mu(\mathbf{A}) \approx n$ and $\mu(\mathbf{B}) \approx m$. In this scenario, we require approximately $s \approx \epsilon^{-2}(m \vee n)\log(m+n)$ samples. The total arithmetic cost for the Monte Carlo approximation is $O((m^2 n \vee mn^2)\log(m+n))$. When $d \gg m \vee n$, the computational cost of the ϵ-approximation is much more efficient compared to the $O(dmn)$ cost of the naïve matrix multiplication.

3.2.2. Monte Carlo matrix products by importance sampling. In the case where columns of $\mathbf{A}$ and $\mathbf{B}$ have drastically different norms, it is natural to adjust the sampling scheme to compensate such inhomogeneity. (We have encountered a similar situation in the development of randomized Karzmacz algorithm in Sec. 1.2) Specifically, we consider the following importance sampling distribution:

$$p_k = \frac{\|\mathbf{a}_k\|^2 + \|\mathbf{b}_k\|^2}{\sum_{k=1}^{d}\left(\|\mathbf{a}_k\|^2 + \|\mathbf{b}_k\|^2\right)} = \frac{\|\mathbf{a}_k\|^2 + \|\mathbf{b}_k\|^2}{\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2} \quad \text{for} \quad k = 1,\dots,d. \tag{31}$$

This construction has a natural interpretation: we sample the k-th term in the decomposition (31) with probability proportional to a measure of its 'energy', which in this case is measured by the sum of the squared norms of the kth columns of the two factor matrices. The choice of the above probabilities is guided by optimizing the statistics that appear in the matrix Mone Carlo bounds in Theorem 3.1.2.

To analyze the performance of matrix Monte Carlo estimation with importance sampling, we recall the definition of the stable rank:

Definition 3.2.6 (Stable rank)**.** The *stable rank* of a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ is defined as the ratio of the squared Frobenius and ℓ_2-norms:

$$\operatorname{srank}(\mathbf{A}) = \frac{\|\mathbf{A}\|_F^2}{\|\mathbf{A}\|_2^2} \in [1, \operatorname{rank}(\mathbf{A})].$$

Exercise 3.2.7 (Stable rank versus coherence)**.** Under the assumption that $\|\mathbf{A}\| = 1$, show that

$$\operatorname{srank}(\mathbf{A}) \le \mu(\mathbf{A}).$$

As before, we will apply the matrix Monte Carlo theroem (Thm. 3.1.2) to obtain error bounds on the Monte Carlo approximation of matrix products in (29) with importance sampling in (31).

Corollary 3.2.8 (Approximating matrix products by importance sampling)**.** *Fix matrices $\mathbf{A} \in \mathbb{C}^{m\times d}$ and $\mathbf{B} \in \mathbb{C}^{n\times d}$, with the inner dimension $d \gg m \wedge n$. Suppose that $\|\mathbf{A}\|_2 = \|\mathbf{B}\|_2 = 1$. Let $\bar{\mathbf{X}}_s$ denote the Monte Carlo estimator for the product $\mathbf{AB}^*$ in (29) where the sampling distribution $(p_1, \dots, p_d)$ is the importance sampling distribution on $[d]$ in (31). Denote the average stable rank*

$$\operatorname{asr}(\mathbf{A}, \mathbf{B}) := \frac{1}{2}\left(\operatorname{srank}(\mathbf{A}) + \operatorname{srank}(\mathbf{B})\right).$$

Then we have the following error bound: for $t \geq 0$,

$$\mathbb{P}\left(\|\bar{\mathbf{X}}_s - \mathbf{AB}^*\|_2 \geq t\right) \leq (m+n)\exp\left(\frac{-st^2/2}{\operatorname{asr}(\mathbf{A},\mathbf{B}) + (2t/3)\operatorname{asr}(\mathbf{A},\mathbf{B})}\right).$$

Furthermore,

$$\mathbb{E}\left[\|\bar{\mathbf{X}}_s - \mathbf{AB}^*\|_2\right] \leq \sqrt{\frac{4\operatorname{asr}(\mathbf{A},\mathbf{B})\log(m+n)}{s}} + \frac{2\operatorname{asr}(\mathbf{A},\mathbf{B})\log(m+n)}{3s}.$$

In particular, we have the following implication:

$$s \geq 4\left(\varepsilon^{-2} \vee (1/6)\varepsilon^{-1}\right)\operatorname{asr}(\mathbf{A},\mathbf{B})\log(m+n) \quad \Longrightarrow \quad \mathbb{E}[\|\bar{\mathbf{X}}_s - \mathbf{AB}^*\|_2] \leq 2\epsilon. \tag{32}$$

PROOF. We compute the per-sample second moment $m_2(\mathbf{X})$ and provide an almost sure upper bound $\|\mathbf{X}\| \leq R$ for the sample matrix $\mathbf{X}$ constructed with the imporatnce sampling distribution.

Recall that $\mathbf{X} = p_k^{-1}\mathbf{a}_k\mathbf{b}_k^*$ with probability p_k in (31) for $1 \leq k \leq d$. Hence, almost surely,

$$\begin{aligned}
\|\mathbf{X}\|_2 &\leq \max_k \|p_k^{-1}\mathbf{a}_k\mathbf{b}_k^*\|_2 \\
&= \left(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2\right)\max_k \frac{\|\mathbf{a}_k\|\|\mathbf{b}_k^*\|}{\|\mathbf{a}_k\|^2 + \|\mathbf{b}_k\|^2} \\
&\leq \frac{1}{2}\left(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2\right) \\
&= \frac{1}{2}\left(\frac{\|\mathbf{A}\|_F^2}{\|\mathbf{A}\|_2} + \frac{\|\mathbf{B}\|_F^2}{\|\mathbf{B}\|_2}\right) =
\end{aligned}$$

where the second inequality uses the inequaliy $a + b \geq 2\sqrt{ab}$ for $a, b \geq 0$ and the equality that follows uses the hypothesis $\|\mathbf{A}\|_2 = \|\mathbf{B}\|_2 = 1$.

Next, let us estimate the per-sample second moment. Note that

$$\begin{aligned}
\mathbb{E}[\mathbf{XX}^*] = \sum_{k=1}^d p_k\left(p_k^{-1}\mathbf{a}_k\mathbf{b}_k^*\right)\left(p_k^{-1}\mathbf{a}_k\mathbf{b}_k^*\right)^* &= \sum_{k=1}^d p_k^{-1}\|\mathbf{b}_k\|^2\mathbf{a}_k\mathbf{a}_k^* \\
&= \left(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2\right)\sum_{k=1}^d \frac{\|\mathbf{b}_k\|^2}{\|\mathbf{a}_k\|_2 + \|\mathbf{b}_k\|_2}\mathbf{a}_k\mathbf{a}_k^* \\
&\preceq \left(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2\right)\mathbf{AA}^*.
\end{aligned}$$

It follows that

$$\|\mathbb{E}[\mathbf{XX}^*]\|_2 \leq \left(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2\right)\|\mathbf{AA}^*\|_2 = \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2,$$

where the last equality is from the hypothesis $\|\mathbf{A}\|_2 = 1$. A similar computation (with $\|\mathbf{B}\|_2 = 1$) shows that

$$\|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\|_2 \leq \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2.$$

Combining the above, we bet

$$m_2(\mathbf{X}) = \|\mathbb{E}[\mathbf{XX}^*]\| \vee \|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\| \leq \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2 = \operatorname{srank}(\mathbf{A}) + \operatorname{srank}(\mathbf{B}).$$

Now invoking Theorem 3.1.2, we obtain the desired bounds. The last implication about the sample complexity follows immediately from (28). □

Remark 3.2.9 (Computational cost)**.** Exercise 3.2.7 shows that

$$\operatorname{srank}(\mathbf{A}) + \operatorname{srank}(\mathbf{B}) \le \mu(\mathbf{A}) + \mu(\mathbf{B}),$$

which leads to the conclusion that the bound (32) for the sample complexity of matrix multiplication using importance sampling always improves upon the uniform sampling bound (30). This improvement is especially notable for matrices with low rank (small stable rank) and non-uniform columns (large coherence). However, while the uniform sampling probabilities are easily obtained without additional computation, calculating the importance sampling probabilities in (31) requires additional computational cost of $O(d(m+n))$ with access to all columns of the factor matrices prior to the approximation.

3.3. Randomized matrix sparsification

Many data analysis tasks involve large, dense matrices that often contain redundant information. For instance, experiments that collect numerous variables across many subjects usually result in low-rank data matrices because subjects tend to be similar to one another (e.g., Netflix user-movie rating matrix[2]). Many of the questions we ask about these data matrices can be tackled using spectral methods, particularly through singular value decomposition.

When a data matrix is approximately low rank, it has much fewer degrees of freedom compared to its full dimension. This allows us to build a simpler approximation that retains most of the matrix's essential information. One approach for this is to replace the dense matrix with a sparse matrix that is close in terms of spectral-norm distance by selecting a subset of entries and turning all other entries to zero. This procedure can also be viewed as a Monte Carlo approximation of the target matrix if we decompose it as the sum of single-entry matrices.

Sparsifying a matrix offers several benefits. First, sparse matrices require significantly less storage than dense matrices. Second, many algorithms designed for spectral computations run more efficiently on sparse matrices.

In this section, we explore a recent approach to randomized sparsification introduced by Kundu and Drineas [KD14].

Let $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{d_1 \times d_2}$ be a fixed target matrix that may be dense. The matrix sparsification problem requires us to find a sparse matrix $\hat{\mathbf{A}}$ that is close to $\mathbf{A}$ in spectral norm. The key to our randomized matrix sparsification is to decompose $\mathbf{A}$ into the sum of single-entry matrices:

$$\mathbf{A} = \sum_{i=1}^{d_1} \sum_{j=1}^{d_2} a_{ij} \mathbf{E}_{ij},$$

where $\mathbf{E}_{ij}$ denotes the $d_1 \times d_2$ matrix whose (i, j) entry equals one and all other entries are zero.

As in the usual matrix Monte Carlo approximation, we would like to sample some number s of the summands above. Here, the sampling probability would directly reflect our preference in the selection. We will use the following non-trivial sampling probabilities:

$$p_{ij} := \frac{1}{2}\left(\frac{|a_{ij}|^2}{\|\mathbf{A}\|_F^2} + \frac{|a_{ij}|}{\|\mathbf{A}\|_{\ell_1}}\right) \qquad \text{for } (i,j) \in [m] \times [n], \tag{33}$$

where $\|\cdot\|_{\ell_1}$ denotes the entrywise ℓ_1-norm of matrices (i.e., the absolute sum of the entries). It is easy to check that (33) indeed defines a probability distribution over the mn entries of $\mathbf{A}$.

Now, define a random matrix $\mathbf{X}$ of size $d_1 \times d_2$, which has exactly one non-zero entry:

$$\mathbb{P}\left(\mathbf{X} = \frac{a_{ij}}{p_{ij}} \mathbf{E}_{ij}\right) = p_{ij}. \tag{34}$$

[2] See https://en.wikipedia.org/wiki/Matrix_completion

Note that $p_{ij} = 0$ if and only if $a_{ij} = 0$. We will adopt the convention of writing $0/0 = 0$, so zero entries do not require special handling. One can easily check that such a random matrix $\mathbf{X}$ is an unbiased estimator of $\mathbf{A}$:

$$\mathbb{E}[\mathbf{X}] = \sum_{i,j} p_{ij} \left(\frac{a_{ij}}{p_{ij}} \mathbf{E}_{ij}, \right) = \sum_{i,j} a_{ij} \mathbf{E}_{ij} = \mathbf{A}.$$

We can then form the following matrix Monte Carlo estimator

$$\bar{\mathbf{X}}_s := \frac{1}{s} \sum_{k=1}^{s} \mathbf{X}_k, \tag{35}$$

where $\mathbf{X}_1, \dots, \mathbf{X}_s$ are i.i.d. copies of $\mathbf{X}$ in (34).

Note that since each $\mathbf{X}_k$ has exactly one non-overlapping possibly nonzero entries, the number of nonzero entries in the approximation $\bar{\mathbf{X}}_s$, i.e., the ℓ_0-norm of it, is at most s. Therefore, in order to make $\bar{\mathbf{X}}_s$ a sparse approximation of $\mathbf{A}$, we would like to keep the sample size s as small as possible. However, we may need to use some good number of samples to bring down the variance of $\bar{\mathbf{X}}_s$, since individual parts $\mathbf{X}_k$ has large variance since it contains a single nonzero entry at an arbitrary position. The challenge again is to keep s as small as possible while making the spectral norm approximation error $\mathbb{E}[\|\mathbf{A} - \bar{\mathbf{X}}_s\|_2]$ small.

Corollary 3.3.1 (Matrix sparsification)**.** *Fix a matrix $\mathbf{A} \in \mathbb{C}^{d_1 \times d_2}$. Let $\bar{\mathbf{X}}_s$ denote the Monte Carlo estimator for $\mathbf{A}$ in (35). Then we have the following error bound: for $t \geq 0$,*

$$\mathbb{P}\left(\|\bar{\mathbf{X}}_s - \mathbf{A}\|_2 \geq t \right) \leq (d_1 + d_2) \exp\left(\frac{-st^2/2}{2(d_1 \vee d_2)\|\mathbf{A}\|_F^2 + (4t/3)\|\mathbf{A}\|_{\ell_1}} \right).$$

Furthermore,

$$\mathbb{E}\left[\|\bar{\mathbf{X}}_s - \mathbf{A}\|_2 \right] \leq \sqrt{\frac{4\|\mathbf{A}\|_F^2 (d_1 \vee d_2) \log(d_1 + d_2)}{s}} + \frac{4\|\mathbf{A}\|_{\ell_1} \log(d_1 + d_2)}{3s}. \tag{36}$$

In particular, we have the following implication: For each $\varepsilon \in [0, 1]$,

$$s \geq 4\varepsilon^{-2} \mathrm{srank}(\mathbf{A})(d_1 \vee d_2) \log(d_1 + d_2) \qquad \Longrightarrow \qquad \frac{\mathbb{E}[\|\bar{\mathbf{X}}_s - \mathbf{A}\|_2]}{\|\mathbf{A}\|_2} \leq \varepsilon + \frac{\varepsilon^2}{3\sqrt{\mathrm{srank}(\mathbf{A})}} \overset{(a)}{\leq} 2\varepsilon. \tag{37}$$

PROOF. We compute the per-sample second moment $m_2(\mathbf{X})$ and provide an almost sure upper bound $\|\mathbf{X}\| \leq L$ for the sample matrix $\mathbf{X}$ constructed with the imporatnce sampling distribution. In our analysis, a crucial fact is the following lower bounds on the sampling probabilities:

$$p_{ij} \geq \frac{1}{2} \frac{|a_{ij}|^2}{\|\mathbf{A}\|_F^2} \qquad \text{and} \qquad p_{ij} \geq \frac{1}{2} \frac{|a_{ij}|}{\|\mathbf{A}\|_{\ell_1}}. \tag{38}$$

Recall that $\mathbf{X} = p_{ij}^{-1} a_{ij} \mathbf{E}_{ij}$ with probability p_{ij} in (33). Hence, almost surely,

$$\|\mathbf{X}\|_2 \leq \max_{i,j} \|p_{ij}^{-1} a_{ij} \mathbf{E}_{ij}\|_2 \leq 2\|\mathbf{A}\|_{\ell_1},$$

where the second inequality uses the second lower bound on p_{ij} in (38) and that $\|\mathbf{E}_{ij}\| = 1$. Thus $\|\mathbf{X}\|_2 \leq R = 2\|\mathbf{A}\|_{\ell_1}$ almost surely.

Next, let us estimate the per-sample second moment. Note that using the first lower bound on p_{ij} in (38),

$$\begin{aligned}\mathbb{E}[\mathbf{X}\mathbf{X}^*] &= \sum_{i,j} p_{ij} \frac{1}{p_{ij}^2} (a_{ij}\mathbf{E}_{ij})(a_{ij}\mathbf{E}_{ij})^* \\ &= \sum_{i,j} \frac{a_{ij}^2}{p_{ij}} \mathbf{E}_{ii}^{d_1 \times d_1} \\ &\preceq 2\|\mathbf{A}\|_F^2 \sum_{i,j} \mathbf{E}_{ii}^{d_1 \times d_1} = 2d_2 \|\mathbf{A}\|_F^2 \mathbf{I}_{d_1 \times d_1}.\end{aligned}$$

It follows that

$$\|\mathbb{E}[\mathbf{X}\mathbf{X}^*]\|_2 \le 2d_2\|\mathbf{A}\|_F^2.$$

A similar computation shows that

$$\|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\|_2 \le 2d_1\|\mathbf{A}\|_F^2$$

Combining the above, we bet

$$m_2(\mathbf{X}) = \|\mathbb{E}[\mathbf{X}\mathbf{X}^*]\| \vee \|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\| \le 2(d_1 \vee d_2)\|\mathbf{A}\|_F^2.$$

Now invoking Theorem 3.1.2, we obtain the desired bounds.

Lastly, we deduce the sample complexity of randomized matrix sparsification claimed in (37). First, we use Cauchy-Schwarz inequality to bound the ℓ_1-norm by the Frobenius norm:

$$\|\mathbf{A}\|_{\ell_1} \le \sqrt{d_1 d_2}\|\mathbf{A}\|_F \le (d_1 \vee d_2)\|\mathbf{A}\|_F.$$

Hence, putting the expected error bound in (36) on a relative scale, we get:

$$\frac{\mathbb{E}\left[\|\tilde{\mathbf{X}}_n - \mathbf{A}\|_2\right]}{\|\mathbf{A}\|_2} \le \underbrace{\frac{\|\mathbf{A}\|_F}{\|\mathbf{A}\|_2}}_{=\sqrt{\operatorname{srank}(\mathbf{A})}} \left(\sqrt{\frac{4(d_1 \vee d_2)\log(d_1+d_2)}{s}} + \frac{4(d_1 \vee d_2)\log(d_1+d_2)}{3s}\right).$$

From this, we can immediately deduce (37). Note that (a) in (37) follows from $\operatorname{srank}(\mathbf{A}) \ge 1$. □

Remark 3.3.2. When can we approximate a dense matrix by a sparse matrix with small spectral norm error? Note that the matrix $\mathbf{A}$, if dense, will have order $d_1 d_2 = (d_1 \wedge d_2) \cdot (d_1 \vee d_2)$ entries. Hence the required sparsity level s should be much less then this to obtain a sparse approximation of $\mathbf{A}$. From the sample complexity in (37), this is the case if $\operatorname{srank}(\mathbf{A}) \ll d_1 \wedge d_2$. Hence in this case, we can achieve a significant reduction in the number of non-zero entries needed to retain the spectral information of $\mathbf{A}$.

3.4. Kernel approximation via random features

As a final example of empirical matrix approximation, we will explore a modern concept from machine learning known as random features. While this technique may seem more advanced than randomized sparsification or matrix multiplication, it follows the same core principles. Ali Rahimi and Ben Recht introduced random feature maps in [RR07]. The analysis discussed here is based on the work of David Lopez-Paz et al. in [LPSS+14].

Let $\mathscr{X}$ be a set of possibly high-dimensional raw features of data points for learning and inference tasks. We define a bounded similarity measure Φ between pairs of points in $\mathscr{X}$ as follows:

Definition 3.4.1 (Kernel)**.** For a set $\mathscr{X}$, a *kernel* (or a similarity measure) Φ is a map $\Phi : \mathscr{X} \times \mathscr{X} \to [-1,1]$ such that $\Phi(\mathbf{x},\mathbf{x}) = 1$ for all $\mathbf{x} \in \mathscr{X}$. We say Φ is *symmetric* if $\Phi(\mathbf{x},\mathbf{y}) = \Phi(\mathbf{y},\mathbf{x})$ for all $\mathbf{x},\mathbf{y} \in \mathscr{X}$. Suppose that $\mathbf{x}_1,\dots,\mathbf{x}_N \in \mathscr{X}$ are observations. The corresponding *kernel matrix* $\mathbf{G} = (g_{ij}) \in \mathbb{R}^{N \times N}$ of Φ is the $N \times N$ matrix regarding the values of Φ function for each pair of data points:

$$g_{ij} = \Phi(\mathbf{x}_i,\mathbf{x}_j) \quad \text{for } i,j \in \{1,\dots,N\}.$$

We say that the kernel Φ is *positive definite* if the kernel matrix $\mathbf{G}$ is positive semidefinite for any choice of observations $\{\mathbf{x}_i\} \subseteq \mathscr{X}$.

The kernel Φ on $\mathscr{X}$ assigns a value of $+1$ when comparing identical points and lower values for dissimilar points.

Example 3.4.2 (Angular similarity)**.** A simple example of a kernel is the angular similarity between a pair of points in Euclidean space:

$$\Phi(\mathbf{x},\mathbf{y}) = \frac{2}{\pi}\arcsin\left(\frac{\langle \mathbf{x},\mathbf{y}\rangle}{\|\mathbf{x}\|\,\|\mathbf{y}\|}\right) = 1 - \frac{2\measuredangle(\mathbf{x},\mathbf{y})}{\pi} \quad \text{for } \mathbf{x},\mathbf{y} \in \mathbb{R}^d,$$

where $\measuredangle(\cdot,\cdot)$ denotes the planar angle between two vectors, measured in radians. As usual, we adopt the convention that $0/0 = 0$. ▲

In Euclidean spaces, many statistical learning algorithms rely solely on inner products between pairs of observed featrue vectors. These methods can be generalized to the kernel setting by substituting each inner product with a kernel evaluation. As a result, kernel matrices are useful for tasks such as classification, regression, and feature selection. Kernels offer a distinct advantage because they can capture data-specific, possibly non-Euclidean geometry in data. This idea, commonly known as the "kernel trick", is a foundational concept in modern machine learning.

A significant challenge for kernel-based algorithms is the size N of the kernel matrix, which is the number of data point used to evaluate the kernel. Constructing kernel matrix G costs $O(dN^2)$ computation, where d is the number of parameters defining each point in the set $\mathscr{X}$.

Nevertheless, there is a potential advantage: large datasets are often redundant, and this redundancy results in the kernel matrix being approximately *low-rank*. As a result, the kernel matrix can be substituted with a low-rank approximation, which, for certain similarity measures, can be achieved through empirical approximation. Utilizing this low-rank structure provides benefits in terms of storing and processing kernel matrices.

3.4.1. Random feature map and construction of Monte Carlo estimator of the kernel matrix. In some instances, a positive-definite kernel can be represented as an expectation, enabling us to construct a Monte Carlo approximation of the kernel matrix based using this formulation. The key to such an expectation representation of kernel matrices is the following notion of 'reproducing property of a random feature map'.

Let $\mathscr{W}$ be a probabiliy space equipped with a probability measure μ. Define a bounded feature map:

$$\psi : \mathscr{X} \times \mathscr{W} \to [-b,+b] \quad \text{where} \quad b \geq 0.$$

Consider a random variable w that takes values in $\mathscr{W}$ and is distributed according to the measure μ. We assume that this random variable satisfies the *reproducing property*:

$$\Phi(\mathbf{x},\mathbf{y}) = \mathbb{E}_{w\sim\mu}\left[\psi(\mathbf{x};w)\cdot\psi(\mathbf{y};w)\right] \quad \text{for all} \quad \mathbf{x},\mathbf{y} \in \mathscr{X}. \tag{39}$$

The pair (ψ, w) is referred to as a random feature map for the kernel Φ.

We aim to approximate the kernel matrix using a set $\{\mathbf{x}_1,\ldots,\mathbf{x}_N\} \subset \mathscr{X}$ of observations. To do so, we draw a random vector $w \in \mathscr{W}$ distributed according to μ. We then sample a random vector $\mathbf{z} \in \mathbb{R}^N$ by applying the feature map to each data point with the *same* choice of the random vector w. That is,

$$\mathbf{z} = \mathbf{z}(w) = \begin{bmatrix} z_1(w) \\ \vdots \\ z_N(w) \end{bmatrix} := \begin{bmatrix} \psi(\mathbf{x}_1;w) \\ \vdots \\ \psi(\mathbf{x}_N;w) \end{bmatrix}. \tag{40}$$

The vector $\mathbf{z}$ is often referred to as a *random feature*. By applying the reproducing property (39) for the random feature map, we have

$$g_{ij} = \Phi(\mathbf{x}_i,\mathbf{x}_j) = \mathbb{E}_{w\sim\mu}\left[\psi(\mathbf{x}_i;w)\cdot\psi(\mathbf{x}_j;w)\right] = \mathbb{E}_w\left[z_i(w)\cdot z_j(w)\right] \quad \text{for } i,j = 1,\ldots,N.$$

That is, the kernel matrix $\mathbf{G}$ is the expectation of the rank-one random matrix $\mathbf{X} := \mathbf{z}\mathbf{z}^*$:

$$\mathbf{G} = \mathbb{E}_{w\sim\mu}\left[\mathbf{z}\mathbf{z}^*\right]. \tag{41}$$

Therefore, the random matrix $\mathbf{X} = \mathbf{z}\mathbf{z}^*$ is an unbiased rank-one estimator for the kernel matrix G. In particular, this formulation of the kernel matrix shows that random feature maps, as defined here, exist only for positive-definite kernels.

As usual, we construct a more refined empirical approximation of the kernel matrix G by averaging several realizations of the simple estimator $\mathbf{X}$:

$$\bar{\mathbf{X}}_s := \frac{1}{s}\sum_{k=1}^{s}\mathbf{X}_k \quad \text{where } \mathbf{X}_1,\dots,\mathbf{X}_n \text{ are i.i.d. copies of } \mathbf{X} = \mathbf{z}\mathbf{z}^*. \tag{42}$$

To form the i.i.d. samples of rank-one gram matrices $\mathbf{X}_1,\dots,\mathbf{X}_s$, we can sample i.i.d. copies of the random variables $w_1,\dots,w_s$ from μ and let $\mathbf{X}_i := \mathbf{z}(w_i)\mathbf{z}(w_i)^*$ for $i = 1,\dots,s$. The key question is how many random features are needed for the estimator to be sufficiently accurate.

The quality of the Monte Carlo estimation of $\mathbf{G}$ above will depend on the 'intrinsic dimension' of $\mathbf{G}$.

Definition 3.4.3 (Intrinsic dimension)**.** For a PSD matrix $\mathbf{A}$, the intrinsic dimension is the quantity

$$\operatorname{intdim}(\mathbf{A}) := \frac{\operatorname{tr}(\mathbf{A})}{\|\mathbf{A}\|_2} = \operatorname{srank}(\mathbf{A}^{1/2}). \tag{43}$$

The last equality in (43) is due to $\|\mathbf{A}^{1/2}\|_2^2 = \|\mathbf{A}\|_2$ (consider SVD of $\mathbf{A}$) and $\|\mathbf{A}^{1/2}\|_F^2 = \operatorname{tr}((\mathbf{A}^{1/2})^*\mathbf{A}^{1/2}) = \operatorname{tr}(\mathbf{A})$. We interpret the intrinsic dimension as a measure of the number of dimensions where $\mathbf{A}$ has significant spectral content. Let us make a few observations that support this view.

Exercise 3.4.4. PSD matrices must be self-adjoint Show that for a PSD matrix $\mathbf{A}$,

$$1 \le \operatorname{intdim}(\mathbf{A}) \le \operatorname{rank}(\mathbf{A}) \le \dim(\mathbf{A}).$$

By expressing the trace and the norm in terms of the eigenvalues, we can verify that

The first inequality is attained precisely when $\mathbf{A}$ has rank one, while the last inequality is attained precisely when $\mathbf{A}$ has full rank. The middle inequality is attained when the nonzero ei

Corollary 3.4.5 (Monte Carlo kernel matrix approximation)**.** *Let Φ be a symmetric positive-definite kernel on a feature space $\mathscr{X}$ and let $\mathbf{G}$ denote its kernel matrix corresponding to fixed data points $\mathbf{x}_1,\dots,\mathbf{x}_N \in \mathscr{X}$. Suppose Φ admits a reproducing random feature map ψ as in (39) with bounded range $[-b,b]$. Let $\bar{\mathbf{X}}_n$ denote the n-sample Monte Carlo estimator of $\mathbf{G}$ in (42). Then we have the following error bound: for $t \ge 0$,*

$$\mathbb{P}\left(\|\bar{\mathbf{X}}_s - \mathbf{G}\|_2 \ge t\right) \le 2N\exp\left(\frac{-st^2/2}{bN\|\mathbf{G}\|_2 + (2t/3)bN}\right). \tag{44}$$

Furthermore,

$$\mathbb{E}\left[\|\bar{\mathbf{X}}_s - \mathbf{A}\|_2\right] \le \sqrt{\frac{2b^2N\|\mathbf{G}\|_2\log(2N)}{s}} + \frac{2b^2N\log(2N)}{3s}. \tag{45}$$

In particular, we have the following implication: For each $\varepsilon \in [0,1]$,

$$s \ge 2b^2\varepsilon^{-2}\operatorname{intdim}(\mathbf{G})\log(2N) \qquad \Longrightarrow \qquad \frac{\mathbb{E}[\|\bar{\mathbf{X}}_n - \mathbf{G}\|_2]}{\|\mathbf{G}\|_2} \le \varepsilon + \varepsilon^2/3. \tag{46}$$

PROOF. As usual, we will compute the per-sample second moment $m_2(\mathbf{X})$ and provide an almost sure upper bound $\|\mathbf{X}\| \le L$ for the sample matrix $\mathbf{X}$ constructed with the imporatnce sampling distribution.

First, recall that $\mathbf{X} = \mathbf{z}\mathbf{z}^*$ where $\mathbf{z}$ is the random feature vector in (40). Hence

$$\|\mathbf{X}\|_2 = \|\mathbf{z}\mathbf{z}^*\| = \|\mathbf{z}\|^2 \le b^2N.$$

Second, let us estimate the per-sample second moment. Note that

$$\mathbb{E}[\mathbf{X}\mathbf{X}^*] = \mathbb{E}[(\mathbf{z}\mathbf{z}^*)(\mathbf{z}\mathbf{z}^*)] = \|\mathbf{z}\|^2 \mathbb{E}[\mathbf{z}\mathbf{z}^*] = \|\mathbf{z}\|^2 \mathbf{G} \preceq b^2 N \mathbf{G},$$

where the last identity is from the fact that $\mathbf{X} = \mathbf{z}\mathbf{z}^*$ is an unbiased estimator fo $\mathbf{G}$ (see (41)) and the semi-definite ordering is from $\|\mathbf{z}\|^2 \le bN$ and that G is PD by the hypothesis. Since $\mathbf{X}^* = \mathbf{X}$, it follows that

$$\|\mathbb{E}[\mathbf{X}^*\mathbf{X}]\|_2 = \|\mathbb{E}[\mathbf{X}\mathbf{X}^*]\|_2 \le b^2 N \|\mathbf{G}\|_2.$$

Hence $m_2(\mathbf{X}) \le b^2 N \|\mathbf{G}\|_2$.

Now invoking Theorem 3.1.2, we obtain (44) and (45) immediately. For the sample complexity bound in (46), note that $\operatorname{tr}(\mathbf{G}) = N$ since $\Phi(\mathbf{x},\mathbf{x}) = 1$ for all $\mathbf{x} \in \mathscr{X}$. Hence dividing both sides of (45) by $\|\mathbf{G}\|_2$,

$$\frac{\mathbb{E}\left[\|\bar{\mathbf{X}}_n - \mathbf{A}\|_2\right]}{\|\mathbf{G}\|_2} \le \sqrt{\frac{2b^2 \operatorname{intdim}(\mathbf{G}) \log(2N)}{s}} + \frac{2b^2 \operatorname{intdim}(\mathbf{G}) \log(2N)}{3s}.$$

Then (46) follows immediately from the above. □

3.4.2. Examples of random feature maps. In this section, we provide some examples of random feature maps.

Example 3.4.6 (Angular similarity)**.** Recall the angular similarity (Ex. 3.4.2) defined on $\mathbb{R}^d$. We can construct a random feature map using a classical result from Euclidean geometry. If we draw w uniformly from the unit sphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$, then

$$\Phi(\mathbf{x},\mathbf{y}) = 1 - \frac{2\measuredangle(\mathbf{x},\mathbf{y})}{\pi} = \mathbb{E}_w\left[\operatorname{sgn}\langle \mathbf{x}, w\rangle \cdot \operatorname{sgn}\langle \mathbf{y}, w\rangle\right] \quad \forall \mathbf{x},\mathbf{y} \in X. \tag{47}$$

The easy proof of this relation can be seen from Figure 3.4.1.

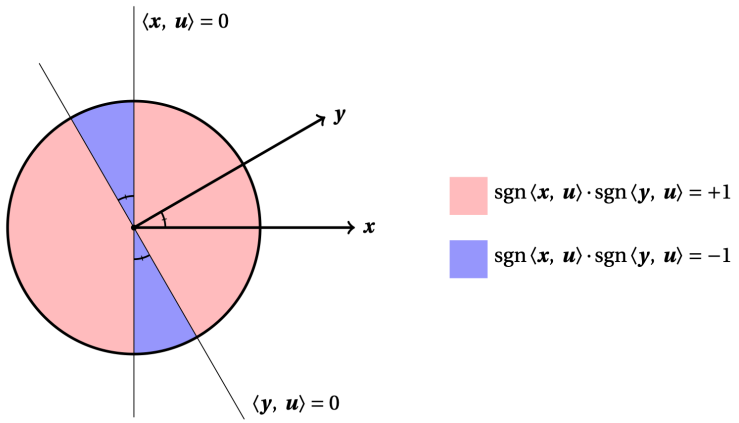


FIGURE 3.4.1. The angular similarity between two vectors. Let $\mathbf{x}$ and $\mathbf{y}$ be nonzero vectors in $\mathbb{R}^2$ with angle $\measuredangle(x,y)$. The red region contains the directions $\mathbf{u}$ where the product $\operatorname{sgn}\langle x,u\rangle \cdot \operatorname{sgn}\langle y,u\rangle = +1$, and the blue region contains the directions where the product equals -1. The blue region subtends a total angle of $2\measuredangle(x,y)$, and the red region subtends $2\pi - 2\measuredangle(x,y)$. Figure and caption excerpted from [T+15].

According to (47), we set $\mathscr{W} = \mathbb{S}^{d-1}$ with the uniform probability measure, and we define the feature map

$$\psi(\mathbf{x}; w) := \operatorname{sgn}\langle \mathbf{x}, w\rangle.$$

The reproducing property (39) follows immediately from (47). Therefore, the pair (ψ, w) forms a random feature map for the angular similarity kernel. ▲

Next, we describe an important class of kernels that can be expressed using random feature maps.

Example 3.4.7 (Translation invariant kernels)**.** A kernel on $\mathbb{R}^d$ is called *translation invariant* if there exists a function $\phi : \mathbb{R}^d \to \mathbb{R}$ such that

$$\Phi(\mathbf{x}, \mathbf{y}) = \phi(\mathbf{x} - \mathbf{y}) \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

Bôchner's Theorem, a classical result from harmonic analysis, provides a representation for each continuous, positive-definite, translation-invariant kernel:

$$\Phi(x, y) = c \int_{\mathbb{R}^d} e^{i\langle \mathbf{x}, w\rangle} \cdot e^{-i\langle \mathbf{y}, w\rangle} \, d\mu(w) \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d, \tag{48}$$

where the positive scaling factor c and the probability measure μ depend only on the function ϕ.

The formula (48) gives rise to a (complex-valued) random feature map:

$$\psi_c(\mathbf{x}; w) := \sqrt{c} e^{i\langle \mathbf{x}, w\rangle} \quad \text{where } w \sim \mu \text{ on } \mathbb{R}^d.$$

This map satisfies a complex version of the reproducing property :

$$\Phi(\mathbf{x}, \mathbf{y}) = \mathbb{E}_w \left[\psi_c(\mathbf{x}; w) \cdot \psi_C(\mathbf{y}; w)^* \right] \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

Assuming the kernel Φ is symmetric, it is in fact not hard to construct a real-valued random feature map from the observation above. Since $\Phi(x, y) = \Phi(y, x)$, the complex exponentials in (48) can be expressed in terms of cosines. This leads to the random feature map using an additional random variable U:

$$\psi(\mathbf{x}; w, U) := \sqrt{2c} \cos(\langle \mathbf{x}, w\rangle + U) \quad \text{where } w \sim \mu \text{ and } U \sim \text{Uniform}([0, 2\pi]). \tag{49}$$

In Exercise (3.4.8), we verify that the above indeed is a random feature reproducing the kernel Φ. ▲

Exercise 3.4.8. Verify that (49) indeed reproduces the translation invariant kernel Φ. (*Hint*: Use angle-sum formula for the cosine.)

We conclude this section with one of the most important examples of a random feature map from the symmetric translation invariant class.

Example 3.4.9 (Gaussian kernel)**.** Consider the *Gaussian radial basis function kernel*:

$$\Phi(x, y) = e^{-\alpha \|\mathbf{x} - \mathbf{y}\|^2 / 2} \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

The positive parameter α controls how close two points must be for them to be considered similar. For the Gaussian kernel, Bôchners Theorem (48) holds with scaling factor $c = 1$ and probability measure $\mu = \mathcal{N}(0, \alpha \mathbf{I}_d)$. Thus, we define the random feature map:

$$\psi(x; w, U) := \sqrt{2} \cos(\langle x, w\rangle + U) \quad \text{where } w \sim \mathcal{N}(0, \alpha \mathbf{I}_d) \text{ and } U \sim \text{Uniform}([0, 2\pi]).$$

By Exercise 3.4.8, this random feature map reproduces the Gaussian radial basis function kernel. ▲

CHAPTER 4

Random Embedding

4.1. Random Embeddings

The purpose of an embedding is to reduce the dimension of the data while preserving its geometry. By operating on the low-dimensional representation, we can develop faster algorithms that give approximate solutions to data analysis problems.

Definition 4.1.1 (Embedding)**.** Consider a set $\mathtt{E} \subset \mathbb{R}^n$. Let $\varepsilon \in (0,1)$ be a tolerance, which we call the *distortion*. A (linear) map $\mathbf{S} : E \to \mathbb{R}^d$ is called an ℓ_2*-embedding* with distortion ε if

$$(1-\varepsilon)\|\mathbf{x}\| \le \|\mathbf{S}\mathbf{x}\| \le (1+\varepsilon)\|\mathbf{x}\| \quad \text{for all } \mathbf{x} \in \mathtt{E}, \tag{50}$$

where $\|\cdot\|$ denotes the ℓ_2-norm of a vector.

The linearity of $\mathbf{S}$ ensures that $\mathbf{S}$ is a metric space embedding of $(\mathtt{E}, \ell_2)$ into $(\mathbb{R}^d, \ell_2)$ in the sense that

$$(1-\varepsilon)\|\mathbf{x}-\mathbf{y}\| \le \|\mathbf{S}\mathbf{x}-\mathbf{S}\mathbf{y}\| \le (1+\varepsilon)\|\mathbf{x}-\mathbf{y}\| \quad \text{for all } \mathbf{x} \in \mathtt{E},$$

In many cases, the embedding dimension d is much smaller than the ambient dimension n, so the map $\mathbf{S}$ performs dimension reduction on $\mathtt{E}$. But how can we construct such a map? Randomness offers a powerful mechanism. Below in this section, we will develop a preliminary approach for constructing a random embedding that has small distortion with high probability.

First, taking squares of the nonnegative terms in (50), we see that it is equivalent to

$$(1-\varepsilon)^2\|\mathbf{x}\|^2 \le \mathbf{x}^\top \mathbf{S}^\top \mathbf{S}\mathbf{x} \le (1+\varepsilon)^2\|\mathbf{x}\|^2 \quad \text{for all } \mathbf{x} \in \mathtt{E}.$$

In turn, the above is implied by the following stronger condition

$$(1-\varepsilon)^2 \le \lambda_{\min}(\mathbf{S}^\top \mathbf{S}) \le \lambda_{\max}(\mathbf{S}^\top \mathbf{S}) \le (1+\varepsilon)^2.$$

Our key idea is to choose a random matrix that is isotropic on average and show that (using matrix concentration inequalities) that the random matrix is very close to its expectation. This leads to the following notion of 'random embedding'.

Definition 4.1.2 (Random embedding)**.** A random matrix $\mathbf{S} \in \mathbb{R}^{d\times n}$ is a *random embedding* for a set $\mathtt{E} \subseteq \mathbb{R}^n$ if preserves the squared norm of vectors in $\mathtt{E}$ on average:

$$\mathbb{E}\left[\|\mathbf{S}\mathbf{x}\|^2\right] = \|\mathbf{x}\|^2 \quad \text{for all } \mathbf{x} \in \mathtt{E}.$$

We say $\mathbf{S}$ is *isotropic* in expectation if [1]

$$\mathbb{E}[\mathbf{S}^\top \mathbf{S}] = \mathbf{I}_n.$$

Observe that $\mathbf{S}$ is a random embedding if it is isotropic on average. Since we have learned that the spectral properties of a large random matrix are close to those in expectation, we should be able to make the embedding dimension d large enough so that the eigenvalues. To verify that $\mathbf{S}$ is an embedding, we need uniform control on the deviations from the average behavior. The question is how large the embedding dimension $d = d(\varepsilon)$ should be to guarantee distortion ε.

Exercise 4.1.3 (Embedding shifted sets subspaces)**.** Let $\mathtt{E}$ be a subset of $\mathbb{C}^n$ and fix a vector $\mathbf{b} \in \mathbb{C}^n$. Show that matrix $\mathbf{S} \in \mathbb{C}^{d\times n}$ is an ε-embedding of $\mathtt{E}$ if and only if it is an ε-embedding of the translation $\mathtt{E} + \mathbf{b}$.

[1] Even if $\mathbf{S}^*\mathbf{S}$ has rank at most $d \le n$ almost surely, its expectation can have full rank.

4.2. Gaussian Embeddings

In this section, we develop and analyze a fundamental example of a random embedding. This framework helps us explore the possibilities for dimension reduction. We will focus on the real number field $\mathbb{R}$ to utilize techniques related to Gaussian processes; however, this limitation is not essential in practical applications.

Definition 4.2.1 (Gaussian Embedding)**.** A *Gaussian embedding* is defined as a random matrix $\mathbf{\Gamma} \in \mathbb{R}^{d\times n}$ with independent and identically distributed entries drawn from the normal distribution $\mathcal{N}(0, d^{-1})$. The terms n and d represent the *ambient* and *embedding* dimensions, respectively.

The Gaussian embedding possesses several straightforward properties. Firstly, it is isotropic on average:

Exercise 4.2.2 (Isotropy of Gaussian Embeddings)**.** Let $\mathbf{\Gamma} \in \mathbb{R}^{d\times n}$ be a random Gaussian matrix with i.i.d. entries that follow the normal distribution $\mathcal{N}(0, d^{-1})$. Demonstrate that

$$\mathbb{E}[\|\mathbf{\Gamma x}\|]^2 = \|\mathbf{x}\|^2 \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

Secondly, the construction is *oblivious* to the set $\mathtt{E}$, meaning that no information about the set is used during the embedding process. However, the embedding dimension d necessary to achieve a distortion ε is influenced by the geometry of the set $\mathtt{E}$.

Next, we introduce some useful notation to describe the embedding characteristics of a linear map. The concept of 'restricted singular values' for a linear map $\mathbf{S} : \mathtt{E} \subseteq \mathbb{R}^n \to \mathbb{R}^d$ captures how $\mathbf{S}$ modifies the size of the set $\mathtt{E}$.

Definition 4.2.3 (Set-restricted Singular Values)**.** Let $\mathtt{E} \subset \mathbb{R}^n$. The *restricted singular values* of a linear map $S : \mathbb{R}^n \to \mathbb{R}^d$ are defined as follows:

$$\sigma_{\min}(\mathbf{S}; \mathtt{E}) := \inf_{\mathbf{x}\in\mathtt{E}} \frac{\|\mathbf{Sx}\|}{\|\mathbf{x}\|} \quad \text{and} \quad \sigma_{\max}(\mathbf{S}; \mathtt{E}) := \sup_{\mathbf{x}\in\mathtt{E}} \frac{\|\mathbf{Sx}\|}{\|\mathbf{x}\|}.$$

To clarify the terminology, note that the restricted singular values for the entire space $E = \mathbb{R}^n$ align with the standard minimum and maximum singular values. In practice, when $\sigma_{\min}(\mathbf{S}; E) = 0$, it indicates that $\mathbf{S}$ maps some points in E to zero, although the exact value of $\sigma_{\max}$ may be less significant. What is essential is that

$$\sigma_{\min}(\mathbf{S}; E) \le \frac{\|\mathbf{Sx}\|}{\|\mathbf{x}\|} \le \sigma_{\max}(\mathbf{S}; E) \quad \text{for all } \mathbf{x} \in E.$$

Consequently, if both $1 - \varepsilon \le \sigma_{\min}(\mathbf{S}; E)$ and $\sigma_{\max}(S; E) \le 1 + \varepsilon$ hold, then the matrix S serves as an embedding of E with distortion ε.

The analysis of a Gaussian embedding involves a parameter that summarizes the geometry of the set E. For this purpose, we introduce the following concept.

Definition 4.2.4 (Gaussian width)**.** Let $\mathtt{E} \subset \mathbb{R}^n$. The *Gaussian width* of $\mathtt{E}$ is defined as

$$w(\mathtt{E}) := \mathbb{E}\left[\sup_{\mathbf{x}\in\mathtt{E}} \langle \mathbf{g}, \mathbf{x}\rangle\right] \quad \text{where } \mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_n).$$

The Gaussian width measures the extent to which a random vector $\mathbf{g}$, sampled from a standard Gaussian distribution, "supports" the set $\mathtt{E}$. It can be interpreted as the average interaction between $\mathtt{E}$ and random directions.

The Gaussian width has the following properties:

(i) (*Monotonicity*): The width is increasing with respect to set inclusion; if $\mathtt{E} \subseteq \mathtt{F}$, then $w(\mathtt{E}) \le w(\mathtt{F})$.

(ii) (*Invariance*): The width is invariant under Euclidean rigid motions; $w(\mathbf{v} + \mathbf{Q}\mathtt{E}) = w(\mathtt{E})$ for all $\mathbf{v} \in \mathbb{R}^n$ and all orthogonal matrices $\mathbf{Q} \in \mathbb{R}^{n\times n}$ (i.e., $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_n$).

(iii) (*Range*): The width lies in the range $0 \le w(\mathtt{E}) \le \sqrt{n} \cdot \text{radius}(\mathtt{E})$.

Here is a simple but important example of a Gaussian width computation.

Example 4.2.5 (Subspace)**.** Let $\mathtt{L} \le \mathbb{R}^n$ be a k-dimensional subspace of $\mathbb{R}^n$. Let us evaluate the Gaussian width $w(\mathtt{E})$ of the spherical set $\mathtt{E} = \mathtt{L} \cap \mathbb{S}^{n-1}$, where $\mathbb{S}^{n-1}$ is the unit sphere in $\mathbb{R}^n$. Denote $P_{\mathtt{L}}$ to be the orthogonal projection from $\mathbb{R}^n$ to the subspace $\mathtt{L}$. Then

$$w(\mathtt{E}) = \mathbb{E}\left[\sup_{\mathbf{x}\in\mathtt{E}} \langle \mathbf{g}, \mathbf{x}\rangle\right] = \mathbb{E}\left[\|P_{\mathtt{L}}\mathbf{g}\|\right] \overset{(a)}{=} \mathbb{E}[\chi_k] \overset{(b)}{\in} [\sqrt{k-1}, \sqrt{k}].$$

Here we have written χ_k^2 for a chi-distributed random variable with k degrees of freedom; $\chi_k^2 =_d Z_1^2 + \cdots + Z_k^2$ where $Z_1, \ldots, Z_k$ are i.i.d. standard normal random variables. The equality (a) needs a bit more justification, see Exc. 4.2.9. In high level, it is true since $P_{\mathtt{L}}\mathbf{g}$ is an orthogonal transform of the standard k-dimensional Gaussian vector, whose norm squared is distributed as χ_k^2.

For (b), by Jensen's inequality, we get the upper bound $\mathbb{E}[\chi_k] \le \sqrt{\mathbb{E}[\chi_k^2]} \le \sqrt{k}$. To lower bound uses the expression

$$\mathbb{E}[\xi_k] = \sqrt{2}\frac{\Gamma((k+1)/2)}{\Gamma(k/2)},$$

where Γ denotes the Gamma function. Gautschi's inequality states that for any positive x and $s \in (0,1)$,

$$x^{1-s} < \frac{\Gamma(x+1)}{\Gamma(x+s)} < (x+1)^{1-s}.$$

Plugging in $x = \frac{k-1}{2}$ and $s = 1/2$, we get

$$\mathbb{E}[\xi_k] = \sqrt{2}\frac{\Gamma((k+1)/2)}{\Gamma(k/2)} \ge \sqrt{2}\sqrt{\frac{k-1}{2}} = \sqrt{k-1}.$$

▲

Example 4.2.6 (Finite set)**.** Let $\mathtt{E} = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \subseteq \mathbb{R}^n$ be a finite subset of $\mathbb{R}^n$ consisting of k unit vectors. Then

$$w(\mathtt{E}) \le \sqrt{2\log|\mathtt{E}|}.$$

To see this, denote $Z := \max\{Z_1, \ldots, Z_k\}$, where $Z_i := \langle \mathbf{g}, \mathbf{u}_i\rangle$ for $i = 1, \ldots, k$. Since each $\mathbf{u}_i$ is a unit vector, $Z_i \sim \mathcal{N}(0,1)$ so it has the gaussian MGF

$$\mathbb{E}[\exp(tZ_i)] = \exp\left(\frac{t^2}{2}\right) \qquad \text{for all } t \in \mathbb{R}.$$

For a fixed $t > 0$, it follows that

$$\exp(t\mathbb{E}[Z]) \le \mathbb{E}[\exp(tZ)] = \mathbb{E}\left[\max_{1\le i\le k} \exp(tZ_i)\right] \le \mathbb{E}\left[\sum_{i=1}^{k} \exp(tZ_i)\right] = k\exp(t^2/2),$$

where the first inequality is from Jensen's inequality and the second inequality is by bounding the maximum of positive numbers of their sum. Now taking $t = \sqrt{2\log N}$,

$$w(\mathtt{E}) = \mathbb{E}[Z] \le \frac{t}{2} + \frac{\log k}{t} = \sqrt{2\log k}.$$

▲

Remark 4.2.7 (Heuristic Interpretation)**.** The squared Gaussian width $w^2(\mathtt{E})$ serves as an indicator of the "dimension" of a subset $\mathtt{E}$ within the unit sphere.

Remark 4.2.8 (Valuation Property)**.** The Gaussian width acts as a *valuation* for the class of convex bodies in $\mathbb{R}^n$, leading to various geometric implications.

Exercise 4.2.9 (Projection of Gaussian Vectors)**.** Consider a standard Gaussian vector $g \sim \mathcal{N}(0, \mathbf{I}_n)$ in $\mathbb{R}^n$. Let $L \subseteq \mathbb{R}^n$ represent a k-dimensional subspace, with P_L denoting the orthogonal projection onto L such that $P_L^2 = P_L$ and $P_L^\top = P_L$.

(i) Demonstrate that $P_{\mathrm{L}} g \sim \mathcal{N}(0, P_{\mathrm{L}})$.

(ii) Let $\mathbf{U} \in \mathbb{R}^{n\times k}$ be a matrix with columns forming an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_k$ for the subspace L. Then, the orthogonal projection P_{L} can be expressed as $P_L = \mathbf{U}\mathbf{U}^\top$. Show that $\tilde{g} := \mathbf{U}^\top g \sim \mathcal{N}(0, \mathbf{I}_k)$ and that

$$P_{\mathrm{L}} g = \mathbf{U}\tilde{g}.$$

(iii) Prove that

$$\|P_{\mathrm{L}} g\|^2 = \|\mathbf{U}\tilde{g}\|^2 = \|\tilde{g}\|^2 \sim \chi^2_k.$$

For any subset of the unit sphere, the behavior of a Gaussian embedding is influenced by the Gaussian width.

Theorem 4.2.10 (Restricted Singular Values of Gaussian Embedding)**.** *Let* $\mathrm{E} \subset \mathbb{S}^{n-1}$. *Consider a Gaussian embedding* $\mathbf{\Gamma} \in \mathbb{R}^{d\times n}$. *The restricted singular values of* $\mathbf{\Gamma}$ *are subject to:*

$$\mathbb{P}\left(\sigma_{\max}(\mathbf{\Gamma}; \mathrm{E}) \geq 1 + \frac{w(\mathrm{E})}{\sqrt{d}} + t\right) \leq \exp\left(-\frac{d t^2}{2}\right),$$
$$\mathbb{P}\left(\sigma_{\min}(\mathbf{\Gamma}; \mathrm{E}) \leq 1 - \frac{w(\mathrm{E})}{\sqrt{d}} - t\right) \leq \exp\left(-\frac{d t^2}{2}\right).$$

In particular, for each $\varepsilon, \delta > 0$,

$$d \geq \left(1 + \frac{\sqrt{2\log\delta^{-1}}}{w(\mathrm{E})}\right)^2 w(\mathrm{E})^2 \varepsilon^{-2} \quad \Longrightarrow \quad \mathbb{P}(1-\varepsilon \leq \sigma_{\min}(\mathbf{\Gamma}; \mathrm{E}) \leq \sigma_{\max}(\mathbf{\Gamma}; \mathrm{E}) \leq 1+\varepsilon) \geq 1-\delta. \tag{51}$$

OUTLINE OF THE PROOF. The two inequalities in Theorem 4.2.10 originate from Chevet and Gordon. A comprehensive discussion of these results can be found in [Ver18]. The paper [TOH14] offers a partial converse for the lower bound on the restricted minimum singular value.

The 'dimension complexity' is a simple consequence of the bounds above. In order to acheive the desired high-probability distortion bound in (51), it is enough to set d and t to satisfy, for any $\alpha \in (0,1)$,

$$t = (1-\alpha)\varepsilon, \qquad \frac{w(E)}{\sqrt{d}} \leq \alpha\varepsilon, \qquad -\frac{d(1-\alpha)^2\varepsilon^2}{2} \leq \log\delta,$$

or equivalentely,

$$t = (1-\alpha)\varepsilon, \qquad d \geq \frac{w(\mathrm{E})^2\varepsilon^{-2}}{\alpha^2}, \qquad d \geq \frac{2}{(1-\alpha)^2}\varepsilon^{-2}\log\delta^{-1}.$$

We can choose α so that the required lower bounds on d coincide:

$$\frac{(1-\alpha)^2}{\alpha^2} = \frac{2\log\delta^{-1}}{w(\mathrm{E})^2}.$$

This gives $\frac{1}{\alpha} = 1 + \sqrt{\frac{2\log\delta^{-1}}{w(\mathrm{E})^2}}$ and the desired result follows. □

Theorem 4.2.10 leads to a significant conclusion:

Corollary 4.2.11. *To achieve an embedding of* $\mathrm{E} \subseteq \mathbb{R}^n$ *with distortion* ϵ *through a Gaussian embedding* $\mathbf{\Gamma} \in \mathbb{R}^{d\times n}$, *it suffices that the embedding dimension* d *satisfies* $d \sim \frac{w^2(\mathrm{E})}{\epsilon^2}$, *with the constant of proportionality being approximately one.*

Moreover, the tail bounds above for $\sigma_{\min}$ and $\sigma_{\max}$ hold for a broad class of random matrix with i.i.d. entries, not necessarily with Gaussian distribution. This phenomenon is called the "universality" in random matrix theory that the local eigenvalue statistics of random matrices depends only on the first few (usually 2 or 4) moments of the entries (see, e.g., Tao and Vu [TV11]). Based on such universality results, [OT18] demonstrates that any random embedding with i.i.d. sub-Gaussian entries behaves similarly to a Gaussian embedding, provided that E is not excessively small. (In more precise terms, the universality condition requires that $w(\mathrm{E}) \sim \mathrm{const} \cdot n$.)

Next, we use Theorem 4.2.10 to develop two well-known results on Gaussian random embedding of linear subspaces and finite sets into a low-dimensional space. We first consider embedding linear subspaces.

Corollary 4.2.12 (Subspace embedding)**.** *Let $\mathtt{L} \leq \mathbb{R}^n$ be a k-dimensional subspace in $\mathbb{R}^n$, and let $\varepsilon \in (0,1)$. Let $\mathbf{\Gamma} \in \mathbb{R}^{d\times n}$ denote a Gaussian embedding. Then for each $\varepsilon, \delta > 0$,*

$$\mathbb{P}\left(1-\varepsilon \leq \inf_{\mathbf{x}\in \mathtt{L}} \frac{\|\mathbf{S}\mathbf{x}\|}{\|\mathbf{x}\|} \leq \sup_{\mathbf{x}\in \mathtt{L}} \frac{\|\mathbf{S}\mathbf{x}\|}{\|\mathbf{x}\|} \leq 1+\varepsilon\right) \geq 1-\delta$$

provided

$$d \geq \left(1+\frac{\sqrt{2\log\delta^{-1}}}{k}\right)^2 k\,\varepsilon^{-2} \approx k\varepsilon^{-2}.$$

PROOF. Define $\mathtt{E} := \mathtt{L} \cap \mathbb{S}^{n-1}$, the set of unit vectors in $\mathbb{L}$. Note that by the result in Example 4.2.5,

$$w(\mathtt{E}) \leq \sqrt{k}.$$

Then by Theorem 4.2.10, for each $\varepsilon, \delta > 0$,

$$d \geq \left(1+\frac{\sqrt{2\log\delta^{-1}}}{k}\right)^2 k\,\varepsilon^{-2} \quad \Longrightarrow \quad \mathbb{P}(1-\varepsilon \leq \sigma_{\min}(\mathbf{\Gamma};\mathtt{E}) \leq \sigma_{\max}(\mathbf{\Gamma};\mathtt{E}) \leq 1+\varepsilon) \geq 1-\delta.$$

This is enough to conclude. □

Common practical choice of the embedding dimmension is $d = k + r$ where $r \in \{5, 10, k\}$.

Next, consider embedding finite subsets into low-dimensional spaces. This result is the celebrated Johnson-Lindenstrauss lemma.

Corollary 4.2.13 (Johnson-Lindenstrauss Lemma)**.** *Let $\mathtt{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\} \subset \mathbb{R}^n$ be a set of N points, and let $\varepsilon \in (0,1)$. Let $\mathbf{\Gamma} \in \mathbb{R}^{d\times n}$ denote a Gaussian embedding. Then for each $\varepsilon, \delta > 0$,*

$$\mathbb{P}\left(1-\varepsilon \leq \inf_{1\leq i,j\leq N} \frac{\|\mathbf{S}\mathbf{x}_i - \mathbf{S}\mathbf{x}_j\|}{\|\mathbf{x}_i - \mathbf{x}_j\|} \leq \sup_{1\leq i,j\leq N} \frac{\|\mathbf{S}\mathbf{x}_i - \mathbf{S}\mathbf{x}_j\|}{\|\mathbf{x}_i - \mathbf{x}_j\|} \leq 1+\varepsilon\right) \geq 1-\delta$$

provided

$$d \geq \left(1+\frac{\sqrt{\log\delta^{-1}}}{\sqrt{2\log N}}\right)^2 4(\log N)\,\varepsilon^{-2} \approx 4\varepsilon^{-2}\log N.$$

PROOF. Define the set of secants

$$\mathtt{E} := \left\{\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|} : i < j\right\} \subset \mathbb{S}^{n-1}.$$

This is a set of at most N^2 unit vectors in $\mathbb{R}^n$. Note that by the result in Example 4.2.6,

$$w(\mathtt{E}) \leq \sqrt{2\log|\mathtt{E}|} \leq 2\sqrt{\log N}.$$

Then by Theorem 4.2.10, for each $\varepsilon, \delta > 0$,

$$d \geq \left(1+\frac{\sqrt{\log\delta^{-1}}}{\sqrt{2\log N}}\right)^2 4\log N\,\varepsilon^{-2} \quad \Longrightarrow \quad \mathbb{P}(1-\varepsilon \leq \sigma_{\min}(\mathbf{\Gamma};\mathtt{E}) \leq \sigma_{\max}(\mathbf{\Gamma};\mathtt{E}) \leq 1+\varepsilon) \geq 1-\delta.$$

This is enough to conclude. □

A classic algorithmic application of this type of embedding is to accelerate approximate nearest neighbor computation.

The JL-lemma can be established by an elementary arguments using Gaussian concentration inequalities and union bound.

Exercise 4.2.14. The following gives a sketch of an elementary proof of the Johnson-Lindenstrauss lemma (Thm. 4.2.13) without appleaing to Theorem 4.2.10. Make it a rigorous proof.

1. For each $i, j \in [N]$, $Z_{ij} := \|\mathbf{\Gamma}(\mathbf{x}_i - \mathbf{x}_j)\| / \|\mathbf{x}_i - \mathbf{x}_j\|$. Show that $Z_{ij}^2 \sim d^{-1}\chi_d^2$, the chi-squared distribution with d degrees of freedom.
2. Using a concentration inequality for chi-squared random variables, we have
$$\mathbb{P}\left(|Z_{ij} - 1| \geq \varepsilon\right) \leq 2\exp\left(-\frac{d\varepsilon^2}{4}\right).$$
Thus, the probability that the random projection distorts the distance by more than a factor of $(1 \pm \varepsilon)$ is exponentially small in d.
3. There are at most N^2 pairs of points. By a union bound, show that
$$\mathbb{P}\left(\max_{1\leq i,j\leq N} |Z_{ij} - 1| \geq \varepsilon\right) \leq 2N^2\exp\left(-\frac{d\varepsilon^2}{4}\right).$$
Note that the upper bound above is $\ll 1$ if $d\varepsilon^2/4 \geq 3\log N$. (The resulting sample complexity has sub-optimal constant multiplicity.)

4.3. Structured random embedding

While Gaussian embeddings can be highly effective, they come with a few practical limitations:

- Constructing a Gaussian embedding requires generating nd independent and identically distributed (i.i.d.) samples from a normal distribution, which can be computationally expensive.
- Gaussian embeddings demand significant storage as they consist of nd entries.
- Gaussian matrices are unstructured, meaning that applying such a matrix to a vector involves $O(nd)$ arithmetic operations.

We noted that, according to the universality phenomenon in the random matrix theory, properties that hold for random Gaussian matrices should also hold for more general class of random matrices with non-Gaussian entries. Hence, in this section, we investigate *random but structured* embeddings. These types of embeddings offer solutions to the drawbacks of Gaussian embeddings and, in some cases, provide improved embedding properties. However, the analysis of structured embeddings is more complex and typically less precise compared to Gaussian embeddings.

We will focus on two specific types of structured embeddings:

(1) **Sparse sign matrices**: These embeddings are based on sparse random matrices. The sparsity reduces the cost of construction, storage, and matrix-vector multiplication.

(2) **Subsampled randomized Fourier transforms (SRFTs)**: These embeddings use a fast Fourier transform (FFT) and benefit from the properties of partial unitaries (isometries). As a result, they can outperform Gaussian embeddings while being more efficient to store and construct.

Both constructions are effective for embedding general sets $\mathtt{E}$. We will specifically study their behavior as subspace embeddings, which is critical for most numerical linear algebra applications such as sketching techniques for solving large least squares problems.

4.3.1. Relation between random embedding and matrix concentration. Let $\mathtt{L} \subseteq \mathbb{R}^n$ be a k-dimensional subspace. Recall that a linear map $\mathbf{S} : \mathbb{R}^n \to \mathbb{R}^d$ is called a subspace embedding with distortion ϵ if
$$(1-\epsilon)\|\mathbf{x}\| \leq \|\mathbf{S}\mathbf{x}\| \leq (1+\epsilon)\|\mathbf{x}\| \quad \text{for all } \mathbf{x} \in \mathtt{L}. \tag{52}$$
In the Gaussian case, we expect the embedding dimension d to scale approximately as k/ϵ^2 (Cor. 4.2.12).

To extend this analysis to random structured embeddings, we can reformulate the embedding condition. Let $\mathbf{U} \in \mathbb{R}^{n\times k}$ be a matrix whose columns form an orthonormal basis for $\mathtt{L}$. Then an arbitrary

element $\mathbf{x} \in \mathtt{L}$ can be written as $\mathbf{x} = \mathbf{U}\mathbf{z}$ for some $\mathbf{z} \in \mathbb{R}^k$. Hence (52) is equvialent to

$$(1-\epsilon)\|\mathbf{U}\mathbf{z}\| \le \|\mathbf{S}\mathbf{U}\mathbf{z}\| \le (1+\epsilon)\|\mathbf{U}\mathbf{z}\| \quad \text{for all } \mathbf{z} \in \mathbb{R}^k. \tag{53}$$

Squaring the nonnegative quantities above and recalling $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_k$, (53) is equivalent to

$$(1-\epsilon)^2\|\mathbf{z}\|^2 \le \mathbf{z}^\top \underbrace{(\mathbf{S}\mathbf{U})^\top(\mathbf{S}\mathbf{U})}_{=:\mathbf{Y}}\mathbf{z} \le (1+\epsilon)^2\|\mathbf{z}\|^2 \quad \text{for all } \mathbf{z} \in \mathbb{R}^k. \tag{54}$$

Dividing through $\|\mathbf{z}\|^2$, we see that the above is equivalent to

$$(1-\epsilon)^2 \le \lambda_{\min}(\mathbf{Y}) \le \lambda_{\max}(\mathbf{Y}) \le (1+\epsilon)^2.$$

Therefore, the candidate random embedding $\mathbf{S}$ has a small distortion if the random matrix

$$\mathbf{Y} = (\mathbf{S}\mathbf{U})^\top(\mathbf{S}\mathbf{U}) \in \mathbb{R}^{k\times k}$$

has eigenvalues close to one.

Note that assuming $\mathbf{S}$ is isotropic on average, we know that the expectation of $\mathbf{Y}$ equals the identity matrix and hence has all eigenvalues equal to one:

$$\mathbb{E}[\mathbf{S}^\top\mathbf{S}] = \mathbf{I}_n \qquad \Longrightarrow \qquad \mathbb{E}[\mathbf{Y}] = \mathbf{U}^\top\mathbb{E}[\mathbf{S}^\top\mathbf{S}]\mathbf{U} = \mathbf{U}^\top\mathbf{U} = \mathbf{I}_k.$$

Thus the desired inequality for $\mathbf{S}$ to be ε-distortion embedding for the subspace $\mathtt{L}$ is satisfied if we replace $\mathbf{Y}$ by its expectation $\mathbb{E}[\mathbf{Y}]$. Therefore, it will be enough to verify that the eigenvalues of $\mathbf{Y}$ are concentrated around the eigenvalues of $\mathbb{E}[\mathbf{Y}]$. This type of result is what is provided by the matrix concentration inequalities we established in Ch. 2. In particular, we will use the matrix Chernoff inequalities in Theorem 2.5.15. In contrast to the matrix Bernstein inequality (Thm. 2.5.13), matrix Chernoff does not require matrix variance bounds so it is easier to use.

4.3.2. Sparse sign matrices. Our first construction of an efficient random embedding is called a 'sparse sign matrix'. The idea is to construct the matrix $S \in \mathbb{R}^{d\times n}$ randomly so that most of its entries are zero. The simplest approach is based on (appropriately rescaled) matrices with iid random ternary entries.

Definition 4.3.1 (Sparse sign matrix)**.** A random matrix $\mathbf{S} \in \mathbb{R}^{d\times n}$ is called a *sparse sign matrix* with parameter $p \in [0,1]$ and $\alpha > 0$ if

$$\mathbf{S} = \alpha \begin{bmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{d1} & \cdots & S_{dn} \end{bmatrix} \in \mathbb{R}^{d\times n},$$

where the entries S_{ij} are i.i.d. with the following distribution:

$$S_{ij} = \begin{cases} +1 & \text{with prob. } p/2, \\ -1 & \text{with prob. } p/2, \\ 0 & \text{with prob. } 1-p. \end{cases}$$

The probability $p \in [0,1]$ is the density of nonzero entries in $\mathbf{S}$ and α is a scaling factor chosen to ensure isotropy. Setting

$$\alpha^2 = \frac{1}{dp}$$

ensures isotropy on average.

Exercise 4.3.2 (Isotropy of sparse sign matrices)**.** Verify that this choice of $\alpha^2 = 1/(dp)$ in the sparse sign matrix in Def. 4.3.1 leads to an isotropic random embedding.

Note that each column of the sparse sign matrix $\mathbf{S}$ in Def. 4.3.1 contains, on average, dp nonzero entries. Therefore, the total storage cost is approximately $O(ndp)$, which is much better than the $O(nd)$ cost of storing a dense matrix when p is small. Applying S to a vector requires about $O(ndp)$ arithmetic operations. Efficient implementation of a sparse random embedding requires the use of high-performance sparse arithmetic libraries.

Theorem 4.3.3 (Concentration of sparse sign matrix)**.** *Fix a linear subspace $\mathtt{L} \le \mathbb{R}^n$ of dimension k. Let $\mathbf{S} \in \mathbb{R}^{d\times n}$ be the random sparse sign matrix in Def. 4.3.1. Fix $\varepsilon, \delta \in [0,1]$. Then*

$$\mathbb{P}\left(\mathbf{S} \text{ is an } \varepsilon\text{-distortion embedding of } \mathtt{L}\right) \ge 1 - 4\delta,$$

provided

$$d \ge C(k + \sqrt{k}) \log n,$$

where $C > 0$ is a constant depending on ε, δ.

PROOF. Let $\mathbf{U} \in \mathbb{R}^{n\times k}$ denote an orthogonal matrix whose columns form an orthonormal basis for $\mathtt{L}$. Let $\mathbf{s}_i \in \mathbb{R}^n$ for $i = 1, \dots, d$ denote the ith row of $\alpha^{-1}\mathbf{S}$. Let $\mathbf{Y} = (\mathbf{SU})^\top(\mathbf{SU})$. By (54), it is enough to show that

$$\mathbb{P}\left((1-\epsilon)^2 \le \lambda_{\min}(\mathbf{Y}) \le \lambda_{\max}(\mathbf{Y}) \le (1+\epsilon)^2\right) \ge 1 - 4\delta$$

provided the proposed embedding dimension d. We will first decompose $\mathbf{Y}$ into the sum of PSD rank-one matrices (similarly as in the randomized matrix multiplication, see Sec. 3.2):

$$\mathbf{Y} = \sum_{i=1}^{d} \underbrace{\frac{1}{dp}\mathbf{U}^\top \mathbf{s}_i^\top \mathbf{s}_i \mathbf{U}}_{=:\mathbf{X}_i}.$$

Note that $\mathbf{X}_i$'s defined above are independent (since $\mathbf{s}_i$'s are) and rank-one. Since $\mathbb{E}[\mathbf{S}^\top\mathbf{S}] = \mathbf{I}_n$, we have $\mathbb{E}[\mathbf{Y}] = \mathbf{I}_k$.

In order to prepare for applying the matrix Chernoff inequality (Thm. 2.5.15), we need to show that $\mathbf{X}_i \preceq R\mathbf{I}_k$ almost surely for some constant R. We will first try to get a naive bound. Observe that

$$\|\mathbf{U}^\top \mathbf{s}_i^\top \mathbf{s}_i \mathbf{U}\|_2 \le \|\mathbf{s}_i^\top \mathbf{s}_i\|_2 = \sqrt{\lambda_{\max}((\mathbf{s}_i^\top \mathbf{s}_i)(\mathbf{s}_i^\top \mathbf{s}_i))} = \|\mathbf{s}_i\| \sqrt{\lambda_{\max}(\mathbf{s}_i^\top \mathbf{s}_i)} = \|\mathbf{s}_i\|^2.$$

Since $\mathbf{s}_i \in \{-1, 0, 1\}^n$, it follows that

$$\|\mathbf{X}_i\|_2 \le \frac{1}{dp}\|\mathbf{s}_i\|^2 = \frac{n}{dp}. \tag{55}$$

Is the bound above good enough? The bound $R = \frac{n}{dp}$ is too large to yield any meaningful result. In order for the matrix Chernoff inequality (Thm. 2.5.15) to be meaningful, we need to have

$$\frac{\mu_{\max}}{R} = \frac{dp}{n} \gg 1.$$

However, the above cannot be true since we desire our embedding dimension d to be much less than n (and recall $p \in [0,1]$). This is the first time that our initial attempt to apply matrix concentration inequalites ever failed!

It turns out that our worst-case bound on the random spectral norm above is too large compared to the expected spectral norm. Indeed, let's compute the expectation of $\|\mathbf{X}_i\|_2$. Recall that for any row vector $\mathbf{v} \in \mathbb{R}^{1\times k}$,

$$\|\mathbf{v}^\top \mathbf{v}\|_2 = \sqrt{\lambda_{\max}(\mathbf{v}^\top \mathbf{v}\mathbf{v}^\top \mathbf{v})} = \|\mathbf{v}\| \sqrt{\lambda_{\max}(\mathbf{v}^\top \mathbf{v})} = \|\mathbf{v}\|^2.$$

Applying this with $\mathbf{v} = \mathbf{s}_i\mathbf{U} \in \mathbb{R}^{1\times k}$, we get

$$\|\mathbf{U}^\top \mathbf{s}_i^\top \mathbf{s}_i \mathbf{U}\|_2 = \|\mathbf{s}_i\mathbf{U}\|^2 = \mathbf{s}_i(\mathbf{U}\mathbf{U}^\top)\mathbf{s}_i^\top. \tag{56}$$

Notice that the RHS above is a chaos (Def. 4.4.1). Since $\mathbf{s}_{i\ell}$'s are mean zero, independent, and $\mathbf{s}_{i\ell}^2 \sim$ Bernoulli(p),

$$\mathbb{E}\left[\mathbf{s}_i(\mathbf{U}\mathbf{U}^\top)\mathbf{s}_i^\top\right] = \sum_{\ell,\ell'} \mathbb{E}[\mathbf{s}_{i\ell}\mathbf{s}_{i\ell'}][\mathbf{U}\mathbf{U}^\top]_{\ell,\ell'} = \sum_{\ell} \underbrace{\mathbb{E}[\mathbf{s}_{i\ell}^2]}_{=p}[\mathbf{U}\mathbf{U}^\top]_{\ell,\ell} = p\,\mathrm{tr}(\mathbf{U}\mathbf{U}^\top) = p\|\mathbf{U}\|_F^2 = pk.$$

This yields

$$\mathbb{E}[\|\mathbf{X}_i\|_2] = \frac{1}{dp}(kp) = \frac{k}{d}.$$

Compared to the expected spectral norm of $\mathbf{X}_i$, the naive almost sure bound in (55) is too large.

In order to leverage the fact that the typical value of $\|\mathbf{X}_i\|_2$ is small, we will use a truncation argument. Namely, define the event

$$E_i := \left\{\|\mathbf{X}_i\|_2 \le \frac{k}{d} + t\right\}$$

for some $t > 0$ to be determined. Then almost surely for all i,

$$\|\mathbf{1}_{E_i}\mathbf{X}_i\|_2 \le \frac{k}{d} + t,$$

so we should be in a much better position to apply matrix Chernoff inequality. Accordingly, we consider the truncated random matrix

$$\mathbf{Y}_{\mathrm{trunc}} := \sum_{i=1}^{d} \mathbf{1}_{E_i}\mathbf{X}_i.$$

For truncation argument to work, one needs to show two things:

(I) Truncation should make actual difference only with a small probability (i.e., $\mathbb{P}(E_i^c) \ll 1$)

(II) Truncation should not induce a large bias on the expectation.

A typical stretagy is to first show (**I**) and then deduce (**II**) from it and a crude uniform bound.

With the truncation, we usually proceed as follows. We fail either if truncation fails or the truncated random matrix fails. Namely, by a union bound,

$$\begin{aligned}
\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2\right) &= \mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2 \,\middle|\, \mathbf{Y} = \mathbf{Y}_{\mathrm{trunc}}\right) \underbrace{\mathbb{P}(\mathbf{Y} = \mathbf{Y}_{\mathrm{trunc}})}_{\le 1} \\
&\quad + \underbrace{\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2 \,\middle|\, \mathbf{Y} = \mathbf{Y}_{\mathrm{trunc}}\right)}_{\le 1} \mathbb{P}(\mathbf{Y} \ne \mathbf{Y}_{\mathrm{trunc}}) \\
&\le \mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2 \,\middle|\, \mathbf{Y} = \mathbf{Y}_{\mathrm{trunc}}\right) + \underbrace{\mathbb{P}(\mathbf{Y} \ne \mathbf{Y}_{\mathrm{trunc}})}_{\le \sum_i \mathbb{P}(E_i^c) = n\mathbb{P}(E_1^c)}.
\end{aligned}$$

We wish to make the above at most 2δ. We will argue similarly to show $\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \le (1-\varepsilon)^2\right) \le 2\delta$, which is enough to conclude.

To ensure $\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2\right) \le 2\delta$, we will first choose t large enough so that (for (**I**))

$$\mathbb{P}(E_1^c) \le \frac{\delta}{n} \wedge \frac{p}{4n^2}. \tag{57}$$

(The requirement p/n^2 is later needed to control the truncation bias.) To do so, we need to appeal to another type of concentration inequality for chaos (Def. 4.4.1). Namely, by (56), we observe that $\|\mathbf{X}_i\|_2$ equals the random quadratic form (chaos) $\mathbf{x}^\top\mathbf{A}\mathbf{x}$ where $\mathbf{x} = \mathbf{s}_i^\top$ and $\mathbf{A} = \frac{1}{dp}\mathbf{U}\mathbf{U}^\top$ and a chaos concentrates

around its mean by Hanson-Wright inequality (Thm. 4.4.5). This gives

$$\begin{aligned}
\mathbb{P}(E_1^c) = \mathbb{P}\left(\|\mathbf{X}_1\|_2 > \frac{k}{d} + t\right) &\le \mathbb{P}\left(\|\mathbf{X}_1\|_2 > \mathbb{E}[\|\mathbf{X}_i\|_2] + t\right) \\
&\le 2\exp\left(-\min\left(\frac{t^2}{K^4\|\mathbf{A}\|_F^2}, \frac{t}{K^2\|\mathbf{A}\|_2}\right)\right) \\
&\le 2\exp\left(-\min\left(\frac{t^2}{K^4\|\mathbf{A}\|_F^2}, \frac{t}{K^2\|\mathbf{A}\|_F}\right)\right) \qquad (\text{Since } \|\mathbf{A}\|_2 \le \|\mathbf{A}\|_F) \\
&\le 2\exp\left(-\frac{t}{K^2\|\mathbf{A}\|_F}\right) \qquad \left(\text{Choosing } t \text{ s.t. } \frac{t}{K^2\|\mathbf{A}\|_F} \ge 1\right)
\end{aligned}$$

where $\mathbf{A} = \frac{1}{dp}\mathbf{U}\mathbf{U}^\top$ and $K > 0$ is a constant[2]. Note that

$$\|\mathbf{A}\|_F^2 = \frac{1}{d^2p^2}\operatorname{tr}((\mathbf{U}\mathbf{U}^\top)^\top(\mathbf{U}\mathbf{U}^\top)) = \frac{1}{d^2p^2}\operatorname{tr}(\mathbf{U}\mathbf{U}^\top) = \frac{1}{d^2p^2}\operatorname{tr}(\mathbf{U}^\top\mathbf{U}) = \frac{k}{d^2p^2}.$$

Thus we get

$$\mathbb{P}(E_1^c) \le 2\exp\left(-\frac{tdp}{K^2\sqrt{k}}\right).$$

In order for the RHS above be $\le \delta/n \wedge p/n^2$, it is enough to choose

$$t = \frac{K^2\sqrt{k}}{dp}\log\frac{8n^2}{\delta \wedge p}.$$

(As long as $\frac{2n^2}{\delta} \ge e$, this choice also satisfies $t \ge K^2\|\mathbf{A}\|_F$.)

Next, by the matrix Chernoff inequality (Thm. 2.5.15),

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2 \,\middle|\, \mathbf{Y} = \mathbf{Y}_{\text{trunc}}\right) \le d\underbrace{\left(\frac{e^{2\varepsilon+\varepsilon^2}}{(1+\varepsilon)^{2(1+\varepsilon)^2}}\right)}_{:=c(\varepsilon)<1}^{\lambda_{\max}(\mathbb{E}\mathbf{Y}_{\text{trunc}})\frac{d}{k+td}}. \tag{58}$$

Note that we now have $\lambda_{\max}(\mathbb{E}\mathbf{Y}_{\text{trunc}})$ instead of $\lambda_{\max}(\mathbb{E}\mathbf{Y}) = 1$ in the Chernoff bound due to the truncation, and this is why we need to control the truncation bias (**II**). To do this, by Weyl's inequality on spectral stability of self-adjoint matrices,

$$\begin{aligned}
|\lambda_{\max}(\mathbb{E}\mathbf{Y}) - \lambda_{\max}(\mathbb{E}\mathbf{Y}_{\text{trunc}})| &\le \|\mathbb{E}\mathbf{Y} - \mathbb{E}\mathbf{Y}_{\text{trunc}}\|_2 \\
&\le \sup\|\mathbf{Y} - \mathbf{Y}_{\text{trunc}}\|_2 \mathbb{P}(\mathbf{Y} \ne \mathbf{Y}_{\text{trunc}}) \\
&\le d\frac{2n}{dp} n\mathbb{P}(E_1^c) \\
&= \frac{2n^2}{p}\mathbb{P}(E_1^c) \le 1/2
\end{aligned}$$

by (57). Since $\lambda_{\max}(\mathbb{E}\mathbf{Y}) = 1$, this yields $\lambda_{\max}(\mathbb{E}\mathbf{Y}_{\text{trunc}}) \ge 1/2$. In turn,

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\varepsilon)^2 \,\middle|\, \mathbf{Y} = \mathbf{Y}_{\text{trunc}}\right) \le dc(\varepsilon)^{\frac{d/2}{k+td}}.$$

With the above choice of t, we turn our attention to (58) and choose d large enough so that its RHS is at most δ. For this, it is enough to satisfy

$$\log d + \frac{d/2}{k + (K^2\sqrt{k}/p)\log(2n/\delta)}\log c(\varepsilon) \le \log\delta.$$

[2] the sub-Gaussian norm of the entries of $\mathbf{S}$

For this, the following choice for some constant $C' > 0$ is enough:

$$d \geq C'(k + \sqrt{k}) \log(2n/\delta) \cdot \log(\delta / c(\varepsilon)).$$

□

4.4. Hanson-Wright inequality

In this section, we study quadratic forms given by a deterministic matrix and a random vector.

Definition 4.4.1 (Chaos)**.** Fix a matrix $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ a random vector $\mathbf{x} = (X_1, \dots, X_n)^\top \in \mathbb{R}^n$ of independent coordinates. The quadratic form

$$\mathbf{x}^\top \mathbf{A}\mathbf{x} = \sum_{i,j=1}^{n} a_{ij} X_i X_j \tag{59}$$

is called a *chaos*.

Computing the expectation of a chaos is easy. For simplicity, let us assume that X_i have zero means and unit variances. Then

$$\mathbb{E}[\mathbf{x}^\top \mathbf{A}\mathbf{x}] = \sum_{i,j=1}^{n} a_{ij} \mathbb{E}[X_i X_j] = \sum_{i=1}^{n} a_{ii} = \operatorname{tr}(\mathbf{A}).$$

The Hanson-Wright inequality is a concentration inequality that bounds the probability that a quadratic form of a random vector with sub-Gaussian entries deviates from its mean. It is useful in probability theory and random matrix theory when dealing with random vectors and quadratic forms.

Compared to its expectation, it is much harder to establish concentration of a chaos. The key difficulty is that the terms of the sum in (59) since many of them share the same random variables. This difficulty can be overcome by the decoupling technique, which we will study now.

The purpose of decoupling is to replace the quadratic form (59) with the bilinear form

$$\sum_{i,j=1}^{n} a_{ij} X_i X'_j = \mathbf{x}^\top \mathbf{A}\mathbf{x}',$$

where $\mathbf{x}' = (X'_1, \dots, X'_n)$ is an independent copy of $\mathbf{x}$. The upshot here is that the bilinear form is easier to analyze than the quadratic form, since it is linear in $\mathbf{x}$. Indeed, if we condition on $\mathbf{x}'$, we may treat the bilinear form as a sum of independent random variables

$$\sum_{i=1}^{n} \left(\sum_{j=1}^{n} a_{ij} X'_j \right) X_i = \sum_{i=1}^{n} c_i X_i$$

with fixed coefficients c_i.

Lemma 4.4.2 (Decoupling lemma)**.** *Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ with zero diagonal entries. Let $\mathbf{x} = (X_1, \dots, X_n)$ be a random vector with independent, mean-zero coordinates X_i. Then, for every convex function $F : \mathbb{R} \to \mathbb{R}$, one has*

$$\mathbb{E}\left[F\left(\mathbf{x}^\top \mathbf{A}\mathbf{x}\right)\right] \leq \mathbb{E}\left[F\left(4\mathbf{x}^\top \mathbf{A}\mathbf{x}'\right)\right]$$

where $\mathbf{x}'$ is an independent copy of $\mathbf{x}$.

PROOF. Omitted. See [Ver18, Thm. 6.1.1]. □

Lemma 4.4.3 (Comparison)**.** *Consider independent, mean zero, sub-gaussian random vectors $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$ with $\|\mathbf{x}\|_{\psi_2} \leq K$ and $\|\mathbf{x}'\|_{\psi_2} \leq K$. Consider also independent random vectors $\mathbf{g}, \mathbf{g}' \sim N(0, I_n)$. Let $\mathbf{A}$ be an $n \times n$ matrix. Then for some absolute constant $C > 0$ the following hold:*

$$\mathbb{E} \exp(\lambda \mathbf{x}^\top \mathbf{A}\mathbf{x}') \leq \mathbb{E} \exp(CK^2 \lambda \mathbf{g}^\top \mathbf{A}\mathbf{g}') \qquad \text{for all } \lambda \in \mathbb{R}.$$

PROOF. Omitted. See [Ver18, Thm. 6.2.3]. □

Lemma 4.4.4 (MGF of Gaussian chaos). *Let $\mathbf{x}, \mathbf{x}' \sim N(0, I_n)$ be independent and let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix. Then for some absolute constants $C, c > 0$,*

$$\mathbb{E} \exp(\lambda \mathbf{x}^\top \mathbf{A} \mathbf{x}') \le \exp(C\lambda^2 \|\mathbf{A}\|_F^2)$$

for all λ satisfying $|\lambda| \le \frac{c}{\|\mathbf{A}\|_2}$.

PROOF. Omitted. See [Ver18, Thm. 6.2.2]. □

Theorem 4.4.5 (Hanson-Wright inequality for concentration of chaos). *Let $\mathbf{x} = (X_1, X_2, \dots, X_n)^\top$ be a random vector with independent, mean-zero, sub-Gaussian entries, and fix a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then, there exist constants C_1 and C_2 such that for every $t \ge 0$,*

$$\mathbb{P}\left(\left|\mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbb{E}\left[\mathbf{x}^\top \mathbf{A} \mathbf{x}\right]\right| \ge t\right) \le 2 \exp\left(-\min\left(\frac{t^2}{K^4 \|\mathbf{A}\|_F^2}, \frac{t}{K^2 \|\mathbf{A}\|_2}\right)\right),$$

where $K = \max_{1 \le i \le n} \|X_i\|_{\psi_2}$ denotes the maximum sub-Gaussian norm of the coordinates of $\mathbf{x}$.

PROOF. Without loss of generality, we may assume that $K = 1$. (Why?) As usual, it is enough to bound the one-sided tail

$$p := \mathbb{P}\left(\mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \ge t\right).$$

Indeed, once we have a bound on this upper tail, a similar bound will hold for the lower tail as well (since one can replace $\mathbf{A}$ with $-\mathbf{A}$). By combining the two tails, we would complete the proof.

In terms of the entries of $\mathbf{A} = (a_{ij})_{i,j=1}^n$, we have

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} X_i X_j \quad \text{and} \quad \mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \sum_i a_{ii} \mathbb{E}[X_i^2],$$

where we used the mean zero assumption and independence. So we can express the deviation as

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \sum_i a_{ii} \left(X_i^2 - \mathbb{E}[X_i^2]\right) + \sum_{i \ne j} a_{ij} X_i X_j.$$

The problem reduces to estimating the diagonal and off-diagonal sums:

$$\mathbb{P}\left(\mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbb{E}\left[\mathbf{x}^\top \mathbf{A} \mathbf{x}\right] \ge t\right) \le \underbrace{\mathbb{P}\left(\sum_i a_{ii} \left(X_i^2 - \mathbb{E}[X_i^2]\right) \ge \frac{t}{2}\right)}_{=:p_1} + \underbrace{\mathbb{P}\left(\sum_{i,j: i \ne j} a_{ij} X_i X_j \ge \frac{t}{2}\right)}_{=:p_2}.$$

Step 1: Diagonal sum.

Since X_i are independent, sub-Gaussian random variables, $X_i^2 - \mathbb{E}[X_i^2]$ are independent, mean-zero, sub-exponential random variables, and

$$\|X^2 - \mathbb{E}[X^2]\| \lesssim \|X^2\| \lesssim \|X\|^2 \lesssim 1.$$

Then Bernsteins inequality for subexponential random variables (Theorem 2.4.5) gives

$$p_1 \le \exp\left(-c \min\left(\sum_i a_{ii}^2, \max_i |a_{ii}|\right)\right) \le \exp\left(-c \min\left(\|\mathbf{A}\|_2, \|\mathbf{A}\|_F\right)\right).$$

Step 2: Off-diagonal sum.

It remains to bound the off-diagonal sum

$$S := \sum_{i \ne j} a_{ij} X_i X_j.$$

Let $\lambda > 0$ be a parameter whose value we will determine later. By Markovs inequality, we have

$$p_2 = \mathbb{P}(S \ge \frac{t}{2}) = \mathbb{P}(\lambda S \ge \lambda \frac{t}{2}) \le \exp\left(-\frac{\lambda t}{2}\right) \mathbb{E}\left[\exp(\lambda S)\right].$$

Now note that

$$\begin{aligned}\mathbb{E}\left[\exp(\lambda S)\right] &\overset{(a)}{\leq} \mathbb{E}\left[\exp(4\lambda \mathbf{x}^\top \mathbf{A}\mathbf{x}')\right] \\ &\overset{(b)}{\leq} \mathbb{E}\left[\exp(C_1\lambda \mathbf{g}^\top \mathbf{A}\mathbf{g}')\right] \quad \text{(by Comparison Lemma 6.2.3)} \\ &\overset{(c)}{\leq} \exp(C\lambda^2\|\mathbf{A}\|_F^2) \quad \text{(by Lemma 6.2.2 for Gaussian chaos),}\end{aligned}$$

where (a) follows from decoupling (Lem. 4.4.2), (b) follows from comparison (Lem. 4.4.3), and (c) uses the Gaussian chaos MGF (Lem. 4.4.4). The above holds for all λ such that $|\lambda| \leq c/\|\mathbf{A}\|_2$, where $c > 0$ is the absolute constant in Lemma 4.4.4. Putting this bound into the earlier inequality, we obtain

$$p_2 \leq \exp\left(-\frac{\lambda t}{2} + C\lambda^2\|\mathbf{A}\|_F^2\right).$$

Optimizing over $0 \leq \lambda \leq c/\|\mathbf{A}\|_2$, we conclude that

$$p_2 \leq \exp\left(-c\min\left(\|\mathbf{A}\|_2^2, \|\mathbf{A}\|_F^2\right)\right).$$

□

4.5. Subsampled randomized Fourier transforms

Our second construction of an efficient random embedding is called a *subsampled randomized Fourier transform* (SRFT). The basic idea is to apply a fast Fourier transform (FFT) to mix vector coordinates quickly and efficiently.

Recall that the *(discrete) Fourier transform matrix* $\mathbf{F}$ is given by:

$$\mathbf{F} = \frac{1}{\sqrt{n}}\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)(n-1)} \end{bmatrix}, \tag{60}$$

where $\omega = e^{-2\pi i/n}$ denotes the primitive n-th root of unity. For a specific example with $n = 4$:

$$\mathbf{F} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

where i represents the imaginary unit $i = \sqrt{-1}$. Since $\omega^n = 1$, $\omega^{(i-j)} = 0$ for $i = j$ and is an n-th root of unity if $1 \leq i < j \leq n$. It follows that the Fourier transform matrix is unitary:

$$\mathbf{F}\mathbf{F}^* = \mathbf{I}.$$

Definition 4.5.1 (Subsampled randomized Fourier transform)**.** The *Subsampled randomized Fourier transform* (SRFT) is the random matrix $\mathbf{S} := \sqrt{\frac{n}{d}}\mathbf{R}\mathbf{F}\mathbf{E} \in \mathbb{C}^{d\times n}$, where

- $\mathbf{R} \in \mathbb{C}^{d\times n}$ is a random restriction (i.e., each row of $\mathbf{R}$ is an i.i.d. uniform row of $\mathbf{I}_n$);
- $\mathbf{F} \in \mathbb{C}^{n\times n}$ is a discrete Fourier transform in (60);
- $\mathbf{E} = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_n) \in \mathbb{C}^{n\times n}$ is a random diagonal sign matrix, where $\varepsilon_i \sim \{\pm 1\}$ i.i.d.. with equal probability.

SRFTs are very efficient in terms of storage and application. One only needs to store $n + d$ numbers, namely the diagonal entries of the matrix $\mathbf{E}$ and the d coordinates chosen by the restriction $\mathbf{R}$. Applying the SRFT to a vector in $\mathbb{C}^n$ can be achieved via a subsampled FFT and requires only $O(n \log d)$ arithmetic operations. However, it is important to use a good FFT library to achieve this performance.

Exercise 4.5.2. Show that SRFT is an isotropy on average: $\mathbb{E}[\mathbf{S}^*\mathbf{S}] = \mathbf{I}_n$.

The design of the SRHT may seem mysterious, but there are clear intuitions to support it [AC09]. Suppose that we want to estimate the energy (i.e., squared ℓ_2 norm) of a fixed vector $\mathbf{x}$ by sampling ℓ of its n entries at random. On average, these random entries carry ℓ/n of the total energy. (The factor $\sqrt{\ell/n}$ reverses this scaling.) When $\mathbf{x}$ has a few large components, the variance of this estimator is very high. On the other hand, when the components of $\mathbf{x}$ have comparable magnitude, the estimator has much lower variance, so it is precise with very high probability.

The purpose of the matrix product $\mathbf{FE}$ is to flatten out input vectors before we sample. When we multiply a vector $\mathbf{x}$ by $\mathbf{FE}$, each entry in $\mathbf{FEx}$ becomes a sum of the entries of $\mathbf{x}$, each weighted by a complex exponential with unit magnitude (since the Fourier matrix has entries $e^{-2\pi ikj/n}$) and i.i.d. equal-probability signs (from $\mathbf{E}$). This operation spreads the influence of each entry of $\mathbf{x}$ across all entries of $\mathbf{FEx}$. Intuitively, each entry of $\mathbf{FEx}$ becomes a sum of many independent or nearly independent terms due to the random signs in $\mathbf{E}$ and the orthogonality of the Fourier basis. By the law of large numbers and concentration inequalities, these sums converge to values of similar magnitude, leading to a near-uniform distribution of magnitudes in the entries of $\mathbf{FEx}$.

More precisely, fix a unit vector $\mathbf{x}$, and examine the first component of $\mathbf{FEx}$:

$$(\mathbf{FEx})_i = \sum_j \mathbf{F}_{ij}\varepsilon_j \mathbf{x}_j.$$

This random sum clearly has zero mean. The random variables $\mathbf{F}_{ij}\varepsilon_j\mathbf{x}_j$ take values from $[-n^{-1}, n^{-1}]$ almost surely, so Hoeffdings inequality (Prop. 2.2.10) shows that

$$\mathbb{P}\left(|(\mathbf{FEx})_i| \geq t\right) \leq 2e^{-nt^2/2}.$$

By union bound,

$$\mathbb{P}\left(\max_i |(\mathbf{FEx})_i| \geq t\right) \leq 2ne^{-nt^2/2}.$$

The right-hand side is at most ε if t is large enough so that $t \geq \sqrt{2/n}\log n\varepsilon^{-1}$, which is slightly larger than the standard deviation $O(n^{-1/2})$ of the entries in the random sum above.

We will prepare some lemmas. The next result states that randomly sampling rows from a matrix with orthonormal columns results in a well-conditioned matrix. The minimum size of the sample depends primarily on the row norms, and the sampling procedure is most efficient when the matrix has uniformly small rows. We state this result in detail because it may have independent interest.

Lemma 4.5.3 (Row sampling, Lem. 3.4 in [Tro11])**.** *Let $\mathbf{W} \in \mathbf{C}^{n\times k}$ be a matrix with orthonormal columns, and define the quantity $M := n \cdot \max_{i=1,\dots,n} \|\mathbf{e}_i^*\mathbf{W}\|^2$. For a positive parameter α, select the sample size*

$$d \geq \alpha M \log k.$$

Draw a random subset T from $\{1,2,\dots,n\}$ by sampling d coordinates without replacement. Then

$$\mathbb{P}\left(\sqrt{\frac{(1-\delta)d}{n}} \leq \sigma_{\min}(\mathbf{R}_T\mathbf{W}) \leq \sigma_{\max}(\mathbf{R}_T\mathbf{W}) \leq \sqrt{\frac{(1+\eta)d}{n}}\right)$$
$$\geq 1 - k\left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\alpha\log k} - k\left(\frac{e^{\eta}}{(1+\eta)^{1+\eta}}\right)^{\alpha\log k},$$

where $\mathbf{R}_T \in \{0,1\}^{|T|\times n}$ is the matrix that projects onto coordinates in T.

PROOF. Let $\mathbf{w}_i^*$ denote the i-th row of $\mathbf{W}$. Then $M = n\cdot\max_i \|w_i\|_2$. We can control the extreme singular values of the random matrix $\mathbf{R}^T\mathbf{W}$ by bounding the extreme eigenvalues of the $k\times k$ Gram matrix

$$\mathbf{Y} := (\mathbf{R}_T\mathbf{W})^*(\mathbf{R}_T\mathbf{W}) = \sum_{j\in T}\mathbf{w}_i\mathbf{w}_i^* = \sum_{i=1}^d \mathbf{X}_i.$$

Recall that T is a uniform random d-subset of $\{1,2,\dots,n\}$. Therefore, we may just as well view $\mathbf{Y}$ as a sum of d random matrices $X_1,\dots,X_l$ sampled *without* replacement from the family

$$\mathscr{T} := \{\mathbf{w}_i\mathbf{w}_i^* : i = 1,2,\dots,n\}$$

of PSD matrices with dimension k. The matrix Chernoff bound (Thm. 2.5.15) with the extension in Exc. 2.5.19 for sampling without replacement, allows us to obtain large deviation bounds for the extreme eigenvalues of $\mathbf{Y}$. For this, we just need to determine the parameters involved in the Chernoff bound. First, note that

$$\lambda_{\max}(\mathbf{w}_i\mathbf{w}_i^*) = \|\mathbf{w}_i\|^2 \le M/n \quad \text{for } j = 1,2,\dots,n.$$

We easily compute the expectation of the first sample using the fact that the columns of $\mathbf{W}$ are orthonormal:

$$\mathbb{E}[\mathbf{Y}] = d\mathbb{E}[\mathbf{X}_1] = \frac{d}{n}\sum_{j=1}^{n}\mathbf{w}_i\mathbf{w}_i^* = \frac{d}{n}\mathbf{W}^*\mathbf{W} = \frac{d}{n}\mathbf{I}.$$

Hence we get

$$\mathbb{P}\left(\lambda_{\min}(\mathbf{Y}) \le (1-\delta)\frac{d}{n}\right) \le k\left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{d/M} \qquad \text{for } 0 < \delta \le 1,$$

$$\mathbb{P}\left(\lambda_{\max}(\mathbf{Y}) \ge (1+\delta)\frac{d}{n}\right) \le k\left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{d/M} \qquad \text{for } \delta > 0.$$

Finally, subsitute $\lambda_{\min}(\mathbf{Y}) = \sigma_k(\mathbf{R}_T\mathbf{W})^2$ and $\lambda_{\max}(\mathbf{Y}) = \sigma_1(\mathbf{R}_T\mathbf{W})^2$. □

Lemma 4.5.4 (Row norms, Lem. 3.3 in [Tro11])**.** *Let* $\mathbf{W}$ *be an* $n \times k$ *matrix with orthonormal columns. Then* $\mathbf{FEW}$ *is an* $n \times k$ *matrix with orthonormal columns and*

$$\mathbb{P}\left(\max_i \|\mathbf{e}_i^*(\mathbf{FEW})\| \ge \sqrt{\frac{k}{n}} + \sqrt{\frac{8\log\beta n}{n}}\right) \le \frac{1}{\beta}.$$

PROOF. Since the columns of $\mathbf{W}$ are orthonormal, $\mathbf{W}^*\mathbf{W} = \mathbf{I}_k$. Therefore, $\|\mathbf{W}\|_2 = 1$ and $\|\mathbf{W}\|_F = \sqrt{k}$. To check that $\mathbf{FEW}$ is also an orthonormal matrix, we note that

$$(\mathbf{FEW})^*(\mathbf{FEW}) = \mathbf{W}^*\mathbf{W} = \mathbf{I}_k,$$

since $\mathbf{E}$ and $\mathbf{F}$ are orthogonal matrices.

Fix a row index $j \in \{1,2,\dots,n\}$ and define the function

$$f(\mathbf{x}) := \|\mathbf{e}_i^*\mathbf{F}\,\mathrm{diag}(\mathbf{x})\mathbf{W}\| = \|\mathbf{x}^*\mathbf{DW}\|,$$

where we denote $\mathbf{D} := \mathrm{diag}(\mathbf{e}_i^*\mathbf{F})$ for the diagonal matrix constructed from the i-th row of $\mathbf{F}$ and we also used the fact that

$$\mathbf{u}^*\mathrm{diag}(\mathbf{v}) = (\mathbf{u}_1^*\mathbf{v}_1,\dots,\mathbf{u}_d^*\mathbf{v}_d) = (\mathbf{v}_1^*\mathbf{u}_1,\dots,\mathbf{v}_d^*\mathbf{u}_d) = \mathbf{v}^*\mathrm{diag}(\mathbf{u}).$$

Then with the Rademacher vector $\boldsymbol{\varepsilon} = (\varepsilon_1,\dots,\varepsilon_n)$,

$$f(\boldsymbol{\varepsilon}) = \|\mathbf{e}_i^*\mathbf{FEW}\| = \|\boldsymbol{\varepsilon}^*\mathbf{DW}\|.$$

Note that $\|\mathbf{D}\|_2 = n^{-1/2}$ since each element of $\mathbf{F}$ has modulus $n^{-1/2}$. Hence

$$\mathbb{E}[f(\boldsymbol{\varepsilon})] \le \mathbb{E}\left[f(\boldsymbol{\varepsilon})^2\right]^{1/2} = \mathbb{E}[\|\boldsymbol{\varepsilon}^*\mathbf{DW}\|^2] = \|\mathbf{DW}\|_F \le \|\mathbf{D}\|_2\|\mathbf{W}\|_F = \sqrt{k/n}.$$

Next, note that

$$|f(\mathbf{x}) - f(\mathbf{y})| \le \|(\mathbf{x}-\mathbf{y})^*\mathbf{DW}\| \le \|\mathbf{x}-\mathbf{y}\|\|\mathbf{D}\|_2\|\mathbf{W}\|_2 = \frac{1}{\sqrt{n}}\|\mathbf{x}-\mathbf{y}\|.$$

Hence f is $n^{-1/2}$-Lipschitz continuous. Invoking McDiarmid's inequality (Prop. 2.2.12),

$$\mathbb{P}\left(\|\mathbf{e}_i^* \mathbf{FEW}\| \geq \sqrt{\frac{k}{n}} + \sqrt{\frac{8\log \beta n}{n}} \right) \leq e^{-\frac{8\log(\beta n)}{8}} = \frac{1}{\beta n}.$$

The above bound holds for all $i = 1, \dots, n$. Then the assertion follows from a union bound. □

We are now ready to state and prove the main theoretical guarantee for SRFT. With large probability, it preserves the geometry up to a constant factor provided the embedding dimension d is the true dimension k times a log factor in n.

Theorem 4.5.5 (Concentration of SRFT)**.** *Fix a linear subspace $\mathsf{L} \leq \mathbb{R}^n$ of dimension k. Let $\mathbf{U} \in \mathbb{R}^{n \times k}$ denote an orthogonal matrix whose columns form an orthonormal basis for L. Let $\mathbf{S} \in \mathbb{R}^{d \times n}$ be the SRFT in Def. 4.5.1. Fix $\varepsilon, \gamma \in [0,1]$. Then*

$$\mathbb{P}\left(1 - \varepsilon \leq \sigma_{\min}(\mathbf{SU}) \leq \sigma_{\max}(\mathbf{SU}) \leq 1 + \varepsilon\right) \geq 1 - \gamma$$

provided

$$d \geq \frac{\log_k(2k\gamma^{-1})}{\varepsilon^2} \left(\sqrt{k} + \sqrt{8\log kn}\right)^2 \log k \qquad \left(= \Omega(\varepsilon^{-2} k \log k)\right).$$

PROOF. Denote $M = n \max_i \|\mathbf{e}_i^* (\mathbf{FEU})\|^2$, which is random variable depending on $\mathbf{E}$. Set the embedding dimension

$$d := \alpha \left(\sqrt{k} + \sqrt{8\log kn}\right)^2 \log k,$$

where $\alpha > 0$ is a constant to be determined. Define events

$$A := \left\{ \sqrt{\frac{(1-\delta)d}{n}} \leq \sigma_{\min}(\mathbf{RFEU}) \leq \sigma_{\max}(\mathbf{RFEU}) \leq \sqrt{\frac{(1+\eta)d}{n}} \right\} \in \sigma(\mathbf{R}, \mathbf{E}),$$
$$B := \{d \geq \alpha M \log k\} \in \sigma(\mathbf{E}),$$

where A depends on the two independent random matrices $\mathbf{R}$ and $\mathbf{E}$ while B is determined by $\mathbf{E}$ only. Since $\mathbf{S} = \sqrt{n/d}\,\mathbf{RFE}$, the event A defined above equals

$$A = \left\{ \sqrt{1-\delta} \leq \sigma_{\min}(\mathbf{SU}) \leq \sigma_{\max}(\mathbf{SU}) \leq \sqrt{1+\eta} \right\}.$$

Choosing $\eta = 2\varepsilon$ and $\delta = \varepsilon$ gives

$$A \subseteq \{1 - \varepsilon \leq \sigma_{\min}(\mathbf{SU}) \leq \sigma_{\max}(\mathbf{SU}) \leq 1 + \varepsilon\}.$$

Thus it suffices to show $\mathbb{P}(A) \geq 1 - \gamma$. Hence we wish to show that A occurs with large probability. By Lemma 4.5.3, we know

$$\mathbb{P}_{\mathbf{R}}(A \,|\, B, \mathbf{E}) \geq 1 - k\left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\alpha \log k} - k\left(\frac{e^{\eta}}{(1+\eta)^{1+\eta}}\right)^{\alpha \log k}$$

for an arbitrary realization of $\mathbf{E}$. We will choose α large enough so that each of the second and the third terms in the right-hand side above is at most $\gamma/2$. For instance,

$$\begin{aligned} k\left(\frac{e^{\eta}}{(1+\eta)^{1+\eta}}\right)^{\alpha \log k} \leq \gamma/2 \qquad &\iff \qquad \alpha (\log k)\left(\eta - (1+\eta)\log(1+\eta)\right) \leq \log(\gamma/2k) \\ &\iff \qquad \alpha \geq \frac{\log_k(2k\gamma^{-1})}{(1+\eta)\log(1+\eta) - \eta} \\ &\impliedby \qquad \alpha \geq \frac{\log_k(2k\gamma^{-1})}{\eta^2}. \end{aligned}$$

For the last implication, we used the following observation. Write $g(\eta) = (1+\eta)\log(1+\eta) - \eta$, we have $g(0) = g'(0) = 0$ and $g''(0) \geq 1$, so second-order Taylor expansion yields $g(\eta) \geq \eta^2/2$. Replacing η with $-\delta$, similar argument gives

$$k\left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\alpha \log k} \leq 1/k \quad \Longleftrightarrow \quad \alpha \geq \frac{\log_k(2k\gamma^{-1})}{(1-\delta)\log(1-\delta)+\delta} \quad \alpha \geq \frac{\log_k(2k\gamma^{-1})}{\delta^2}.$$

Thus it suffices to choose

$$\alpha \geq \max\left\{\frac{2}{(1+\eta)\log(1+\eta)-\eta}, \frac{2}{(1-\delta)\log(1-\delta)+\delta}\right\} \quad \text{or} \quad \alpha \geq \frac{\log_k(2k\gamma^{-1})}{(\eta \wedge \delta)^2}.$$

In particular, for $\delta = 5/6$ and $\eta = 7/6$, $\alpha = 4$ satisfies the first condition above.

Now by Lemma 4.5.4 (with $\beta = k$),

$$\mathbb{P}_{\mathbf{E}}(B^c) = \mathbb{P}\left(\max_i \|\mathbf{e}_i^*(\mathbf{FEU})\| \geq \sqrt{\frac{k}{n}} + \sqrt{\frac{8\log kn}{n}}\right) \leq k^{-1}.$$

Thus it follows that

$$\begin{aligned}
\mathbb{P}(A^c) &= \mathbb{P}(A^c \mid B)\mathbb{P}(B) + \mathbb{P}(A^c \mid B^c)\mathbb{P}(B^c) \\
&\leq \mathbb{P}(A^c \mid B) + \mathbb{P}(B^c) \\
&= \mathbb{E}_{\mathbf{E}}[\mathbb{P}_{\mathbf{R}}(A^c \mid B, \mathbf{E})] + \mathbb{P}(B^c) \\
&\leq 3k^{-1}.
\end{aligned}$$

This is enough to conclude. □

Exercise 4.5.6. Modify the proof of Theorem 4.5.5 and show the following under the same setting,

$$\mathbb{P}\left(\sqrt{1/6} \leq \sigma_{\min}(\mathbf{SU}) \leq \sigma_{\max}(\mathbf{SU}) \leq \sqrt{13/6}\right) \geq 1 - 3k^{-1},$$

provided

$$d \geq 4(\sqrt{k} + \sqrt{8\log kn})^2 \log k.$$

CHAPTER 5

Randomized Least Squares

Let $\mathbf{A}$ be a matrix in $\mathbb{C}^{m\times n}$ with linearly independent columns, and let $\mathbf{b}$ be a vector in $\mathbb{C}^m$. The ordinary least squares problem with respect to these parameters is to compute

$$\mathbf{x}^\star = \arg\min_{\mathbf{x}\in\mathbb{C}^n} \|\mathbf{A}\mathbf{x}-\mathbf{b}\|^2. \tag{61}$$

The assumption that $\mathbf{A}$ has linearly independent columns ensures that (61) has a unique minimizer. In particular, by taking the gradient of the objective and setting it equal to zero, we find that (see Exc. 1.3.9) $\mathbf{x}^\star$ solves the normal equations

$$\mathbf{A}^*\mathbf{A}\mathbf{x}^\star = \mathbf{A}^*\mathbf{b}. \tag{62}$$

In this chapter, we are interested in least squares in the *overdetermined* regime $n \ll m$, where the number number m of linear equations to be satisfied are much larger than the number n of the variables (i.e., the coordinates of $\mathbf{x}^\star$), so it is impossible to find an exact solution.

5.1. Classical algorithms for least squares

5.1.1. A direct method. In linear algebra, a direct method is an algorithm that produces an exact solution to a given problem using a finite number of arithmetic operations. Modulo issues with finite-precision arithmetic, direct methods in numerical linear algebra generally produce extremely accurate solutions. Direct methods exhibit almost no variation in runtime, unless they attempt to account for sparsity.

Directly solving least-squares problem (*total complexity*= $O(mn^2)$)

Goal: Given $\mathbf{A}\in\mathbb{C}^{m\times n}$ with full rank and $\mathbf{b}\in\mathbb{C}^m$, find $\mathbf{x}^*\in\arg\min_{\mathbf{x}\in\mathbb{C}^n}\|\mathbf{A}\mathbf{x}-\mathbf{b}\|^2$, or equivalently, (62).

1. *Decompose* $\mathbf{A}=\mathbf{Q}\mathbf{R}$ *s.t.*$\mathbf{Q}\in\mathbb{C}^{m\times n}$ *orthonormal* ($\mathbf{Q}^*\mathbf{Q}=\mathbf{I}_n$) *and* $\mathbf{R}\in\mathbb{C}^{n\times n}$ *upper-triangular*
   - This uses Gram-Schmidt process and takes $O(mn^2)$ time.
   - The columns of $\mathbf{Q}$ forms an orthonormal basis for range($\mathbf{A}$) and the columns of $\mathbf{R}$ are the coordinates of the respective columns of $\mathbf{A}$ expressed on this orthonormal basis.
2. *The normal equation* (62) *becomes* $\mathbf{R}^*\mathbf{R}\mathbf{x}^\star=\mathbf{R}^*\mathbf{Q}\mathbf{b}$, *so we set* $\mathbf{x}^\star=\mathbf{R}^{-1}\mathbf{Q}\mathbf{b}$.
   - The computation above involves two matrix-vector products, in total $O(mn)$ arithmetic operations.

5.1.2. Conjugate gradient method. When the matrix $\mathbf{A}$ is very large but sparse, then computing the gram matrix $\mathbf{A}^*\mathbf{A}$ can be extremely costly. Likewise, its QR-decomposition $O(mn^2)$ is also very expensive. For $\mathbf{A}$ sparse, we can instead leverage the fact that matrix-vector multiplications of the form $\mathbf{A}\mathbf{y}$ are very cheap at $O(mn)$ and does not require reading and storing the entire $\mathbf{A}$.

An *iterative method* is an algorithm that generates a sequence of parameters $\mathbf{x}_k$ where $\lim_{k\to\infty}\mathbf{x}_k=\mathbf{x}^\star$ for some value $\mathbf{x}^\star$ with desired properties. Iterative methods are essential for vast diversity of optimization problems. In partigular, they also play an important role in solving linear equations. A classical example of iterated algorithm for solving overdetermined least squares is the *conjugate gradient* (CG) algorithm. The conjugate gradient method is typically introduced as a technique for solving positive-definite linear systems

$$\mathbf{G}\mathbf{z}^\star=\mathbf{h} \qquad \text{where } \mathbf{G} \text{ is symmetric positive definite.} \tag{63}$$

The advantage of CG over matrix factorization methods is that CG only requires access to $\mathbf{G}$ via the matrixvector multiplication primitive: $\mathbf{z} \mapsto \mathbf{Gz}$.

The key concept behind CG is the following notion of conjugacy. We say that two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are *conjugate* (with respect to $\mathbf{G}$) if

$$\mathbf{u}^\top \mathbf{G}\mathbf{v} = 0.$$

Since $\mathbf{G}$ is symmetric and positive-definite, the left-hand side defines an inner product:

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{G}} := \mathbf{u}^\top \mathbf{G}\mathbf{v} = \langle \mathbf{G}\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{G}^\top \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{G}\mathbf{v} \rangle.$$

Hence two vectors are conjugate if and only if they are orthogonal with respect to this inner product.

Now suppose that

$$\mathscr{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_n\} \subseteq \mathbb{C}^n$$

is a set of n mutually conjugate vectors with respect to $\mathbf{G}$, i.e., $\langle \mathbf{p}_i, \mathbf{p}_j \rangle_{\mathbf{G}} = 0$ for all $i \neq j$. Then $\mathscr{P}$ forms a basis for $\mathbb{C}^n$, and we may express the solution $\mathbf{z}^\star$ of $\mathbf{G}\mathbf{z}^\star = \mathbf{h}$ in this basis:

$$\mathbf{z}^\star = \sum_{i=1}^{n} \alpha_i \mathbf{p}_i \qquad \Longrightarrow \qquad \mathbf{G}\mathbf{z}^\star = \sum_{i=1}^{n} \alpha_i \mathbf{G}\mathbf{p}_i.$$

Then left-multiplying the equation $\mathbf{G}\mathbf{z}^\star = \mathbf{h}$ by the vector $\mathbf{p}_k^\top$ gives

$$\mathbf{p}_k^\top \mathbf{h} = \mathbf{p}_k^\top \mathbf{G}\mathbf{z}^\star = \sum_{i=1}^{n} \alpha_i \mathbf{p}_k^\top \mathbf{G}\mathbf{p}_i = \sum_{i=1}^{n} \alpha_i \langle \mathbf{p}_k, \mathbf{p}_i \rangle_{\mathbf{G}} = \alpha_k \langle \mathbf{p}_k, \mathbf{p}_k \rangle_{\mathbf{G}}$$

and so $\mathbf{z}^\star$ can be computed from the coefficients below:

$$\alpha_k = \frac{\langle \mathbf{p}_k, \mathbf{h} \rangle}{\langle \mathbf{p}_k, \mathbf{p}_k \rangle_{\mathbf{G}}}. \tag{64}$$

This provides the following method for solving the equation $\mathbf{G}\mathbf{x} = \mathbf{h}$: *Find a sequence of n conjugate directions, and then compute the coefficients α_k* by (64).

In order to from the above sketch of CG as an iterative algorithm, we make the following observation that solving the equation $\mathbf{G}\mathbf{z}^\star = \mathbf{h}$ is equivalent to solving an unconstrained quadratic problem:

$$\mathbf{G}\mathbf{z}^\star = \mathbf{h} \qquad \Longleftrightarrow \qquad \mathbf{z}^\star = \arg\min_{\mathbf{z}} \left(f(\mathbf{z}) := \frac{1}{2} \mathbf{z}^* \mathbf{G}\mathbf{z} - \mathbf{h}^* \mathbf{z} \right).$$

CG is overall a descent method for the quadratic minimization in the right-hand side but with additional reqirement that the search directions $\mathbf{p}_k = \mathbf{z}_k - \mathbf{z}_{k-1}$ are conjugate. For instance, the gradient of the quadratic function in the right-hand side equals the residual:

$$\nabla f(\mathbf{z}) := \mathbf{G}\mathbf{z} - \mathbf{h}.$$

Hence, given $\mathbf{z}_k$, we may want to move toward $-\nabla f(\mathbf{z}_k) = \mathbf{h} - \mathbf{G}\mathbf{z}^k$. But this negative gradient direction may not be conjugate to all previous directions $\mathbf{p}_1, \ldots, \mathbf{p}_{k-1}$, so we may need to apply Gram-Schmidt and leave only the $\mathbf{G}$-orthogonal directions in $\mathbf{p}_k$. The standard implementation of CG for solving positive definite equation (63) is stated in Algorithm 2 below.

Algorithm 2 Conjugate Gradient Method for Solving $\mathbf{Gz} = \mathbf{h}$

Goal: Given $\mathbf{G} \in \mathbb{R}^{n\times n}$ (symmetric positive definite) and $\mathbf{h} \in \mathbb{R}^n$, find $\mathbf{z}^* \in \mathbb{R}^n$ such that $\mathbf{Gz}^* = \mathbf{h}$.
Input: Initial guess $\mathbf{z}_0$; number of iterations T (or tolerance)
Initialize the residual: $\mathbf{r}_0 \leftarrow \mathbf{h} - \mathbf{Gz}_0$
Initialize the search direction: $\mathbf{p}_0 \leftarrow \mathbf{r}_0$
for $k = 0, \ldots, T-1$ **do**
  Compute step size: $\alpha_k \leftarrow \|\mathbf{r}_k\|^2 / \langle \mathbf{p}_k, \mathbf{p}_k \rangle_{\mathbf{G}}$
  Update the solution: $\mathbf{z}_{k+1} \leftarrow \mathbf{z}_k + \alpha_k \mathbf{p}_k$
  Update the residual: $\mathbf{r}_{k+1} \leftarrow \mathbf{r}_k - \alpha_k \mathbf{G}\mathbf{p}_k$
  if $\|\mathbf{r}_{k+1}\|$ is sufficiently small **then break**
  Update conjugate direction coefficient: $\beta_k \leftarrow \frac{\|\mathbf{r}_{k+1}\|^2}{\|\mathbf{r}_k\|^2}$
  Update search direction: $\mathbf{p}_{k+1} \leftarrow \mathbf{r}_{k+1} + \beta_k \mathbf{p}_k$
end for
Output: $\mathbf{z}_T$ (or $\mathbf{z}_{k+1}$ if terminated early)

Now for solving the normal equation (62) in least squares, we may take $\mathbf{G} = \mathbf{A}^*\mathbf{A}$ and $\mathbf{h} = \mathbf{A}^*\mathbf{b}$ and apply CG.

Below are benefits for CG for solving least squares problems over the direct method:

1. (*Fast Convergence*): In exact arithmetic, CG converges in at most n iterations for an n-dimensional problem.
2. (*Memory Efficient*): Unlike direct methods (e.g., QR or LU decomposition), CG does not require storing large intermediate matrices since it only requires matrix-vector products $\mathbf{z} \mapsto \mathbf{Az}$.

In addtion, CG enjoyes the following exponential convergence guarantee:

Theorem 5.1.1 (Convergence of CG)**.** *Let $\mathbf{e}_k = \mathbf{z}_k - \mathbf{z}^\star$ denote the error of CG in Alg. 2 after k iterations. Then we have the bound*

$$\|\mathbf{G}^{1/2}\mathbf{e}_k\| \le 2\left(\frac{\sqrt{\kappa(\mathbf{G})}-1}{\sqrt{\kappa(\mathbf{G})}+1}\right)^k \|\mathbf{G}^{-1/2}\mathbf{h}\|, \tag{65}$$

where $\kappa(\mathbf{G}) := \sigma_{\max}(\mathbf{G})/\sigma_{\min}(\mathbf{G})$ is the condition number of $\mathbf{G}$.

Remark 5.1.2 (Preconditioned CG)**.** As one can see from (65), the bound on the convergence rate for CG becomes very slow if $\mathbf{G}$ is ill-conditioned (i.e., $\kappa(\mathbf{G}) \gg 1$). This is not just the problem of the upper bound getting large, but practically CG is known to suffer for ill-conditioned problems.

One remedy is to employ the 'preconditioned conjugate gradient' (PCG) method. The idea behind PCG is to form a preconditioner $\mathbf{C}$ for which we can efficiently evaluate both $\mathbf{C}$ and $\mathbf{C}^{-1}$, and the condition number of the matrix $\hat{\mathbf{G}} := \mathbf{C}^{-1}\mathbf{G}(\mathbf{C}^{-1})^*$ is $O(1)$. We then solve $\hat{\mathbf{G}}\mathbf{y} = \mathbf{C}^{-1}\mathbf{h}$ with just a few iterations of CG. Finally, we report $\mathbf{z} = (\mathbf{C}^*)^{-1}\mathbf{y}$. We will revisit this idea when we discuss the 'sketch-and-precondition' method later in Section 5.4.

5.2. Three randomized algorithms for least-squares

This section introduces and analyzes three methods for subspace embeddings to address problem (62). Each method requires selecting an appropriate family of embeddings, such as Gaussian embeddings, sparse sign matrices, or SRFTs (Subsampled Randomized Fourier Transforms), with the choice ideally tailored to leverage the structure of the problem.

Given that $\mathbf{A}$ is dense, both sparse sign matrices and SRFTs serve as suitable options here. In this discussion, we focus on SRFT embeddings, and designate $\mathbf{S} \in \mathbb{C}^{d\times m}$ as an SRFT embedding matrix, with d determined by the context. We rely on the fact that $\mathbf{S}$ can be stored in $O(m)$ space and applied to a vector in $\mathbb{C}^m$ in $O(m \log d)$ time. Additionally, we make use of Theorem 4.5.5, which says the following:

(Thm. 4.5.5 and Exc. 4.1.3) A $d \times n$ SRFT is an ε-embedding of any fixed n-dim affine subspace $\mathtt{L}$ of $\mathbb{R}^m$ ($n \le m$) w.h.p. if $d = \Omega(\varepsilon^{-2} k \log k)$.

For applications requiring high-precision $\varepsilon \ll 1$, the factor ε^{-2} in the embedding dimension d can be prohibitively large.

5.2.1. One-shot Sketch and Solve. Given a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$, a vector $\mathbf{b} \in \mathbb{C}^m$, and an embedding dimension $d \le m$, the "Sketch-and-Solve" approach first constructs a subspace embedding $\mathbf{S} \in \mathbb{C}^{d\times m}$. It then finds

$$\hat{\mathbf{x}} \in \arg\min_{\mathbf{x}\in\mathbb{C}^n} \|\mathbf{SAx} - \mathbf{Sb}\| \qquad \Longleftrightarrow \qquad (\mathbf{SA})^*(\mathbf{SA})\hat{\mathbf{x}} = (\mathbf{SA})^*\mathbf{b} \tag{66}$$

and returns $\hat{\mathbf{x}}$ as an approximate solution to (62). The hope is that $\hat{\mathbf{x}}$ is close to $\mathbf{x}^\star$ in the sense that they both give similar residuals:

$$\|\underbrace{\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}}_{=:\hat{\mathbf{r}}}\| \approx \|\underbrace{\mathbf{A}\mathbf{x}^\star - \mathbf{b}}_{=:\mathbf{r}^\star}\|.$$

Corollary 5.2.1 (One-shot sketch-and-solve)**.** *Let $\mathbf{x}^\star$ denote the solution of the overdetermined LS in (61). Let $\mathbf{S}$ denote the $d \times n$ SRFE. Choose $d = \Omega(\varepsilon^{-2} n \log n)$ so that $\mathbf{S}$ is an ε-embedding of the affine subspace $\mathtt{L} := \mathrm{range}(\mathbf{A}) - \mathbf{b}$ with high probability. On the event that $\mathbf{S}$ is an ε-embeding of $\mathtt{L}$, we have*

$$\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\| \le \frac{1+\varepsilon}{1-\varepsilon}\|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|. \tag{67}$$

PROOF. Assuming $\mathbf{S}$ is an ϵ-embedding for the affine subspace $\mathtt{L} = \mathrm{range}(\mathbf{A}) - \mathbf{b}$, note that

$$(1-\epsilon)\|\hat{\mathbf{r}}\| \le \|\mathbf{S}\hat{\mathbf{r}}\| \le \|\mathbf{S}\mathbf{r}^\star\| \le (1+\epsilon)\|\mathbf{r}^\star\|,$$

where first and third inequalities use the ϵ-embedding property of $\mathbf{S}$ and that $\hat{\mathbf{r}}, \mathbf{r}^\star \in \mathtt{L}$, while the middle inequality relies on the optimality of $\hat{\mathbf{x}}$ for (66). Comparing the first and the last quantities, this gives (67), as desired. □

Remark 5.2.2 (Computational cost of Sketch-and-Solve)**.** Deriving the data $\mathbf{SA}$ and $\mathbf{Sb}$ to form the sketched problem (66) takes $O(mn\log d)$ time (recall the fast matrix-vector multiplication $\mathbf{Sy}$ in $O(m\log d)$ times; we do this n times for each column of $\mathbf{A}$). After this, we can solve (66) directly in $O(dn^2)$ time (see Sec. 5.1.1). If accuracy is not a concern, the total runtime $O(mn\log d + n^2 d)$ might seem efficient. However, achieving the performance guarantee (67) with high probability requires choosing d proportional to $\varepsilon^{-2} n\log n$ (Thm. 4.5.5). With this choice, the runtime becomes $O(mn\log(\varepsilon^{-2}n) + n^3\log(n)/\epsilon^2)$, where the dependence on $1/\epsilon^2$ makes Sketch-and-Solve impractical for high accuracy demands.

5.3. Iterative sketching

In this section, we study the "Iterative-Sketch" algorithm. A key element to this algorithms success is sketching in a way that allows for successive refinement. Observe that (61) is equivalent to the following quadratic minimization problem

$$\min_{\mathbf{x}\in\mathbb{C}^n} \left[f(\mathbf{x}) := \frac{1}{2}\|\mathbf{Ax}\|^2 - (\mathbf{A}^*\mathbf{b})^*\mathbf{x} \right]. \tag{68}$$

We will view it as minimizing the linear function $-(\mathbf{A}^*\mathbf{b})^*\mathbf{x}$ along with the quadratic penalizaiton $\frac{1}{2}\|\mathbf{Ax}\|^2$.

The key idea in iterative sketching is that when we solve (68), we only apply sketching for the quadratic panelty term and consider

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}\in\mathbb{C}^n} \left[\frac{1}{2}\|\mathbf{SAx}\|^2 - (\mathbf{A}^*\mathbf{b})^*\mathbf{x} \right].$$

This is to reduce the extra bias coming from sketching the target vector $\mathbf{b}$ as $\mathbf{Sb}$. Since sketching is only applied on the quadratic term that determines the Hessian, this method is also called the iterative Hessian sketching [PW16].

Suppose we compute the solution $\hat{\mathbf{x}}$ to the above sketched problem by solving the corresponding normal equation (eq. for critical point)

$$(\mathbf{SA})^*(\mathbf{SA})\hat{\mathbf{x}} = \mathbf{A}^*\mathbf{b}.$$

We could return $\hat{\mathbf{x}}$ as an approximate solution to the original least squares problem (61), but we can improve it *iteratively*. To see this, let $\boldsymbol{\varepsilon}$ denote the error of this approximation:

$$\mathbf{x}^\star = \hat{\mathbf{x}} + \boldsymbol{\varepsilon}.$$

If we can figure out what is the error $\boldsymbol{\varepsilon}$, then we could improve our initial approximation $\hat{\mathbf{x}}$ to $\hat{\mathbf{x}} + \boldsymbol{\varepsilon}$ and obtain the exact solution. In order to do so, observe that

$$\mathbf{A}\boldsymbol{\varepsilon} = \mathbf{A}\mathbf{x}^\star - \mathbf{A}\hat{\mathbf{x}} \approx \mathbf{b} - \mathbf{A}\hat{\mathbf{x}} =: \tilde{\mathbf{b}}.$$

Thus, we can in principle compute the approximation error $\boldsymbol{\varepsilon}$ by solving the following derived least-squares problem

$$\boldsymbol{\varepsilon} = \arg\min_{\mathbf{z}} \|\mathbf{A}\mathbf{z} - \tilde{\mathbf{b}}\|.$$

The above is the same least squares problem $\mathbf{A}\mathbf{x}^\star \approx \mathbf{b}$ that we initally tried to solve but with $\mathbf{b}$ replaced with $\tilde{\mathbf{b}} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$. Thus we can re-apply our sketching technique to the above derived problem. This amounts to solving

$$(\mathbf{SA})^*(\mathbf{SA})\boldsymbol{\varepsilon} = \mathbf{A}^*\tilde{\mathbf{b}}.$$

Once such $\boldsymbol{\varepsilon}$ is found, we update our approximation solution as $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \boldsymbol{\varepsilon}$. Repeating this procesure is exactly the iterative sketching method.

Algorithm 3 Iterative Sketching for Least-Squares

Goal: Find an approximate solution $\hat{\mathbf{x}}$ for the least-squares problem $\min_{\mathbf{x}\in\mathbb{F}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|$
Input: Matrix $\mathbf{A} \in \mathbb{F}^{m\times n}$, vector $\mathbf{b} \in \mathbb{F}^m$, sketching matrix $\mathbf{S} \in \mathbb{F}^{d\times m}$, number of iterations K
Sample a random embedding $\mathbf{S} \in \mathbb{C}^{d\times m}$
Initialize $\hat{\mathbf{x}} \leftarrow \mathbf{0}$ (initial solution guess)
Set the initial residual: $\mathbf{r} \leftarrow \mathbf{b}$
Form $\mathbf{G} = (\mathbf{SA})^*(\mathbf{SA})$ and compute its Cholskey factorization $\mathbf{G} = \mathbf{L}\mathbf{L}^*$ ($\triangleright$ $\mathbf{L}$ *lower-triangular*)
for $k = 1, \ldots, K$ **do**
    Solve for $\mathbf{u}$ in $\mathbf{G}\mathbf{u} = \mathbf{A}^*\mathbf{r}$
    Update solution: $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \mathbf{u}$
    Update residual: $\mathbf{r} \leftarrow \mathbf{r} - \mathbf{A}\mathbf{u}$
end for
Output: Approximate solution $\hat{\mathbf{x}}$

In each step, the major computation is done when we solve the linear system

$$\mathbf{G}\mathbf{u} = \mathbf{L}\mathbf{L}^*\mathbf{u} = \mathbf{A}^*\mathbf{r}. \tag{69}$$

Note that we need to form $\mathbf{G} = (\mathbf{SA})^*(\mathbf{SA})$ at a cost $O(mn\log d + dn^2)$ and then performing a Cholesky factorization $\mathbf{G} = \mathbf{L}\mathbf{L}^*$ at an additional cost of $O(n^3)$, both only once before the for loop. Since $n < d$, the total complexity of forming $\mathbf{G}$ and computing the Cholesky factorization is $O(mn\log d + dn^2)$. Then in each iteration inside the for loop, we must solve (69) with a new vector $\mathbf{r}$, which requires forming $\mathbf{h} = \mathbf{A}^*\mathbf{r}$ in $O(nm)$ time (using forward and backward substitution) and then solving $\mathbf{G}\mathbf{u} = \mathbf{h}$ in $O(n^2)$ time, which is much less than $O(nm)$.

Now note that the complexity of the first iteration of the Iterative-Sketch approach matches that of the entire Sketch-and-Solve paradigm, provided d is the same. However, Sketch-and-Solve requires $d = \Omega(n\log n/\epsilon^2)$ to yield an ϵ-accurate solution, while the inner steps of Iterative-Sketch require only $d \in \Omega(n\log n)$. The upshot is that, in the iterative approach, the *dependence on ϵ appears only in the number of iterations*. The following proposition provides the reason for this.

Lemma 5.3.1 (Recursive improvement of Iterative Sketching)**.** *Set* $\mathbf{P} \in \mathbb{C}^{m\times m}$ *as the orthogonal projector onto* range$(\mathbf{A})$*. Suppose* $\mathbf{S}$ *is a subspace embedding for* range$(\mathbf{A})$ *with distortion* $\delta \in (0, 0.25]$*. Let* $\mathbf{x}_k$ *denote the estimated parameter after* k *steps of Alg. 3. Then*

$$\|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{P}\mathbf{b}\| \leq \left[\frac{(2-\delta)\delta}{(1-\delta)^2}\right]^k \|\mathbf{P}\mathbf{b}\|. \tag{70}$$

Furthermore,

$$\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 \leq \|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2 + \left[\frac{(2-\delta)\delta}{(1-\delta)^2}\right]^{2k} \|\mathbf{P}\mathbf{b}\|^2. \tag{71}$$

(c.f., Note that $\left[\frac{(2-\delta)\delta}{(1-\delta)^2}\right]^2 \leq 3\delta$ *for* $\delta \leq 1/4$*.)*

PROOF. We first show (71) assuming (70). First note that, for any $\mathbf{x} \in \mathbb{C}^n$,

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \|\mathbf{A}\mathbf{x} - \mathbf{P}\mathbf{b} - (\mathbf{I}-\mathbf{P})\mathbf{b}\|^2 = \|\mathbf{A}\mathbf{x} - \mathbf{P}\mathbf{b}\|^2 + \|(\mathbf{I}-\mathbf{P})\mathbf{b}\|^2 \tag{72}$$

since $(\mathbf{I}-\mathbf{P})\mathbf{b}$ is perpendicular to the range of $\mathbf{A}$. Since $\mathbf{x}^*$ minimizes the LHS and since the minimum value of the RHS is $\|(\mathbf{I}-\mathbf{P})\mathbf{b}\|^2$ that is achieved when $\mathbf{A}\mathbf{x} = \mathbf{P}\mathbf{b}$ (note that $\mathbf{P}\mathbf{b} \in$ range$(\mathbf{A})$), we have

$$\mathbf{A}\mathbf{x}^\star = \mathbf{P}\mathbf{b} \quad \text{and} \quad \|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2 = \|(\mathbf{I}-\mathbf{P})\mathbf{b}\|^2. \tag{73}$$

Using (72) with $\mathbf{x} = \hat{\mathbf{x}}_k$, this yields (71) as

$$\begin{aligned}\|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{b}\|^2 &= \|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{P}\mathbf{b}\|^2 + \|(\mathbf{I}-\mathbf{P})\mathbf{b}\|^2 \\ &\leq \left[\frac{(2-\delta)\delta}{(1-\delta)^2}\right]^{2k} \|\mathbf{P}\mathbf{b}\|^2 + \|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2,\end{aligned}$$

where the inequality uses the first inequality (70).

It remains to show (70). We will show this by an induction on k. Recall that $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + \mathbf{u}_k$, where

$$\mathbf{u}_k = \arg\min_{\mathbf{u}} \left[\frac{1}{2}\|\mathbf{S}\mathbf{A}\mathbf{u}\|^2 - (\mathbf{A}^*\tilde{\mathbf{b}}_k)^*\mathbf{z}\right]. \qquad \text{where} \qquad \tilde{\mathbf{b}}_k := \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}_{k-1}. \tag{74}$$

By QR-decomposition, write

$$\mathbf{S}\mathbf{A} = \mathbf{Q}\mathbf{R}.$$

This gives an implicit Cholesky decomposition

$$(\mathbf{S}\mathbf{A})^*(\mathbf{S}\mathbf{A}) = \mathbf{R}^*\mathbf{R}.$$

The critical point equation for the problem defining $\hat{\mathbf{u}}_k$ in (74) is $(\mathbf{S}\mathbf{A})^*(\mathbf{S}\mathbf{A})\hat{\mathbf{u}}_k = \mathbf{A}\tilde{\mathbf{b}}_k$, which can be re-written as

$$\mathbf{R}^*\mathbf{R}(\mathbf{x}_k - \mathbf{x}_{k-1}) = \mathbf{A}^*(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}_{k-1}).$$

Multiplying both sides by $(\mathbf{R}^*\mathbf{R})^{-1}$ and rearranging, we obtain the following recursive update equation

$$\mathbf{x}_k = \left[\mathbf{I} - (\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\mathbf{A}\right]\hat{\mathbf{x}}_{k-1} + (\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\mathbf{b}. \tag{75}$$

Importantly, notice that the optimal solution $\mathbf{x}^\star$ for the least squares problem is a fixed point of the above update due to the normal equation $\mathbf{A}^*\mathbf{A}\mathbf{x}^\star = \mathbf{A}^*\mathbf{b}$, so we get

$$\mathbf{x}^\star = \left[\mathbf{I} - (\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\mathbf{A}\right]\hat{\mathbf{x}}^\star + (\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\mathbf{b}. \tag{76}$$

Subtracting (76) from (75), we get

$$\hat{\mathbf{x}}_k - \mathbf{x}^\star = \left[\mathbf{I} - (\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\mathbf{A}\right](\hat{\mathbf{x}}_{k-1} - \mathbf{x}^\star).$$

Multiplying both sides by $\mathbf{A}$, we get (recall that $\mathbf{R}$ is invertible since $\mathbf{A}$ has full rank)

$$\begin{aligned}\mathbf{A}(\hat{\mathbf{x}}_k - \mathbf{x}^\star) &= \left[\mathbf{A} - \mathbf{A}(\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\mathbf{A}\right](\hat{\mathbf{x}}_{k-1} - \mathbf{x}^\star) \\ &= \left[\mathbf{I} - \mathbf{A}(\mathbf{R}^*\mathbf{R})^{-1}\mathbf{A}^*\right]\mathbf{A}(\hat{\mathbf{x}}_{k-1} - \mathbf{x}^\star) \\ &= \Big[\underbrace{\mathbf{I} - (\mathbf{A}\mathbf{R}^{-1})(\mathbf{A}\mathbf{R}^{-1})^*}_{:=\mathbf{B}}\Big]\mathbf{A}(\hat{\mathbf{x}}_{k-1} - \mathbf{x}^\star).\end{aligned}$$

Taking spectral norms,

$$\|\mathbf{A}(\hat{\mathbf{x}}_k - \mathbf{x}^\star)\| \le \|\mathbf{B}\|_2 \|\mathbf{A}(\hat{\mathbf{x}}_{k-1} - \mathbf{x}^\star)\|.$$

Note that since $\mathbf{B}$ is self-adjoint, using (1),

$$\begin{aligned}\|\mathbf{B}\| &= \max\{\lambda_{\max}(\mathbf{B}), \lambda_{\max}(-\mathbf{B})\} \\ &= \max\left\{1 - \sigma_{\min}(\mathbf{A}\mathbf{R}^{-1})^2, -1 + \sigma_{\max}(\mathbf{A}\mathbf{R}^{-1})^2\right\} \\ &\overset{(a)}{\le} \max\left\{1 - \left(\frac{1}{1+\delta}\right)^2, -1 + \left(\frac{1}{1-\delta}\right)^2\right\} \\ &= -1 + \left(\frac{1}{1+\delta}\right)^2 = \frac{(2-\delta)\delta}{(1-\delta)^2}.\end{aligned}$$

For (a) we have used the following observation

$$\sigma_{\max}(\mathbf{A}\mathbf{R}^{-1}) \le \frac{1}{1-\delta}\sigma_{\max}(\mathbf{S}\mathbf{A}\mathbf{R}^{-1}) = \frac{1}{1-\delta}\sigma_{\max}(\mathbf{Q}) = \frac{1}{1-\delta}, \quad \text{and similarly,} \quad \sigma_{\min}(\mathbf{A}\mathbf{R}^{-1}) \ge \frac{1}{1+\delta}. \tag{77}$$

This shows

$$\|\mathbf{A}(\hat{\mathbf{x}}_k - \mathbf{x}^\star)\| \le \left[\frac{(2-\delta)\delta}{(1-\delta)^2}\right]\|\mathbf{A}(\hat{\mathbf{x}}_{k-1} - \mathbf{x}^\star)\|.$$

Repeating this inequality with recalling $\hat{\mathbf{x}}_0 = \mathbf{0}$ and $\mathbf{A}\mathbf{x}^\star = \mathbf{P}\mathbf{b}$, we deduce (70), as desired. □

Note that (71) is the key to the success of the Iterative Sketch paradigm. We can set δ as a modest constant (e.g., $\delta = \frac{1}{4}$) and choose the embedding dimension $d = \Theta(n\log n)$. In terms of the 'dynamic range' $D := \|\mathbf{P}\mathbf{b}\|/\|(\mathbf{I}-\mathbf{P})\mathbf{b}\| > 0$ of the least-squares problem, we only need to run for $O(\log(D/\epsilon))$ iterations to find an ϵ-accurate solution $\hat{\mathbf{x}}$.

Theorem 5.3.2 (Iterative Sketching)**.** *Let $\mathbf{A} \in \mathbb{C}^{m\times n}$ and $\mathbf{b} \in \mathbb{C}^m$ specify an instance of problem (61). Set $D = \|\mathbf{P}\mathbf{b}\|/\|(\mathbf{I}-\mathbf{P})\mathbf{b}\|$ and assume $D > 0$. Let $\mathbf{x}_k$ denote the estimated parameter after k steps of the SRFT-based Iterative Sketch (Alg. 3) with embedding dimension $d = Cn\log n$ for some $C > 0$ a large enough constant. Then we have*

$$\mathbb{P}\left(\|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{b}\|^2 \le (1 + (7/9)^k D^2)\|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2\right) \ge 1 - n^{-1}.$$

In particular, with $k = \log_{15/7}(D^2/\varepsilon)$ iterations, we obtain and ε-approximate solution of (61) with probability at least $1 - n^{-1}$ and the total computational cost is

$$O\left(mn\log n + n^3\log n + mn\log\left(\frac{D}{\epsilon}\right)\right).$$

PROOF. Note that the quantity $\frac{(2-\delta)\delta}{(1-\delta)^2}$ equals $7/9$ when $\delta = 1/4$. Hence by (73) and (71) in Lem. 5.3.1, on the event that $\mathbf{S}$ is a $(1/4)$-embedding of Range($\mathbf{A}$),

$$\begin{aligned}\|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{b}\|^2 &\le \|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2 + (7/9)^k\|\mathbf{P}\mathbf{b}\|^2 \\ &= \|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2 + (7/9)^k D^2\|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2 \\ &= (1 + (7/9)^k D^2)\|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2.\end{aligned}$$

Thus,

$$\mathbb{P}\Big(\|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{b}\|^2 \le (1+(7/9)^k D^2)\|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2\Big) \ge \mathbb{P}(\mathbf{S} \text{ is an } 1/4\text{-embedding of Range}(\mathbf{A})).$$

By Theorem 4.5.5 with $k = n$ and $\gamma = n^{-1}$, we see that if $d = Cn\log n$ for some $C > 0$ large enough, the probability in the RHS above is at least $1 - n^{-1}$. The second part of the statement follows immediately from the first part and Remark 5.2.2. □

5.4. Sketch-and-Precondition

The Sketch-and-Solve and Iterative-Sketch paradigms use subspace embeddings to directly transform the problem data for (61). This section introduces the Sketch-and-Precondition approach, which operates in a different way: the subspace embedding is only used to accelerate the classical conjugate gradient method in Sec. 5.1.2

We proceed by sampling a δ-distortion subspace embedding $\mathbf{S}$ (e.g., SRFT) for range($\mathbf{A}$), where δ is a modest constant. By Theorem 4.5.5, taking $d = \Theta(n\log n)$ allows us to sample such $\mathbf{S}$ with high probability. We then form the dense matrix $\mathbf{Y} = \mathbf{SA}$ in $O(mn\log n)$ time (using fast SRFT multiplication), and perform the QR factorization $\mathbf{Y} = \mathbf{QR}$ in an additional $O(n^3\log n)$ time. Lastly, we solve the least-squares problem (61) using the preconditioned conjugate gradient method, where the matrix $\mathbf{R}^*$ serves as the preconditioner, and then transform the solution by multiplying by $\mathbf{R}^{-1}$. These steps are summerized below.

1. Sample a δ-distortion embedding $\mathbf{S}$ for Range($\mathbf{A}$);
2. Form $\mathbf{SA} = \mathbf{QR}$;.
3. Obtain an approximate solution $\hat{\mathbf{y}}$ for the equation $(\mathbf{AR}^{-1})^*(\mathbf{AR}^{-1})\mathbf{y} = (\mathbf{AR}^{-1})^*\mathbf{b}$ using CG;
4. Transform $\hat{\mathbf{x}} = \mathbf{R}^{-1}\hat{\mathbf{y}}$.

Since $\mathbf{R}$ is upper-triangular, we can compute matrix-vector products with $\mathbf{G} = (\mathbf{AR}^{-1})^*(\mathbf{AR}^{-1})$ in $O(nm)$ time. Thus, the complexity of each preconditioned conjucate gradient (PCG) (see Rmk 5.1.2) iteration is only $O(mn)$. The postprocessing step of transforming the preconditioned solution back to the original domain is done in negligible $O(n^2)$ time. The total arithmetic cost of the Sketch-and-Precondition approach with k iterations is $O(mn\log n + n^3\log n + mnk)$.

Now we bound the number k of iterations in terms of a desired precision ϵ.

Exercise 5.4.1 (Preconditioning by subspace embedding)**.** Let $\mathbf{S}$ be a δ-embedding of Range($\mathbf{A}$) and write $\mathbf{SA} = \mathbf{QR}$. Show that

$$(1-\delta)^2\mathbf{A}^*\mathbf{A} \preceq \mathbf{R}^*\mathbf{R} \preceq (1+\delta)^2\mathbf{A}^*\mathbf{A}. \tag{78}$$

Continuing from the exercise, conjugate the relations in (78) by $\mathbf{R}^{-1}$, and substitute the definition of $\mathbf{G} = (\mathbf{R}^{-1})^*\mathbf{A}^*\mathbf{AR}^{-1} = (\mathbf{AR}^{-1})^*(\mathbf{AR}^{-1})$ to obtain[1]

$$(1-\delta)^2\mathbf{G} \preceq \mathbf{I} \preceq (1+\delta)^2\mathbf{G}.$$

This implies that the condition number $\kappa(\mathbf{G})$ of $\mathbf{G}$ is bounded above by $(1+\delta)/(1-\delta)$. We can easily take $\delta = 1/7$ so that $\kappa(\mathbf{G}) \le 2$.

Theorem 5.4.2 (Sketch and precondition)**.** *Let $\mathbf{A} \in \mathbb{C}^{m\times n}$ and $\mathbf{b} \in \mathbb{C}^m$ specify an instance of problem (61). Set $D = \|\mathbf{Pb}\|/\|(\mathbf{I}-\mathbf{P})\mathbf{b}\|$ and assume $D > 0$. Let $\mathbf{x}_k$ denote the estimated parameter after k steps of the SRFT-based Sketch-and-Precondition with embedding dimension $d = Cn\log n$ for some $C > 0$ a large enough constant. Then we have*

$$\mathbb{P}\Big(\|\mathbf{A}\hat{\mathbf{x}}_k - \mathbf{b}\|^2 \le (1+2(1/4)^k D^2)\|\mathbf{A}\mathbf{x}^\star - \mathbf{b}\|^2\Big) \ge 1 - n^{-1}.$$

[1] We have shown above directly in (77).

In particular, with $k = \log_4(2D^2/\varepsilon)$ iterations for the conjugate gradient, we obtain and ε-approximate solution of (61) with probability at least $1 - n^{-1}$ and the total computational cost is

$$O\left(mn\log n + n^3\log n + mn\log\left(\frac{D}{\epsilon}\right)\right).$$

PROOF. Denote $\mathbf{G} = (\mathbf{AR}^{-1})^*(\mathbf{AR}^{-1})$. In preconditioned conjugate gradient (PCG), the majority of the computational effort is devoted to computing

$$\hat{\mathbf{y}} \approx \mathbf{y} := \mathbf{G}^{-1}\mathbf{h} \qquad \text{where} \qquad \mathbf{h} = (\mathbf{AR}^{-1})^*\mathbf{b}$$

iteratively using conjugate gradient. Our goal is to understand the quality of this approximation and translate it into an approximation for the final solution $\hat{\mathbf{x}}$. For simplicity, set

$$\mathbf{B} := \mathbf{AR}^{-1} \text{ so that } \mathbf{G} = \mathbf{B}^*\mathbf{B} \text{ and } \mathbf{h} = \mathbf{B}^*\mathbf{b}.$$

Also, let $\mathbf{P}$ denote the orthogonal projector onto range$(\mathbf{B})$ = range$(\mathbf{A})$.

Theorem 5.1.1 provides a bound on the error $E := |\mathbf{G}^{1/2}(\hat{\mathbf{y}} - \mathbf{y})|$. We seek to re-express E in a form that aligns more closely with the theorems statement. First, observe that

$$E = \|\mathbf{G}^{1/2}(\hat{\mathbf{y}} - \mathbf{y})\| = \|\mathbf{B}(\hat{\mathbf{y}} - \mathbf{y})\|.$$

Next, recall that $\mathbf{y}$ satisfies the normal equation $\mathbf{B}^*\mathbf{By} = \mathbf{B}^*\mathbf{b}$ for the following least-square problem

$$\mathbf{y} = \arg\min_{\mathbf{z}} \|\mathbf{Bz} - \mathbf{b}\|.$$

From this, it folllows that (recall (73))

$$\mathbf{By} = \mathbf{Pb}.$$

This gives

$$\underbrace{\|\mathbf{B}\hat{\mathbf{y}} - \mathbf{Pb}\|^2}_{=\|\mathbf{B}(\hat{\mathbf{y}}-\mathbf{y})\|^2 = E^2} + \underbrace{\|(\mathbf{I}-\mathbf{P})\mathbf{b}\|^2}_{=\|\mathbf{Ax}^\star - \mathbf{b}\|^2} = \underbrace{\|\mathbf{B}\hat{\mathbf{y}} - \mathbf{b}\|^2}_{=\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2}.$$

Hence we deduce

$$\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 = \|\mathbf{Ax}^\star - \mathbf{b}\|^2 + E^2.$$

Next, we derive a suitable bound on E^2. Let $\kappa = \kappa(\mathbf{G})$. Theorem 5.1.1 provides that

$$E \le 2\rho^k \|\mathbf{G}^{-1/2}\mathbf{h}\|, \qquad \text{where } \rho = \rho(\mathbf{S}) = \frac{\sqrt{\kappa(\mathbf{G})} - 1}{\sqrt{\kappa(\mathbf{G})} + 1}.$$

The quantity ρ is a random variable depending on the embedding matrix $\mathbf{S}$ since $\mathbf{G}$ is determined by $\mathbf{A}$ and $\mathbf{R}$, where $\mathbf{R}$ was obtained by the QR decomposition of the sketched matrix $\mathbf{SA} = \mathbf{QR}$. Also note that

$$\mathbf{Pb} = \mathbf{By} = \mathbf{BG}^{-1}\mathbf{h} = \mathbf{BG}^{-1}\mathbf{B}^*\mathbf{b},$$

which yields

$$\|\mathbf{G}^{-1/2}\mathbf{h}\|^2 = \mathbf{b}^*(\mathbf{BG}^{-1}\mathbf{B}^*\mathbf{b}) = \mathbf{b}^*\mathbf{Pb} = (\mathbf{Pb})^*(\mathbf{Pb}) = \|\mathbf{Pb}\|^2.$$

This gives

$$\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 \le \|\mathbf{Ax}^\star - \mathbf{b}\|^2 + 2\rho^k \underbrace{\|\mathbf{Pb}\|^2}_{=D\|\mathbf{A}\hat{\mathbf{x}}-\mathbf{b}\|^2\|}.$$

Now, recall that we have observed that $\kappa(\mathbf{G}) \le \frac{1+\delta}{1-\delta}$ on the event that $\mathbf{S}$ is a δ-embedding of Range$(\mathbf{A})$. If $\delta = 1/7$, then $\kappa(\mathbf{G}) \le 2$, so $\rho(\mathbf{S}) \le \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2 \le 0.42^2 \le 1/4$. This gives

$$\mathbb{P}\left(\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 \le \|\mathbf{Ax}^\star - \mathbf{b}\|^2 + 2(1/4)^k D^2\|\mathbf{Pb}\|^2\right) \ge \mathbb{P}(\mathbf{S} \text{ is a } (1/3)\text{-embedding of Range}(\mathbf{A})).$$

The probability on the RHS is at least $1 - n^{-1}$ when $d = \Omega(n\log n)$ by Theorem 4.5.5. The second part follows easily from the earlier discussion. □

CHAPTER 6

Randomized SVD

6.1. Spectral and Singular Value Decompositions

In this section, we review some basic facts about the spectral decomposition for self-adjoint matrices and the singular value decompsition for arbitrary matrices.

Self-adjoint matrices admit a spectral decomposition $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ where $\mathbf{V}$ is unitary (i.e., $\mathbf{V}^*\mathbf{V} = \mathbf{I}$) and $\mathbf{\Lambda}$ is a diagonal matrix with diagonal entries $\lambda_1(\mathbf{A}), \dots, \lambda_n(\mathbf{A})$ the *eigenvalues* of $\mathbf{A}$ (see Exc. 6.1.2). Writing $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ into columns, $\mathbf{u}_i$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda_i(\mathbf{A})$ (i.e., $\mathbf{A}\mathbf{v}_i = \lambda_i(\mathbf{A})\mathbf{v}_i$).

Let $\mathbf{B} \in \mathbb{C}^{m\times n}$ be a rectangualr matrix. Write its (compact) SVD as $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ with $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$ (see Exc. 6.1.1). The diagonal entries of $\mathbf{\Sigma}$ are called the *singular values* of $\mathbf{B}$.

Exercise 6.1.1 (Constructing SVD)**.** Let $\mathbf{B} \in \mathbb{C}^{m\times n}$ and $r := \operatorname{rank}(\mathbf{B})$. We will show that it admits a *singular value decomposition*, $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$, where

- $\mathbf{U} \in \mathbb{C}^{m\times r}$ and $\mathbf{V} \in \mathbb{C}^{n\times r}$ have orthonormal columns and $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is a diagonal matrix with positive diagonal entries, say, $\sigma_1(\mathbf{B}), \dots, \sigma_r(\mathbf{B}) > 0$.

The columns of $\mathbf{V}$ (resp., $\mathbf{U}$) are called the *right (*resp., *left) singular vectors*. The decomposition $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ is called a *compact SVD* and it can also be written as the additive rank-one decomposition $\mathbf{B} = \sum_{i=1}^r \sigma_i(\mathbf{B})\mathbf{u}_i\mathbf{v}_i^*$. Such a decomposition can be constructed as follows.

(i) Apply spectral decomposition for the self-adjoint matrix $\mathbf{B}^*\mathbf{B} \in \mathbb{C}^{n\times n}$ to write

$$\mathbf{B}^*\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^*$$

Here $\mathbf{\Lambda} \in \mathbb{R}^{n\times n}$ is the diagonal matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ of $\mathbf{B}^*\mathbf{B}$ and the columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ of $\mathbf{V}$ are the corresponding (orthonormal) eigenvectors of $\mathbf{B}^*\mathbf{B}$. WLOG assume $\lambda_1 \geq \cdots \geq \lambda_n$.

(ii) Show that $\lambda_i > 0$ for all $1 \leq i \leq r$. Define $\sigma_i := \sqrt{\lambda_i}$ and $\mathbf{u}_i := \frac{1}{\sigma_i}\mathbf{B}\mathbf{v}_i$ for $1 \leq i \leq r$. Show that $\mathbf{u}_1, \dots, \mathbf{u}_r$ are orthonormal. Note that

$$[\mathbf{u}_1, \dots, \mathbf{u}_r] \operatorname{diag}(\sigma_1, \dots, \sigma_r)[\mathbf{v}_1, \dots, \mathbf{v}_r]^* = \sum_{i=1}^r \sigma_i \left(\frac{1}{\sigma_i}\mathbf{B}\mathbf{v}_i\right)\mathbf{v}_i^* = \mathbf{B}\underbrace{\left(\sum_{i=1}^r \mathbf{v}_i\mathbf{v}_i^*\right)}_{=:\mathbf{Q}}.$$

Denote $\tilde{\mathbf{V}} := [\mathbf{v}_1, \dots, \mathbf{v}_r] \in \mathbb{C}^{n\times r}$. Then $\mathbf{Q} = \tilde{\mathbf{V}}\tilde{\mathbf{V}}^*$ is the orthogonal projection matrix onto the range of $\tilde{\mathbf{V}}$. Show that this subspace is the column space of $\mathbf{B}^*$, so $\mathbf{Q}\mathbf{B}^* = \mathbf{B}^*$. Denoting $\tilde{\mathbf{U}} := [\mathbf{u}_1, \dots, \mathbf{u}_r]$ and $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$, conclude that

$$\mathbf{B} = \tilde{\mathbf{U}}\mathbf{\Sigma}\tilde{\mathbf{V}}.$$

The compact SVD above can be extended to a 'full SVD' by suitably completing orthonormal bases and appending zeros to $\mathbf{\Sigma}$.

Exercise 6.1.2 (Spectral decomposition for self-adjoint matrices)**.** Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be a self-adjoint matrix, i.e., $\mathbf{A}^* = \mathbf{A}$. Then the spectral theorem for real symmetric matrices states that the following holds:

(1) $\mathbf{A}$ has n real eigenvalues (counting multiplicities);
(2) There exists a unitary matrix $\mathbf{U} \in \mathbb{C}^{n\times n}$, $\mathbf{U}^*\mathbf{U} = \mathbf{I}$ and a diagonal matrix $\mathbf{D} \in \mathbb{R}^{n\times n}$ such that

$$\mathbf{A} = \mathbf{U}^*\mathbf{D}\mathbf{U};$$

(3) In the above, the unitary matrix $\mathbf{U}$ consists of eigenvectors of $\mathbf{A}$ and the diagonal entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$.

In this exercise, we will prove this result by an induction on the dimension n.

(i) Show that all of the eigenvalues of $\mathbf{A}$ are real.

(ii) Let $\mathbf{v}_1$ be an eigenvector of $\mathbf{A}$ and let λ_1 be its associated eigenvalue. By **(i)**, λ_1 is real. Extend $\mathbf{v}_1$ to an orthonormal basis $\mathscr{B} := \{\mathbf{v}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ for $\mathbb{C}^n$ and let $\mathbf{Q} := [\mathbf{v}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n] \in \mathbb{C}^{n\times n}$. Show that

$$\mathbf{Q}^*\mathbf{A}\mathbf{Q} = \left[\begin{array}{c|ccc} \lambda_1 & 0 & \ldots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{A}' & \\ 0 & & & \end{array}\right],$$

where $\mathbf{A}' \in \mathbb{C}^{(n-1)\times(n-1)}$ is a self-adjoint matrix.

(iii) By the induction hypothesis, there exists a unitary matrix $\mathbf{V} \in \mathbb{C}^{(n-1)\times(n-1)}$ and a diagonal matrix $\mathbf{D}' \in \mathbb{R}^{(n-1)\times(n-1)}$ such that $\mathbf{A}' = \mathbf{V}^*\mathbf{D}'\mathbf{V}$. Define

$$\overline{\mathbf{V}} := \left[\begin{array}{c|ccc} 1 & 0 & \ldots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{V} & \\ 0 & & & \end{array}\right] \in \mathbb{C}^{n\times n}.$$

Then show that

$$\mathbf{A} = \mathbf{Q}\overline{\mathbf{V}}^* \left[\begin{array}{c|ccc} \lambda_1 & 0 & \ldots & 0 \\ \hline 0 & & & \\ \vdots & & \mathbf{D}' & \\ 0 & & & \end{array}\right] \overline{\mathbf{V}}\mathbf{Q}^*.$$

Also show that $\mathbf{U} := \overline{\mathbf{V}}\mathbf{Q}^* \in \mathbb{C}^{n\times n}$ is unitary.

6.2. The Truncated SVD

Every matrix $\mathbf{B} \in \mathbb{C}^{m\times n}$ has a singular value decomposition:

$$\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*,$$

where $\mathbf{U} \in \mathbb{C}^{m\times m}$ and $\mathbf{V} \in \mathbb{C}^{n\times n}$ are unitary, and $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \sigma_2, \ldots)$ is a nonnegative, diagonal, and weakly decreasing matrix. The columns of $\mathbf{U}$ are called left singular vectors, the columns of $\mathbf{V}$ are called right singular vectors, and the numbers $\sigma_1, \sigma_2, \ldots$ are the singular values. For concreteness, we assume that $m \geq n$.

Now, we consider a related matrix decomposition. For a parameter $k \leq n$, we write

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma}_1 & 0 \\ 0 & \mathbf{\Sigma}_2 \end{bmatrix},$$

where $\mathbf{\Sigma}_1 \in \mathbb{R}^{k\times k}$ lists the largest k singular values, and $\mathbf{\Sigma}_2$ contains the remaining singular values. We conformally decompose the singular vector matrices as

$$\mathbf{U} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 \end{bmatrix} \quad \text{and} \quad \mathbf{V} = \begin{bmatrix} \mathbf{V}_1 & \mathbf{V}_2 \end{bmatrix},$$

where $\mathbf{U}_1$ and $\mathbf{V}_1$ each have k columns. The k-truncated SVD is the matrix approximation

$$\mathbf{B}_k := \mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^*.$$

Here, the matrices $\mathbf{U}_1 \in \mathbb{F}^{m\times k}$ and $\mathbf{V}_1 \in \mathbb{C}^{n\times k}$ have orthonormal columns, and $\mathbf{\Sigma}_1 \in \mathbb{R}^{k\times k}$ is nonnegative, diagonal, and weakly decreasing.

Remark 6.2.1 (Uniqueness of the SVD)**.** The singular values are always determined uniquely. However, the SVD itself is not unique. Indeed, we can always switch column signs to obtain a different factorization. When there is a duplicated singular value, we can also choose singular vectors to form arbitrary bases for the singular subspaces associated with that singular value. The truncated SVD inherits this lack of uniqueness.

Applications of the Truncated SVD. The truncated SVD plays a fundamental role in matrix approximation, as shown by the following classic theorem.

The k-truncated SVD gives the best rank-k approximation of a matrix B with respect to both the Frobenius norm and the spectral norm. More precisely,

$$\min_{\text{rank}(\mathbf{B}')=k} \|\mathbf{B}-\mathbf{B}'\|_F = \|\mathbf{B}-\mathbf{B}_k\|_F = \sqrt{\sum_{j>k}\sigma_j^2} \quad \text{(Eckart-Young)};$$
$$\min_{\text{rank}(\mathbf{B}')=k} \|\mathbf{B}-\mathbf{B}'\| = \|\mathbf{B}-\mathbf{B}_k\| = \sigma_{k+1} \quad \text{(Mirsky)} \tag{79}$$

A similar statement holds for every unitarily invariant norm.

The truncated SVD has a wide range of applications in linear algebra, statistics, and related fields, including:

- Principal component analysis (PCA), total least-squares (TLS), proper orthogonal decomposition (POD), KarhunenLoève expansions;
- Computations for ordinary least-squares, regularized least-squares problems (ridge regression or Tikhonov regularization), and regularized solutions of linear systems;
- Data reduction, feature extraction, compression, and visualization.

Classical SVD Algorithms and Runtime Comparison. The truncated SVD is challenging to compute in the dense case. By combining column-pivoted QR, bidiagonal reduction, and bidiagonal SVD algorithms, we can compute the k-truncated SVD of an $m \times n$ matrix with $O(mnk)$ arithmetic cost.

For sparse matrices, algorithms based on Lanczos bidiagonalization require k matrix-vector multiplies in sequence plus $O(k(m+n))$ extra arithmetic. Proper implementation is essential for these methods.

Randomized SVD algorithms do not achieve better complexity, as they require a small number of $m \times n \times k$ matrix-matrix multiplies, plus $O((m+n)k)$ extra arithmetic. This computational profile resembles an algorithm for sparse SVD computation. However, matrix-matrix multiplies are often more efficient on most platforms, especially with optimized routines and GPU support, making randomized SVD methods practically faster than classical SVD algorithms.

6.3. Two-Phase Randomized SVD Algorithms

Randomized SVD algorithms follow a two-phase approach:

1. Find a subspace aligned with the span of the dominant left-singular vectors, i.e., range($\mathbf{U}_1$).
2. Compress the matrix to this subspace and compute its SVD.

The underlying philosophy of the above approach can be phrased as: "*figure out where to look, and go look there.*"

The first step is the key building block in randomized SVD and this problem is is known as the *rangefinder problem.*

Problem 6.3.1 (The rangefinder problem)**.** *Given a matrix $\mathbf{B} \in \mathbb{C}^{m\times n}$, a target rank k, and a number of samples $l = k+p$, find an orthonormal matrix $\mathbf{Q} \in \mathbb{C}^{m\times l}$ such that range($\mathbf{Q}$) $\approx$ range($\mathbf{U}_1$).*

Heuristically, the solution $\mathbf{Q}$ of the rangefinder problem safistifes

$$\|\mathbf{B}-\mathbf{Q}\mathbf{Q}^*\mathbf{B}\| \approx \sigma_{k+1}.$$

The key idea is to choose $l = k + p$, slightly larger than the target rank k. Note that the new matrix $\mathbf{QQ}^*\mathbf{B}$ has rank at most l. According to Mirskys theorem (79), there are fundamental restrictions on this procedure: the approximation error must be at least σ_{l+1}. By allowing $l > k$ and aiming for an error close to σ_{k+1}, we can develop highly reliable rangefinder algorithms.

Given a solution $\mathbf{Q}$ to the rangefinder problem, we can approximate the k-truncated SVD of the matrix $\mathbf{B}$.

Randomized k-SVD: $\mathbf{B} \approx \hat{\mathbf{B}} = \hat{\mathbf{U}}_1\hat{\mathbf{\Sigma}}_1\hat{\mathbf{V}}_1$

1. Form $\mathbf{Q}^*\mathbf{B} \in \mathbb{C}^{l\times n}$ using one $l \times m \times n$ matrix-matrix multiplication.
2. Compute the dense $l \times n$ full SVD:
$$\mathbf{Q}^*\mathbf{B} = \hat{\mathbf{U}}\hat{\mathbf{\Sigma}}\hat{\mathbf{V}}^*.$$
This step requires $O(l^2 n)$ arithmetic operations, but $l \ll m$.
3. Obtain an approximate truncated SVD of the original matrix in factored form:
$$\hat{\mathbf{B}} = (\mathbf{Q}\hat{\mathbf{U}})\hat{\mathbf{\Sigma}}\hat{\mathbf{V}}^*,$$
using a single $m \times l \times l$ matrix multiplication.
4. Note that $\hat{\mathbf{B}}$ is actually a rank-l approximate SVD. We can truncate $\hat{\mathbf{\Sigma}}$ to $\hat{\mathbf{\Sigma}}_k$ to obtain a rank-k approximation of the k-truncated SVD. The cost of this optional truncation is negligible, though the corresponding analysis is more involved.

The total cost includes one $l \times m \times n$ matrixmatrix multiplication, plus $O((m+n)l)$ additional arithmetic. Modulo rounding errors, the error in the approximate SVD $\hat{\mathbf{B}}$ is the same as the approximation error in the rangefinder step:

$$\|\mathbf{B} - \hat{\mathbf{B}}\| = \|\mathbf{B} - \mathbf{QQ}^*\mathbf{B}\|.$$

Therefore, solving the rangefinder problem is sufficient.

6.4. Randomized Rangefinders

In this section, we discuss how we solve the rangefinder problem in Problem 6.3.1.

6.4.1. Basic randomized rangefinder. We start with a basic randomized rangefinder. The idea is extremely simple and elegant. We let $\mathbf{B}$ act on a set of random vectors $\mathbf{x}_1, \dots, \mathbf{x}_\ell$ and look at the subspace $\mathtt{L}$ generated by $\mathbf{Bx}_1, \dots, \mathbf{Bx}_\ell$. If ℓ is comparable to k, then $\mathtt{L}$ should be very similar to Range($\mathbf{U}_1$), so we can take $\mathbf{Q}$ to be the orthogonal basis for $\mathtt{L}$.

The underlying geometric idea is that the left singular vectors of $\mathbf{B}$. Recall that $\mathbf{B} \in \mathbb{R}^{m\times n}$ sends the unit sphere in $\mathbb{R}^n$ to an ellipsoid in $\mathbb{R}^m$, and the longest k semiaxes are exactly the top k left-singular vectors of $\mathbf{B}$, which are exactly the columns of $\mathbf{U}_1$. So if we through a few vectors into $\mathbf{B}$, it is most likely that the output vectors align with the dominant axes. See Fig. 6.4.1.

We can translate this intuition into a computational procedure, called the *randomized rangefinder*:

Randomized Rangefinder:

1. Draw a random test matrix $\mathbf{\Omega} \in \mathbb{R}^{n\times l}$ (with l samples).
2. Compute $\mathbf{Y} = \mathbf{B\Omega}$ (one $m \times n \times l$ matrix multiplication).
3. Orthogonalize: $\mathbf{Y} = \mathbf{QR}$ (one $m \times l$ economic QR decomposition).

The total cost is dominated by the single $m \times n \times l$ matrix multiplication. Typically, the additional cost $O(ml^2)$ is much smaller by comparison.

This simple description of the randomized rangefinder masks a number of significant decisions about the implementation, which we list below:

(*What Test Matrix?*) It turns out standard Gaussian matrices work fabulously. But sometimes, one can use an SRFT (which works even better), or a sparse sign matrix (which works roughly the same), and other candidates for random embeddings. Gaussians are actually pretty natural in this setting.

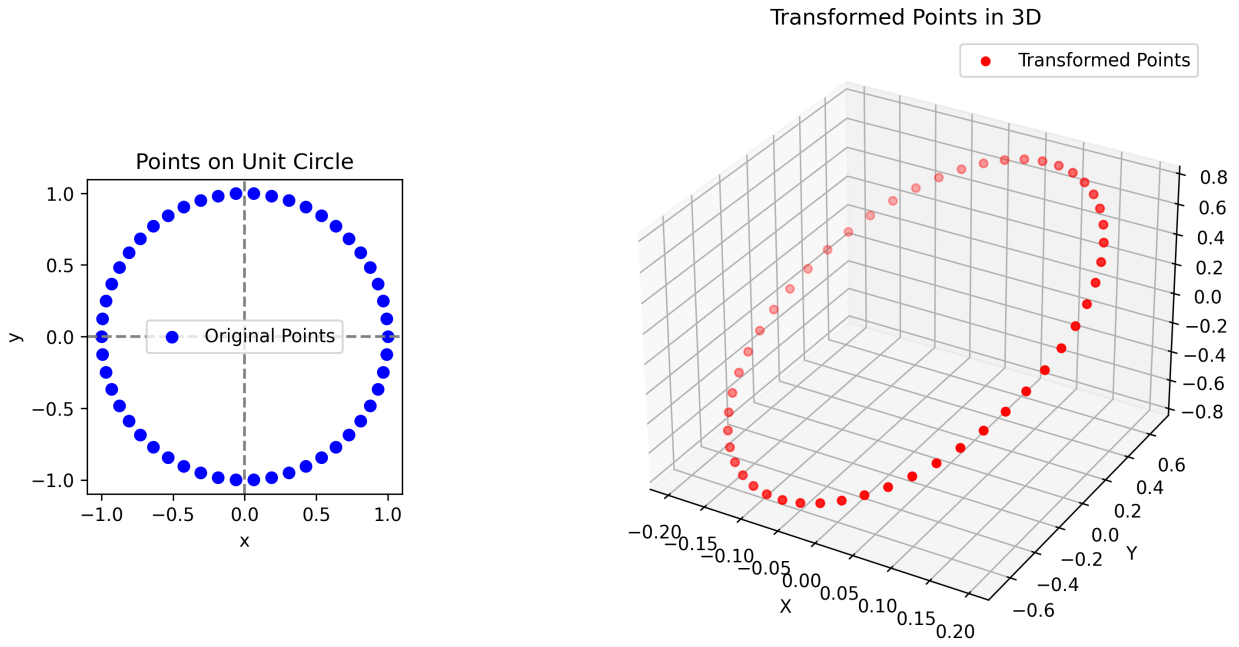


FIGURE 6.4.1. Illustration of the action of a 3×2 matrix $\mathbf{B}$ on sample points in the unit sphere.

(*What Matrix Multiplication?*) Exploit advantageous structure in $\mathbf{B}$ (e.g., sparsity or a fast multiply) whenever possible. There are instances where structure in $\mathbf{\Omega}$ can be exploited as well. Unfortunately, the randomized rangefinder second phase in the randomized SVD procedure involves the basis $\mathbf{Q}$, which is typically dense and unstructured.

(*How Should We Orthogonalize?*) It is important to use stable procedures. Most of the vectors $\mathbf{B}\mathbf{x}_i$ will be aligned in similar directions. Use Householder orthogonalization, or double Gram-Schmidt, or another reliable method. Do not use (modified) Gram-Schmidt.

(*How Much Oversampling?*) If k is known, it often suffices to take $l = k + p$, where $p = 5$ or $p = 10$ or maybe even $p = k$. The theory gives detailed justification for these choices. If k is not known, use randomized error estimation (based on randomized trace estimation), and adaptively increase l until you get satisfactory results.

We will present a detailed analysis of the randomized rangefinder problem in the next lecture. This argument leads to an informative result [HMT11].

Theorem 6.4.1 (Randomized Rangefinder)**.** *Let $\mathbf{B} \in \mathbb{R}^{m \times n}$ be a matrix with singular values $\sigma_1 \geq \sigma_2 \geq \dots$. Let $\mathbf{\Omega} \in \mathbb{R}^{n \times l}$ be a Gaussian test matrix (i.i.d. entries from $\mathcal{N}(0, n^{-1})$). For each $k < l - 1$, the randomized rangefinder produces an orthogonal matrix $\mathbf{Q}$ with the following property:*

$$\mathbb{E}\left[\left\|\mathbf{B} - \mathbf{Q}\mathbf{Q}^*\mathbf{B}\right\|_2\right] \leq \left(1 + \sqrt{\frac{k}{\ell - k - 1}}\right)\sigma_{k+1} + \frac{e\sqrt{\ell}}{\ell - k}\sqrt{\sum_{j>k}\sigma_j^2}.$$

This error bound features two constituents. The first term resembles the spectral-norm error from Mirskys theorem; it shrinks slowly to zero as l increases. The second term resembles the Frobenius-norm error from the EckartYoung theorem; it decreases more quickly as the oversampling increases. If l is very close to k, both contributions can be large. However, if we allow l to be just a bit larger than k, the second factor moderates quickly.

6.4.2. Randomized Subspace Iteration. The bounds in Theorem 6.4.1 suggest that the rangefinder procedure is most effective when the matrix $\mathbf{B}$ exhibits strong spectral decay; that is,

$$\sum_{j>k} \sigma_j^2 \text{ is comparable with } \sigma_{k+1}^2.$$

This insight is correct. Moreover, the converse is also true: the rangefinder tends to produce large errors when the spectral tail has a lot of energy. In this case, we can improve the performance of the rangefinder by augmenting it with powering. The following method is called randomized subspace iteration. It is based on a classic approach from numerical linear algebra. Let l be the number of samples, and let q be a power parameter.

(1) Draw a random test matrix $\mathbf{\Omega} \in \mathbb{R}^{m\times l}$, and set $\mathbf{Q} = \mathbf{\Omega}$. We have changed the dimension for reasons that will become clear in a moment.
(2) Form $\mathbf{Y} = (\mathbf{B}\mathbf{B}^*)^q\mathbf{B}\mathbf{\Omega}$ by alternating application of $\mathbf{B}$ and $\mathbf{B}^*$.
(3) Compute QR decomposition $\mathbf{Y} = \mathbf{Q}\mathbf{R}$.

The idea behind this approach is the following insight: Matrix powers suppress small eigenvalues very quickly. This procedure has the effect of reducing the energy in the spectral tail. Namely, if we use SVD to write $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$, we have

$$(\mathbf{B}\mathbf{B}^*)^q\mathbf{B} = (\mathbf{U}\mathbf{\Sigma}^2\mathbf{U}^*)^q(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^*) = \mathbf{U}\mathbf{\Sigma}^{2q+1}\mathbf{V}^*.$$

It follows that

$$\sigma_j((\mathbf{B}\mathbf{B}^*)^q\mathbf{B}) = \sigma_j(\mathbf{B})^{2q+1}. \tag{80}$$

The analysis of randomized subspace iteration follows as a corollary of the analysis of the randomized rangefinder. We need the following lemma.

Lemma 6.4.2 (Powering)**.** *Let $\mathbf{B}$ be an $m\times n$ matrix, and let $\mathbf{\Omega}$ be an $n\times\ell$ matrix. Fix a nonnegative integer q, form $\mathbf{A} = (\mathbf{B}\mathbf{B}^*)^q\mathbf{B}$, and compute the sample matrix $\mathbf{Z} = \mathbf{A}\mathbf{\Omega}$. Then*

$$\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{B}\|_2 \le \|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{A}\|_2^{1/(2q+1)}.$$

PROOF. Note that

$$\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{B}\|_2 \le \|(\mathbf{I}-\mathbf{P}_\mathbf{Z})(\mathbf{B}\mathbf{B}^*)^q\mathbf{B}\|_2^{1/(2q+1)} = \|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{A}\|_2^{1/(2q+1)}$$

as a direct consequence of Exc. 6.4.3. □

Exercise 6.4.3. Show the following:

(i) If $\mathbf{R}$ is an orthogonal projector and if $\mathbf{D}$ is a nonnegative diagonal matrix, then

$$\|\mathbf{R}\mathbf{D}\mathbf{R}\|_2^t \le \|\mathbf{R}\mathbf{D}^t\mathbf{R}\|_2.$$

(ii) If $\mathbf{R}$ is an orthogonal projector and if $\mathbf{M}$ is a matrix of compatible size,

$$\|\mathbf{R}\mathbf{M}\|_2^{2q+1} \le \|\mathbf{R}(\mathbf{M}\mathbf{M}^*)^q\mathbf{M}\|_2.$$

(*Hint*: Write $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ and use **(i)**.)

As a consequence of Theorem 6.4.1 and Lemma 9.5, we immediately arrive at the following result.

Corollary 6.4.4 (Randomized Subspace Iteration)**.** *Under the hypotheses of Theorem 6.4.1, let $\mathbf{Q}$ be the orthonormal basis computed by the randomized subspace iterations after q steps (i.e., $\mathbf{Q}\mathbf{R} = (\mathbf{B}\mathbf{B}^*)^q\mathbf{B}\mathbf{\Omega}$). Then*

$$\mathbb{E}\left[\left\|\mathbf{B}-\mathbf{Q}\mathbf{Q}^*\mathbf{B}\right\|_2\right] \le \left[\left(1+\sqrt{\frac{k}{\ell-k-1}}\right)\sigma_{k+1}^{2q+1} + \frac{e\sqrt{\ell}}{\ell-k}\sqrt{\sum_{j>k}\sigma_j^{2(2q+1)}}\right]^{\frac{1}{2q+1}}.$$

PROOF. Denote $\mathbf{A} = (\mathbf{B}\mathbf{B}^*)^q\mathbf{B}$. By Jensen's inequality and Theorem 6.4.1,

$$\mathbb{E}\,[\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{B}\|_2] \le \left(\mathbb{E}\left[\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{B}\|_2^{2q+1}\right]\right)^{\frac{1}{2q+1}} \le (\mathbb{E}\,[\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{A}\|_2])^{\frac{1}{2q+1}}.$$

Now invoke Theorem 6.4.1 to bound the right-hand side, noting that $\sigma_j(\mathbf{A}) = \sigma_j(\mathbf{B})^{2q+1}$ from (80). □

The true message of Corollary 10.10 emerges if we bound the series using its largest term σ_{k+1}^{4q+2} and draw the factor σ_{k+1} out of the bracket:

$$\mathbb{E}\,[\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{A}\|_2] \le \left[1 + \sqrt{\frac{k}{\ell-k-1}} + \frac{e\sqrt{\ell}}{\ell-k}\sqrt{(m\wedge n)-k}\right]^{\frac{1}{2q+1}} \sigma_{k+1} \tag{81}$$

In words, as we increase the exponent q, the power scheme drives the multiplicative factor in the error to one exponentially fast. As soon as $q \sim \log(m\wedge n)$, we have $\mathbb{E}\|(\mathbf{I}-\mathbf{P}_\mathbf{Z})\mathbf{A}\|_2 \sim \sigma_{k+1}$, which is the baseline for the spectral norm.

In most situations, the error bound given by Corollary 6.4.4 is substantially better than the crude bound we discussed in (81), where the dominant error term $(m\wedge n)-k$ came from bounding all subsequent singular values by the $(k+1)$st one. When singular values decay quickly so that the series in Corollary 6.4.4 is comparable with its largest term (i.e., $\sum_{j>k}\sigma_j^{2(2q+1)} = O(\sigma_{k+1}^{4q+2})$), then we can remove the dimensional factor $\min\{m,n\}$ from the error bound. In practice, $q=2$ or $q=3$ is entirely adequate for most examples.

6.4.3. Randomized Krylov method. To achieve very high accuracy, randomized subspace iteration may not be sufficient. Instead, we can use a *randomized block Krylov method*. We give a brief summary of the ideas here, but implementations require further attention.

The block Krylov method collects all of the information acquired during the execution of randomized subspace iteration:

$$\mathbf{Y} = \begin{bmatrix}\boldsymbol{\Omega} & \mathbf{B}\mathbf{B}^*\boldsymbol{\Omega} & \cdots & (\mathbf{B}\mathbf{B}^*)^q\boldsymbol{\Omega}\end{bmatrix}.$$

Then, we compute a QR factorization of the sample matrix:

$$\mathbf{Y} = \mathbf{Q}\mathbf{R}.$$

The matrix $\mathbf{Q}$ is very well aligned with the dominant left singular subspace of $\mathbf{B}$. We can use this matrix $\mathbf{Q}$ to compute an approximate k-truncated SVD $\hat{\mathbf{B}}$ of the input matrix.

A detailed analysis of this approach is beyond the scope of this course. The singular values of the approximation satisfy heuristic error bounds:

$$\frac{|\sigma_j(\mathbf{B}) - \sigma_j(\hat{\mathbf{B}})|}{\sigma_j(\mathbf{B})} \lesssim q^2,$$

for $j < l$. In contrast, randomized subspace iteration only yields a bound like:

$$\frac{|\sigma_j(\mathbf{B}) - \sigma_j(\hat{B})|}{\sigma_j(\mathbf{B})} \sim q.$$

This comparison highlights the advantage of the randomized block Krylov method over randomized subspace iteration for computing the maximum eigenvalue. It indicates that we can obtain much better relative accuracy by means of a Krylov method.

6.5. Theoretical analysis of randomized rangefinder

We will provide theoretical derivation of Theorem 6.4.1 on basic randomized rangefinder with Gaussian test matrix. We aim to prove the following two results, which are drawn from [HMT11].

Theorem 6.5.1 (Randomized rangefinder: Frobenius-norm error)**.** *Suppose that the test matrix $\mathbf{\Omega} \in \mathbb{R}^{n\times \ell}$ is standard Gaussian matrix (i.i.d. entries from $\mathcal{N}(0,1)$). For all $k < l - \alpha$, the error in the randomized rangefinder approximation obeys*

$$\mathbb{E}_{\mathbf{\Omega}} \left[\|(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B}\|_F^2 \right] \leq \left(1 + \frac{k}{l-k-1}\right) \sum_{j>k} \sigma_j^2.$$

Furthermore, assuming $p = \ell - k \geq 4$, for each $u, t \geq 1$,

$$\mathbb{P}\left(\|(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B}\|_F \geq \left[(1 + t\sqrt{12k/p}) \sqrt{\sum_{j>k} \sigma_j^2} \right] + ut \frac{e\sqrt{\ell}}{p+1} \sigma_{k+1} \right) \leq 5t^{-p} + 2e^{-u^2/2}. \tag{82}$$

Theorem 6.5.2 (Randomized Rangefinder: Spectral-norm error)**.** *Let $\mathbf{B} \in \mathbb{R}^{m\times n}$ be a matrix with singular values $\sigma_1 \geq \sigma_2 \geq \dots$. Let $\mathbf{\Omega} \in \mathbb{R}^{n\times l}$ be standard Gaussian matrix (i.i.d. entries from $\mathcal{N}(0,1)$) and let $\mathbf{QR} = \mathbf{B\Omega}$ denote the QR decomposition. Then for each $k < l - 1$,*

$$\mathbb{E}\left[\|(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B}\|_2 \right] \leq \left(1 + \sqrt{\frac{k}{\ell - k - 1}}\right) \sigma_{k+1} + \frac{e\sqrt{\ell}}{\ell - k} \sqrt{\sum_{j>k} \sigma_j^2}.$$

Furthermore, assuming $p = \ell - k \geq 4$, for each $u, t \geq 1$,

$$\mathbb{P}\left(\|(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B}\|_2 \geq \left[(1 + t\sqrt{12k/p})\sigma_{k+1} + t\frac{e\sqrt{\ell}}{p+1} \sqrt{\sum_{j>k} \sigma_j^2} \right] + ut \frac{e\sqrt{\ell}}{p+1} \sigma_{k+1} \right) \leq 5t^{-p} + e^{-u^2/2}. \tag{83}$$

Specializing t and u in Theorem 6.5.2, we obtain the following simplified error bounds.

Corollary 6.5.3 (Simplified tail bounds for Randomized Rangefinder: Spectral-norm error)**.** *Under the setting of Theorem 6.5.2 with $p \geq 4$, we have (specializing with $t = e$ and $u = \sqrt{2p}$)*

$$\mathbb{P}\left(\|(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B}\|_2 \geq \left(1 + 17\sqrt{1 + (k/p)}\right) \sigma_{k+1} + \frac{8\sqrt{k+p}}{p+1} \sqrt{\sum_{j>k} \sigma_j^2} \right) \leq 6e^{-p}.$$

Also, we have (specializing with $t = p$ and $u = \sqrt{2p\log p}$)

$$\mathbb{P}\left(\|(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B}\|_2 \geq \left(1 + 8\sqrt{(k+p)p\log p}\right) \sigma_{k+1} + 3\sqrt{k+p} \sqrt{\sum_{j>k} \sigma_j^2} \right) \leq 6p^{-p}.$$

The overall proof strategy is simple:

(1) Understand the form of the error and single out extreme target matrices.
(2) Compute a deterministic bound on the error.
(3) Use random matrix theory to obtain explicit results for Gaussian test matrices.

We will phrase most of the argument for Frobenius norm errors, not the operator norm. The analysis is slightly simpler than the spectral norm case, and the two results are similar in spirit.

6.5.1. Schur complements and extreme cases. The rangefinder procedure finds its most natural expression using the language of Schur complements. For the moment, we will work in either the real or complex field.

Definition 6.5.4 (Schur complement)**.** Let $\mathbf{A} \in \mathbb{C}^n$ be a psd matrix and let $\mathbf{X} \in \mathbb{C}^{n\times k}$ be a test matrix. The Schur complement is the matrix

$$\mathbf{A}/\mathbf{X} := \mathbf{A} - (\mathbf{AX})(\mathbf{X}^*\mathbf{A}^*\mathbf{X})^\dagger(\mathbf{AX})^*, \tag{84}$$

where † denotes the pseudo-inverse.

There are many things to say about Schur complements and their properties (e.g., computing the inverse of a 2×2 block matrix). Schur complements occur naturally in block Gaussian elimination and Cholesky decompositions. Other applications include partial least squares, as well as Nyström decompositions.

We can write the error in the rangefinder procedure in terms of a Schur complement.

Proposition 6.5.5 (Rangefinder: Error)**.** *Fix a target $\mathbf{B} \in \mathbb{C}^{m\times n}$ and let $\mathbf{\Omega} \in \mathbb{C}^{n\times l}$ be a test matrix. Define the (matrix) error in the rangefinder:*

$$\mathbf{E} := \mathbf{E}(\mathbf{B}, \mathbf{\Omega}) = (\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B},$$

Then the squared error matrix takes the form

$$\mathbf{E}^*\mathbf{E} = (\mathbf{B}^*\mathbf{B})/\mathbf{\Omega}.$$

PROOF. Recall that the orthogonal projector onto the range of $\mathbf{Y}$ can be written as $\mathbf{P}_\mathbf{Y} = \mathbf{Y}(\mathbf{Y}^*\mathbf{Y})^\dagger\mathbf{Y}^*$. Hence,

$$\mathbf{P}_{\mathbf{B\Omega}} = (\mathbf{B\Omega})(\mathbf{\Omega}^*\mathbf{B}^*\mathbf{B\Omega})^\dagger(\mathbf{B\Omega})^*.$$

Computing the squared absolute value reveals a Schur complement:

$$\mathbf{E}^*\mathbf{E} = \mathbf{B}^*(\mathbf{I} - \mathbf{P}_{\mathbf{B\Omega}})\mathbf{B} = \mathbf{B}^*\mathbf{B} - \mathbf{B}^*\mathbf{P}_{\mathbf{B\Omega}}\mathbf{B} = \mathbf{B}^*\mathbf{B} - (\mathbf{B}^*\mathbf{B\Omega})(\mathbf{\Omega}^*\mathbf{B}^*\mathbf{B\Omega})^\dagger(\mathbf{B}^*\mathbf{B\Omega})^* = (\mathbf{B}^*\mathbf{B})/\mathbf{\Omega}.$$

This is the required result. □

Corollary 6.5.6 (Rangefinder: Monotonicity)**.** *Suppose that $\mathbf{B}^*\mathbf{B} \preceq \mathbf{C}^*\mathbf{C}$. Then*

$$\mathbf{E}(\mathbf{B}, \mathbf{\Omega})^*\mathbf{E}(\mathbf{B}, \mathbf{\Omega}) \preceq \mathbf{E}(\mathbf{C}, \mathbf{\Omega})^*\mathbf{E}(\mathbf{C}, \mathbf{\Omega}).$$

PROOF. The Schur complement with respect to a fixed test matrix $\mathbf{\Omega}$ is operator monotone. □

This monotonicity result allows us to identify extremals, or worst-case target matrices, for the rangefinder.

Corollary 6.5.7 (Rangefinder: Worst-case)**.** *Let $\mathbf{B} = \mathbf{U\Sigma V}^*$, where $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \sigma_2, \ldots)$. Fix k and define $\mathbf{C} = \mathbf{U\bar{\Sigma}V}^*$, where $\bar{\mathbf{\Sigma}} := \operatorname{diag}(\sigma_1, \ldots, \sigma_1, \sigma_{k+1}, \sigma_{k+2}, \ldots)$. Then*

$$\mathbf{E}(\mathbf{B}, \mathbf{\Omega})^*\mathbf{E}(\mathbf{B}, \mathbf{\Omega}) \preceq \mathbf{E}(\mathbf{C}, \mathbf{\Omega})^*\mathbf{E}(\mathbf{C}, \mathbf{\Omega}).$$

PROOF. By construction, $\mathbf{B}^*\mathbf{B} \preceq \mathbf{C}^*\mathbf{C}$. Apply Corollary 6.5.6. □

We conclude this section with another fact about Schur complements that will play a role in the analysis.

Exercise 6.5.8 (Schur complements: Inclusion)**.** Fix a PSD matrix $\mathbf{A}$. Let $\mathbf{X}, \mathbf{Y}$ be test matrices. If $\operatorname{range}(\mathbf{Y}) \subseteq \operatorname{range}(\mathbf{X})$, then $\mathbf{A}/\mathbf{X} \preceq \mathbf{A}/\mathbf{Y}$.

In other words, if $\mathbf{Y}$ is smaller than $\mathbf{X}$, then the Schur complement with respect to $\mathbf{Y}$ removes less information from $\mathbf{A}$ than the Schur complement with respect to $\mathbf{X}$.

6.5.2. Deterministic test matrices. In this section, we will develop a deterministic bound for the rangefinder that operates for an arbitrary test matrix. In the next section, we will instantiate this result using a random test matrix.

This approach requires a fair amount of notation. Let $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ be the singular value decomposition (SVD) of the target matrix. For a parameter $k \le l$, write

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{\Sigma}_2 \end{bmatrix},$$

where $\mathbf{\Sigma}_1 = \operatorname{diag}(\sigma_1, \dots, \sigma_k) \in \mathbb{R}^{k\times k}$ and $\mathbf{\Sigma}_2 = \operatorname{diag}(\sigma_{k+1}, \sigma_{k+2}, \dots) \in \mathbb{R}^{(n-k)\times(n-k)}$. We conformally partition the singular vector matrices as

$$\mathbf{U} = \begin{bmatrix} \underbrace{\mathbf{U}_1}_{m\times k} & \underbrace{\mathbf{U}_2}_{m\times(n-k)} \end{bmatrix} \quad \text{and} \quad \mathbf{V} = \begin{bmatrix} \underbrace{\mathbf{V}_1}_{n\times k} & \underbrace{\mathbf{V}_2}_{n\times(n-k)} \end{bmatrix} \tag{85}$$

and we extend this partition to the test matrix $\mathbf{\Omega}$:

$$\mathbf{Y} = \mathbf{B}\mathbf{\Omega} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*\mathbf{\Omega} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 \end{bmatrix} \begin{bmatrix} \mathbf{\Sigma}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{\Sigma}_2 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1^*\mathbf{\Omega} \\ \mathbf{V}_2^*\mathbf{\Omega} \end{bmatrix} = \mathbf{U}_1\mathbf{\Sigma}_1 \underbrace{\mathbf{V}_1^*\mathbf{\Omega}}_{=:\mathbf{\Omega}_1} + \mathbf{U}_2\mathbf{\Sigma}_2 \underbrace{\mathbf{V}_2^*\mathbf{\Omega}}_{=:\mathbf{\Omega}_2}. \tag{86}$$

Note that the first matrix $\mathbf{\Omega}_1$ tells us how much the test matrix is aligned with the k dominant right singular vectors of $\mathbf{B}$; this is the important part of $\mathbf{B}$. The second matrix $\mathbf{\Omega}_2$ tells us how well $\mathbf{\Omega}$ is aligned with the tail singular vectors of $\mathbf{B}$; this is the unimportant part of $\mathbf{B}$.

From now on, we assume that $\mathbf{\Omega}_1 \in \mathbb{C}^{k\times l}$ has full *row* rank. This assumption is natural (why?) and essential.

The following deterministic theorem relates the rangefinder to the interplay between $\mathbf{\Omega}_1$ and $\mathbf{\Omega}_2$. This statement and proof are essentially due to Halko et al. [HMT11]

Theorem 6.5.9 (Deterministic error bound)**.** *Let $\mathbf{A}$ be an $m \times n$ matrix with singular value decomposition $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$, and fix $k \ge 0$. Choose a test matrix $\mathbf{\Omega}$, and construct the sample matrix $\mathbf{Y} = \mathbf{B}\mathbf{\Omega}$. Partition $\mathbf{\Sigma}$ as specified in (9.1), and define $\mathbf{\Omega}_1$ and $\mathbf{\Omega}_2$ via (9.2). Assuming that $\mathbf{\Omega}_1$ has full row rank, the approximation error satisfies*

$$\|(\mathbf{I} - \mathbf{P}_\mathbf{Y})\mathbf{B}\|_\xi^2 \le \|\mathbf{\Sigma}_2\|_\xi^2 + \|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_\xi^2,$$

where $\|\cdot\|_\xi$ denotes either the spectral norm or the Frobenius norm.

Note that the first term exactly reproduces the error in the best rank-k approximation of the matrix $\mathbf{B}$. The second term reflects the energy of $\mathbf{\Omega}$ contained in the tail subspace range($\mathbf{V}_2$), weighted by the associated singular values listed in $\mathbf{\Sigma}_2$. The pseudo-inverted matrix reflects the conditioning of $\mathbf{\Omega}_1$.

The proof of Theorem 6.5.9 is based on perturbation theory.

PROOF OF THEOREM 6.5.9. We first isolate an ideal test matrix case and treat the concrete instance as a perturbation of the ideal. Recall that our idea for rangefinder is to test the action of $\mathbf{B}$ on some select vectors (i.e., columns of the test matrix $\mathbf{\Omega} \in \mathbb{C}^{n\times\ell}$) and take the span of their image under $\mathbf{B}$ (i.e., Range($\mathbf{B}\mathbf{\Omega}$)) as an approximate range of the left singular vector fo $\mathbf{B}$ (i.e., $\mathbf{U}_1 \in \mathbb{C}^{m\times k}$). From the decomposition in (86), the ideal case would be when $\mathbf{\Omega} = \mathbf{V}_1$, in which case

$$\mathbf{B}\mathbf{\Omega} = \mathbf{U}_1\mathbf{\Sigma}_1.$$

In this case, note that

$$(\mathbf{I} - \mathbf{P}_{\mathbf{B}\mathbf{\Omega}})\mathbf{B} = (\mathbf{I} - \mathbf{U}_1\mathbf{U}_1^*)\mathbf{B} = \left(\mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^* + \mathbf{U}_2\mathbf{\Sigma}_2\mathbf{V}_2^*\right) - \mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^* = \mathbf{U}_2\mathbf{\Sigma}_2\mathbf{V}_2^*$$

and

$$\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger = \mathbf{\Sigma}_2 \underbrace{(\mathbf{V}_2^*\mathbf{V}_1)}_{=\mathbf{O}} \mathbf{\Omega}_1^\dagger = \mathbf{O},$$

so

$$\|(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}\|_\xi^2 = \|\mathbf{\Sigma}_2\|_\xi^2.$$

Next, we consider the general case. By Prop. 6.5.5, we know that

$$[(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}]^*\,[(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}] = \mathbf{E}(\mathbf{B},\mathbf{\Omega})^*\mathbf{E}(\mathbf{B},\mathbf{\Omega}) = (\mathbf{B}^*\mathbf{B})/\mathbf{\Omega}.$$

We will make the following reductions without loss of generality:

(1) $\mathbf{V} = \mathbf{I}_n$ (by a change of basis for $\mathbb{R}^n$);
(2) $\sigma_1 = 1$ (by rescaling)
(3) $\sigma_1 = \cdots = \sigma_k$ (by considering the worst case in Cor. 6.5.7).

After these reductions, we have

$$\mathbf{B}^*\mathbf{B} = \mathbf{\Sigma}^2 = \begin{bmatrix}\mathbf{I}_k & \mathbf{O}\\ \mathbf{O} & \mathbf{\Sigma}_2^2\end{bmatrix} =: \mathbf{I}_k \oplus \mathbf{\Sigma}_2^2 \qquad \text{where} \qquad \Sigma_2^2 = \operatorname{diag}(\sigma_{k+1},\sigma_{k+2},\ldots).$$

Therefore, it is now sufficient to produce a semidefinite upper bound for the matrix

$$(\mathbf{I}_k \oplus \mathbf{\Sigma}_2)/\mathbf{\Omega}.$$

For intuition about how to proceed, we consult the ideal case: $\mathbf{\Omega} = \mathbf{V}_1$. In this case

$$\mathbf{\Omega} = \mathbf{V}^*\mathbf{\Omega} = \begin{bmatrix}\mathbf{\Omega}_1\\ \mathbf{\Omega}_2\end{bmatrix} = \begin{bmatrix}\mathbf{I}_k\\ \mathbf{0}\end{bmatrix}.$$

We will consider the general case as a perturbation of the above ideal situation. For this, define

$$\mathbf{Z} := \mathbf{\Omega}\mathbf{\Omega}_1^\dagger = \begin{bmatrix}\mathbf{\Omega}_1\mathbf{\Omega}_1^\dagger\\ \mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\end{bmatrix} = \begin{bmatrix}\mathbf{I}_k\\ \mathbf{F}\end{bmatrix}, \quad \text{where} \quad \mathbf{F} := \mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger.$$

For the last equality above, we have used the assumption that $\mathbf{\Omega}_1$ has full row rank so that $\mathbf{\Omega}_1\mathbf{\Omega}_1^* = \mathbf{I}_k$. This construction ensures Range($\mathbf{Z}$) $\subseteq$ Range($\mathbf{\Omega}$). Hence by Exc. 6.5.8, we deduce

$$(\mathbf{I}_k \oplus \mathbf{\Sigma}_2^2)/\mathbf{\Omega} \preceq (\mathbf{I}_k \oplus \mathbf{\Sigma}_2^2)/\mathbf{Z}.$$

Below we will explicitly compute the Schur complement on the RHS to finish the proof.

In the next calculation, we use the symbol $*$ to indicate a block matrix that does not play a role in the argument. Note that

$$\mathbf{Z}^*(\mathbf{I}_k \oplus \mathbf{\Sigma}_2^2)^*\mathbf{Z} = \begin{bmatrix}\mathbf{I}_k & \mathbf{F}^*\end{bmatrix}\begin{bmatrix}\mathbf{I}_k & \mathbf{0}\\ \mathbf{0} & \mathbf{\Sigma}_2^2\end{bmatrix}\begin{bmatrix}\mathbf{I}_k\\ \mathbf{F}\end{bmatrix} = \mathbf{I}_k + \mathbf{F}^*\mathbf{\Sigma}_2^2\mathbf{F},$$

which is positive definite, so its pseudo-inverse is exactly its inverse. Then using the definition of Schur complement in (84), we compute

$$\begin{aligned}(\mathbf{I}_k \oplus \mathbf{\Sigma}_2^2)/\mathbf{Z} &= \begin{bmatrix}\mathbf{I}_k & \mathbf{0}\\ \mathbf{0} & \mathbf{\Sigma}_2^2\end{bmatrix} - \begin{bmatrix}\mathbf{I}_k\\ \mathbf{\Sigma}_2^2\mathbf{F}^*\end{bmatrix}(\mathbf{I}_k + \mathbf{F}^*\Sigma_2^2\mathbf{F})^{-1}\begin{bmatrix}\mathbf{I}_k & \mathbf{F}\mathbf{\Sigma}_2^2\end{bmatrix}\\ &= \begin{bmatrix}\mathbf{I}_k - (\mathbf{I}_k + \mathbf{F}^*\mathbf{\Sigma}_2^2\mathbf{F})^{-1} & \triangle\\ \triangle^* & \mathbf{\Sigma}_2\Big[\mathbf{I} - \underbrace{\mathbf{\Sigma}_2^2\mathbf{F}^*(\mathbf{I}_k + \mathbf{F}^*\Sigma_2^2\mathbf{F})^{-1}\mathbf{F}\mathbf{\Sigma}_2}_{\text{drop}}\Big]\mathbf{\Sigma}_2\end{bmatrix}\\ &\preceq \begin{bmatrix}\mathbf{F}^*\mathbf{\Sigma}_2^2\mathbf{F} & \triangle\\ \triangle^* & \mathbf{\Sigma}_2^2\end{bmatrix}.\end{aligned}$$

To see the PSD ordering, note that we can dominate the $(2,2)$ block in the middle line by dropping the PSD matrix that is being subtracted from $\mathbf{I}$. For the $(1,1)$ block, we can apply the numerical fact $1-(1+a)^{-1} \le a$, valid for $a \ge 0$, to its eigenvalues, and noting that $\mathbf{I} - (\mathbf{I}+\mathbf{A})^{-1}$ with PSD $\mathbf{A}$ has a power-series expansion in $\mathbf{A}$ so it shares the same eigenvectors with $\mathbf{A}$.

So far, we have shown that

$$[(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}]^*\,[(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}] \preceq \begin{bmatrix} \mathbf{F}^*\boldsymbol{\Sigma}_2^2\mathbf{F} & \triangle \\ \triangle^* & \boldsymbol{\Sigma}_2^2 \end{bmatrix}.$$

Taking trace, the Frobenius norm bound follows readily:

$$\|(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}\|_F^2 \le \operatorname{tr}(\boldsymbol{\Sigma}_2^2) + \operatorname{tr}(\mathbf{F}^*\boldsymbol{\Sigma}_2^2\mathbf{F}) = \|\boldsymbol{\Sigma}_2\|_F^2 + \|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\|_F^2.$$

The spectral norm bound follows from the above with and additional observation on the spectral norm of 2×2 block PSD matrices (see Exc. 6.5.10). □

Exercise 6.5.10. Show that $\|\mathbf{M}\|_2 \le \|\mathbf{A}\|_2 + \|\mathbf{C}\|_2$ where $\mathbf{M}$ is the following 2×2 PSD matrix:

$$\mathbf{M} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{bmatrix}.$$

6.5.3. Expectation bound with Gaussian test matrices. Theorem 6.5.9 provides us with crucial insights on the rangefinder performance for any test matrix $\boldsymbol{\Omega}$. It turns out that using a random test matrix leads to near-optimal bounds, regardless of the singular-value spectrum of the target matrix $\mathbf{B}$. In this section, we work with a standard normal test matrix $\boldsymbol{\Omega} \in \mathbb{R}^{n\times l}$. We exploit the fact that $\boldsymbol{\Omega}_1 = \mathbf{V}_1^*\boldsymbol{\Omega}$ and $\boldsymbol{\Omega}_2 = \mathbf{V}_2^*\boldsymbol{\Omega}$ are statistically independent standard normal matrices.

We will need a series of preliminary results on Gaussian random matrices.

Exercise 6.5.11. Let $\boldsymbol{\Omega} \in \mathbb{R}^{n\times \ell}$ be a standard Gaussian matrix (i.e., i.i.d. entries from $\mathcal{N}(0,1)$) and fix deterministic matrices $\mathbf{A},\mathbf{B}$ of compatible sizes. Show that

$$\mathbb{E}[\operatorname{tr}(\mathbf{A}\mathbf{X}^*\mathbf{B}\mathbf{X})] = \operatorname{tr}(\mathbf{A})\operatorname{tr}(\mathbf{B}).$$

(*Hint*: First show $\mathbb{E}[(\mathbf{X}^*\mathbf{B}\mathbf{X})_{i,j}] = \operatorname{tr}(\mathbf{B})\,\mathbf{1}(i=j)$.)

Exercise 6.5.12. For real matrices $\mathbf{A},\mathbf{B}$ and a standard Gaussian matrix $\boldsymbol{\Omega}$ of compatible sizes, show that

$$\mathbb{E}[\|\mathbf{A}\boldsymbol{\Omega}\mathbf{B}\|_2] \le \|\mathbf{A}\|_2\|\mathbf{B}\|_F + \|\mathbf{A}\|_F\|\mathbf{B}\|_2.$$

(c.f. This is a non-trivial result due to Gordon [Gor85, Gor88].)

Lemma 6.5.13 (Tail bound on the spectral norm of inverse Gaussian matrix)**.** *Let $\boldsymbol{\Omega} \in \mathbb{R}^{m\times n}$ be a standard Gaussian matrix with $n \ge m \ge 2$. Then for each $t > 0$,*

$$\mathbb{P}(\|\boldsymbol{\Omega}^\dagger\|_2 \ge t) \le \frac{1}{\sqrt{2\pi(n-m+1)}}\left(\frac{e\sqrt{n}}{n-m+1}\right)^{n-m+1} t^{-(n-m+1)}.$$

PROOF. Omitted. See [CD05, Lem. 4.1]. □

Exercise 6.5.14 (Bound on the expected spectral norm of inverse Gaussian matrix)**.** Let $\boldsymbol{\Omega} \in \mathbb{R}^{m\times n}$ be a standard Gaussian matrix with $n-m \ge 1$ and $m \ge 2$. Show that

$$\mathbb{E}[\|\boldsymbol{\Omega}^\dagger\|_2] \le \frac{e\sqrt{n}}{n-m}.$$

(*Hint:* Use the tail bound in Lem. 6.5.13 and $\mathbb{E}[Z] \le a + \mathbb{E}[Z\mathbf{1}(Z \ge a)]$ for a random variable Z and scalar a.)

Lemma 6.5.15 (Expected Frobenius norm of inverse Gaussian matrix)**.** *Let $\boldsymbol{\Omega} \in \mathbb{R}^{m\times n}$ be a standard Gaussian matrix with $n-m \ge 2$. Then*

$$\mathbb{E}[\|\boldsymbol{\Omega}^\dagger\|_F^2] = \frac{m}{n-m-1}.$$

PROOF. Recall that $\mathbf{W} := \mathbf{\Omega}\mathbf{\Omega}^*$ is an $m \times m$ Wishart matrix with n degrees of freedom. It is known that $\mathbf{W}$ is invertible almost surely and

$$\mathbb{E}[\mathbf{W}] = \frac{\mathbf{\Sigma}^{-1}}{n-m-1},$$

where $\mathbf{\Sigma}$ denotes the $m \times m$ covariance matrix for the columns of $\mathbf{\Omega}$, which is $\mathbf{I}_m$. Since trace and expectation commutes,

$$\mathbb{E}[\|\mathbf{\Omega}^\dagger\|_F^2] = \mathbb{E}[\mathrm{tr}((\mathbf{\Omega}^\dagger)^*\mathbf{\Omega}^\dagger)] = \mathbb{E}[\mathrm{tr}((\mathbf{\Omega}\mathbf{\Omega}^*)^{-1})] = \mathrm{tr}\left(\mathbb{E}[(\mathbf{\Omega}\mathbf{\Omega}^*)^{-1}]\right) = \frac{m}{n-m-1}.$$

□

We can now prove the expectation bounds in Theorems 6.5.2 and 6.5.1.

PROOF OF THE EXPECTATION BOUND IN THEOREM 6.5.2 (SPECRAL). We begin with the bound from Theorem 6.5.9: (Recall that $\mathbf{\Omega}_1$ is a $k \times \ell$ standard Gaussian matrix with $\ell \geq k$ so it has full row rank almost surely; why?)

$$\mathbb{E}\|(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}\|_2 \leq \mathbb{E}\left[\|\mathbf{\Sigma}_2\|_2^2 + \|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_2^2\right]^{1/2} \leq \|\mathbf{\Sigma}_2\|_2 + \mathbb{E}\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_2,$$

where the last inequality is from Jensen's inequality with $\sqrt{a^2+b^2} \leq a+b$ for $a, b \geq 0$. Recall that $\mathbf{\Omega}_1 = \mathbf{V}_1^*\mathbf{\Omega}$ and $\mathbf{\Omega}_2 = \mathbf{V}_2^*\mathbf{\Omega}$ are independent random matrices with i.i.d. standard Gaussian entries since $\mathbf{V}_1$, $\mathbf{V}_2$ are orthogonal and standard Gaussian distribution is rotationally invariant. In particular, $\mathbf{\Omega}_2$ is still standard Gaussian conditioned on $\mathbf{\Omega}_1$. Hence using Exc. 6.5.12

$$\mathbb{E}_{\mathbf{\Omega}_2}\left[\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_2^2 \,\middle|\, \mathbf{\Omega}_1\right] \leq \|\mathbf{\Sigma}_2\|_2\,\|\mathbf{\Omega}_1^\dagger\|_F + \|\mathbf{\Sigma}_2\|_F\,\|\mathbf{\Omega}_1^\dagger\|_2.$$

Thus by iterated expectation with Exc 6.5.14 and Lem. 6.5.15,

$$\begin{aligned}
\mathbb{E}\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_2 &= \mathbb{E}_{\mathbf{\Omega}_1}\left[\mathbb{E}_{\mathbf{\Omega}_2}\left[\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_2^2 \,\middle|\, \mathbf{\Omega}_1\right]\right] \\
&\leq \|\mathbf{\Sigma}_2\|_2 \underbrace{\mathbb{E}\|\mathbf{\Omega}_1^\dagger\|_F}_{\leq\sqrt{\mathbb{E}\|\mathbf{\Omega}_1^\dagger\|_F^2}\leq\sqrt{k/p}} + \|\mathbf{\Sigma}_2\|_F \underbrace{\mathbb{E}\|\mathbf{\Omega}_1^\dagger\|_2}_{\leq\frac{e\sqrt{\ell}}{p}} \\
&\leq \sqrt{\frac{k}{p-1}}\|\mathbf{\Sigma}_2\| + e\frac{\sqrt{\ell}}{p}\|\mathbf{\Sigma}_2\|_F.
\end{aligned}$$

Since $\|\mathbf{\Sigma}_2\|_2 = \sigma_{k+1}$ and $\|\mathbf{\Sigma}_2\|_F = \sqrt{\sum_{j>k}\sigma_j^2}$, this is enough to conclude. □

PROOF OF THE EXPECTATION BOUND IN THEOREM 6.5.1 (FROBENIUS). We begin with the bound from Theorem 6.5.9.

$$\mathbb{E}_{\mathbf{\Omega}}\left[\|(\mathbf{I}-\mathbf{P}_\mathbf{Y})\mathbf{B}\|_F^2\right] \leq \|\mathbf{\Sigma}_2\|_F^2 + \mathbb{E}_{\mathbf{\Omega}}\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_F^2.$$

We compute the expectation by iterated expectation. Recall that $\mathbf{\Omega}_1 = \mathbf{V}_1^*\mathbf{\Omega}$ and $\mathbf{\Omega}_2 = \mathbf{V}_2^*\mathbf{\Omega}$ are indepndent random matrices with i.i.d. standard Gaussian entries since $\mathbf{V}_1$, $\mathbf{V}_2$ are orthogonal and standard Gaussian distribution is rotationally invariant. Note that using cyclic property of trace and Exc. 6.5.11),

$$\begin{aligned}
\mathbb{E}_{\mathbf{\Omega}_2}\left[\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_F^2 \,\middle|\, \mathbf{\Omega}_1\right] &= \mathbb{E}_{\mathbf{\Omega}_2}\left[\mathrm{tr}\left((\mathbf{\Omega}_1^\dagger)^*\mathbf{\Omega}_2^*\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\right) \,\middle|\, \mathbf{\Omega}_1\right] \\
&= \mathbb{E}_{\mathbf{\Omega}_2}\left[\mathrm{tr}\left(\mathbf{\Omega}_1^\dagger(\mathbf{\Omega}_1^\dagger)^*\mathbf{\Omega}_2^*\mathbf{\Sigma}_2^2\mathbf{\Omega}_2\right) \,\middle|\, \mathbf{\Omega}_1\right] \\
&= \mathrm{tr}(\mathbf{\Omega}_1^\dagger(\mathbf{\Omega}_1^\dagger)^*)\,\mathrm{tr}(\mathbf{\Sigma}_2^2) \\
&= \|\mathbf{\Omega}_1^\dagger\|_F^2\,\|\mathbf{\Sigma}_2\|_F^2.
\end{aligned}$$

Integrating out $\mathbf{\Omega}_1$, we get

$$\mathbb{E}\left[\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_F^2\right] = \mathbb{E}_{\mathbf{\Omega}_1}\left[\mathbb{E}_{\mathbf{\Omega}_2}\left[\|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_F^2 \,\middle|\, \mathbf{\Omega}_1\right]\right] = \|\mathbf{\Sigma}_2\|_F^2\,\mathbb{E}\left[\|\mathbf{\Omega}_1^\dagger\|_F^2\right] = \|\mathbf{\Sigma}_2\|_F^2\frac{k}{\ell-k-1},$$

where for the last equality, we used Lemma 6.5.15 for the standard Gaussian matrix $\mathbf{\Omega}_1 \in \mathbb{R}^{k\times\ell}$. This is enough to conclude. □

6.5.4. Tail bounds with Gaussian test matrices. Next, we seek to prove the tail bounds in Theorems 6.5.1 and 6.5.2.

Lemma 6.5.16 (Concentration for functions of a Gaussian matrix)**.** *Suppose that h is a Lipschitz function on matrices satisfying:*

$$|h(\mathbf{X}) - h(\mathbf{Y})| \le L\|\mathbf{X}-\mathbf{Y}\|_F \quad \text{for all } \mathbf{X},\mathbf{Y} \in \mathbb{R}^{m\times n}.$$

Let $\mathbf{G}$ be a standard Gaussian matrix. Then $h(\mathbf{G})$ is concentrated at its mean:[1]

$$\mathbb{P}\left(h(\mathbf{G}) - \mathbb{E}[f(\mathbf{G})] \ge Lt\right) \le e^{-t^2/2}.$$

PROOF. Omitted. See [Bog98, Thm. 4.5.7]. □

Lemma 6.5.17 (Tail bound on the Frobenius norm of inverse Gaussian matrix)**.** *Let $\mathbf{\Omega} \in \mathbb{R}^{m\times n}$ be a standard Gaussian matrix with $n - m \ge 4$. Then*

$$\mathbb{P}\left(\|\mathbf{\Omega}^\dagger\|_F^2 \ge \frac{12m}{n-m}t\right) \le 4t^{-(n-m)/2}$$

PROOF. This result is due to Halko, Martinsson, and Tropp [HMT11, Thm. A.6]. The key observation is the following striking property of Gaussian matrices: A $m \times n$ ($n \ge m$) standard Gaussian matrix $\mathbf{G}$ is orthogonally equivalent to the following random bidiagonal matrix

$$\mathbf{L} = \begin{bmatrix} X_n & & & & & \\ Y_{m-1} & X_{n-1} & & & & \\ & Y_{m-2} & X_{n-2} & & & \\ & & \ddots & \ddots & & \\ & & & Y_1 & X_{n-m+1} \end{bmatrix}_{m\times n}$$

where for each i, X_i^2 and Y_i^2 follows χ^2-distribution with i degrees of freedom and these random variables are mutually independent. Note that

$$\mathbf{L}^\dagger = \begin{bmatrix} X_n^{-1} & & & & \\ \frac{-Y_{m-1}}{X_n X_{n-1}} & X_{n-1}^{-1} & & & \\ & \frac{-Y_{m-2}}{X_{n-1}X_{n-2}} & X_{n-2}^{-1} & & \\ & & \ddots & \ddots & \\ & & & \frac{-Y_1}{X_{n-(m-2)}X_{n-(m-1)}} & X_{n-m+1}^{-1} \end{bmatrix}_{n\times m}$$

Since Frobenius norm is invariant under orthogonal conjugation,

$$\|\mathbf{G}^\dagger\|_F^2 \overset{d}{=} \|\mathbf{L}^\dagger\|_F^2 \le \sum_{j=0}^{m-1} \underbrace{\frac{1}{X_{n-j}^2}\left(1 + \frac{Y_{m-j}^2}{X_{n-j+1}^2}\right)}_{=:W_j}.$$

Using properties of chi-squared distribution, we can bound

$$\mathbb{E}[W_j^q]^{1/q} \le \frac{12}{n-j} \qquad \text{for } j = 0, 1, \ldots, m-1.$$

[1] (c.f., apply the same bound for $-h(\mathbf{G})$ to deduce that $|h(\mathbf{G}) - \mathbb{E}[h(\mathbf{G})]| \ge Lt$ with probability at most $2e^{-t^2/2}$.

Now by union bound and Markov's inequality,

$$\begin{aligned}\mathbb{P}\left(\|\boldsymbol{\Omega}^\dagger\|_F^2 \geq \frac{12m}{n-m}t\right) &\leq \sum_{j=0}^{m-1} \mathbb{P}\left(W_j \geq \frac{12}{n-m}t\right) \\ &\leq \sum_{j=0}^{m-1} \left(\frac{n-m}{12t}\right)^q \mathbb{E}[W_j^q] \\ &\leq t^{-q} \sum_{j=0}^{m-1} \left(\frac{n-m}{n-j}\right)^q.\end{aligned}$$

Note that

$$\begin{aligned}(n-m)^q \sum_{k=n-(m-1)}^{n} k^{-q} &\leq (n-m)^q \int_{n-m}^{\infty} x^{-q}\,dx \\ &\leq (n-m)^q \frac{1}{q-1}(n-m)^{-q+1} = t^{-q}\frac{n-m}{q-1}.\end{aligned}$$

We now specialize $q = (n-m)/2$ so that the last expression is $\frac{2(n-m)}{n-m-2} \leq 4$ since $n-m \geq 4$. This is enough to conclude. □

PROOF OF THE TAIL BOUND IN THEOREM 6.5.2 (SPECRAL). In this proof, we denote $\|\cdot\|_2 = \|\cdot\|$ for brevity. Since $\boldsymbol{\Omega}_1$ and $\boldsymbol{\Omega}_2$ are independent of each other, we can study how the error depends on the matrix $\boldsymbol{\Omega}_2$ by conditioning on the event that $\boldsymbol{\Omega}_1$ is not too irregular. To that end, we define a (parameterized) event on which the spectral and Frobenius norms of the matrix $\boldsymbol{\Omega}_1^\dagger$ are both controlled. For $t \geq 1$, let

$$E_t = \left\{\boldsymbol{\Omega}_1 : \|\boldsymbol{\Omega}_1^\dagger\|_2 \leq A \quad \text{and} \quad \|\boldsymbol{\Omega}_1^\dagger\|_F \leq B\right\} \quad \text{where} \quad A := e\sqrt{\frac{k+p}{p+1}}\,t \quad \text{and} \quad B := \sqrt{\frac{12k}{p+1}}\,t.$$

By a union bound with the tail bounds in Lemmas 6.5.13 and 6.5.17, we find that

$$\mathbb{P}(E_t^c) \leq t^{-(p+1)} + 4t^{-p} \leq 5t^{-p}.$$

Now, consider the function $h(\mathbf{X}) = \|\boldsymbol{\Sigma}_2\mathbf{X}\boldsymbol{\Omega}_1^\dagger\|$. We compute its Lipschitz constant L using the reverse triangle inequality and standard norm estimates:

$$\|h(\mathbf{X}) - h(\mathbf{Y})\| \leq \|\boldsymbol{\Sigma}_2(\mathbf{X}-\mathbf{Y})\boldsymbol{\Omega}_1^\dagger\| \leq \|\boldsymbol{\Sigma}_2\|\|\mathbf{X}-\mathbf{Y}\|\|\boldsymbol{\Omega}_1^\dagger\| \leq \|\boldsymbol{\Sigma}_2\|\|\mathbf{X}-\mathbf{Y}\|_F\|\boldsymbol{\Omega}_1\|_2^\dagger.$$

Thus, h is L-Lipschitz continuous with $L = \|\boldsymbol{\Sigma}_2\|\|\boldsymbol{\Omega}_1^\dagger\|$ for both the spectral and the Frobenius norms.

Using Exc. 6.5.12, we have

$$\mathbb{E}[h(\boldsymbol{\Omega}_2) \mid \boldsymbol{\Omega}_1] = \mathbb{E}[\|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\| \mid \boldsymbol{\Omega}_1] \leq \|\boldsymbol{\Sigma}_2\|\|\boldsymbol{\Omega}_1^\dagger\|_F + \|\boldsymbol{\Sigma}_2\|_F\|\boldsymbol{\Omega}_1^\dagger\|.$$

Applying the concentration inequality for Lipschitz function of Gaussian matrices (Lem. 6.5.16) conditionally on $\boldsymbol{\Omega}_1$, we obtain:

$$\mathbb{P}\Big(\|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\| \geq \underbrace{\|\boldsymbol{\Sigma}_2\|\|\boldsymbol{\Omega}_1^\dagger\|_F + \|\boldsymbol{\Sigma}_2\|_F\|\boldsymbol{\Omega}_1^\dagger\|}_{\geq \mathbb{E}[\|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\| \mid \boldsymbol{\Omega}_1]} + \underbrace{\|\boldsymbol{\Sigma}_2\|\|\boldsymbol{\Omega}_1^\dagger\|}_{=L} \cdot u \,\Big|\, E_t\Big) \leq e^{-u^2/2}.$$

On the event E_t, we have explicit bounds on the norms of $\boldsymbol{\Omega}_1^\dagger$, so:

$$\mathbb{P}\left(\|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\| \geq B\|\boldsymbol{\Sigma}_2\| + A\|\boldsymbol{\Sigma}_2\|_F + A\|\boldsymbol{\Sigma}_2\|u \,\middle|\, E_t\right) \leq e^{-u^2/2}.$$

Using the fact that $\mathbb{P}(E_t^c) \leq 5t^{-p}$, we can remove the conditioning by partitioning the sample space:

$$\begin{aligned}\mathbb{P}\left(\|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\| \geq B\|\boldsymbol{\Sigma}_2\| + A\|\boldsymbol{\Sigma}_2\|_F + A\|\boldsymbol{\Sigma}_2\|u\right) &\leq \mathbb{P}\left(\|\boldsymbol{\Sigma}_2\boldsymbol{\Omega}_2\boldsymbol{\Omega}_1^\dagger\| \geq B\|\boldsymbol{\Sigma}_2\| + A\|\boldsymbol{\Sigma}_2\|_F + A\|\boldsymbol{\Sigma}_2\|u \,\middle|\, E_t\right) + \mathbb{P}(E_t^c) \\ &\leq 5t^{-p} + e^{-u^2/2}.\end{aligned}$$

Finally, substitute the expressions for the norms of $\mathbf{\Sigma}_2$ into this result to complete the tail bound in (83). □

Exercise 6.5.18. Prove the tail bound (82) on the error of the randomized range finder in Frobenius norm.

6.5.5. Tail bound with SRFT test matrices. In this section, we analye the performance of randomized rangefinder with SRFT test matrix. This test matrix should be preferred over the Gaussian one when one seeks for faster matrix-vector multiplication.

Theorem 6.5.19 (Error bounds for SRFT)**.** *Fix an $m \times n$ matrix $\mathbf{B}$ with singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots$. Draw an $n \times l$ SRFT matrix $\mathbf{\Omega}$, where*

$$\ell \geq \frac{\log_k(2k\gamma^{-1})}{\varepsilon^2}\left(\sqrt{k}+\sqrt{8\log kn}\right)^2 \log k \qquad \left(=\Omega(\varepsilon^{-2}k\log k)\right)$$

for $\varepsilon, \gamma \in (0,1)$. Construct the sample matrix $\mathbf{Y} = \mathbf{B\Omega}$. Then for $\xi \in \{2, F\}$,

$$\mathbb{P}\left(\|(\mathbf{I}-\mathbf{P_Y})\mathbf{B}\|_\xi \leq \|\mathbf{\Sigma}_2\|_\xi \sqrt{1+(1-\varepsilon)^{-2}(n/\ell)}\right) \geq 1-\gamma. \tag{87}$$

PROOF. We will denote by $\|\cdot\|_\xi$ either the spectral or the Frobenius norm. Write $\mathbf{B} = \mathbf{U\Sigma V}^*$ and write $\mathbf{V} = [\mathbf{V}_1, \mathbf{V}_2]$ as in (85), where $\mathbf{V}_1$ has k columns. Recall that

$$\mathbf{\Omega}_1 = \mathbf{V}_1^*\mathbf{\Omega} \quad \text{and} \quad \mathbf{\Omega}_2 = \mathbf{V}_2^*\mathbf{\Omega},$$

where $\mathbf{\Omega}$ is the conjugate transpose of an SRFT. By Theorem 4.5.5 we have

$$\mathbb{P}\left(1-\varepsilon \leq \sigma_{\min}(\mathbf{\Omega}^*\mathbf{V}_1^*) \leq \sigma_{\max}(\mathbf{\Omega}^*\mathbf{V}_1^*) \leq 1+\varepsilon\right) \geq 1-\gamma.$$

Call the event in the left-hand side E. Then on E, $\mathbf{\Omega}_1^* = \mathbf{\Omega V}_1^*$ has full column rank, so $\mathbf{\Omega}_1$ has full row rank. Hence we can apply Theorem 6.5.9 conditional on on E. This gives that, on E,

$$\begin{aligned}\|(\mathbf{I}-\mathbf{P_Y})\mathbf{B}\|_\xi &\leq \sqrt{\|\mathbf{\Sigma}_2\|_\xi^2 + \|\mathbf{\Sigma}_2\mathbf{\Omega}_2\mathbf{\Omega}_1^\dagger\|_\xi^2} \\ &\leq \|\mathbf{\Sigma}_2\|_\xi \cdot \sqrt{1+\|\mathbf{\Omega}_2\|_2^2\|\mathbf{\Omega}_1^\dagger\|_2^2} \qquad (\text{use } \|\mathbf{AB}\|_\xi \leq \|\mathbf{A}\|_\xi\|\mathbf{B}\|_2)\end{aligned}$$

On E, we also have

$$\|\mathbf{\Omega}_1^\dagger\|_2^2 = \frac{1}{\sigma_{\min}(\mathbf{\Omega}_1)^2} \leq \frac{1}{(1-\varepsilon)^2}.$$

We may bound the spectral norm of $\mathbf{\Omega}_2$ deterministically:

$$\|\mathbf{\Omega}_2\|_2 = \|\mathbf{V}_2^*\mathbf{\Omega}\|_2 \leq \|\mathbf{V}_2^*\|_2\|\mathbf{\Omega}\|_2 \leq \sqrt{\frac{n}{\ell}},$$

since both $\mathbf{V}_2$ is an orthonormal matrix and $\|\sqrt{\ell/n}\mathbf{\Omega}\|_2 = \|\mathbf{RFE}\|_2 \leq 1$. Combining these results we can then deduce (88). □

The likelihood of error with an SRFT test matrix is substantially worse than in the Gaussian case. The failure probability γ can be taken to be polynomial in k: k^{-a} for $a \geq 1$ at the cost of scaling the embedding dimension $d \geq \varepsilon^{-2}(a+1)k\log k$ proportional to a. In the Gaussian case, the failure probability is roughly $e^{-(\ell-k)}$. This qualitative difference is not an artifact of the analysis; discrete sampling techniques inherently fail with higher probability.

Theorem 6.5.20 (Error bounds for sparse sgin matrix)**.** *Fix an $m \times n$ matrix $\mathbf{B}$ with singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots$ and $\varepsilon, \delta > 0$. Draw an $n \times l$ sparse sign matrix $\mathbf{\Omega}$ with probability p of nonzero entries (see Def. 4.3.1) and construct the sample matrix $\mathbf{Y} = \mathbf{B\Omega}$. Then for $\xi \in \{2, F\}$,*

$$\mathbb{P}\left(\|(\mathbf{I}-\mathbf{P_Y})\mathbf{B}\|_\xi \leq \|\mathbf{\Sigma}_2\|_\xi \sqrt{1+(1-\varepsilon)^{-2}(n^2/(\ell p))}\right) \geq 1-4\delta. \tag{88}$$

provided

$$\ell \geq C(k+\sqrt{k})\log n,$$

for $C = C(\varepsilon, \delta) > 0$ a sufficiently large constant.

PROOF. We will denote by $\|\cdot\|_\xi$ either the spectral or the Frobenius norm. Write $\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ and write $\mathbf{V} = [\mathbf{V}_1, \mathbf{V}_2]$ as in (85), where $\mathbf{V}_1$ has k columns. Recall that

$$\mathbf{\Omega}_1 = \mathbf{V}_1^* \mathbf{\Omega} \quad \text{and} \quad \mathbf{\Omega}_2 = \mathbf{V}_2^* \mathbf{\Omega},$$

where $\mathbf{\Omega}$ is the conjugate transpose of an SRFT. By Theorem 4.3.3 we have

$$\mathbb{P}\left(1 - \varepsilon \le \sigma_{\min}(\mathbf{\Omega}^* \mathbf{V}_1^*) \le \sigma_{\max}(\mathbf{\Omega}^* \mathbf{V}_1^*) \le 1 + \varepsilon\right) \ge 1 - 4\delta.$$

Call the event in the left-hand side E. Then on E, $\mathbf{\Omega}_1^* = \mathbf{\Omega}\mathbf{V}_1^*$ has full column rank, so $\mathbf{\Omega}_1$ has full row rank. Hence we can apply Theorem 6.5.9 conditional on on E. This gives that, on E,

$$\begin{aligned}
\|(\mathbf{I} - \mathbf{P}_\mathbf{Y})\mathbf{B}\|_\xi &\le \sqrt{\|\mathbf{\Sigma}_2\|_\xi^2 + \|\mathbf{\Sigma}_2 \mathbf{\Omega}_2 \mathbf{\Omega}_1^\dagger\|_\xi^2} \\
&\le \|\mathbf{\Sigma}_2\|_\xi \cdot \sqrt{1 + \|\mathbf{\Omega}_2\|_2^2 \|\mathbf{\Omega}_1^\dagger\|_2^2} \qquad (\text{use } \|\mathbf{AB}\|_\xi \le \|\mathbf{A}\|_\xi \|\mathbf{B}\|_2)
\end{aligned}$$

On E, we also have

$$\|\mathbf{\Omega}_1^\dagger\|_2^2 = \frac{1}{\sigma_{\min}(\mathbf{\Omega}_1)^2} \le \frac{1}{(1-\varepsilon)^2}.$$

We may bound the spectral norm of $\mathbf{\Omega}_2$ deterministically:

$$\|\mathbf{\Omega}_2\|_2 = \|\mathbf{V}_2^* \mathbf{\Omega}\|_2 \le \|\mathbf{V}_2^*\|_2 \|\mathbf{\Omega}\|_2 \le \frac{n}{\sqrt{\ell p}},$$

since both $\mathbf{V}_2$ is an orthonormal matrix and $\|\mathbf{\Omega}\|_2 \le \|\mathbf{\Omega}\|_\infty = n/\sqrt{\ell p}$. Combining these results we can then deduce (88). □

6.6. Streaming SVD

In this section, we consider the problem of computing an approximate truncated SVD in scenarios where the matrix can only be observed once or is dynamically evolving. This is referred to as the *streaming model*, which arises in numerous practical applications. It might initially appear counterintuitive that such a task is feasible, but it is indeed possible to approximate the SVD without accessing the matrix multiple times.

6.6.1. Turnstile Streaming Problem. Consider a matrix $\mathbf{B} \in \mathbb{F}^{m \times n}$, which is presented as a sequence of linear updates:

$$\mathbf{B} = \mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_3 + \cdots,$$

where $\mathbf{H}_i \in \mathbb{F}^{m \times n}$ is termed the i-th *innovation*. These innovations $\mathbf{H}_i$ are often simple structures, such as sparse or low-rank matrices. We assume the dimensions m and n are very large, making it impractical to store a complete representation of $\mathbf{B}$. Algorithmically, this implies processing each innovation and discarding it after use. This setup is referred to as the *turnstile streaming model*, distinguishing it from simpler streaming paradigms.

Notable examples of updates within this model include:

(1) $\mathbf{H}_j = \mathbf{b}_j \mathbf{e}_j^*$: updates consist of individual matrix columns.
(2) $\mathbf{H}_j = \mathbf{e}_i \mathbf{b}_j^*$: updates consist of individual matrix rows.
(3) $\mathbf{H}_{ij} = \mathbf{h}_{ij} \mathbf{E}_{ij}$: updates consist of individual matrix entries.

Furthermore, these updates can occur in any order, and specific rows, columns, or entries may be updated multiple times.

This model is often motivated by *inventory systems*, where the quantity of items can increase (e.g., returns or restocks) or decrease (e.g., sales). Recently, the turnstile streaming model has also been applied to *one-pass data access scenarios*, such as:

- When a large matrix $\mathbf{B}$ is stored on a sequential-access device like a tape drive, where random or repeated access incurs significant costs.

- In *scientific simulations*, where solving PDEs iteratively generates intermediate solution fields that are large and computationally expensive to handle repeatedly.

The objective in this section is to design an algorithm for approximate SVD computation within the turnstile streaming framework. Specifically, given a rank parameter k (known in advance), the algorithm should compute an approximate k-truncated SVD of $\mathbf{B}$ at a specified time T, while efficiently managing working storage, update time, and SVD computation time.

Instead of storing the entire matrix $\mathbf{B}$, the algorithm maintains a randomized linear sketch of $\mathbf{B}$. Formally, let:

$$\mathscr{S} : \mathbb{F}^{m\times n} \to \mathbb{F}^d$$

be a linear mapping from matrices to vectors. The sketch is computed as:

$$\mathscr{S}(\mathbf{B}) = \mathscr{S}(\mathbf{H}_1) + \mathscr{S}(\mathbf{H}_2) + \mathscr{S}(\mathbf{H}_3) + \cdots.$$

The linearity of $\mathscr{S}$ allows the sketch to be updated incrementally, processing each innovation $\mathbf{H}_i$ as it arrives and then discarding it.

The embedding dimension d is a tunable parameter. If $d \ll mn$, then $\mathscr{S}$ will have a substantial null space. However, with a suitably randomized $\mathscr{S}$, the sketch is highly likely to preserve the essential structure of $\mathbf{B}$ with high probability, provided $\mathbf{B}$ is statistically independent of $\mathscr{S}$. Once the data stream has been processed, the sketch $\mathscr{S}(\mathbf{B})$ can be used to approximate the SVD of $\mathbf{B}$, ensuring high-probability guarantees on the accuracy of the approximation at a given time T.

6.6.2. The SketchySVD Algorithm: IDEA. With randomized linear sketching at hand, the obvious question becomes: how do we design a streaming SVD algorithm? Let us start with the randomized SVD algorithm in its simplest form and see where it breaks.

First we review the Randomized SVD and see why it fails to for the streaming setting. The randomized rangefinder to find a basis $\mathbf{Q}$ that captures the range of $\mathbf{B}$:

$$\mathbf{B\Omega} = \mathbf{Y} = \mathbf{QR},$$

where the matrix $\mathbf{\Omega} \in \mathbb{F}^{m\times l}$ is random. The second step is to compress $\mathbf{B}$ to the range of $\mathbf{Q}$. Then, we perform a dense SVD computation:

$$\mathbf{C} = \mathbf{Q}^*\mathbf{B} = \widehat{\mathbf{U}}\widehat{\mathbf{\Sigma}}\widehat{\mathbf{V}}^*.$$

Finally, we obtain a factored SVD:

$$[\![\mathbf{B}]\!]_k \approx \hat{\mathbf{B}} = (\mathbf{Q}\widehat{\mathbf{U}})\widehat{\mathbf{\Sigma}}\widehat{\mathbf{V}}^*.$$

However, this approach *requires two passes over the matrix* $\mathbf{B}$. Since we cannot avoid sketching $\mathbf{B}$, we should think about how we can avoid the second pass of $\mathbf{B}$ when we product it on the range of $\mathbf{Q}$. After all, this can be thought of as another sketch of $\mathbf{B}$ but from the left – sketchign the 'co-range' of $\mathbf{B}$.

Now, the key idea is to sketch the matrix $\mathbf{B}$ on the left too. More precisely, we maintain:

$$\mathbf{Y} = \mathbf{B\Omega} \in \mathbb{F}^{m\times l}, \quad \mathbf{X} = \mathbf{\Upsilon B} \in \mathbb{F}^{l\times m}.$$

These are sketches for the range and co-range, respectively. To compute an SVD, we first perform two QR decompositions:

$$\mathbf{Y} = \mathbf{QR}, \quad \mathbf{X}^* = \mathbf{PR},$$

where $\mathbf{Q} \in \mathbb{F}^{m\times l}$ and $\mathbf{P} \in \mathbb{F}^{n\times l}$ are orthonormal. The columns of $\mathbf{Q}$ capture the range of $\mathbf{B}$, while the columns of $\mathbf{P}$ capture the co-range. Heuristically,

$$\mathbf{B} \approx \mathbf{QQ}^*\mathbf{BPP}^* = \mathbf{Q}\underbrace{(\mathbf{Q}^*\mathbf{BP})}_{=:\mathbf{C}}\mathbf{P}^*.$$

Intuitively, the matrix $\mathbf{B}$ is well-approximated by its projection onto the range of $\mathbf{Q}$ on the left and its projection onto the range of $\mathbf{P}$ on the right. We have also defined the 'core matrix' $\mathbf{C} = \mathbf{Q}^*\mathbf{BP} \in \mathbb{R}^{l\times l}$.

If we had direct access to $\mathbf{C}$, we would be done. But this is not the case because we can only query $\mathbf{B}$ once, and we already used $\mathbf{B}$ to compute $\mathbf{Q}$ and $\mathbf{P}$. To bypass this problem, let us tabulate the information we already have:

$$\mathbf{Y} = \mathbf{B}\boldsymbol{\Omega} \approx \mathbf{Q}\mathbf{C}\mathbf{P}^*\boldsymbol{\Omega},$$
$$\mathbf{X} = \boldsymbol{\Upsilon}\mathbf{B} \approx \boldsymbol{\Upsilon}\mathbf{Q}\mathbf{C}\mathbf{P}^*.$$

The matrices on the left-hand side are known, and the only unknown on the right-hand sides is the core matrix $\mathbf{C}$. Hence, we can use these formulas to find $\mathbf{C}$ by solving a least-squares problem. The following is a useful observation about the least-square reduction.

Lemma 6.6.1.

$$\mathbf{A}^\dagger\mathbf{Y}\mathbf{B}^\dagger \in \operatorname*{arg\,min}_{\mathbf{X}} \|\mathbf{Y} - \mathbf{A}\mathbf{X}\mathbf{B}\|_F^2.$$

PROOF. Using Exc. 1.3.10,

$$\begin{aligned}\nabla_{\mathbf{X}}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\mathbf{B}\|_F^2 &= \nabla_{\mathbf{X}} \operatorname{tr}\left((\mathbf{Y} - \mathbf{A}\mathbf{X}\mathbf{B})^*(\mathbf{Y} - \mathbf{A}\mathbf{X}\mathbf{B})\right) \\ &= \nabla_{\mathbf{X}}\left[\operatorname{tr}\left(\mathbf{B}^*\mathbf{X}^*\mathbf{A}^*\mathbf{A}\mathbf{X}\mathbf{B}\right) - \operatorname{tr}\left(\mathbf{Y}^*\mathbf{A}\mathbf{X}\mathbf{B}\right) - \operatorname{tr}\left((\mathbf{A}\mathbf{X}\mathbf{B})^*\mathbf{Y}\right)\right] \\ &= 2\mathbf{A}^*\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{B}^* - 2\mathbf{A}^*\mathbf{Y}\mathbf{B}^*.\end{aligned}$$

Following the analysis in Exc. 1.3.9, we can also show that the objective function is convex. Hence any critical point is a global minimizer. Thus it suffices to show that $\mathbf{X} = \mathbf{A}^\dagger\mathbf{Y}\mathbf{B}^\dagger$ is a critical point. To see this, write $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^*$ and observe

$$\mathbf{A}^*\mathbf{A}\mathbf{A}^\dagger = \mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^*\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^*\mathbf{V}\boldsymbol{\Sigma}^\dagger\mathbf{U}^* = \mathbf{V}\boldsymbol{\Sigma}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\dagger\mathbf{U}^* = \mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^* = \mathbf{A}^*.$$

Similarly, we can verify $\mathbf{B}^\dagger\mathbf{B}\mathbf{B}^* = \mathbf{B}^*$. It follows that the gradient of the objective above vanishes at $\mathbf{X} = \mathbf{A}^\dagger\mathbf{Y}\mathbf{B}^\dagger$, as desired. □

6.6.3. The SketchySVD Algorithm: Details. Now we are going to pursue a more elaborate algorithm that collects additional information to approximate the core matrix $\mathbf{C}$ more accurately. Draw and fix two additional random matrices $\boldsymbol{\Phi} \in \mathbb{F}^{s\times m}$ and $\boldsymbol{\Psi} \in \mathbb{F}^{m\times s}$, where $s \geq l$. We use these additional matrices to maintain an $s \times s$ *core sketch*:

$$\mathbf{Z} = \boldsymbol{\Phi}\mathbf{B}\boldsymbol{\Psi} \in \mathbb{F}^{s\times s}.$$

Heuristically, we expect:

$$\mathbf{Z} \approx (\boldsymbol{\Phi}\mathbf{Q})\mathbf{C}(\mathbf{P}^*\boldsymbol{\Psi}).$$

By Lemma 6.6.1, we can use this relation to approximate the core matrix as follows:

$$\widehat{\mathbf{C}} = (\boldsymbol{\Phi}\mathbf{Q})^\dagger\mathbf{Z}(\mathbf{P}^*\boldsymbol{\Psi})^\dagger.$$

This approach has a crucial advantage: the sketch for the core is statistically independent from the sketches for the row and column spaces. If the test matrices are Gaussian, the approximated core matrix is even an unbiased estimator for the true core matrix. The idea of adding a core sketch is due to Upadhyay [Upa16], with subsequent improvements by Tropp et al. [TYUC17].

SketchySVD:

(1) Draw and fix random matrices: $\boldsymbol{\Omega} \in \mathbb{F}^{n\times l}$, $\boldsymbol{\Psi} \in \mathbb{F}^{n\times s}$, $\boldsymbol{\Upsilon} \in \mathbb{F}^{l\times m}$, and $\boldsymbol{\Phi} \in \mathbb{F}^{s\times m}$, with the only requirement that $s \geq l$.

(2) Draw and fix random matrices:

$$\mathbf{Y} = \mathbf{B}\boldsymbol{\Omega} \quad \text{(range sketch)}$$
$$\mathbf{X} = \boldsymbol{\Upsilon}\mathbf{B} \quad \text{(co-range sketch)}$$
$$\mathbf{Z} = \boldsymbol{\Phi}\mathbf{B}\boldsymbol{\Psi} \quad \text{(core sketch)}.$$

These sketches are all linear, so they can be maintained in the turnstile streaming model.

(3) Perform two QR factorizations: $\mathbf{Y} = \mathbf{Q}\mathbf{R}$, $\mathbf{X}^* = \mathbf{P}\mathbf{R}$.

(4) Compute a core approximation: $\widehat{\mathbf{C}} = (\mathbf{\Phi Q})^{\dagger}\mathbf{Z}(\mathbf{P}^*\mathbf{\Psi})^{\dagger}$.
(5) Form an SVD of the core: $\widehat{\mathbf{C}} = \widehat{\mathbf{U}}\widehat{\mathbf{\Sigma}}\widehat{\mathbf{V}}^* \in \mathbb{F}^{s\times s}$.
(6) Set $\widehat{\mathbf{B}} = (\mathbf{Q}\widehat{\mathbf{U}})\widehat{\mathbf{\Sigma}}(\mathbf{P}\widehat{\mathbf{V}})^*$.
(7) Truncate $\widehat{\mathbf{\Sigma}} \to [\![\widehat{\mathbf{\Sigma}}]\!]_k$ with truncation rank k.

Storage, Runtime, and Analysis. The storage cost of this procedure is $O(s^2 + l(m+n))$. One can use structured random matrices to nearly eliminate the cost of storing the sketching operators themselves. The arithmetic cost is $O(s^3 + l^2(m+n))$ for the SVD computation; this is dominated by the cost of the QR decompositions.

A serious drawback of such methods is that we can only query $\mathbf{B}$ once. If we make a mistake, we cannot hope to rectify it by revisiting $\mathbf{B}$. The implications of this limitation are two-fold: 1. We need good parameter choices (s, l). This requires serious theory. 2. A posteriori validation is important.

We will quickly discuss these points in the following subsections.

The following rigorous theorem provides guidance for choosing sketching parameters s and l.

Theorem 6.6.2. *Assume $\mathbb{F} = \mathbb{C}$. If the test matrices are i.i.d. standard normal and $s \geq 2l$, then:*

$$\mathbb{E}\left[\|\mathbf{B} - \mathbf{Q}\hat{\mathbf{C}}\mathbf{P}^*\|_F^2\right] \leq \frac{s}{s-\ell} \min_{k<\ell} \frac{\ell+k}{\ell-k} \left(\sum_{j>k} \sigma_j^2(\mathbf{B}) \right).$$

PROOF. See [TYUC17]. □

CHAPTER 7

Online Matrix Factorization

The matrials in this chapter are based on [LNB20, Lyu24].

7.1. Basics on Matrix Factorization

Matrix factorization is one of the fundamental tools in dictionary learning problems. Given a large data matrix $\mathbf{X}$, can we find some small number of "dictionary vectors" so that we can represent each column of the data matrix as a linear combination of dictionary vectors? More precisely, given a data matrix $\mathbf{X} \in \mathbb{R}^{d\times n}$ and a rank parameter $r \in \mathbb{N}$, we wish to factorize X into the product of $\mathbf{W} \in \mathbb{R}^{d\times r}$ and $\mathbf{H} \in \mathbb{R}^{r\times n}$ by solving the following optimization problem

$$\text{(Matrix Factorization)} \qquad \inf_{\mathbf{W}\in\mathbb{R}^{d\times r},\mathbf{H}\in\mathbb{R}^{r\times n}} \|\mathbf{X}-\mathbf{W}\mathbf{H}\|_F^2. \tag{89}$$

Here $\mathbf{W}$ is called the *dictionary* and $\mathbf{H}$ is the *code* of data $\mathbf{X}$ using dictionary $\mathbf{W}$. A solution of such matrix factorization problem is illustrated in Figure 7.1.1.

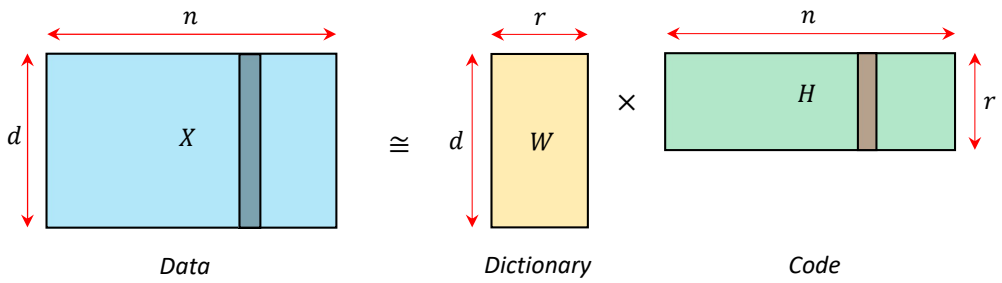


FIGURE 7.1.1. Illustration of matrix factorization. Each column of the data matrix is approximated by the linear combination of the columns of the dictionary matrix with coefficients given by the corresponding column of the code matrix. Figure excerpted from [LNB20].

Recall the method of least squared we studied for polynomial regression: (See also Exercise 1.3.9).

$$\hat{\mathbf{H}} = \operatorname*{arg\,min}_{\mathbf{H}\in\mathbb{R}^{r\times n}} \|\underbrace{\mathbf{X}}_{\text{given}} - \underbrace{\mathbf{W}}_{\text{given}}\underbrace{\mathbf{H}}_{\text{TBD}}\|_F^2 + \lambda\|\underbrace{\mathbf{H}}_{\text{TBD}}\|_F^2. \tag{90}$$

One can think of (90) as a matrix factorization problem where the first factor (dictionary) is known and we are only looking for the code $\mathbf{H}$. Becase of this reason, we also call (90) as a *coding* problem, which is also called a *sparse coding* problem if the regularizer λ is positive[1].

While the optimization problem (89) is an *unconstrained* problem in the sense that we are free to choose any dictionary $\mathbf{W} \in \mathbb{R}^{d\times r}$ and code $\mathbf{H} \in \mathbb{R}^{r\times n}$, We will be more interested in its *constrained* versions, in particular using nonnegativity constraints. Consider the following modifications:

$$\text{(Dictionary Learning)} \qquad \inf_{\mathbf{W}\in\mathbb{R}^{d\times r},\mathbf{H}\in\mathbb{R}^{r\times n}_{\geq 0}} \|\mathbf{X}-\mathbf{W}\mathbf{H}\|_F^2, \tag{91}$$

$$\text{(NMF)} \qquad \inf_{\mathbf{W}\in\mathbb{R}^{d\times r}_{\geq 0},\mathbf{H}\in\mathbb{R}^{r\times n}_{\geq 0}} \|\mathbf{X}-\mathbf{W}\mathbf{H}\|_F^2. \tag{92}$$

[1] It is more effective to use an L_1-regularizer $\lambda\|\mathbf{H}\|_1$ for the purpose of sparse coding. This is also closely related to (Lagrangian version of) the popular algorithm of LASSO in statistics.

In (91), we imposed an additional constraint that the code $\mathbf{H} \in \mathbb{R}^{r\times n}$ should only take nonngative entries. This allows us to interpret the columns of dictionary $\mathbf{W}$ as 'features' that directly contribute to composing the columns of $\mathbf{X}$ (samples)[2]. We can go even further and also impose the same nonnegativity constraint on the dictioanry matrix $\mathbf{W} \in \mathbb{R}^{r\times n}$ as in (92). This problem is called *nonnegative matrix factorization* (NMF), which is an extremely popular technique in modern machine learning for its high interpretibility and mathematical simplicity.

Exercise 7.1.1. Show the following.

(i) Show that the function $\mathbf{H} \mapsto \|\mathbf{X}-\mathbf{W}\mathbf{H}\|_F^2$ is convex.
(ii) Show that the function $\mathbf{W} \mapsto \|\mathbf{X}-\mathbf{W}\mathbf{H}\|_F^2 = \|\mathbf{X}^T-\mathbf{H}^T\mathbf{W}^T\|_F^2$ is convex.
(iii) Show that the function $(\mathbf{W},\mathbf{H}) \mapsto \|\mathbf{X}-\mathbf{W}\mathbf{H}\|_F^2$ is non-convex.

According to Exercise 7.1.1, this is a non-convex optimization problem so it is difficult to find its global minima. However, if we fix one of the factors, then ℓ is convex with respect to the other factor (see Exercise 7.1.1), which is in fact a least squares problem (see Exercise 1.3.9). Hence one can iteratively fix one factor and optimize the other, each time solving a least squares problem. This is called *Alternating Least Squares* (ALS) algorithm for matrix factorizaiton, which we state in Algorithm 4 for MF with nonnegative constraints (Nonnegative Matrix Factorization) []

Algorithm 4 Alternating Least Squares for Matrix Factorization

Input: $\mathbf{X} \in \mathbb{R}^{d\times n}_{\geq 0}$ (data matrix); $r \geq 1$ (rank parameter); $[\mathbf{W}_0,\mathbf{H}_0] \in \mathbb{R}^{d\times r}_{\geq 0} \times \mathbb{R}^{r\times n}_{\geq 0}$ (initial estimate); N (number of iterations)
  for $n = 1,\dots,N$ **do**:
    $\mathbf{H}_n \leftarrow \arg\min_{\mathbf{H}\in\mathbb{R}^{r\times n}_{\geq 0}} \|\mathbf{X}-\mathbf{W}_{t-1}\mathbf{H}\|_F^2$
    $\mathbf{W}_n \leftarrow \arg\min_{\mathbf{W}\in\mathbb{R}^{d\times r}_{\geq 0}} \|\mathbf{X}-\mathbf{W}\mathbf{H}_n\|_F^2$
  end for
output: $[\mathbf{W}_N,\mathbf{H}_N]$.

The ALS algorithm in Algorithm 4 is a special case of *block coordinate descent* (BCD) algorithms, which is to iteratively optimize the objective function with respect to a block coordinate (group of individual coordinates). The general algorithm is stated in Algorithm 5. For more backgrounds and its convergence properties, see [LL20].

Algorithm 5 Block Coordinate Descent

Requirement: $\boldsymbol{\theta} = [\theta_1,\dots,\theta_m] \mapsto f(\theta_1,\dots,\theta_m)$ is convex in each block-coordinate θ_i, $i = 1,\dots,m$.
Input: $\boldsymbol{\theta}_0 = (\theta_0^{(1)},\cdots,\theta_0^{(m)}) \in \Theta^{(1)}\times\cdots\times\Theta^{(m)}$ (initial estimate); N (number of iterations)
  for $n = 1,\dots,N$ **do**:
    Update estimate $\boldsymbol{\theta}_n = [\theta_n^{(1)},\cdots,\theta_n^{(m)}]$ by
      For $i = 1,\cdots,m$ **do**:

$$\theta_n^{(i)} \in \arg\min_{\theta\in\Theta^{(i)}} f\left(\theta_n^{(1)},\cdots,\theta_n^{(i-1)},\theta,\theta_{n-1}^{(i+1)},\cdots,\theta_{n-1}^{(m)}\right);$$

      end for
  end for
output: $\boldsymbol{\theta}_N$

[2] A general formulation of dictionary learning may also impose a convex constraint $\mathscr{C} \subseteq \mathbb{R}^{d\times r}$ for dictionary $\mathbf{W}$ along with the nonnegativity constraint on the code $\mathbf{H}$.

7.2. Online Matrix Factorization: Problem Formulation

Suppose now, instead of factorizing a fixed matrix $\mathbf{X}$, we are given with a sequence of data matrices $\mathbf{X}_1, \mathbf{X}_2, \dots$ each arriving sequentially. The goal of *online matrix factorization* (OMF) is to construct a sequence of dictionary matrices $(\mathbf{W}_t)_{t\geq 1}$ so that $\mathbf{W}_t$ provides a reasonable factorization for *all* previous data matrices $\mathbf{X}_1, \dots, \mathbf{X}_t$. We formulate this as an 'empirical loss minimization' problem, which we introduce briefly here.

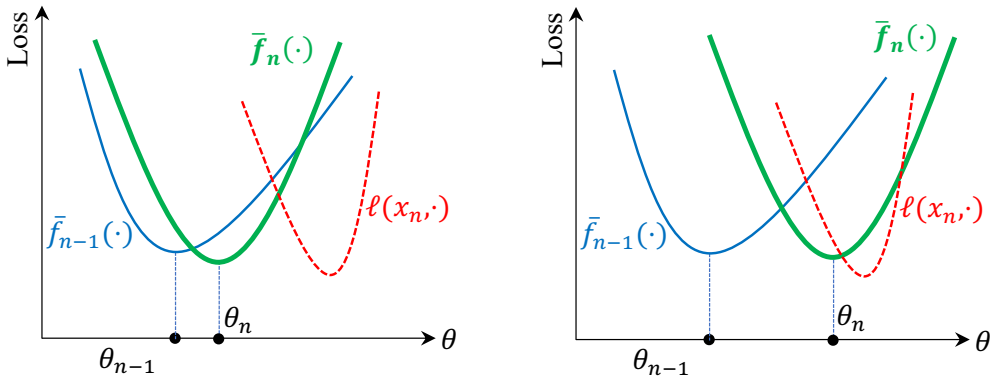
Suppose we have a loss function $\ell : \mathfrak{X} \times \boldsymbol{\Theta} \to [0, \infty)$ that measures the fitness of a parameter $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subseteq \mathbb{R}^p$ with respect to an observed data $\mathbf{x} \in \mathfrak{X}$. Consider a sequence of newly observed data $(\mathbf{x}_n)_{n\geq 1}$ in $\mathfrak{X}$, where $\mathfrak{X}$ can be a general topological space. We would like to estimate a sequence of parameters $(\boldsymbol{\theta}_n)_{n\geq 1}$ such that $\boldsymbol{\theta}_n$ is in some sense the best fit to all data $\mathbf{x}_1, \cdots, \mathbf{x}_n$ up to time n. In the *empirical risk minimization* (ERM) framework, one seeks to minimize a recursively defined empirical loss function as new data arrives:

$$\text{Upon arrival of } \mathbf{x}_n\text{:} \quad \boldsymbol{\theta}_n \in \operatorname*{arg\,min}_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \big(\bar{f}_n(\boldsymbol{\theta}) := (1-w_n)\bar{f}_{n-1}(\boldsymbol{\theta}) + w_n \ell(\mathbf{x}_n, \boldsymbol{\theta}) \big), \tag{93}$$

where $(w_n)_{n\geq 1}$ is a sequence of *adaptivity weights* in $(0,1]$ and the function $\bar{f}_n$ is the *empirical loss function* recursively defined by the weighted average as in (93) with $f_0 \equiv 0$. More explicitly, we can write

$$\bar{f}_n(\boldsymbol{\theta}) = \sum_{k=1}^{n} \ell(\mathbf{x}_k, \boldsymbol{\theta})\, w_k^n, \qquad w_k^n := w_k \prod_{i=k+1}^{n} (1-w_i).$$

The adaptivity weight w_n in (93) controls how much we want our new estimate $\boldsymbol{\theta}_n$ deviate from minimizing the previous empirical loss $\bar{f}_{n-1}$ to adapting to the newly observed tensor data $\mathbf{x}_n$. In the extreme case of $w_n \equiv 1$, $\boldsymbol{\theta}_n$ is a minimizer of the time-n loss $\ell(\mathbf{x}_n, \cdot)$ and ignores the past $\bar{f}_{n-1}$. If $w_n \equiv \alpha \in (0,1)$ then the history is forgotten exponentially fast, that is, $\bar{f}_n(\cdot) = \sum_{k=1}^{n} \alpha(1-\alpha)^{n-k} \ell(\mathbf{x}_k, \cdot)$. On the other hand, the 'balanced weight' $w_n = 1/n$ makes the empirical risk be the arithmetic mean: $\bar{f}_n(\cdot) = \frac{1}{n}\sum_{k=1}^{n} \ell(\mathbf{x}_s, \cdot)$, which is the canonical choice in the literature including maximum likelihood estimation and online NMF problem in [MBPS10]. Hence, one may choose the sequence of weights $(w_n)_{n\geq 1}$ in (93) to decay fast for learning average features and decay slow (or keep it constant) for learning trending features). See Figure 7.2.1 for illustration.



Slow adaptation Fast adaptation

FIGURE 7.2.1. Schematic plot of empirical loss $\bar{f}_n(\cdot) = (1-w_n)\bar{f}_{n-1}(\cdot) + w_n \ell(\mathbf{x}_n, \cdot)$ when the adaptivity weight w_n is small ('slow adaptation' regime) or large ('fast adaptation' regime). Excerpted from [Lyu24].

Now apply the framework of ERM for the OMF problem. Suppose we have a sequence of data matrices $(\mathbf{X}_t)_{t\geq 1}$, $\mathbf{X}_t \in \mathbb{R}^{p\times n}$ and a sequence of weights $(w_t)_{t\geq 1}$. At each time $t \geq 0$, the per-sample loss function f_t for OMF is defined by

$$f_t(\mathbf{W}) = \inf_{\mathbf{H}\subseteq\mathbb{R}^{r\times n}} \|\mathbf{X}_t - \mathbf{W}\mathbf{H}\|_F^2, \tag{94}$$

where one may empose additional constraints for the 'code matrix' $\mathbf{H}$ in the infemum above (e.g., non-negativity, norm constraints) and one can also add suitable regularization for $\mathbf{W}$ or $\mathbf{H}$ to (94), but we will keep our discussion in the simpleast setting. The corresponding ERM problem is

$$\mathbf{W}_t \in \operatorname*{arg\,min}_{\mathbf{W}\subseteq\mathbb{R}^{p\times r}} \left[\bar{f}_t(\mathbf{W}) := (1-w_t)\bar{f}_{t-1}(\mathbf{W}) + w_n \ell(\mathbf{X}_t, \mathbf{W})\right].$$

Next, we discuss a class of online algorithms for solving ERM (93) and apply it to the OMF problem and derive the famous OMF algorithm due to Mairal, Bach, Sapirro, and Ponce [MBPS10].

Recall that in ERM (93), instead of a single objective function f, we have a sequence of empirical loss functions $\bar{f}_t$ to be minimized online. If the per-sample loss function $f_t(\boldsymbol{\theta}) := \ell(\mathbf{x}_t, \boldsymbol{\theta})$ is nonconvex (which is the case for OMF due to the infemum in $\mathbf{H}$), then so are their weighted averages $\hat{f}_t$. Hence it is impossible or out of reach to directly minimize the empirical loss function at each iteration. Instead, we will construct a *convex upper bounding function* g_t of f_t, which is call the *majorizing surrogate* of f_t, and create a sequence of convex surrogates of $\hat{f}_t$ in the same recursive manner.

It would be natural to first find a majorizing surrogate g_n of the one-point loss function $f_n(\cdot) := \ell(\mathbf{x}_n, \cdot)$ at the current estimate $\boldsymbol{\theta}_n$ and then undergo the same recursive averaging step as in (93) to compute an averaged surrogate $\bar{g}_n$ so that $\bar{g}_n$ is a majorizing surrogate of the current objective $\bar{f}_n$. We can then minimize $\bar{g}_n$ to compute the next estimate $\boldsymbol{\theta}_n$. This method is called the *stochastic marjorization-minimization* (SMM) [Mai13], which we state concisely as follow:

$$\textbf{(SMM)}\quad \text{Upon arrival of } \mathbf{x}_t\text{:} \quad \begin{cases} g_t \leftarrow \text{Convex majorizing surrogate of } f_t(\cdot) = \ell(\mathbf{x}_t, \cdot) \\ \qquad\qquad (\triangleright \quad g_t(\cdot) \geq f_t(\cdot) \text{ and } g_t(\boldsymbol{\theta}_{t-1}) = f_t(\boldsymbol{\theta}_{t-1})) \\ \boldsymbol{\theta}_t \in \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \left(\bar{g}_t(\boldsymbol{\theta}) := (1-w_t)\bar{g}_{t-1}(\boldsymbol{\theta}) + w_t\, g_t(\boldsymbol{\theta})\right). \end{cases} \tag{95}$$

Now we apply SMM for the OMF setting. Denote

$$\mathbf{H}_t \in \operatorname*{arg\,min}_{\mathbf{H}} \|\mathbf{X}_t - \mathbf{W}_{t-1}\mathbf{H}\|_F^2.$$

which is an optimal code for factorizing $\mathbf{X}_t$ using the previous dictionary $\mathbf{W}_{t-1}$. Then the function

$$g_t(\mathbf{W}) = \|\mathbf{X}_t - \mathbf{W}\mathbf{H}_t\|_F^2$$

is a majorizing surrogate of f_t since

$$f(t)(\mathbf{W}) = \|\mathbf{X}_t - \mathbf{W}_{t-1}\mathbf{H}\|_F^2 \leq \|\mathbf{X}_t - \mathbf{W}\mathbf{H}_t\|_F^2 = g_t(\mathbf{W})$$

and also

$$f_(t)(\mathbf{W}_{t-1}) = \|\mathbf{X}_t - \mathbf{W}_{t-1}\mathbf{H}\|_F^2 = g_t(\mathbf{W}_{t-1}).$$

Expanding out the Frobenius norm using trace,

$$g_t(\mathbf{W}) = \operatorname{tr}\left(\mathbf{W}\mathbf{H}_t\mathbf{H}_t^\top\mathbf{W}\right) - 2\operatorname{tr}\left(\mathbf{W}\mathbf{H}_t\mathbf{X}_t^\top\right) + \operatorname{tr}\left(\mathbf{X}_t^\top\mathbf{X}_t\right),$$

which is a quadractic function in $\mathbf{W}$. It follows that the resulting averaged surrogate function $\bar{g}_t$ becomes

$$\bar{g}_t(\mathbf{W}) = \operatorname{tr}(\mathbf{W}\mathbf{A}_t\mathbf{W}^\top) - 2\operatorname{tr}(\mathbf{W}\mathbf{B}_t) + \mathbf{C}_t,$$

where the matrices A_t, B_t, C_t are recursively defined as

$$\begin{aligned} \mathbf{A}_t &= (1-w_t)\mathbf{A}_{t-1} + w_t\mathbf{H}_t\mathbf{H}_t^T, \\ \mathbf{B}_t &= (1-w_t)\mathbf{B}_{t-1} + w_t\mathbf{H}_t\mathbf{X}_t^T, \\ \mathbf{C}_t &= (1-w_t)\bar{\mathbf{C}}_{t-1} + w_t\mathbf{X}_t\mathbf{X}_t^T. \end{aligned}$$

Thus the SMM algorithm (95) in this case reduces to

$$\textit{Upon arrival of } \mathbf{X}_t\textit{:} \quad \begin{cases} \mathbf{H}_t = \arg\min_{\mathbf{H}} \|\mathbf{X}_t - \mathbf{W}_{t-1}\mathbf{H}\|_F^2 \\ \mathbf{A}_t = (1-w_t)\mathbf{A}_{t-1} + w_t \mathbf{H}_t \mathbf{H}_t^T \\ \mathbf{B}_t = (1-w_t)\mathbf{B}_{t-1} + w_t \mathbf{H}_t \mathbf{X}_t^T \\ \mathbf{W}_t = \arg\min_{\mathbf{W}} \left(\operatorname{tr}(\mathbf{W}\mathbf{A}_t\mathbf{W}^T) - 2\operatorname{tr}(\mathbf{W}\mathbf{B}_t)\right). \end{cases} \tag{96}$$

This is the celebrated OMF algorithm in [MBPS10]. The authors in [MBPS10] analyzed this algorithm when the data matrices $\mathbf{X}_t$ are i.i.d. from some population distribution. Lyu, Balzano, and Needell [LNB20] extended the analysis for data sequences generated by some underlying Markov chain. The theoretical results obtained in these works are almost sure asymptotic convergence to stationary points. Lyu [Lyu24] extended the algorithm and the analysis further and obtained first rate of convergence results for SMM-type algorithms.

7.3. Analysis

Theorem 7.3.1 (Lyu '24 [Lyu24])**.** *Suppose each data matrix $\mathbf{X}_t$ is a function of some underlying geometrically ergodic Markov chain. Then the OMF algorithm (96) satisfies*

$$\min_{1\le k\le t}\left[\sup_{\mathbf{W}\in\mathbf{\Theta},\|\mathbf{W}-\mathbf{W}_k\|\le 1} \left\langle -\nabla \bar{f}_k(\mathbf{W}_k), \mathbf{W}-\mathbf{W}_k\right\rangle\right] = O\left(\frac{1}{\sqrt{\sum_{k=1}^t w_k}}\right)$$

asymptotically almost surely as $t\to\infty$ where $\mathbf{\Theta}\subseteq\mathbb{R}^{p\times r}$ denotes a convex constraint set for $\mathbf{W}$.

Sketch of Proof. Here we will sketch how we analyze SMM (95). This analysis is a simplification of the one given in [Lyu24].

First, by adding and subtracting $w_{t+1}\ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t)$ to the recursive equation for $\bar{g}_{t+1}$, we get

$$\bar{g}_{t+1}(\boldsymbol{\theta}_t) = (1-w_{t+1})\bar{g}_t(\boldsymbol{\theta}_t) + w_{t+1}\ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t) + w_{t+1}(g_{t+1}(\boldsymbol{\theta}_t) - \ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t)).$$

Then since $\bar{g}_{t+1}$ is minimized at $\boldsymbol{\theta}_{t+1}$,

$$\begin{aligned} \bar{g}_{t+1}(\boldsymbol{\theta}_{t+1}) - \bar{g}_t(\boldsymbol{\theta}_t) &= \underbrace{\bar{g}_{t+1}(\boldsymbol{\theta}_{t+1}) - \bar{g}_{t+1}(\boldsymbol{\theta}_t)}_{\le 0} + \bar{g}_{t+1}(\boldsymbol{\theta}_t) - \bar{g}_t(\boldsymbol{\theta}_t) \\ &\le w_{t+1}(\ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)) + w_{t+1}(\bar{f}_t(\boldsymbol{\theta}_t) - \bar{g}_t(\boldsymbol{\theta}_t)) + w_{t+1}\underbrace{(g_{t+1}(\boldsymbol{\theta}_t) - \ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t))}_{=0} \\ &= w_{t+1}(\ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)) + w_{t+1}(\bar{f}_t(\boldsymbol{\theta}_t) - \bar{g}_t(\boldsymbol{\theta}_t)). \end{aligned}$$

Rearranging,

$$w_{t+1}(\bar{g}_t(\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)) \le \bar{g}_t(\boldsymbol{\theta}_t) - \bar{g}_{t+1}(\boldsymbol{\theta}_{t+1}) + w_{t+1}(\ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)).$$

Telescoping both sides gives

$$\sum_{t=0}^{T-1} w_{t+1}(\bar{g}_t(\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)) \le \bar{g}_0(\boldsymbol{\theta}_0) - \underbrace{\bar{g}_T(\boldsymbol{\theta}_T)}_{\ge \inf_{\mathbf{x},\boldsymbol{\theta}}\ell(\mathbf{x},\boldsymbol{\theta})} + \sum_{t=0}^{T-1} w_{t+1}(\ell(\mathbf{x}_{t+1},\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)), \tag{97}$$

where the underbrace follows from $\bar{g}_t(\boldsymbol{\theta}) \ge \inf_{\mathbf{x}}\ell(\boldsymbol{\theta})$ (why?).

A key indsight in first-order analysis of SMM is that, since $\boldsymbol{\theta}_t$ minimizes $\bar{g}_t$, we have $\nabla g_t(\boldsymbol{\theta}_t) = 0$ if $\boldsymbol{\theta}_t$ lies in the interior of the constraint set $\mathbf{\Theta}$; otherwise $-\nabla\bar{g}_t(\boldsymbol{\theta}_t)$ lies in the normal cone of $\mathbf{\Theta}$ at $\boldsymbol{\theta}_t$. Thus if $\nabla\bar{f}_t(\boldsymbol{\theta}_t)$ aligns well with $\nabla\bar{g}_t(\boldsymbol{\theta}_t)$, then we know $\boldsymbol{\theta}_t$ is approximately a first-order optimal point of $\bar{f}_t$.

To relate the gradient difference with the function value difference, denote the surrogate gap $h_t := g_t - f_t \ge 0$. Assume that they are uniformly L-smooth for some constant $L > 0$. Then their weighted

average $\bar{h}_t := \bar{g}_t - \bar{h}_t \geq 0$ is also L-smooth, so

$$-\bar{h}_t(\boldsymbol{\theta}_t) \leq \bar{h}_t(\boldsymbol{\theta}) - \bar{h}_t(\boldsymbol{\theta}_t) \leq \langle \nabla \bar{h}_t(\boldsymbol{\theta}_t), \boldsymbol{\theta} - \boldsymbol{\theta}_t \rangle + \frac{L}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_t\|^2.$$

Minimizing the RHS w.r.t. $\boldsymbol{\theta}$ gives the optimal choice $\boldsymbol{\theta} = \boldsymbol{\theta}_t - \frac{1}{L}\nabla \bar{h}_t(\boldsymbol{\theta}_t)$ and plugging this in gives

$$\frac{1}{2L}\|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2 \leq \bar{h}_t(\boldsymbol{\theta}) = \bar{g}_t(\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t).$$

Combining with (97),

$$\frac{1}{2L}\sum_{t=0}^{T-1} w_{t+1}\|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2 \leq \bar{g}_0(\boldsymbol{\theta}_0) - \inf \ell + \underbrace{\sum_{t=0}^{T-1} w_{t+1}(\ell(\mathbf{x}_{t+1}, \boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t))}_{=\mathscr{A}(T)}. \tag{98}$$

The key is now bound the expectation of the term $\mathscr{A}(T)$. We will sketch the analysis for the special case when $\mathbf{x}_1, \mathbf{x}_2, \ldots$ are i.i.d. from some data distribution π. Write $f(\boldsymbol{\theta}) := \mathbb{E}_{\mathbf{x}\sim\pi}[\ell(\mathbf{x}, \boldsymbol{\theta})]$, the expected loss function. Then $\mathbf{x}_{t+1}$ conditional on $\mathscr{F}_t = \sigma(\mathbf{x}_1, \ldots, \mathbf{x}_t)$ is still independent from π so

$$\begin{aligned}\mathbb{E}\left[\ell(\mathbf{x}_{t+1}, \boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)\right] &= \mathbb{E}\left[\mathbb{E}\left[\ell(\mathbf{x}_{t+1}, \boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t) \,\middle|\, \mathscr{F}_t\right]\right] \\ &= \mathbb{E}\left[f(\boldsymbol{\theta}_t) - \bar{f}_t(\boldsymbol{\theta}_t)\right].\end{aligned}$$

By a weighted version of uniform law of large numbers (see [Lyu24, Lem. 42]),

$$\sup_{\boldsymbol{\theta}} |f(\boldsymbol{\theta}) - \bar{f}_t(\boldsymbol{\theta})| = O(w_{t+1})$$

as long as $\boldsymbol{\theta}$ is confined in a compact set. This shows the expectation above is $O(w_{t+1})$, so

$$\mathbb{E}[\mathscr{A}(T)] = O\left(\sum_{t=1}^{T} w_t^2\right).$$

The RHS above is finite as long as the weights w_t are square-summable.

Now taking $T \to \infty$ and expectation in (98),

$$\frac{1}{2L}\sum_{t=0}^{\infty} w_{t+1}\mathbb{E}[\|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2] \leq \bar{g}_0(\boldsymbol{\theta}_0) - \inf \ell + O\left(\sum_{t=1}^{\infty} w_t^2\right).$$

The RHS is a finite deterministic quantity. Now using Fubini's theorem for infinite sum of nonnegative random variables,

$$\mathbb{E}\left[\frac{1}{2L}\sum_{t=0}^{\infty} w_{t+1}\|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2\right] \leq \bar{g}_0(\boldsymbol{\theta}_0) - \inf \ell + O\left(\sum_{t=1}^{\infty} w_t^2\right) < \infty.$$

Therefore, the nonnegative RV in the LHS must be almost surely finite:

$$M := \sum_{t=0}^{\infty} w_{t+1}\|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2 < \infty \quad \text{almost surely.}$$

We can deduce all sorts of convergence rate bounds from the above. A typical treatment is to consider the best iterates: For all $t \geq 1$,

$$\min_{1\leq k\leq t} \|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2 \left(\sum_{k=1}^{t} w_k\right) \leq \sum_{k=0}^{T} w_{k+1}\|\nabla \bar{h}_k(\boldsymbol{\theta}_t)\|^2 \leq M.$$

Dividing both sides by the sum of weights give

$$\min_{0\leq k<t} \|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\|^2 = O\left(\frac{1}{\sum_{k=1}^{t} w_k}\right). \tag{99}$$

How do we interpret the above result? If $\boldsymbol{\theta}_t$ is in the interior of $\boldsymbol{\Theta}$, we have $\nabla \bar{g}_t(\boldsymbol{\theta}_t) = \mathbf{0}$ since $\boldsymbol{\theta}_t = \arg\min_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \bar{g}_t$. Hence

$$\|\nabla \bar{h}_t(\boldsymbol{\theta}_t)\| = \|\nabla \bar{g}_t(\boldsymbol{\theta}_t) - \nabla \bar{f}_t(\boldsymbol{\theta}_t)\| = \|\nabla \bar{f}_t(\boldsymbol{\theta}_t)\|.$$

So (99) yields iteration complexity in terms of the gradient norm of the empirical loss function

$$\min_{0\le k<t} \|\nabla \bar{f}_k(\boldsymbol{\theta}_k)\|^2 = O\left(\frac{1}{\sum_{k=1}^t w_k}\right).$$

In general, $\boldsymbol{\theta}_t$ may lie on the boundary of $\boldsymbol{\Theta}$ so the gradient $\nabla \bar{g}_t(\boldsymbol{\theta}_t)$ is not necessarily zero. In order to assess how close a given parameter to satisfying the first-order optimality condition, we look the variational optimality measure

$$\sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\, \|\boldsymbol{\theta}-\boldsymbol{\theta}_t\|\le 1} \langle -\nabla \bar{f}_t(\boldsymbol{\theta}_t), \boldsymbol{\theta} - \boldsymbol{\theta}_t\rangle$$

on the worst-case first-order increase in function value $\bar{f}_t$ when we move away from the current iterate $\boldsymbol{\theta}_t$. The above quantity is nonnegative since we can let $\boldsymbol{\theta} = \boldsymbol{\theta}_t$. If it equals zero, then $\boldsymbol{\theta}_t$ is a stationary point by definition. If $\boldsymbol{\theta}_t$ is deep inside the interior of $\boldsymbol{\Theta}$, then we can let $\boldsymbol{\theta} - \boldsymbol{\theta}_t = -\nabla \bar{f}(\boldsymbol{\theta}_t)/\|\nabla \bar{f}(\boldsymbol{\theta}_t)\|$ and the above equals $\|\bar{f}_t(\boldsymbol{\theta}_t)\|$. To relate it to $\|\bar{h}_t(\boldsymbol{\theta}_t)\|$ that we can control for the general case, use Cauchy-Schwarz inequality to get

$$\sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\, \|\boldsymbol{\theta}-\boldsymbol{\theta}_t\|\le 1} \langle -\nabla \bar{f}_t(\boldsymbol{\theta}_t), \boldsymbol{\theta} - \boldsymbol{\theta}_t\rangle \le \sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\, \|\boldsymbol{\theta}-\boldsymbol{\theta}_t\|\le 1} \underbrace{\langle -\nabla \bar{g}_t(\boldsymbol{\theta}_t), \boldsymbol{\theta} - \boldsymbol{\theta}_t\rangle}_{\le 0} + \|\nabla \bar{g}_t(\boldsymbol{\theta}_t) - \nabla \bar{f}_t(\boldsymbol{\theta}_t)\|,$$

where ≤ 0 comes from the first-order optimality condition of $\boldsymbol{\theta}_t$. This and (99) then yield the desired conclusion. □

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